



Ain Shams University

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zGate Gateway

A Zero Trust Database Access Proxy

GRADUATION PROJECT DOCUMENTATION

Computer Engineering

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Abstract

Traditional database access management relies on shared credentials and static permissions, creating significant security vulnerabilities including credential sprawl, lack of individual accountability, and insider threat exposure. Existing solutions such as VPNs, bastion hosts, and privileged access management systems fail to address the fundamental problem: users still handle production database credentials, which can be extracted, shared, or misused.

This project presents zGate, a Zero Trust Database Access Gateway that fundamentally transforms database security by eliminating direct credential exposure. zGate operates as an intelligent proxy layer between users and databases, intercepting traffic at the wire protocol level to enforce identity-based access control, generate session-specific credentials, and provide comprehensive audit logging with full user attribution.

The system comprises three integrated components: a high-performance Go-based gateway server implementing MySQL wire protocol support with transparent credential injection; a cross-platform command-line interface with secure keyring integration for macOS, Windows, and Linux; and a React-based web administration dashboard for managing users, roles, databases, and monitoring active sessions.

Key security features include JWT-based authentication with short-lived access tokens and automatic refresh, role-based access control with database-level permissions, a composable interceptor pipeline for query safety enforcement and dynamic data masking, and immutable audit trails capturing every database operation.

The Term 1 prototype demonstrates functional Zero Trust database access where users authenticate with organizational identities and connect to MySQL databases without ever seeing or handling production credentials. The architecture validates that Zero Trust principles—never trust, always verify, assume breach—can be practically implemented at the database protocol level while maintaining developer productivity, establishing a foundation for production deployment in Term 2.

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Chapter 1

Team Information

This part covers:

- Team members
- Contact information and LinkedIn profiles

THIS section introduces the dedicated team behind the zGate Gateway project, comprising four Computer Engineering students from Ain Shams University. Each member brings unique skills and perspectives to the project, working collaboratively under expert supervision to deliver a comprehensive Zero Trust database security solution. The team's combined expertise spans software architecture, security systems, full-stack development, and database technologies.

Team Members

Name	ID	LinkedIn
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Kareem Ehab	2100913	View Profile
Hana Shamel	2100468	View Profile
Karen Maurice	2100748	View Profile
Michael George	2100709	View Profile
Mayar Walid	2100953	View Profile
Rodina Mohammed	2100754	View Profile

Table 1.1: Team Members Information

Chapter 2

Introduction & Problem Definition

This part covers:

- Database access vulnerabilities and credential exposure
- Traditional security solution limitations
- Zero Trust principles for database security
- Identity-based access control implementation

IN modern enterprises, databases store critical assets yet traditional access management creates fundamental security vulnerabilities. This chapter examines how static credentials, shared accounts, and legacy authentication patterns expose organizations to breaches and compliance failures. We explore the security imperative driving Zero Trust database access control and establish the foundational problems that zGate addresses.



2.1 Introduction

In the modern data-driven enterprise landscape, databases serve as the foundation for critical business operations, storing sensitive customer information, financial records, intellectual property, and operational data. However, the traditional approaches to database access management have not evolved at the same pace as the sophistication of cyber threats and the complexity of organizational structures. Development teams, database administrators, data analysts, and various other technical personnel routinely require direct database access to perform their duties, creating significant security challenges that existing solutions fail to adequately address.

zGate is a Zero Trust database access gateway designed to fundamentally transform how organizations manage, secure, and audit database access. Built on the principles of Zero Trust security architecture, zGate operates as an intelligent intermediary layer between users and database systems, enforcing identity-based access control, implementing dynamic query-level authorization, and ensuring complete auditability of all database operations. The system comprises three integrated components: a high-performance gateway server that intercepts and controls all database traffic, a command-line interface for streamlined user interactions, and a comprehensive web-based administration dashboard for policy management and monitoring.

By eliminating the need for developers and analysts to possess or handle production database credentials directly, zGate addresses the critical security gap that has led to numerous data breaches and compliance failures across industries. The system enforces the principle of least privilege through role-based access control policies, generates transient session-specific credentials, and provides real-time query filtering and data masking capabilities to protect sensitive information even when legitimate users are accessing the database.

2.2 Problem Statement

2.2.1 The Current State of Database Access Management

Contemporary organizations face a fundamental security dilemma in database access management. On one hand, operational efficiency demands that developers, data engineers, DevOps teams, and analysts have timely access to databases for development, troubleshooting, analytics, and maintenance activities. On the other hand, granting such access using traditional methods introduces severe security vulnerabilities that expose organizations to data breaches, insider threats, and regulatory non-compliance.

2.2.2 Critical Security Vulnerabilities

The prevailing practices in database access management present several critical vulnerabilities:

Static Credential Proliferation: Organizations typically rely on shared or long-lived static credentials that are distributed among team members. These credentials often appear in configuration files, environment variables, documentation, and even code repositories, creating numerous attack vectors. Once compromised, these credentials provide unrestricted access until manually rotated—a process that is infrequent and operationally disruptive.

Lack of Accountability and Auditability: When multiple users share the same database credentials, individual accountability becomes impossible. Security teams cannot determine which specific user executed a particular query, making post-incident investigation extremely difficult and enabling malicious insiders to operate with impunity. Traditional database audit logs capture the database username but not the actual human identity behind the action.

Excessive Privilege and Unrestricted Access: Developers and technical personnel are often granted broader database permissions than necessary for their specific tasks. A developer needing read-only access to a single table might receive full database access simply because granular permission management is too complex or time-consuming to implement properly. This violates the principle of least privilege and dramatically expands the attack surface.

Inadequate Protection of Sensitive Data: Even legitimate users with proper authorization may inadvertently expose sensitive information such as personally identifiable information (PII), financial data, or health records. Current systems lack the capability to dynamically mask or redact sensitive fields based on user identity and context, forcing organizations to choose between operational efficiency and data protection.

Insider Threat Vulnerability: Trusted insiders with legitimate database access

represent one of the most significant security risks. Whether through malicious intent, negligence, or social engineering, insiders can exfiltrate sensitive data, manipulate records, or cause operational disruptions with minimal detection risk under current access paradigms.

2.2.3 Impact and Consequences

These vulnerabilities have tangible consequences:

- **Data Breaches:** Compromised credentials or malicious insiders lead to unauthorized data exfiltration
- **Regulatory Non-Compliance:** Failure to meet requirements of GDPR, HIPAA, PCI-DSS, and other frameworks
- **Financial Losses:** Direct costs from breaches, regulatory fines, and operational disruptions
- **Reputational Damage:** Loss of customer trust and competitive disadvantage
- **Operational Inefficiency:** Cumbersome manual processes for credential management and access provisioning

2.2.4 The Need for Zero Trust Database Access

The transition to Zero Trust architecture in network and application security has demonstrated the effectiveness of “never trust, always verify” principles. However, database access has remained largely unchanged, still operating under implicit trust models. There is an urgent need for a solution that:

- Eliminates direct credential exposure by ensuring users never handle production database credentials
- Enforces identity-based access control tied to organizational identity management systems
- Implements dynamic, session-specific authorization rather than static permissions
- Provides query-level policy enforcement to control what operations each user can perform
- Ensures complete auditability with full traceability of every database operation to individual users
- Supports data-level security through dynamic masking and filtering of sensitive information

2.3 Gaps In Existing Solutions

Organizations have historically relied on several security approaches to protect database access, each with significant limitations that fail to address the core vulnerabilities outlined previously.

2.3.1 Perimeter-Based Security

Traditional perimeter security operates on the assumption that threats exist outside the network boundary while everything inside is trustworthy. Firewalls, network segmentation, and IP whitelisting control which systems can reach database servers.

Limitations:

- **Lateral Movement:** Once an attacker breaches the perimeter (through phishing, compromised endpoints, or insider access), they can move freely within the network and access databases directly
- **No Identity Verification:** Perimeter controls verify network location, not user identity. Anyone on an authorized network or VPN can access databases
- **Coarse-Grained:** Controls apply at the network level, not at the query or data level. A user with network access has unrestricted database access
- **Cloud Incompatibility:** Modern cloud architectures and remote work arrangements render network perimeters increasingly porous and difficult to define

2.3.2 VPN-Based Access

Virtual Private Networks extend the corporate network to remote users, creating an encrypted tunnel that makes remote devices appear as if they're on the internal network.

Limitations:

- **All-or-Nothing Access:** VPN grants network-level access to all resources within its scope. A user connected via VPN can potentially access any database on that network segment
- **Shared Credentials Still Required:** VPN only solves the network connectivity problem; users still need database credentials, perpetuating the static credential problem
- **No Query-Level Control:** VPN cannot inspect, filter, or control database queries based on content or context

- **Session Persistence:** VPN sessions often remain active for extended periods, providing prolonged access windows for potential compromise
- **No Audit Trail:** VPN logs show connection events but provide no visibility into actual database operations performed

2.3.3 Bastion Hosts and Jump Servers

Organizations deploy intermediate servers that users must connect to before accessing databases, providing a centralized access point and audit logging.

Limitations:

- **Credential Exposure:** Users still retrieve and use actual database credentials, even if through a bastion host
- **Limited Policy Enforcement:** Bastion hosts log connections but typically cannot enforce query-level policies or filter sensitive data
- **Operational Overhead:** Requires maintaining additional infrastructure and managing access to the bastion itself
- **Session Recording Limitations:** While some bastion solutions record sessions, they provide after-the-fact forensics rather than real-time policy enforcement
- **Circumvention Risk:** Technical users can potentially bypass bastion hosts if they obtain credentials through other means

2.3.4 Database Native Access Controls

Modern databases include built-in authentication, authorization, and audit logging capabilities.

Limitations:

- **Complex Management:** Managing granular permissions across multiple databases and numerous users becomes administratively prohibitive at scale
- **Static Permissions:** Database roles and privileges are typically static and don't adapt to context (time, location, purpose)
- **Shared Account Pattern:** The complexity of per-user account management often leads organizations to share credentials anyway
- **Limited Masking Capabilities:** While some databases support row-level security and column masking, these features are database-specific, complex to configure, and inflexible

- **No Centralized Policy:** Each database system has its own permission model, preventing consistent policy enforcement across heterogeneous environments

2.3.5 Privileged Access Management (PAM) Solutions

PAM systems manage and audit privileged account credentials, often providing password vaulting, session recording, and credential rotation.

Limitations:

- **Still Credential-Based:** PAM distributes credentials to users, even if temporarily. Users still handle and potentially misuse actual database passwords
- **Session-Level, Not Query-Level:** PAM typically operates at the session level, recording entire sessions but not enforcing policies on individual queries
- **Limited Data Protection:** PAM cannot dynamically mask sensitive data fields based on user identity or query context
- **Operational Friction:** The check-out/check-in process for credentials adds significant overhead to developer workflows
- **PostgreSQL, MySQL, and MSSQL Limitations:** Many PAM solutions were designed for privileged OS access and offer limited database protocol support

2.3.6 Database Activity Monitoring (DAM)

DAM solutions monitor and alert on database activity by analyzing network traffic or database logs.

Limitations:

- **Reactive, Not Preventive:** DAM detects suspicious activity after it occurs rather than preventing unauthorized actions proactively
- **No Access Control:** DAM cannot prevent users from executing queries; it only observes and reports
- **Alert Fatigue:** Organizations receive numerous alerts but lack the ability to block malicious activity in real-time
- **Identity Blindness:** DAM sees database usernames but often cannot tie actions to actual human identities when credentials are shared

2.3.7 The Fundamental Gap

All existing approaches share a common fundamental flaw: they separate authentication/authorization from the actual data access point.

Users authenticate to VPNs, bastion hosts, or PAM systems, but ultimately receive raw database credentials and connect directly to databases. This creates an uncontrolled gap where policy enforcement, audit logging, and data protection cannot be reliably applied.

Additionally, none of these solutions adequately address the credential exposure problem. Whether stored in password vaults, configuration files, or manually entered, database credentials exist outside the security boundary and can be extracted, shared, or misused by authorized users.

2.4 Why Zero Trust for Databases is Different

Zero Trust database access represents a paradigm shift that fundamentally reimagines how database security should operate. Rather than attempting to secure the perimeter or audit after the fact, Zero Trust embeds security directly into the data access path itself.

2.4.1 Core Zero Trust Principles Applied to Databases

Never Trust, Always Verify: Every database access request is authenticated and authorized in real-time, regardless of network location or previous access history. There is no concept of “trusted internal network.”

Least Privilege Access: Users receive the minimum necessary permissions for their specific task at a specific moment. Access rights are dynamically evaluated based on identity, role, context, and policy.

Assume Breach: The architecture assumes that credentials may be compromised and that internal threats exist. Therefore, every query is inspected and controlled, and no user ever possesses credentials that could be misused outside the controlled gateway.

2.4.2 The Zero Trust Database Gateway Architecture

Unlike traditional solutions that operate adjacent to database access, a Zero Trust gateway like zGate becomes the exclusive access point for all database operations. This architectural position enables capabilities impossible with peripheral solutions.

Identity-Based Authentication: Users authenticate using their organizational identity (JWT tokens, SSO integration) rather than database credentials. Authentication is tied to the specific human or service, eliminating shared accounts and enabling true accountability.

Credential Abstraction: The gateway maintains actual database credentials internally. Users never see, handle, or transmit production database passwords. Even if a user’s authentication token is compromised, the attacker gains no direct database access—they must still pass through the gateway’s policy enforcement.

Protocol-Aware Interception: By implementing native database protocols (MySQL, PostgreSQL, MSSQL, etc.), the gateway can parse and inspect every query at the SQL level. This enables surgical policy enforcement that perimeter tools cannot achieve:

- Block specific SQL commands (DROP, DELETE) based on user role
- Restrict queries to specific tables or schemas
- Prevent unauthorized joins or subqueries

- Control result set size and query execution time

Session-Specific Access: Each connection through the gateway represents a distinct, auditable session tied to a specific user identity. Sessions are short-lived, context-aware, and can be terminated immediately if suspicious activity is detected.

Dynamic Policy Enforcement: Policies are evaluated in real-time for every query based on:

- User identity and assigned roles
- Target database and table
- Query type and structure
- Time of day, day of week
- Historical behavior patterns
- Data classification and sensitivity

2.4.3 Query-Level Data Protection

Zero Trust database access enables data protection at the query level, something impossible with network-based or credential-based solutions:

Dynamic Data Masking: Sensitive fields (credit card numbers, social security numbers, personal health information) are automatically masked or redacted based on the requesting user's clearance level. A developer sees masked data while an authorized analyst sees plaintext—from the same query.

Row-Level Filtering: The gateway can inject WHERE clauses or modify queries to restrict which rows a user can access, enforcing data boundaries without requiring database-native row-level security configuration.

Column-Level Restrictions: Certain columns can be completely hidden from specific roles, preventing even the awareness of sensitive data's existence.

2.4.4 Complete Auditability and Traceability

Zero Trust database access provides audit logging that captures not just that an action occurred, but the complete context:

- **Who:** Actual human or service identity, not just database username
- **What:** Exact SQL query executed, including results and data accessed
- **When:** Precise timestamp with session context

- **Where:** Source location, network details, client application
- **Why:** Request context, approval workflows if applicable
- **Outcome:** Success, failure, policy denials, data returned

This audit trail is immutable, centralized, and sufficient for forensic investigation, compliance reporting, and threat detection.

2.4.5 What zGate Implements

The zGate architecture embodies these Zero Trust principles through its three-component design:

Gateway Server: Implements protocol handlers for PostgreSQL, MySQL and MSSQL, intercepting all database traffic at the wire protocol level. The gateway's policy engine evaluates every query against configured rules, the dispatcher manages connection routing, and session managers track user activity in real-time. Users connect to zGate using standard database clients, but the gateway mediates all communication with backend databases.

Command-Line Interface: Provides developers and analysts with a streamlined authentication flow. Users authenticate with their organizational identity, receive time-limited JWT tokens, and establish database sessions without ever handling production credentials. The CLI manages token storage and renewal transparently.

Web Administration Dashboard: Enables security teams to define role-based access control policies, configure database connections, manage user permissions, and monitor active sessions and query logs. Administrators visualize access patterns, audit historical activity, and respond to security events through a comprehensive management interface.



Figure 2.1: what zGate offers

2.4.6 Operational Advantages

Beyond security improvements, Zero Trust database access delivers operational benefits:

- **Faster Onboarding:** New developers gain database access through role assignment in minutes, not days of credential provisioning
- **Reduced Credential Rotation Burden:** Database passwords change infrequently since users never access them



- **Simplified Compliance:** Centralized policy enforcement and comprehensive audit logs satisfy regulatory requirements
- **Cross-Database Consistency:** Single policy framework applies uniformly across MySQL, PostgreSQL, MSSQL, and other database types
- **Developer Experience:** Legitimate users experience minimal friction—authentication is transparent and access is granted immediately upon authorization

2.4.7 Addressing the Gaps

Zero Trust database access directly addresses every gap in existing solutions:

Limitation	Zero Trust Solution
Perimeter breach enables full access	Gateway enforces identity verification regardless of network position
VPN grants network-wide access	Access is scoped per-database, per-session, per-query
Bastion hosts still expose credentials	Users never possess or see database credentials
Static database permissions	Dynamic policy evaluation per query
PAM credentials can be misused	Tokens are gateway-specific and cannot directly access databases
DAM is reactive only	Real-time policy enforcement prevents unauthorized queries
Shared credentials prevent accountability	Every action is tied to individual user identity

Table 2.1: Comparison of Traditional Limitations vs. Zero Trust Solutions

Chapter 3

Project Definition

This part covers:

- Gateway server architecture and components
- Command-line interface for database access
- Web administration dashboard
- Project scope and deliverables

Z GATE is a Zero Trust database access gateway platform eliminating credential exposure through identity-based access control. This chapter defines the three-tier architecture comprising the gateway server for protocol-aware proxying, the CLI for secure user access, and the web dashboard for centralized administration. We detail each component's capabilities and establish clear project scope boundaries.

3.1 Project Definition and Scope

zGate is a comprehensive Zero Trust database access gateway platform designed to eliminate credential exposure, enforce identity-based access control, and provide complete auditability for database operations in enterprise environments. The project encompasses the design, implementation, and deployment of a three-tier architecture that intercepts, authenticates, authorizes, and audits all database access in real-time.

3.1.1 System Components

The platform consists of three integrated components working in concert:

Gateway Server (zGate Core)

- **Protocol Handlers:** Native implementation of MySQL, PostgreSQL, and MSSQL wire protocols, enabling transparent protocol-level interception and inspection of database traffic
- **Authentication System:** JWT-based authentication with configurable token expiration, eliminating the need for users to possess database credentials
- **Policy Engine:** Real-time policy evaluation engine that performs fresh permission lookups for every database operation, supporting role-based and custom permission models
- **Dynamic Proxy Manager:** On-demand creation and management of database proxies with automatic port allocation, eliminating static listener configurations
- **Session Management:** Per-user, per-database session tracking with context preservation and graceful cleanup mechanisms
- **API Server:** RESTful API exposing authentication, database listing, connection management, and session control endpoints

Command-Line Interface (zGate-CLI)

- **Secure Authentication Flow:** Interactive login process with secure credential input and token management
- **Token Storage & Management:** Cross-platform secure token storage using OS keyring with encrypted file fallback, automatic token refresh before expiration
- **Database Discovery:** List all databases accessible to the authenticated user based on their assigned roles

- **Connection Management:** Establish database connections through the gateway with automatic credential handling
- **Session Status:** Real-time visibility into active sessions, token expiration, and authentication state
- **Logout & Cleanup:** Proper session termination with server-side revocation and local credential cleanup

Web Administration Dashboard (zGate-WebUI)

Built with Next.js 16, React 19, and Radix UI, the dashboard provides comprehensive management capabilities:

- **Overview Dashboard:** Real-time system metrics, active sessions, and access patterns visualization
- **Database Management:** Configure database connections, credentials, and connection pooling parameters
- **User Management:** Create, modify, and deactivate user accounts with role assignments
- **Administrator Management:** Dedicated administrative user management with elevated privileges
- **Access Control:** Define and manage roles with granular permission specifications including database scoping and privilege levels (read-only, read-write, admin)
- **Active Sessions Monitoring:** View all active database sessions with user context, connection duration, and query activity
- **Query Execution:** Direct query interface with policy enforcement and audit logging
- **Shared Account Pools:** Manage reusable database account credentials for backend systems
- **Per-User Database Accounts:** Personal database account management for individual user needs
- **Activity Audit:** Comprehensive audit log viewer with filtering, search, and export capabilities
- **Connected Databases:** Real-time status of database connectivity and health

3.1.2 Scope Boundaries

In Scope:

- MySQL, Postgres and MSSQL protocol support with complete authentication and query interception
- Role-based access control with hierarchical permission models
- Temporary database user lifecycle management
- JWT-based authentication with automatic token refresh
- Dynamic proxy allocation and connection pooling
- Comprehensive audit logging of all database operations
- RESTful API for programmatic access
- Cross-platform CLI with secure credential storage
- Web-based administration interface
- Configuration-driven deployment (YAML-based)
- Session-level access control and monitoring

Out of Scope (Current Phase):

- Oracle, and other database protocol support (planned for future phases)
- Integration with external identity providers (SAML, LDAP, Active Directory)
- Multi-factor authentication (MFA) enforcement
- Database activity replay and forensic analysis tools
- High-availability and clustering capabilities
- Advanced query rewriting and optimization
- Automated compliance reporting generation

3.2 Objectives

The zGate project aims to achieve the following measurable technical and security objectives:

3.2.1 Core Security Objectives

Eliminate Direct Credential Exposure

- Implement JWT-based authentication system where users never handle production database passwords
- Develop automatic temporary database user creation with session-specific, cryptographically secure credentials
- Ensure temporary credentials are purged immediately upon session termination, preventing credential accumulation
- Achieve zero instances of production credentials appearing in user configuration files, environment variables, or CLI history

Enforce Identity-Based Access Control

- Build a role-based access control (RBAC) engine supporting hierarchical roles and custom permissions
- Implement real-time permission evaluation that reflects role changes immediately without requiring user re-authentication
- Support per-database, per-user authorization with multiple privilege levels (read-only, read-write, admin)
- Enable policy decisions based on fresh configuration lookups rather than stale token claims

Implement Protocol-Level Interception

- Develop native protocol handlers for PostgreSQL, MySQL and MSSQL wire protocols
- Enable transparent database proxy functionality allowing standard client tools (MySQL Workbench, SSMS) to function without modification
- Intercept and route all database traffic through the gateway with sub-100ms latency overhead for typical operations

- Maintain protocol compatibility ensuring 100% of standard database operations work through the proxy

Provide Comprehensive Audit Logging

- Log all authentication attempts with user identity, timestamp, success/failure status, and failure reasons
- Record complete session lifecycle: establishment, duration, database target, and termination circumstances
- Capture connection metadata sufficient for forensic investigation and compliance verification
- Generate structured logs compatible with standard log aggregation and SIEM tools

3.2.2 Operational Objectives

Deliver Multi-Component Integration

- Build RESTful API server exposing authentication, database listing, connection management, and session control
- Develop cross-platform CLI with secure token storage using OS keyring and automatic token refresh
- Create responsive web administration dashboard supporting all management functions
- Ensure seamless data flow between all three components (Gateway, CLI, WebUI)

Enable Centralized Administration

- Provide web-based interface for managing users, roles, database connections, and access policies
- Support real-time monitoring of active sessions with ability to view connection details and terminate sessions
- Allow administrators to modify permissions and configurations without system restarts
- Implement user and admin account separation with appropriate privilege boundaries

Support Multi-Database Environments

- Design modular architecture allowing support for MySQL, PostgreSQL, and MSSQL through unified interfaces
- Enable addition of new database protocols through implementation of standard handler and manager interfaces
- Maintain per-database configuration for credentials, connection parameters, and access policies
- Support simultaneous connections to multiple database instances of different types

Ensure Production-Ready Stability

- Implement graceful error handling, connection cleanup, and resource management
- Support concurrent sessions from multiple users without resource conflicts
- Provide configuration validation and startup checks to prevent misconfiguration
- Enable zero-downtime configuration reloads for non-breaking changes

3.2.3 Academic and Technical Learning Objectives

Demonstrate Applied Security Architecture

- Apply Zero Trust principles specifically to database access management
- Implement secure authentication flows with proper token lifecycle management
- Design authorization systems with clear separation between authentication, policy evaluation, and enforcement
- Practice secure coding including input validation, SQL injection prevention, and secure credential generation

Develop Full-Stack System Integration Skills

- Build high-performance network services in Go with concurrent connection handling
- Implement database wire protocol parsers and connection proxies
- Create modern React-based web interfaces with real-time data updates
- Develop CLI applications with cross-platform compatibility and user experience focus
- Design and document RESTful APIs for inter-component communication

Chapter 4

Requirements Engineering

This part covers:

- Functional requirements for system components
- Non-functional requirements and performance targets
- Permission system and access control
- System actors and use cases

REQUIREMENTS engineering forms the foundation of successful software development. This chapter systematically documents the functional requirements across authentication, proxying, and administration, alongside non-functional requirements for security, performance, and scalability. We define system actors, comprehensive use cases, and establish measurable quality attributes that guide architectural decisions.

4.1 Functional Requirements

4.1.1 FR-1: User Authentication & Authorization

- User login with bcrypt password hashing
- JWT token management (access & refresh tokens)
- Role-based access control (RBAC)
- Custom user permissions

4.1.2 FR-2: Database Connection Management

- Multi-database support (MSSQL, MySQL, PostgreSQL)
- Dynamic proxy creation with auto port allocation
- Temporary database users with auto-cleanup
- Secure credential generation

4.1.3 FR-3: Policy Engine

- Real-time access control validation
- Permission-based database filtering
- Multi-level permissions (read, write, admin)

4.1.4 FR-4: Command Line Interface

- Auto access token refresh (access & refresh tokens)
- Login methods: username/password (local) and OAuth2/OIDC (external IdP)
- Database listing and connection
- Session status monitoring

4.1.5 FR-5: Web User Interface

- Database configuration management
- User and role management

- Admin account management
- Active session monitoring
- Shared account pools
- Personal database accounts
- Comprehensive audit trail

4.1.6 FR-6: Protocol Handling

- MSSQL, MySQL and Postgres protocol support with temp user creation

4.1.7 FR-7: Audit & Logging

- Security event logging
- User context in all logs
- Structured logging with configurable levels

4.2 Non-Functional Requirements

4.2.1 NFR-1: Security

- Zero trust architecture principles
- Bcrypt password hashing
- Encrypted connections
- Token-based session security

Key Metrics:

- Access token expiry: 5 minutes
- Refresh token expiry: 1 day
- Session isolation with temporary users

4.2.2 NFR-3: Scalability Targets

- 100 concurrent sessions
- 1000 auth requests/minute
- Memory: < 512MB at normal load

4.2.3 NFR-4: Reliability

- 99.5% uptime during business hours
- Graceful error handling
- Data integrity guarantees
- Automatic resource cleanup

4.2.4 NFR-5: Usability

- Unix-style CLI conventions
- Responsive WebUI design
- Helpful error messages
- Comprehensive documentation

4.2.5 NFR-6: Maintainability

- Go best practices
- Configuration persisted in SQLite (not YAML)
- Structured logging
- Health check endpoints

4.2.6 NFR-7: Portability

- Cross-platform CLI (Linux, macOS, Windows)
- Modern browser support
- Docker deployment support

4.2.7 NFR-8: Compatibility

- MSSQL Server 2016+
- MySQL 5.7+
- API versioning
- Backward compatibility

4.3 Actors & Use Cases

4.3.1 System Actors

Actor descriptions

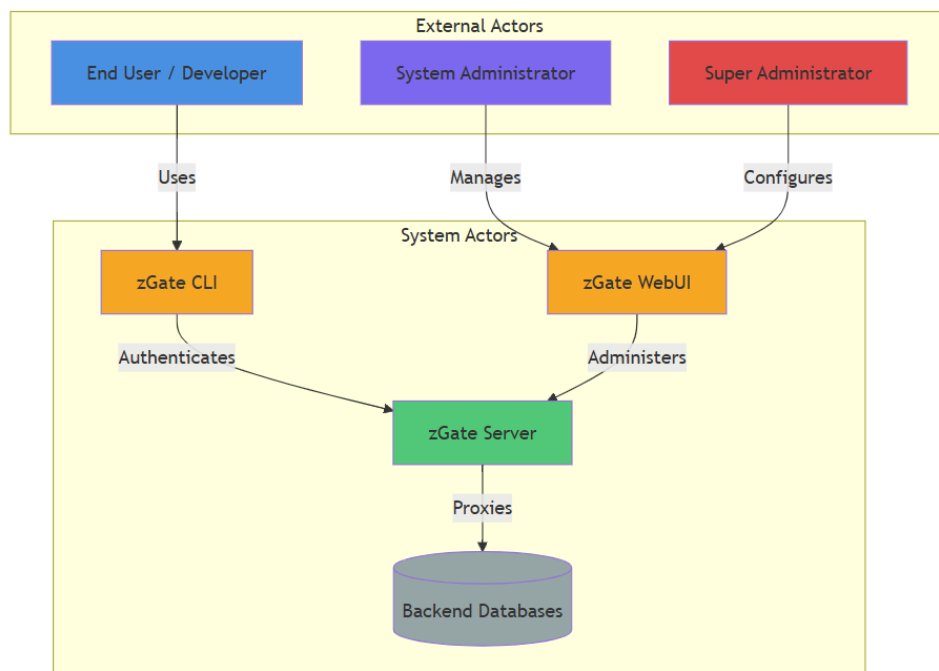


Figure 4.1: Overview of system actors: End Users, System Administrators, Super Administrators, and Backend Databases

End User / Developer

- **Description:** A regular user who needs to access databases for development, analytics, or operations
- **Responsibilities:**
 - Authenticate with the system using CLI
 - Request access to authorized databases
 - Execute database queries within granted permissions
- **Technical Level:** Intermediate (familiar with command line and database clients)

System Administrator

- **Description:** An admin user responsible for day-to-day management of the zGate platform

- **Responsibilities:**
 - Manage user accounts and role assignments
 - Configure database connections
 - Monitor active sessions
 - Review audit logs
 - Manage shared account pools
- **Technical Level:** Advanced (understands database administration and security)

Super Administrator

- **Description:** A privileged admin with full control over the system
- **Responsibilities:**
 - Create and manage admin accounts
 - Configure system-wide settings
 - Manage roles and permissions
 - Handle security incidents
- **Technical Level:** Expert (deep understanding of security and database systems)

zGate Server

- **Description:** The core backend system providing authentication, authorization, and proxy services
- **Responsibilities:**
 - Authenticate users and issue tokens
 - Enforce access policies
 - Create and manage temporary database users
 - Proxy database connections
 - Maintain audit logs

zGate CLI

- **Description:** Command-line client application for end users
- **Responsibilities:**
 - Manage token storage and refresh
 - Display available databases
 - Establish database connections

zGate WebUI

- **Description:** Web-based administration dashboard
- **Responsibilities:**
 - Provide graphical interface for system administration
 - Display real-time system status
 - Facilitate configuration management
 - Present audit trail visualization

Backend Databases

- **Description:** Target database systems (MSSQL, MySQL, PostgreSQL) that users need to access
- **Responsibilities:**
 - Store application data
 - Accept connections from zGate proxy
 - Enforce database-level permissions

4.3.2 Use Case Catalog

UC-1: User Authentication via CLI

Actor: End User

Preconditions: User has valid credentials

Main Flow:

1. User executes `zgate login` command
2. CLI prompts for username and password
3. CLI sends credentials to zGate Server
4. Server validates credentials against user configuration
5. Server generates access and refresh tokens
6. Server returns tokens to CLI
7. CLI stores tokens securely in OS keyring

Postconditions: User is authenticated and tokens are stored

Alternative Flows:

- 3a: Network connection fails → CLI displays error and retries
- 4a: Invalid credentials → Server returns 401, CLI displays error

UC-2: List Available Databases

Actor: End User

Preconditions: User is authenticated

Main Flow:

1. User executes `zgate list` command
2. CLI retrieves stored access token
3. CLI sends list request to zGate Server with token
4. Server validates token signature and expiry
5. Server retrieves user permissions from configuration
6. Server filters database list based on user permissions
7. Server returns list of accessible databases with permission levels
8. CLI displays formatted database list

Postconditions: User sees their accessible databases

Alternative Flows:

- 2a: Token expired → CLI auto-refreshes using refresh token
- 4a: Token invalid → CLI prompts for re-login

UC-3: Connect to Database

Actor: End User

Preconditions: User is authenticated and has permission to target database

Main Flow:

1. User executes `zgate connect <db_name>`
2. CLI sends connection request to zGate Server
3. Server validates user has permission for requested database
4. Server retrieves database configuration
5. Server allocates dynamic local port for proxy

6. Server retrieves protocol handler for database type
7. Server creates temporary database user with appropriate permissions
8. Server establishes connection to backend database
9. Server starts proxy listener on allocated port
10. Server returns connection details (host, port, temp credentials) to CLI
11. CLI displays connection information to user
12. User connects using database client with provided credentials

Postconditions: Secure proxy connection established, temporary user created

Alternative Flows:

- 3a: User lacks permission → Server returns 403, CLI displays error
- 4a: Database not found → Server returns 404
- 7a: Backend database unavailable → Server returns error, cleanup any partial state
- 7b: Temporary user creation fails → Server aborts and returns error

UC-4: Auto Disconnect on Session End

Actor: System (zGate Server)

Trigger: User terminates database connection or network fails

Main Flow:

1. Server detects connection closure
2. Server drops temporary database user from backend
3. Server stops proxy listener
4. Server releases allocated port
5. Server logs disconnect event with user context

Postconditions: All session resources cleaned up, temporary user deleted

UC-5: Admin - Add Database Configuration

Actor: System Administrator

Preconditions: Admin is logged into WebUI

Main Flow:

1. Admin navigates to Databases page
2. Admin clicks "Add Database" button
3. WebUI displays database creation form
4. Admin enters database details (name, type, backend address, admin credentials, permissions)
5. WebUI validates form inputs
6. WebUI sends create request to zGate Server API
7. Server validates admin token
8. Server tests connection to backend database
9. Server adds database configuration
10. Server persists configuration to databases.yaml
11. Server returns success response
12. WebUI displays updated database list

Postconditions: New database is available in configuration

Alternative Flows:

- 5a: Validation fails → Display errors on form
- 8a: Backend connection test fails → Return error, don't persist
- 9a: Duplicate database name → Return error

UC-6: Admin - Create User with Roles

Actor: System Administrator

Preconditions: Admin is logged into WebUI, roles exist

Main Flow:

1. Admin navigates to Users page
2. Admin clicks "Create User" button

3. WebUI displays user creation form
4. Admin enters username, password, and selects roles
5. WebUI validates form inputs
6. WebUI sends create user request to zGate Server API
7. Server validates admin token
8. Server hashes password using bcrypt
9. Server adds user configuration with assigned roles
10. Server persists configuration to users.yaml
11. Server returns success response
12. WebUI displays updated user list

Postconditions: New user can authenticate with assigned permissions

Alternative Flows:

- 5a: Validation fails → Display errors
- 9a: Username already exists → Return error

UC-7: Admin - Define Role with Permissions

Actor: System Administrator

Preconditions: Admin is logged into WebUI, databases exist

Main Flow:

1. Admin navigates to Access Control page
2. Admin clicks "Create Role" button
3. WebUI displays role creation form
4. Admin enters role name and description
5. Admin selects databases and permission levels for each
6. WebUI validates inputs
7. WebUI sends create role request to zGate Server API
8. Server validates admin token
9. Server adds role configuration

10. Server persists configuration to roles.yaml
11. Server returns success response
12. WebUI displays updated role list

Postconditions: New role is available for assignment to users

Alternative Flows:

- 6a: Invalid permission level → Display error
- 9a: Role name already exists → Return error

UC-8: Admin - Monitor Active Sessions

Actor: System Administrator

Preconditions: Admin is logged into WebUI

Main Flow:

1. Admin navigates to Sessions page
2. WebUI sends request to fetch active sessions
3. Server validates admin token
4. Server retrieves all active proxy sessions
5. Server returns session details (user, database, start time, connection info)
6. WebUI displays sessions in table format
7. Admin reviews active connections

Postconditions: Admin has visibility into current system usage

Alternative Flows:

- Admin terminates a session:
 1. Admin clicks "Terminate" on a session
 2. WebUI confirms action
 3. Server closes proxy connection
 4. Server deletes temporary database user
 5. WebUI updates session list

UC-9: Admin - Review Audit Logs

Actor: System Administrator

Preconditions: Admin is logged into WebUI

Main Flow:

1. Admin navigates to Activity Audit page
2. Admin optionally applies filters (user, date range, action type)
3. WebUI sends filtered log request to server
4. Server validates admin token
5. Server retrieves matching audit log entries
6. Server returns paginated log data
7. WebUI displays logs in chronological order with search/filter capabilities

Postconditions: Admin can track user activities and security events

UC-10: Auto Token Refresh

Actor: System (CLI)

Trigger: Access token nearing expiry

Main Flow:

1. CLI detects access token will expire within 1 minute
2. CLI retrieves refresh token from secure storage
3. CLI sends refresh request to zGate Server
4. Server validates refresh token
5. Server generates new access token
6. Server returns new access token
7. CLI updates stored access token
8. CLI proceeds with original request

Postconditions: User session continues seamlessly

Alternative Flows:

- 4a: Refresh token expired → CLI prompts user to login again

UC-11: Logout and Token Revocation

Actor: End User

Preconditions: User is authenticated

Main Flow:

1. User executes `zgate logout` command
2. CLI retrieves stored tokens
3. CLI sends logout request to zGate Server with refresh token
4. Server invalidates refresh token
5. Server returns success response
6. CLI deletes tokens from secure storage
7. CLI confirms logout to user

Postconditions: User session is terminated, tokens are invalid

UC-12: Configure Shared Account Pool

Actor: System Administrator

Preconditions: Admin is logged into WebUI, database exists

Main Flow:

1. Admin navigates to Shared Accounts page
2. Admin clicks "Add Shared Account"
3. WebUI displays configuration form
4. Admin selects database, permission level, enters credentials, sets max concurrent users
5. WebUI validates inputs
6. WebUI sends create request to server
7. Server validates admin token
8. Server tests credentials against database
9. Server creates shared account pool configuration
10. Server persists configuration

11. WebUI displays updated shared account list

Postconditions: Shared account pool is available for session allocation

Alternative Flows:

- 8a: Credential test fails → Return error, don't persist

4.4 Use Case Diagrams

4.4.1 End User Case Diagrams

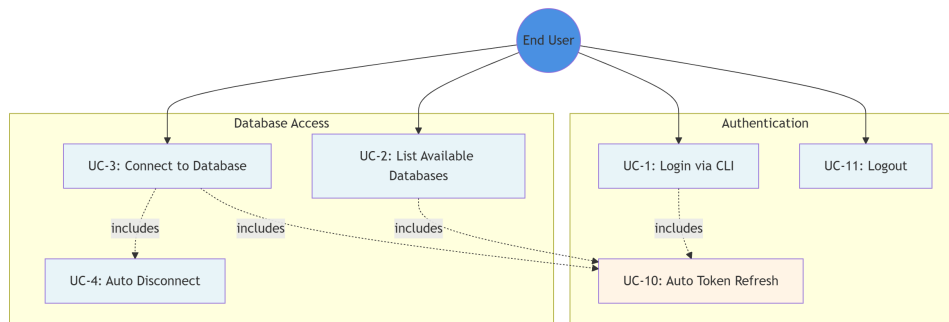


Figure 4.2: End user workflows: authentication, database listing, connection, session management, and logout

4.4.2 Administrator Use Case Diagrams

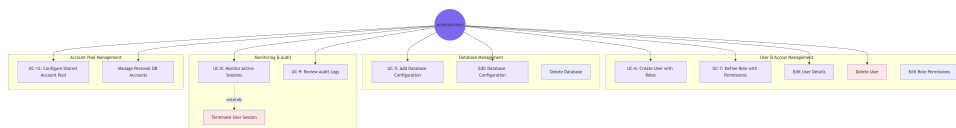


Figure 4.3: Administrator functions: database configuration, user and role management, session monitoring, and audit review

4.4.3 Complete System Use Case Diagrams

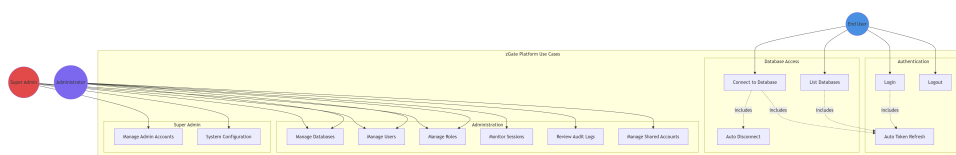


Figure 4.4: Complete system use case diagram showing all actor interactions within the zGate Gateway

4.4.4 System Components Interaction

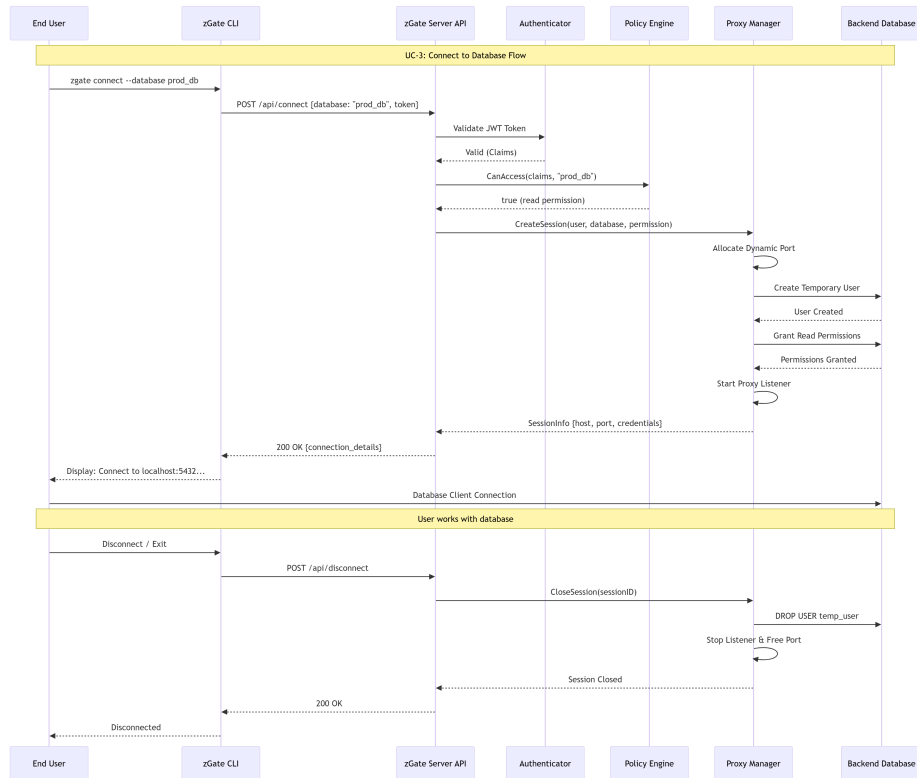


Figure 4.5: System component interactions showing data flow between CLI, Gateway Server, Web Dashboard, Policy Engine, and Backend Databases

4.5 User Stories

4.5.1 End User Stories

Epic 1: Secure Database Access

US-1.1: Login with Credentials

As an end user,

I want to

login using my username and password via the CLI

So that

I can securely authenticate and access authorized databases

Acceptance Criteria:

- CLI prompts for username and password
- Credentials are validated against the server
- Access and refresh tokens are generated on success
- Meaningful error message shown for invalid credentials
- Login completes within 500ms

US-1.2: View My Accessible Databases

As an authenticated user,

I want to

list all databases I have permission to access

So that

I know which databases I can connect to

Acceptance Criteria:

- Command `zgate list` shows my accessible databases
- Each database shows: name, type, my permission level, status, description
- Only databases I have permission for are displayed
- Offline databases are clearly marked
- List loads within 200ms

US-1.3: Connect to a Database

As an authenticated user,

I want to

connect to a specific database using the CLI

So that

I can access data within my permissions

Acceptance Criteria:

- Command `zgate connect <db_name>` initiates connection
- System validates my permission before allowing access
- Temporary credentials are auto-generated
- Connection details (host, port, username, password) are displayed
- I can use any database client with the provided credentials
- I receive clear error if I lack permission
- Connection establishes within 2 seconds

US-1.4: Automatic Session Cleanup

As an end user,

I want

my temporary database access to be automatically cleaned up when I disconnect

So that

I don't have to worry about security hygiene

Acceptance Criteria:

- Temporary database user is deleted when I close my connection
- Proxy resources are released automatically
- Cleanup happens within 30 seconds of disconnect
- No manual intervention required

US-1.5: Seamless Token Refresh

As an authenticated user,

I want

my session to continue without interruption

So that

I don't have to re-login frequently

Acceptance Criteria:

- Access tokens are automatically refreshed before expiry
- I don't experience any interruption during refresh
- I'm only prompted to re-login if my refresh token expires
- Refresh happens transparently in the background

US-1.6: Check Session Status

As an authenticated user,

I want to

check my current session status

So that

I know if I'm logged in and when my session expires

Acceptance Criteria:

- Command `zgate status` shows my current session
- Display includes: username, token expiry, server connection status
- Clear indication if I'm not logged in

US-1.7: Secure Logout

As an authenticated user,

I want to

logout and revoke my session

So that

my credentials are invalidated when I'm done

Acceptance Criteria:

- Command `zgate logout` terminates my session

- Tokens are revoked on the server
- Local token storage is cleared
- Confirmation message is displayed

4.5.2 Administrator Stories

Epic 2: User & Access Management

US-2.1: Create User Accounts

As a system administrator,

I want to

create new user accounts with assigned roles

So that

team members can access databases based on their responsibilities

Acceptance Criteria:

- I can access user creation form in WebUI
- I can enter username and initial password
- I can assign one or more roles to the user
- Password is securely hashed before storage
- User can login immediately after creation
- Duplicate usernames are prevented with clear error

US-2.2: Define Roles with Granular Permissions

As a system administrator,

I want to

define roles with specific database permissions

So that

I can implement least-privilege access control

Acceptance Criteria:

- I can create named roles (e.g., "data_analyst", "developer")
- For each role, I can specify permissions per database
- Permission levels include: read, write, admin

- Roles can be assigned to multiple users
- Changing role permissions affects all assigned users immediately

US-2.3: Manage User-Role Assignments

As a system administrator,

I want to

add or remove roles from existing users

So that

I can adjust access as team responsibilities change

Acceptance Criteria:

- I can view current role assignments for any user
- I can add new roles to a user
- I can remove roles from a user
- Changes take effect immediately for new connections
- Audit log captures all role changes

Epic 3: Database Configuration

US-3.1: Add Database Configurations

As a system administrator,

I want to

add new database connections to the system

So that

users can access additional data sources

Acceptance Criteria:

- I can provide: name, type (MSSQL/MySQL), backend address, admin credentials
- I can specify available permission levels for the database
- Connection is tested before saving
- Configuration is persisted to databases.yaml
- Database appears in user lists immediately
- Clear error shown if connection test fails

US-3.2: Edit Database Settings

As a system administrator,

I want to

update existing database configurations

So that

I can reflect infrastructure changes

Acceptance Criteria:

- I can modify connection details (address, credentials)
- I can update available permissions
- Changes are validated before saving
- Active sessions using old config are not disrupted
- New connections use updated configuration

US-3.3: Remove Decommissioned Databases

As a system administrator,

I want to

delete database configurations that are no longer needed

So that

users don't see outdated options

Acceptance Criteria:

- I can delete a database configuration
- System confirms deletion to prevent accidents
- Database is immediately removed from user lists
- Deletion is logged in audit trail

Epic 4: Monitoring & Audit

US-4.1: Monitor Active Sessions

As a system administrator,

I want to

see all active database sessions in real-time

So that

I can monitor system usage and detect anomalies

Acceptance Criteria:

- I can view list of all active connections
- Each session shows: user, database, connection time, status
- List updates in real-time or on refresh
- I can see connection details (proxy port, temp username)

US-4.2: Terminate Suspicious Sessions

As a system administrator,

I want to

forcefully terminate active sessions

So that

I can respond to security incidents

Acceptance Criteria:

- I can select any active session and terminate it
- System confirms action before executing
- Session proxy is closed immediately
- Temporary database user is deleted
- Action is logged in audit trail
- User receives connection closed error

US-4.3: Review Comprehensive Audit Logs

As a system administrator,

I want to

review all user activities and system events

So that

I can investigate issues and maintain compliance

Acceptance Criteria:

- I can view chronological audit log in WebUI

- Each entry includes: timestamp, user, action, outcome, context
- I can filter by: user, date range, action type, success/failure
- I can search logs by keywords
- Logs include: logins, connection requests, admin actions, errors
- Logs are paginated for performance

US-4.4: Export Audit Logs

As a system administrator,

I want to

export audit logs for compliance reporting

So that

I can provide evidence of access controls

Acceptance Criteria:

- I can export filtered logs to CSV or JSON
- Export includes all relevant fields
- Large exports don't timeout or crash

Epic 5: Advanced Access Control

US-5.1: Configure Shared Account Pools

As a system administrator,

I want to

create pools of shared database credentials

So that

multiple users can share backend accounts efficiently

Acceptance Criteria:

- I can create shared account pool for a database
- I can specify: database, permission level, credentials, max concurrent users
- System allocates shared account when user connects
- System prevents exceeding max concurrent limit
- User transparently gets shared credentials
- Account is returned to pool on disconnect

US-5.2: Assign Personal Database Accounts

As a system administrator,

I want to

assign personal database accounts to specific users

So that

some users have dedicated credentials for auditing

Acceptance Criteria:

- I can create personal account for user-database pair
- I can provide specific database credentials
- User always gets their personal account when connecting
- Personal accounts override shared pools
- I can revoke personal accounts

US-5.3: Define Custom User Permissions

As a system administrator,

I want to

grant custom permissions to individual users beyond their roles

So that

I can handle special cases without creating new roles

Acceptance Criteria:

- I can add database permissions directly to a user
- Custom permissions combine with role-based permissions
- I can see both role and custom permissions in user view
- Custom permissions can be removed independently

4.5.3 Super Administrator Stories

Epic 6: Platform Administration

US-6.1: Manage Admin Accounts

As a super administrator,

I want to

create and manage admin user accounts

So that

I can grant administrative access to trusted personnel

Acceptance Criteria:

- I can create new admin accounts with secure passwords
- I can view all existing admin accounts
- I can disable or delete admin accounts
- Regular users cannot access admin functions
- Admin account changes are logged

US-6.2: Configure System Settings

As a super administrator,

I want to

configure system-wide settings

So that

I can tune the platform for our environment

Acceptance Criteria:

- I can set token expiry durations (access and refresh)
- I can configure backend connection timeouts
- I can set log levels (DEBUG, INFO, WARN, ERROR)
- Settings are validated before applying
- Configuration changes are persisted

US-6.3: Perform System Health Checks

As a super administrator,

I want to

check the health of all system components

So that

I can proactively identify issues

Acceptance Criteria:

- I can view status of: API server, proxy manager, backend databases

- Each database shows: online/offline status, last check time
- I can manually trigger connectivity tests
- Failed health checks generate alerts

4.5.4 System Stories (Non-Interactive)

Epic 7: Automated Operations

US-7.1: Automatic Resource Cleanup

As the zGate system,

I need to

automatically clean up temporary resources

So that

database systems don't accumulate orphaned users

Acceptance Criteria:

- Temporary database users are deleted within 30s of disconnect
- Proxy listeners are stopped immediately on session close
- Allocated ports are freed for reuse
- Cleanup happens even if client disconnects ungracefully
- Failed cleanup attempts are retried with exponential backoff
- Persistent cleanup failures are logged as errors

US-7.2: Real-Time Permission Enforcement

As the zGate system,

I need to

evaluate permissions at request time using current configuration

So that

permission changes take effect immediately

Acceptance Criteria:

- Each connection request checks latest roles and permissions
- Configuration file changes are detected and reloaded
- Users with revoked permissions cannot establish new connections

- Permission checks complete within 50ms
- Permission evaluation is thread-safe

US-7.3: Graceful Error Handling

As the zGate system,

I need to

handle errors gracefully without leaking sensitive information

So that

users get helpful feedback while maintaining security

Acceptance Criteria:

- User-facing errors don't expose internal paths or stack traces
- Authentication failures return generic "invalid credentials" message
- Network errors suggest retry with backoff
- All errors are logged with full context
- Critical errors trigger alerts to administrators

US-7.4: Comprehensive Audit Logging

As the zGate system,

I need to

log all security-relevant events

So that

administrators can audit access and investigate incidents

Acceptance Criteria:

- All authentication attempts are logged (success and failure)
- All database access requests are logged with user context
- All administrative actions are logged
- Logs include: timestamp, user, action, outcome, IP address, session ID
- Logs are structured (JSON format option)
- Log rotation prevents disk exhaustion
- Sensitive data (passwords, credentials) are never logged

Chapter 5

Proposed Solution

This part covers:

- Zero Trust architecture and credential abstraction
- Dynamic proxy management
- JWT-based authentication and token life-cycle
- Policy enforcement and audit logging
- Cross-platform CLI with secure storage

THE zGate solution addresses database security vulnerabilities through Zero Trust architecture. This chapter presents our comprehensive approach to eliminating credential exposure through dynamic proxy management, JWT-based authentication, and real-time policy enforcement. We explore how the gateway server, CLI, and web dashboard integrate to provide identity-based access control while maintaining operational efficiency.

5.1 Solution Overview

The proposed solution is **zGate**, a comprehensive Zero-Trust Database Access Platform that fundamentally changes how organizations manage database security and access control. Rather than relying on shared credentials and static permissions, zGate introduces a dynamic, policy-driven approach where temporary credentials are created for each user session, eliminating the risks associated with credential sprawl and unauthorized access.

5.1.1 The Core Problem

Traditional database access models suffer from several critical security and operational challenges:

1. **Shared Credentials:** Multiple users share common database accounts, making it impossible to audit who performed which actions
2. **Static Permissions:** Once granted, permissions remain indefinitely, increasing the attack surface
3. **Credential Sprawl:** Database passwords stored in configuration files, wikis, and password managers
4. **Lack of Visibility:** No centralized view of who has access to which databases
5. **Manual Overhead:** Granting and revoking access requires manual database administration
6. **Compliance Gaps:** Difficult to prove least-privilege access and maintain audit trails

5.1.2 The zGate Solution

zGate addresses these challenges through a multi-layered architecture built on three core principles:

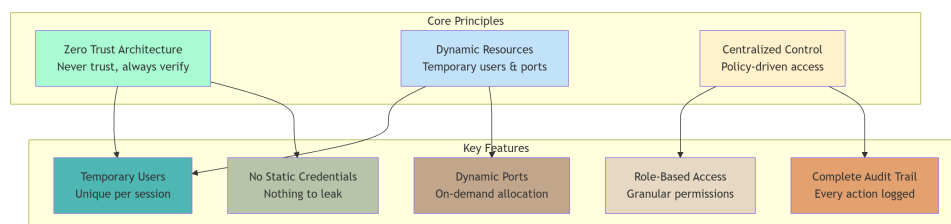


Figure 5.1: zgate proposed solution architecture

5.2 Core Components

5.2.1 zGate Server (Backend)

The server is the heart of the platform, handling all security, policy enforcement, and proxy operations.

Backend Layers

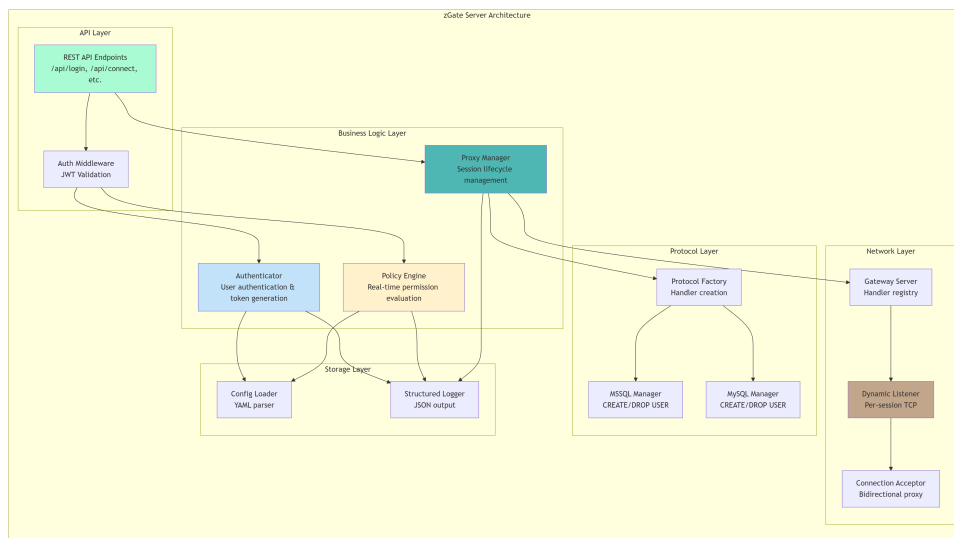


Figure 5.2: zGate backend architecture showing authentication, policy engine, and proxy layers

Key Features

1. JWT-Based Authentication

- Access tokens (15-minute TTL) for API requests
- Refresh tokens (7-day TTL) for session persistence
- Secure token generation with HMAC-SHA256 signatures
- Automatic token refresh in CLI

2. Real-Time Policy Enforcement

- Permissions evaluated from current configuration on every request
- Role-based access control (RBAC) with support for multiple roles per user
- Custom permissions at user level for exceptions

- Immediate effect when permissions are changed (no cache invalidation needed)

3. Dynamic Proxy Management

- On-demand creation of proxy sessions
- Automatic port allocation from ephemeral range
- Temporary database user creation with appropriate grants
- Automatic cleanup on disconnect or crash recovery

4. Protocol Abstraction

- Pluggable architecture for database types
- Current support: MSSQL, MySQL
- Easy extensibility for PostgreSQL, Oracle, etc.
- Database-specific SQL for user management

5. Comprehensive Logging

- Structured logging with key-value pairs
- Every security event logged with user context
- Support for JSON output for log aggregation
- Configurable log levels (DEBUG, INFO, WARN, ERROR)

5.2.2 zGate CLI (Client)

The command-line interface provides developers with a simple, secure way to access databases.

CLI Workflow

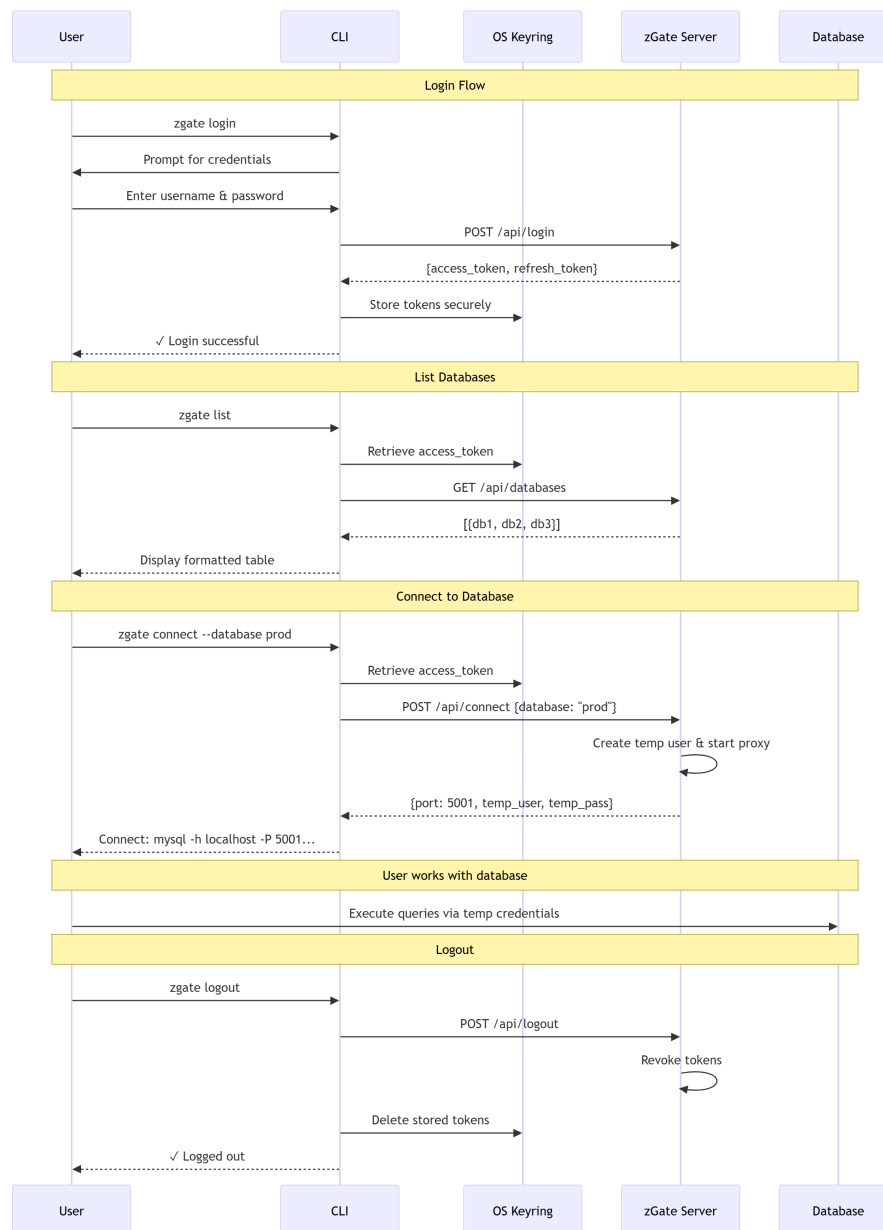


Figure 5.3: zGate CLI workflow illustrating token management and database access process

CLI Features

1. Secure Token Storage

- OS-native keyring integration (Keychain on macOS, Credential Manager on Windows, Secret Service on Linux)
- Encrypted file fallback for headless environments
- Never stores plaintext credentials

2. Automatic Token Refresh

- Detects token expiry proactively
- Refreshes access token using refresh token
- Seamless user experience without re-authentication

3. User-Friendly Interface

- Simple, intuitive commands following Unix conventions
- Colored output for better readability
- Helpful error messages with troubleshooting hints
- Progress indicators for long operations

4. Cross-Platform Support

- Single binary for Windows, macOS, Linux
- No runtime dependencies
- Portable and lightweight (~10MB)

5.2.3 zGate WebUI (Admin Dashboard)

The web interface provides administrators with a comprehensive control panel for managing the entire platform.

Dashboard Architecture

Admin Features

1. Database Management

- Add/edit/delete database configurations
- Test database connectivity before saving
- View all configured databases with status
- Manage available permissions per database

2. User & Role Management

- Create users with bcrypt-hashed passwords
- Assign multiple roles to users

- Define roles with database-specific permissions
- Add custom permissions to individual users

3. Session Monitoring

- Real-time view of active database connections
- See: user, database, port, temporary username, duration
- Terminate sessions if needed (security incidents)
- Filter and search sessions

4. Audit Trail

- Comprehensive logs of all user activities
- Filter by user, date range, action type
- Search within log entries
- Export for compliance reporting

5. Advanced Features

- Shared account pools for efficient credential management
- Personal database accounts for specific users
- Query execution interface (future)
- Connected databases view



5.3 Key Technologies & Design Decisions

Technology Stack

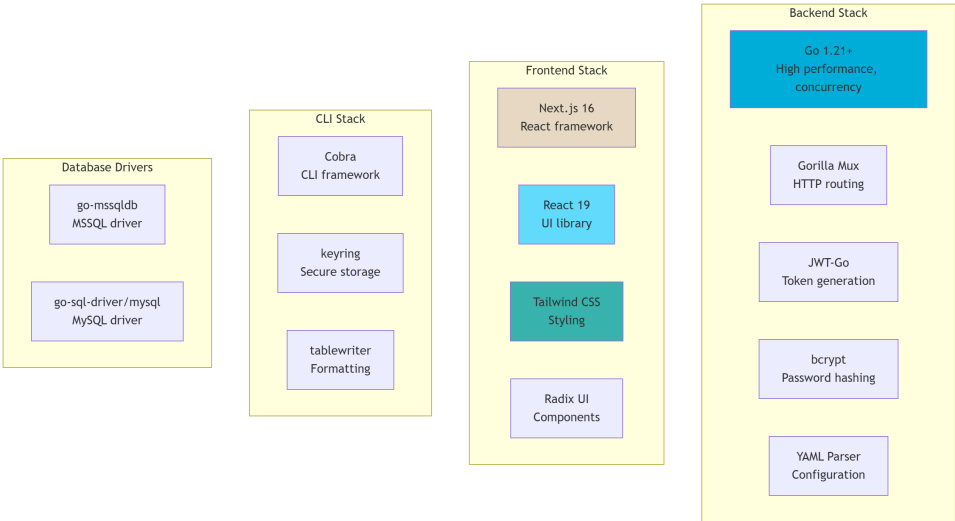


Figure 5.4: zGate technology stack

5.4 Security Architecture

Security is woven into every layer of the zGate platform through multiple defense mechanisms.

5.4.1 Security Layers

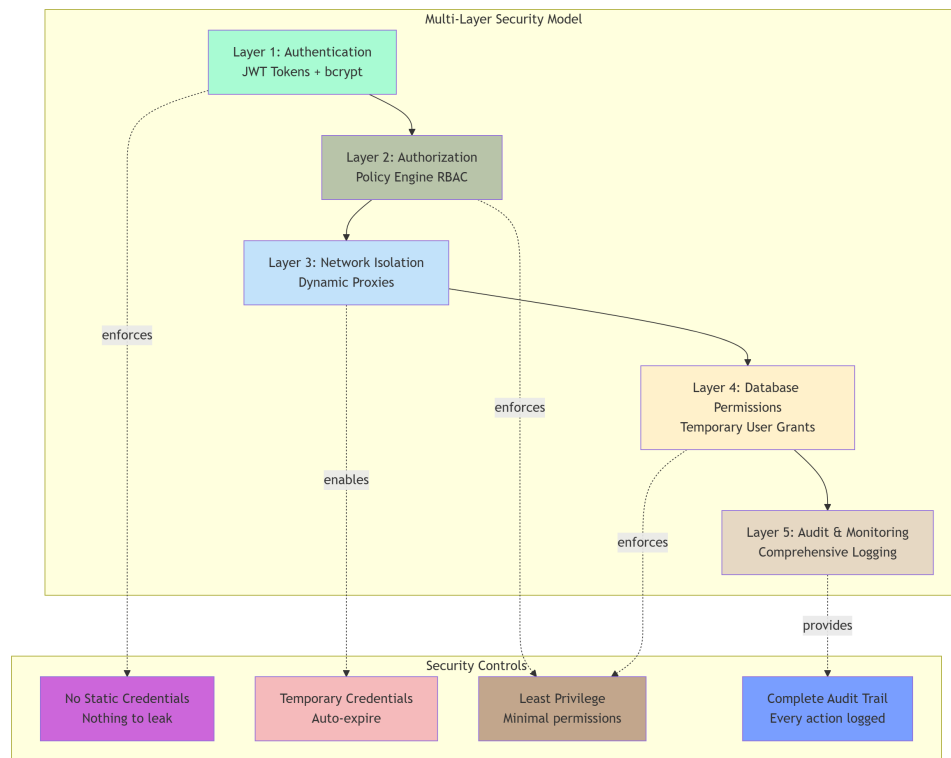


Figure 5.5: Multi-layered security architecture in zGate platform

5.4.2 Security Guarantees

Threat	Mitigation	Implementation
Credential Theft	No static credentials exposed to users	Temporary users with auto-generated passwords
Privilege Escalation	Database-level permission enforcement	Grants limited to user's role (read/write/admin)
Unauthorized Access	Real-time policy evaluation	Every connection request checked against current config
Session Hijacking	Short-lived JWT tokens	15-minute access token expiry
Password Cracking	Strong password hashing	bcrypt with work factor 10
Audit Evasion	Mandatory logging	Every API call, connection, disconnect logged
Resource Exhaustion	Automatic cleanup	Temporary users deleted within 30 seconds of disconnect
Man-in-the-Middle	Encrypted connections	TLS support for backend database connections

Table 5.1: Security threats and their mitigations in zGate



5.5 Advantages of the Proposed Solution

Key Benefits

Enhanced Security

- Eliminates shared credentials
- Temporary users auto-deleted
- No credential sprawl
- Cryptographically secure random passwords
- Short-lived JWT tokens
- Database-level permission enforcement

Operational Efficiency

- Automated user provisioning
- Self-service for developers (via CLI)
- Centralized access management
- No manual database administration
- Quick onboarding/offboarding

Compliance & Audit

- Complete audit trail
- User-attributed actions
- Provable least-privilege
- Real-time access reporting
- Exportable logs

Developer Experience

- Simple CLI commands
- Works with existing tools
- No VPN/bastion required
- Automatic token refresh
- Cross-platform support



Scalability

- Handles thousands of concurrent sessions
- Stateless API (horizontal scaling)
- Lightweight memory footprint
- Dynamic resource allocation

Chapter 6

Alignment with International Standards

This part covers:

- PCI DSS compliance requirements
- HIPAA security alignment
- GDPR data protection principles
- ISO 27001 security controls

COMPLIANCE with international security standards is a fundamental design principle for zGate. This chapter demonstrates how the system aligns with PCI DSS access control requirements, HIPAA security rules, GDPR data protection principles, and ISO 27001 information security controls. We establish that zGate's architecture inherently satisfies regulatory obligations while maintaining robust security postures.

6.1 PCI DSS

- **Access Control (Req. 7):** Role-based access with least privilege per database
- **Authentication (Req. 8):** Unique user identification, no shared credentials, time-limited sessions
- **Audit Logging (Req. 10):** Complete tracking of all database access with user identity, timestamp, and session details

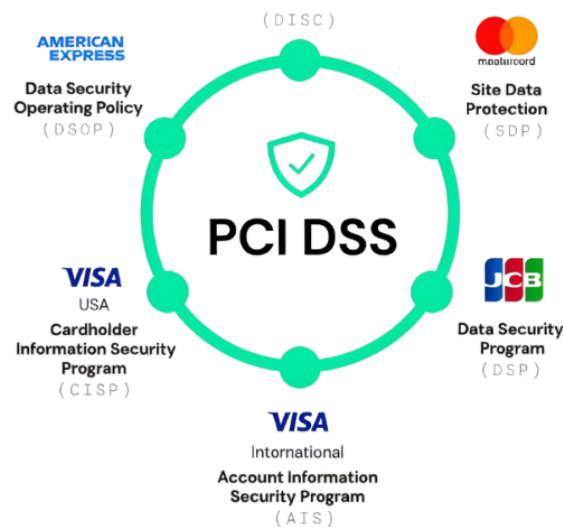


Figure 6.1: zGate alignment with PCI DSS requirements for access control, authentication, and audit logging



6.2 HIPAA

- **Access Control:** Unique user authentication and role-based database access to ePHI
- **Audit Controls:** Comprehensive logs of who accessed what database and when
- **Authentication:** JWT-based identity verification before database access
- **Automatic Log-off:** Session expiration and token timeout mechanisms

6.3 GDPR

- **Data Security (Art. 32):** Identity-based authentication, authorization, and encrypted credentials
- **Access Control (Art. 5):** Role-based permissions limiting data access scope
- **Audit Records (Art. 30):** Complete logging of processing activities with user context
- **Breach Detection (Art. 33):** Real-time monitoring and immediate session termination capabilities



Figure 6.2: zGate alignment with GDPR requirements for data security, access control, audit records, and breach detection

6.4 ISO 27001

- **User Access Management (A.9.2):** Formal registration, role assignment, and access revocation
- **Access Control (A.9.4):** Secure authentication, temporary credentials, session timeouts
- **Logging & Monitoring (A.12.4):** Event logging for all activities with protected audit trails
- **Compliance Reviews (A.18):** Audit logs and monitoring dashboard for compliance verification

Chapter 7

Competitor & Market Analysis

This part covers:

- Existing solution classification and comparison
- Competitive analysis of major platforms
- Market gap identification
- Zero Trust market growth and trends
- Industry breach costs and compliance drivers

THE database security market is experiencing rapid evolution with breach costs reaching \$4.88 million in 2024. This chapter analyzes existing solutions including Teleport, HashiCorp Boundary, and StrongDM, evaluating their technical strengths and limitations. We identify the specific market gap for lightweight, protocol-aware, open-source database governance and validate zGate's commercial viability within the expanding Zero Trust security market.

7.1 Introduction

The landscape of database access control is evolving rapidly in response to an increasingly hostile threat environment. With the global average cost of a data breach reaching a record high of **\$4.88 million** in 2024 [23], organizations are abandoning traditional perimeter-based security models in favor of Zero Trust architectures. The healthcare sector, in particular, faces unique challenges, with the average cost of a breach soaring to **\$7.42 million** in 2025, marking the 14th consecutive year it has been the costliest industry for data breaches.

This chapter categorizes the existing solutions in the market, analyzes their technical strengths and limitations compared to zGate, and identifies the specific research gap this project addresses. Furthermore, it presents a detailed market analysis underpinned by 2024-2025 industry statistics to validate the commercial viability and necessity of the proposed solution.

7.2 Classification of Existing Solutions

To understand where zGate fits, we must classify existing tools into three distinct categories:

1. **General Infrastructure Access Platforms:** Comprehensive suites designed to secure access to servers (SSH), Kubernetes, and databases simultaneously. (e.g., Teleport, HashiCorp Boundary).
2. **Identity-Aware Proxies (IAP):** Tools that focus on authentication and tunnel management but often lack deep protocol-level awareness. (e.g., Google IAP, Cloudflare Access).
3. **Developer-Focused Gateways:** Approaches designed for "ChatOps" or developer convenience rather than strict compliance. (e.g., Hoop.dev).
4. **Traditional Connection Poolers:** Performance-focused tools that aggregate connections but provide minimal security features. (e.g., PgBouncer, ProxySQL).

7.3 Detailed Competitor Analysis

7.3.1 Teleport

Teleport is arguably the market leader in the open-source infrastructure access space. It replaces SSH keys and database passwords with short-lived X.509 certificates.

Technical Comparison: While Teleport is a robust "heavyweight" solution, its broad scope is also its primary limitation for specific database workflows.

- **Protocol Awareness:** Teleport supports database protocol parsing to some extent but focuses primarily on *audit logging* rather than *active content filtering*. It records "who ran what," but its ability to block specific SQL commands (e.g., "DROP TABLE") based on granular policies is less developed compared to its SSH capabilities.
- **Complexity:** Deploying Teleport requires a significant infrastructure investment (Auth Service, Proxy Service, multiple agents), often serving as a replacement for the entire VPN layer. zGate, in contrast, is designed as a lightweight, focused middleware for database teams.

7.3.2 HashiCorp Boundary

Boundary is a modern identity-aware proxy that facilitates access to private hosts based on user identity.

Technical Comparison: Boundary operates primarily at the TCP layer. It authenticates a user and then creates a TCP tunnel to the target service.

- **Lack of Data Awareness:** Boundary treats the database connection as an opaque byte stream. It cannot inspect the SQL query inside the tunnel. Therefore, it cannot enforce policies like "Mask the credit_card column" or "Block DELETE queries."
- **Role:** Boundary is an excellent *connective* tool but not a *data governance* tool. zGate fills this layer by understanding the wire protocol itself.

7.3.3 StrongDM

StrongDM is a popular commercial solution that acts as a proxy for all infrastructure types.

Technical Comparison:

- **Closed Source:** StrongDM is a proprietary SaaS product. This makes it unsuitable for organizations requiring full code sovereignty or on-premise air-gapped deployments without external vendor dependencies.

- **Cost:** It targets the enterprise market with a pricing model that can be prohibitive for smaller teams or educational use cases.

7.3.4 Hoop.dev

Hoop.dev is a "Command Runner" and access gateway focused on developer experience.

Technical Comparison: Hoop is the closest functional equivalent to zGate in terms of acting as an intermediary.

- **Focus:** Hoop emphasizes "ChatOps" and allowing developers to run snippets of code/SQL via web or Slack interfaces.
- **Workflow:** zGate differentiates itself by being a transparent wire-protocol proxy that works seamlessly with *existing* native database tools (like DBeaver, Tableau, or 'psql') without requiring users to switch to a web terminal or chat interface.

7.4 Comparative Feature Matrix

Table 7.1 provides a rigorous comparison of zGate against these competitors. Note that zGate now supports **PostgreSQL** in addition to MySQL and MSSQL, broadening its applicability.

Feature	zGate (Ours)	Teleport	Boundary	StrongDM	Hoop.dev	PgBouncer
License	Open Source	Open Core	Open Source	Proprietary	Open Source	Open Source
Wire Protocol	Deep (AST)	Partial	None (TCP)	Yes	Yes	Minimal
Supported DBs	MySQL/MSSQL/PG	Many	Any (TCP)	Many	Many	Postgres
Data Masking	Yes (Dynamic)	No	No	No	Yes	No
Query Blocking	Yes (Policy)	No	No	No	Yes	No
Zero Trust	Yes	Yes	Yes	Yes	Yes	No

Table 7.1: Comparative Feature Matrix of Database Access Solutions



7.5 Research Gap Analysis

Based on the analysis above, a clear gap emerges in the current ecosystem:

*There is no lightweight, open-source solution dedicated to **granular database governance** that specifically targets the native wire protocols of MySQL, MSSQL, and PostgreSQL for deep inspection (masking/filtering) while retaining compatibility with standard desktop clients.*

7.6 Market Need and Security Challenges

7.6.1 Critical Vulnerabilities in Healthcare and Legacy Systems

The need for advanced database security is most acute in critical sectors. In 2024, the healthcare sector accounted for **23% of all data breaches**, with over 133 million records exposed. Legacy systems remain a primary vulnerability; as of late 2024, **73% of health-care providers** still rely on legacy information systems.

- **Legacy Vulnerabilities:** Older systems often lack modern security features like encryption and MFA, making them easy targets. "Easy Back-Door Entry" via unsupported systems contributes to over **50% of network server breaches**.
- **Cost of Inaction:** The financial impact is staggering, with the average healthcare data breach cost reaching **\$7.42 million** in 2025.
- **Human Factor:** Human error and misdelivery of sensitive data remain top causes of incidents. zGate's ability to enforce data masking directly at the protocol level provides a fail-safe against such inadvertent exposures.

7.6.2 Compliance Pressure

Regulatory frameworks are tightening globally. Both **GDPR** (Europe) and **CCPA** (California) now mandate strict data protection standards. However, readiness is lagging:

- Only **45% of firms** are projected to fully comply with rigorous regulations by 2025.
- **HIPAA Updates:** The impending 2025 updates to the HIPAA Security Rule are expected to mandate stronger protections for legacy systems, creating a direct compliance necessity for tools like zGate that can wrap legacy databases in a secure Zero Trust layer.

7.6.3 Third-Party and AI Threats

The threat landscape is expanding beyond internal staff. Third-party vendors are involved in a majority of breaches, highlighting the need for secure, ephemeral access for contractors. Furthermore, the weaponization of AI has accelerated zero-day exploits, requiring defense mechanisms that do not rely solely on static signatures.



7.7 Database Security Market Landscape

7.7.1 Market Size and Segmentation

The global database security market is experiencing robust growth, with a Compound Annual Growth Rate (CAGR) of **18.9%** projected from 2024 to 2030.

- **Regional Leadership:** North America currently leads the global market in sales, followed by Asia-Pacific and Europe.
- **Component Share:** Software solutions dominate the market share, with services (support/consulting) comprising a smaller portion.
- **Sector Usage:** Usage is highest in Marketing departments, followed by Sales, Operations, and Finance, reflecting the intense data-dependency of these business units.

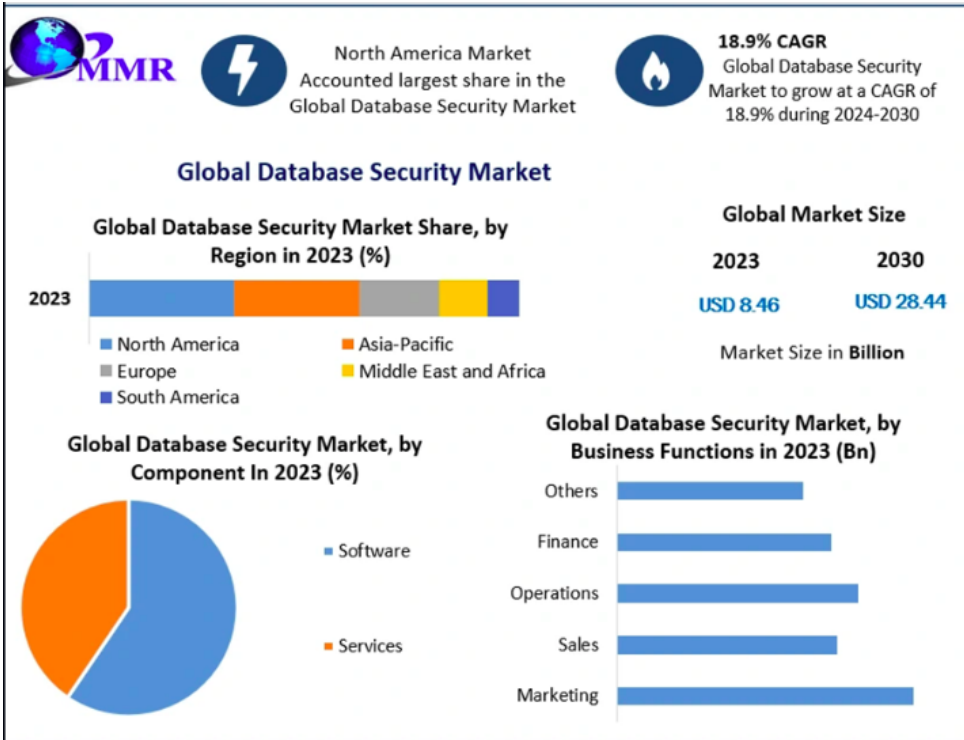


Figure 7.1: Database Security Market Share by Region/Sector

7.8 Zero Trust: Market Demand and Trends

7.8.1 Explosive Market Growth

The adoption of Zero Trust is a primary driver for database security innovation. Industry surveys forecast the Zero Trust market to reach approximately **\$38–39 billion in 2025**, growing to **\$86.6 billion by 2030** (a CAGR of $\approx 17.7\%$).

7.8.2 Universal Adoption Intent

The shift to Zero Trust is now mainstream:

- **96%** of organizations favor a Zero Trust approach.
- **81%** plan to implement it within the next 12 months.
- In a survey of over 2,200 IT leaders, **43%** had already adopted Zero Trust, and another **46%** were actively moving toward it, leaving only $\sim 11\%$ with no plans.

7.8.3 The Database Gap

Crucially for this project, while **89%** of security teams say they are developing Zero Trust policies for database access, only **43%** report having robust controls in place today. This **46%** gap represents the prime market opportunity for zGate.

7.9 Emerging Trends and Growth Opportunities

- **Microsegmentation: 78% of hospitals** utilize microsegmentation, cutting ransomware spread by 40%. zGate applies this concept to the database layer (Database Microsegmentation).
- **Quantum Encryption:** JPMorgan's implementation of quantum encryption reduced credential stuffing attacks by **92%** in 2024, signaling a future trend for high-security proxies.
- **AI Defense:** AI is increasingly used to speed up zero-day attack detection, a potential future roadmap item for zGate's anomaly detection.

7.10 Challenges and Barriers to Adoption

Despite the clear benefits, organizations face hurdles:

- **Complexity and Cost:** Implementing comprehensive Zero Trust often requires expensive overhauls.
- **Legacy Systems:** Integration gaps with older mainframes or databases remain a blocker.
- **Skill Fragmentation:** Teams often lack the specific skills to manage fragmented security tools.

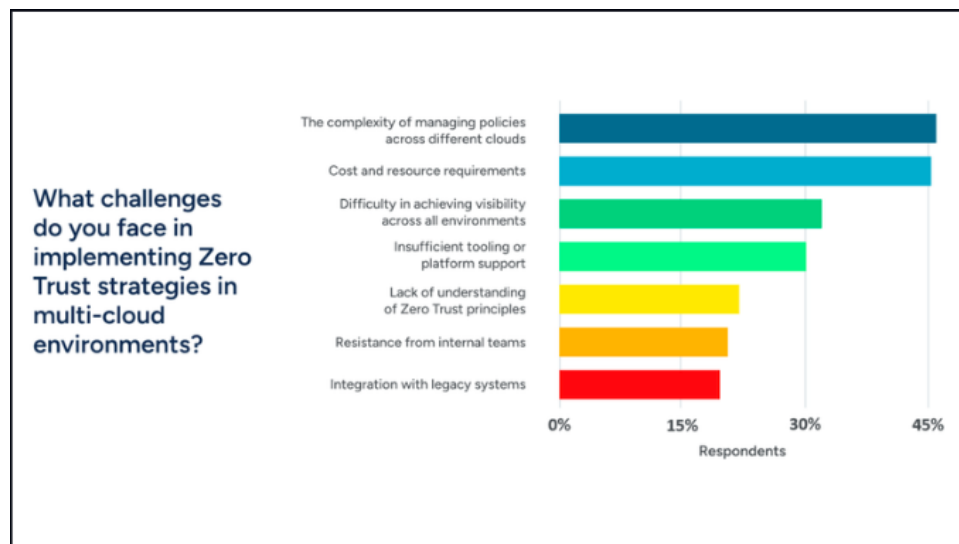


Figure 7.2: Key Challenges in Zero Trust Adoption



7.11 Conclusion

zGate enters a market characterized by high demand but significant technical gaps. By providing a protocol-aware, open-source solution that supports MySQL, MSSQL, and PostgreSQL, it directly addresses the compliance, security, and usability challenges that currently leave over half of the market underserved.

Chapter 8

Scientific Research & Literature Review

This part covers:

- Generative AI in cybersecurity frameworks
- AI integration with Zero Trust architectures
- Blockchain-based authentication systems
- Database driver lifecycle management
- Zero-trust database security research
- Privacy-preserving access control mechanisms

SCIENTIFIC research provides the theoretical foundation for zGate's design. This chapter reviews six seminal papers covering AI-enhanced cybersecurity frameworks, Zero Trust architectures integrated with intelligent authentication, blockchain-based identity management, and privacy-preserving access control mechanisms. We examine how emerging research in generative AI, database driver management, and advanced security systems informs our implementation.

8.1 Generative AI-Enhanced Cybersecurity Framework for Enterprise Data Privacy Management

8.1.1 Purpose of the Study

The paper addresses the growing need for organizations to secure sensitive enterprise data (e.g., financial transactions, patient records, IoT data) while still enabling advanced detection of cyber threats. Traditional security controls and anomaly detection methods often:

- Miss new/unknown attack patterns
- Require direct use of real sensitive data, creating privacy and compliance risks

Goal: The authors propose a Generative AI-enhanced framework that combines Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), and traditional machine learning (ML/DL) anomaly detection with strong privacy-preserving methods (Differential Privacy, encryption, masking). The aim is to balance data privacy, detection accuracy, and computational efficiency.

8.1.2 Framework Overview

The proposed framework works as an end-to-end pipeline with the following main components:

- **Data ingestion:** Collect logs (network, system, application) via tools like Splunk/ELK
- **Generative AI layer (GANs & VAEs):** Create synthetic, privacy-safe data that mimic real-world patterns without exposing identities
- **Privacy layer:** Apply Differential Privacy ($\epsilon = 0.1$ for highly sensitive data), AES-256 encryption, TLS 1.3, and masking
- **Anomaly detection engine:** Train models like Random Forest, SVM, and LSTM on synthetic + sanitized data to detect unusual activity
- **Monitoring & alerting:** Real-time detection with dashboards and alert systems

Analogy: Instead of training guards with real customer data (which is risky), the system uses highly realistic "actors" (synthetic data) to train them — ensuring the guards learn effectively without ever seeing the real people.

8.1.3 Implementation & Experiments

The framework was tested in three simulated enterprise domains:

- **Finance (transaction logs):** 94% accuracy, 95% recall, $\sim 1.2\text{--}1.5$ seconds per transaction
- **Healthcare (EHR access logs):** 96% accuracy, 93% precision, ~ 1.5 seconds per event
- **Smart City/IoT (sensor data):** 91% accuracy, $F1 \approx 90\%$, latency ≤ 100 ms at the edge

Performance trade-offs:

- GAN framework: $\sim 96\%$ accuracy, moderate compute (4GB GPU, 2.5h training)
- LSTM: $\sim 97\%$ accuracy but higher GPU needs (6GB)
- Traditional ML (RF/SVM): lower accuracy ($\sim 92\text{--}94\%$) but lighter
- Very high-accuracy CNN ($\geq 99\%$): impractical resource usage

8.1.4 Privacy & Security Features

- **Differential Privacy:** Adds noise to hide individual user data, ensuring compliance with GDPR/HIPAA
- **Encryption:** AES-256 for data at rest, TLS 1.3 for data in transit
- **Access control:** Role-based restrictions
- **Data masking:** Obscures identifiers in logs

8.1.5 Contributions to the Paper

- First comprehensive framework integrating Generative AI + privacy techniques + anomaly detection
- Provides balanced performance: strong accuracy without extreme computational demands
- Applicable across finance, healthcare, and IoT
- Offers implementation guidance with practical tools (TensorFlow, PyTorch, Scikit-learn, PySyft)

8.1.6 Advantages & Limitations

Advantages:

- Protects privacy while enabling effective training
- Can detect novel/rare attacks better by augmenting datasets with synthetic samples
- Works across domains, modular and adaptable
- More resource-efficient than some deep CNN methods

Limitations:

- Results are simulated, not from live production environments
- Quality of synthetic data can affect detection accuracy
- Managing GANs, VAEs, DP, and anomaly detectors is operationally complex
- Differential Privacy trade-off: stronger privacy (smaller ϵ) may reduce model accuracy

8.1.7 Relevance to Our Project

This study is directly relevant because:

- Our project focuses on secure access and monitoring of sensitive databases
- The paper's synthetic-data + anomaly detection pipeline is a practical approach to train models without exposing real database queries/records
- Techniques like Differential Privacy, RBAC, AES-256 encryption overlap with Zero Trust principles (least privilege, continuous monitoring, encryption everywhere)
- Their results show that real-time detection with privacy is feasible, which strengthens the foundation for our Zero Trust access model

8.2 The Significance of Artificial Intelligence in Zero Trust Technologies: A Comprehensive Review

8.2.1 Problem Addressed

Traditional security models assume anything inside a company's network can be trusted. With today's cloud, remote work, and hybrid environments, this assumption no longer holds. Attackers exploit cloud resources, lateral movement inside networks, and slow manual controls. The study addresses how Artificial Intelligence (AI) can enhance the Zero Trust (ZT) model to meet these modern challenges.

8.2.2 Methodology

- Literature review (20+ studies examined)
- Synthesized how AI is applied across ZT building blocks (IAM, MFA, EDR, ZTNA, SASE, Network Analytics)

8.2.3 Key Contributions of AI to Zero Trust

Identity & Access Management (IAM)

- **Authentication:** Adaptive and continuous (AI monitors typing, device, location; flags anomalies)
- **Authorization:** Intelligent Role-Based Access Control (AI suggests roles, prevents "over-privilege")
- **Administration:** Automated onboarding/offboarding, policy adjustments
- **Audit/Compliance:** AI generates audit trails, suggests policies, detects compliance gaps

Adaptive Multi-Factor Authentication (AMFA)

- AI adjusts authentication strength based on risk (low → password; medium → OTP; high → biometrics)
- Balances usability with security

Endpoint Detection & Response (EDR)

- AI baselines device behavior, detects anomalies, reduces false positives
- Automates containment (isolate compromised laptops)

Zero Trust Network Access (ZTNA) & Secure Access Service Edge (SASE)

- ZTNA grants application-level access (not full network like VPN)
- SASE combines SD-WAN + ZTNA + CASB + FWaaS; AI analyzes telemetry, recommends segmentation
- AI enables dynamic microsegmentation and automated policy creation

8.2.4 Findings

- AI strengthens Zero Trust by making it continuous, adaptive, and automated
- AI reduces human error, speeds up detection, and scales across large organizations
- AI integration is critical for modern cloud and hybrid infrastructures

8.2.5 Comparison with Traditional Methods

- Traditional perimeter security = trust anyone inside
- Zero Trust with AI = checkpoint at every request making context-based decisions in real time

8.2.6 Relevance to Our Project

- IAM insights → directly applicable for database user role mining & continuous verification
- Adaptive MFA → useful for database login protection
- EDR concepts → extend to database clients/endpoints
- ZTNA & SASE → inspire database-level microsegmentation (grant per-query or per-app access)
- Network analytics → parallels database traffic analysis for anomaly detection

8.3 Securing Digital Identity in the Zero Trust Architecture: A Blockchain Approach to Privacy-Focused Multi-Factor Authentication

8.3.1 Problem Addressed

- Traditional MFA depends on centralized servers, which are vulnerable to outages and breaches
- Zero Trust architectures require continuous, strong identity verification, but current MFA approaches are limited in resilience and privacy

The study addresses these issues by designing a decentralized, privacy-focused MFA mechanism that eliminates single points of failure and ensures secrets remain private.

8.3.2 Research Goals

- Build a decentralized authentication system aligned with Zero Trust principles
- Ensure privacy-preserving verification so users never expose OTPs
- Provide auditability and traceability of authentication events
- Deliver a realistic proof-of-concept that demonstrates feasibility and performance

8.3.3 Proposed System

- **Distributed Authentication Mechanism (DAM):** validator nodes collectively handle authentication
- **Distributed OTP Generation:** each validator contributes a random partial secret; combined into full OTP
- **Privacy via zk-SNARKs:** users prove they know the OTP without revealing it
- **Authentication Token:** successful proof leads to issuance of a non-transferable NFT (digital badge) valid for a period

8.3.4 Experimental Results

- **Performance:** comparable to real-world MFA timings (~20 seconds average)
- **Security:** analysis shows probability of attack success is negligible due to distribution, cryptographic verification, and non-transferability

8.3.5 Key Findings and Contributions

- Decentralized authentication reduces single points of failure
- OTP secrets are never exposed; verification occurs through zk-SNARK proofs
- Immutable, auditable on-chain logs improve accountability
- Non-transferable NFTs prevent token theft or resale

8.3.6 Real-World Applications

- **Banking & Fintech:** customers receive distributed OTPs and prove knowledge via zk-SNARK, receiving session tokens for access
- **Corporate IT (Zero Trust):** employees authenticate and receive short-lived NFTs, ensuring continuous verification without exposing passwords
- **Developer Platforms:** integration with existing blockchain and web authentication frameworks for higher resilience

8.3.7 Relevance to Our Project

This research is directly relevant because it shows how decentralized, privacy-preserving multi-factor authentication can be integrated into a Zero Trust architecture; a blueprint for secure identity verification that we can adapt for controlling database access in our Zero Trust Gateway.

8.3.8 Conclusion

This research provides a comprehensive, innovative, and feasible approach to MFA in Zero Trust environments. By using distributed OTP generation, zk-SNARK verification, and non-transferable authentication tokens, it resolves critical weaknesses in traditional MFA. While some limitations exist (cost, setup trust, validator reputation), the overall contribution strongly supports the feasibility of implementing privacy-focused, decentralized identity verification in real-world systems.

8.4 Drivolution: Rethinking the Database Driver Lifecycle

8.4.1 Problem Addressed

Traditional database driver management creates significant operational burdens in large production environments. The study identifies four major challenges:

- **Distribution complexity:** Drivers are distributed separately from database engines, leading to version mismatches and incompatibilities
- **Manual deployment:** Driver installation requires manual operations on each client machine, which doesn't scale well
- **Disruptive upgrades:** Updating drivers requires stopping applications, reconfiguring them, and restarting—causing downtime
- **Security vulnerabilities:** Malicious applications can exploit outdated drivers or use crafted drivers to attack database servers

These issues are amplified in heterogeneous environments where multiple database versions, platforms (63+ for MySQL alone), and client applications coexist. The problem becomes even more acute in replicated database environments where upgrades must account for the Cartesian product of drivers and databases.

8.4.2 Research Methodology

The authors propose and implement Drivolution, an alternative architecture for database driver management. The methodology includes:

- Design of a new driver lifecycle model inspired by OS bootloaders and DHCP protocols
- Implementation for JDBC API integrated with Sequoia database clustering middleware
- Multiple case studies demonstrating real-world applicability
- Performance evaluation in simulated production environments

8.4.3 Drivolution Architecture

The proposed system fundamentally reimagines driver management through several key innovations:

Core Components

- **Driver Storage:** Drivers are stored in the database itself (in regular tables) or in standalone Drivolution servers, treating them as integral parts of the database schema
- **Bootloader:** A small, stable client-side component that rarely needs updating—it downloads and executes driver code from the database
- **Drivolution Protocol:** A DHCP-inspired protocol with three messages (REQUEST, OFFER, ERROR) for driver negotiation
- **Lease System:** Time-limited driver validity periods that enable automatic updates

Technical Implementation

The system uses two key database tables in the information schema:

- **Drivers table:** Stores driver binaries (as BLOBs), API versions, platform specifications, and version information
- **Driver_permission table:** Defines access rights, update policies, lease times, and expiration policies per user/client/database combination

The bootloader intercepts API connection calls, contacts the Drivolution server, downloads appropriate drivers based on client platform and requirements, and dynamically loads them into application memory—all transparently to the application.

8.4.4 Key Innovations

Simplified Lifecycle

Traditional approach (10 steps for upgrade per client):

1. Stop application
2. Uninstall old driver
3. Download new driver package
4. Install driver
5. Configure application
6. Restart application
7. (Plus 4 more verification steps)

Drivolution approach (1 step for all clients):

1. Insert new driver into Drivolution server database

Transparent Updates

Three update policies accommodate different operational needs:

- **AFTER_CLOSE:** Wait for application to close connections naturally
- **AFTER_COMMIT:** Close connections after current transactions complete
- **IMMEDIATE:** Force immediate termination and upgrade

Security Features

- SSL-encrypted, authenticated transfer channels prevent man-in-the-middle attacks
- Digital signature verification ensures driver authenticity
- Standard database access controls limit who can retrieve which drivers
- Centralized management reduces risk of outdated, vulnerable drivers

8.4.5 Case Studies and Results

Heterogeneous DBMS Administration

For DBAs managing multiple database versions, Drivolution reduced:

- Accessing new database: from 6 manual steps to 1 automatic connection
- Driver upgrade: from 6 steps per DBA workstation to 2 centralized operations

Master/Slave Failover

Drivolution enables transparent client reconfiguration during failover:

- Pre-configured drivers (DB_master, DB_slave) distributed based on current topology
- Failover accomplished by marking old driver expired and offering new one
- All clients automatically reconfigure without manual intervention

Sequoia Clustering Middleware

Multiple deployment configurations demonstrated:

- **Standalone server:** Centralized control with hot-standby replication for availability
- **Embedded in controllers:** Drivolution servers replicated across cluster nodes, eliminating single point of failure
- Seamless upgrades of both Sequoia drivers and backend database drivers

Customized Driver Delivery

- **On-demand assembly:** Delivers only required components (e.g., NLS packages for specific languages, GIS extensions only to geographic applications)
- **License management:** Dynamic distribution of per-user licenses (e.g., IBM DB2 licensing model)

8.4.6 Performance Characteristics

- Bootloader overhead: minimal (simple connection interception)
- One-time download per lease period (hours to days)
- No performance impact on queries after driver loaded
- Drivolution server can be replicated for availability without complex consistency requirements (infrequent updates)

8.4.7 Advantages and Contributions

- **Operational simplicity:** Centralized management reduces complexity exponentially in large deployments
- **Zero downtime:** Applications continue running during driver upgrades
- **Version consistency:** Guaranteed compatibility between drivers and databases
- **Security improvement:** Faster deployment of security patches, elimination of forgotten/outdated drivers
- **Legacy compatibility:** Works with existing databases and applications without modifications
- **Platform neutrality:** Single bootloader implementation per API works across all databases
- **Flexibility:** Supports multiple deployment models (in-database, external, standalone service)

8.4.8 Limitations and Considerations

- **Bootloader dependency:** Initial bootloader installation still required (though this is one-time per API/platform)

- **Dynamic loading requirement:** Not all languages/platforms support secure dynamic code loading
- **API stability:** Bootloaders are API-specific; major API changes require new bootloaders
- **Testing discipline:** Ease of updates might tempt skipping rigorous testing (though the paper notes testing is actually easier with short leases for staged rollouts)
- **Drivolution server availability:** Becomes a dependency, though this is mitigated through replication
- **Trust requirements:** Initial bootloader and SSL certificates must be established securely

8.4.9 Relevance to Our Zero Trust Database Access Project

This research provides several valuable insights for our project:

Zero Trust Alignment

- **Continuous verification:** The lease system embodies “never trust, always verify”—clients must regularly re-authenticate to receive driver updates
- **Least privilege:** The `driver_permission` table enables fine-grained control over which clients access which databases with which drivers
- **Explicit trust zones:** Each client-database interaction is mediated through the Drivolution server, creating an explicit trust boundary

Practical Implementation Lessons

- **Centralized policy enforcement:** Storing access policies in the database (`driver_permission` table) parallels our need for centralized Zero Trust policy management
- **Transparent security:** The bootloader approach shows how security enhancements can be added without modifying applications—applicable to our gateway design
- **Gradual rollout:** Short initial leases with gradual expansion demonstrate safe deployment of security updates—relevant for our anomaly detection model updates
- **Backward compatibility:** Supporting both Drivolution and legacy connections shows how to introduce Zero Trust incrementally

Architectural Patterns

- **Interception layer:** The bootloader pattern (intercept, verify, mediate) directly parallels our Zero Trust gateway architecture
- **Dynamic configuration:** Delivering pre-configured drivers is analogous to delivering context-aware access policies
- **Lease-based validity:** Time-limited driver validity maps to session tokens with expiration in Zero Trust
- **Distributed servers:** Replicating Drivolution servers across cluster nodes provides a model for high-availability Zero Trust policy enforcement

Security Considerations

- **Encrypted channels:** SSL/TLS requirements reinforce the importance of encrypting all database traffic in Zero Trust
- **Signature verification:** Driver signing parallels the need to verify integrity of all components in our system
- **Audit trails:** The ability to track which drivers were distributed to which clients informs our logging and compliance requirements
- **Attack surface reduction:** Preventing outdated drivers parallels preventing outdated/vulnerable access patterns in Zero Trust

8.4.10 Implications for zGate Proxy Architecture

While Drivolution addresses driver lifecycle management through client-side bootloaders and centralized driver distribution, our Zero Trust gateway requires a fundamentally different architectural approach. The Drivolution model operates as a transparent intermediary that facilitates driver delivery but does not actively mediate database communication protocols.

For zGate proxy to effectively implement Zero Trust principles—including continuous authentication, fine-grained access control, query-level authorization, and real-time anomaly detection—we must adopt a protocol translation architecture. This necessitates implementing native database protocol handlers for each supported database technology (PostgreSQL, MySQL, Oracle, SQL Server, etc.), enabling zGate to function as a bidirectional protocol mediator.

In this architecture, zGate presents itself as a database server to client applications, accepting connections through database-native protocols and performing authentication,

authorization, and security checks. Simultaneously, zGate maintains authenticated connection pools to backend database servers, acting as a client that forwards validated queries and returns results. This dual-role architecture provides several critical capabilities:

- **Deep packet inspection:** Full visibility into query content enables semantic analysis and policy enforcement at the query level
- **Protocol-level security:** Independent verification of authentication credentials without relying on client-provided drivers
- **Centralized policy enforcement:** All database traffic traverses zGate, ensuring no requests bypass security controls
- **Connection pooling and optimization:** Backend connection management independent of client connection lifecycle
- **Multi-database support:** Protocol handlers for different database technologies enable consistent security across heterogeneous environments
- **Audit and compliance:** Complete transaction logs captured at the protocol level with full context

The implementation of database-specific protocol handlers represents significant engineering complexity, as each database vendor implements proprietary wire protocols with distinct authentication mechanisms, query formats, and result set encodings. However, this approach is essential for achieving the granular control and visibility required in a Zero Trust architecture, where trust must be continuously verified and access decisions made based on comprehensive contextual analysis.

8.4.11 Conclusion

Drivolution demonstrates that fundamental database infrastructure components can be redesigned to reduce operational complexity, improve security, and enable zero-downtime operations. The architecture's emphasis on centralized management, transparent operation, and compatibility with legacy systems provides a valuable blueprint for introducing Zero Trust principles into database access patterns. The successful implementation in production middleware (Sequoia) validates that such architectural changes are not merely theoretical but practically achievable in real-world systems.

However, the requirements of Zero Trust security demand more than transparent driver management—they necessitate active protocol mediation. This insight directly

motivates the zGate proxy architecture, where implementing backend drivers and connection handlers for each database technology enables the system to serve as an intelligent intermediary, presenting a server interface to clients while maintaining authenticated client connections to database backends. This bidirectional translation capability forms the foundation upon which comprehensive Zero Trust security controls can be built.

8.5 Zero-trust database systems: The new frontier in data security

8.5.1 Paradigm Shift in Data Security

Traditional database security models operate on a perimeter-based assumption: once users authenticate to a network or application, they gain broad access to database resources. This "trust-but-verify" approach creates fundamental vulnerabilities, especially in environments with lateral movement attacks, insider threats, and cloud-native architectures where network perimeters are increasingly porous.

Zero-trust database systems represent a fundamental shift: assume no implicit trust at any layer. Every query, transaction, and data access request—regardless of source—must be continuously authenticated, authorized, and audited before execution. This paper examines how zero-trust principles, originally developed for network security, are being adapted to database systems to address modern threat landscapes.

8.5.2 Core Zero-Trust Database Principles

The paper identifies five foundational principles:

- **Verify explicitly:** Every database request requires fresh authentication and authorization checks, even within established sessions
- **Least privilege access:** Users receive the minimum permissions necessary for specific tasks, enforced at row/column granularity
- **Assume breach:** Design systems to contain and mitigate damage from compromised credentials or insider threats
- **Continuous monitoring:** Real-time analysis of query patterns, data access behaviors, and anomalous activities
- **Context-aware policies:** Access decisions incorporate user identity, device posture, query intent, data sensitivity, time, and location

8.5.3 Key Architectural Components

Identity-Centric Authentication

Traditional database authentication relies on static credentials (username/password) valid for session duration. Zero-trust systems implement:

- **Continuous authentication:** Re-verify identity at transaction or query boundaries
- **Multi-factor integration:** Require MFA for sensitive operations or high-risk patterns
- **Identity federation:** Integrate with enterprise identity providers (Okta, Azure AD) for centralized credential management
- **Cryptographic assertions:** Use tokens, JWTs, or certificates with short validity periods

Fine-Grained Authorization

Zero-trust authorization moves beyond traditional role-based access control (RBAC) to implement:

- **Attribute-based access control (ABAC):** Policies that evaluate user attributes, data attributes, environmental context
- **Row-level security:** Dynamic data filtering based on who is requesting and what they can see
- **Column-level masking:** Automatic redaction/encryption of sensitive fields for unauthorized users
- **Query-level validation:** Analysis of SQL statements to prevent unauthorized data exfiltration (e.g., SELECT * queries on sensitive tables)

Real-Time Anomaly Detection

Machine learning models embedded in the database access layer continuously analyze:

- Query frequency and timing (detecting unusual batch downloads)
- Data access patterns (identifying privilege escalation or reconnaissance)
- Schema exploration (flagging excessive metadata queries)
- Behavioral deviations (comparing current actions to historical baselines)

Detected anomalies trigger graduated responses: additional authentication, query rejection, session termination, or administrative alerts.

Encryption Everywhere

Zero-trust databases enforce encryption at multiple layers:

- **In-transit:** TLS 1.3 for all client-database connections
- **At-rest:** Transparent data encryption (TDE) for storage
- **In-use:** Emerging technologies like confidential computing and homomorphic encryption for processing encrypted data without decryption

Comprehensive Audit Logging

Every database interaction generates immutable audit trails including:

- User/service account identity and authentication method
- Query text and parameters (with sensitive data redacted)
- Authorization decisions and policy evaluations
- Result set metadata (rows affected, columns accessed)
- Anomaly detection scores and triggered policies
- Timestamps and source context (IP, device, application)

8.5.4 Real-World Applications

Healthcare: HIPAA Compliance

Hospitals use zero-trust databases to:

- Enforce need-to-know access to patient records (doctors see only assigned patients)
- Detect unauthorized attempts to access celebrity/VIP patient data
- Implement break-glass access with full audit trails for emergencies
- Comply with HIPAA audit requirements through comprehensive logging

Financial Services: PCI-DSS and Fraud Prevention

Banks and payment processors deploy zero-trust to:

- Isolate cardholder data environments (CDE) with query-level controls
- Detect insider threats attempting mass credit card number exports
- Enforce separation of duties (developers cannot access production customer data)
- Meet PCI-DSS requirements for data minimization and access tracking

Cloud-Native SaaS: Multi-Tenancy Security

SaaS providers use zero-trust databases to:

- Prevent tenant data leakage through row-level security
- Detect compromised service accounts attempting cross-tenant access
- Implement dynamic tenant isolation policies
- Audit compliance for SOC 2 and ISO 27001 certifications

8.5.5 Benefits and Advantages

- **Reduced attack surface:** Continuous verification prevents lateral movement from compromised credentials
- **Insider threat mitigation:** Least-privilege and anomaly detection reduce risks from malicious employees
- **Compliance simplification:** Automated audit trails and policy enforcement meet regulatory requirements (GDPR, HIPAA, PCI-DSS)
- **Cloud readiness:** Decoupled identity and context-aware policies work across hybrid and multi-cloud environments
- **Operational visibility:** Real-time monitoring provides security teams with actionable intelligence

8.5.6 Implementation Challenges

- **Performance overhead:** Continuous authentication and query analysis can add latency (typically 5-50ms depending on policy complexity)
- **Integration complexity:** Requires changes to database access layers, identity providers, and monitoring systems
- **Policy management:** Fine-grained policies require careful design to avoid authorization bottlenecks
- **False positives:** Anomaly detection may flag legitimate unusual activities, requiring tuning and feedback loops
- **Legacy compatibility:** Older applications may not support modern authentication mechanisms

8.5.7 Relevance to Our Project

This research is foundational to our Zero Trust gateway because:

- **Validation of approach:** The paper’s industry examples (healthcare, finance, SaaS) demonstrate that zero-trust databases solve real-world security problems
- **Architectural guidance:** The five core principles provide a checklist for our zGate implementation:
 - Continuous verification → Re-authenticate on session boundaries
 - Least privilege → Query-level authorization based on user context
 - Assume breach → Anomaly detection and threat response
 - Continuous monitoring → Real-time logging and analytics
 - Context-aware policies → Dynamic access decisions based on user, device, query, and risk
- **Technical blueprints:** The described authentication, authorization, and audit mechanisms map directly to zGate components (proxy interceptor, policy engine, audit logger)
- **Performance benchmarks:** Reported 5-50ms latency overhead sets realistic expectations for our proxy architecture
- **Implementation challenges:** The identified obstacles (performance, integration, policy management) guide our design decisions for zGate’s interceptor pipeline and policy evaluation engine

The paper validates that zero-trust database access is not a theoretical concept but a deployable security model with proven benefits in regulated industries—directly supporting the core thesis of our project.

8.6 Privacy-Preserving Attribute-Based Access Control Using Homomorphic Encryption

8.6.1 Problem Context

Attribute-Based Access Control (ABAC) enables fine-grained authorization by evaluating policies based on attributes of users, resources, actions, and environment. However, traditional ABAC implementations expose plaintext attributes to policy decision points (PDPs), creating privacy risks:

- User attributes (roles, clearances, project assignments) reveal organizational structure
- Resource attributes (classification levels, sensitivity labels) disclose information architecture
- Centralized policy evaluation creates a high-value target for attackers
- Compliance frameworks (GDPR, CCPA) demand minimization of attribute disclosure

The study addresses how to perform ABAC policy evaluation while keeping all attributes encrypted, ensuring that even the PDP cannot access plaintext sensitive information.

8.6.2 Research Goals

- Design an ABAC system where policy evaluation occurs over encrypted attributes
- Preserve privacy of user, resource, and environmental attributes throughout access decisions
- Maintain functional equivalence to plaintext ABAC (same policies and outcomes)
- Demonstrate feasibility through prototype implementation and performance evaluation

8.6.3 Proposed Solution: Homomorphic ABAC

Core Idea

The research proposes using homomorphic encryption to enable computation on encrypted data without decryption. Policy evaluations execute as homomorphic operations

over ciphertext attributes, producing encrypted access decisions that only authorized parties can decrypt.

System Architecture

- **Attribute authorities:** Encrypt user and resource attributes before distribution
- **Policy Decision Point (PDP):** Evaluates encrypted policies over encrypted attributes using homomorphic operations
- **Policy Enforcement Point (PEP):** Decrypts final decision (permit/deny) and enforces it
- **Homomorphic scheme:** Partially homomorphic encryption (PHE) for basic comparisons or fully homomorphic encryption (FHE) for complex policies

Technical Implementation

1. **Attribute encryption:** All user/resource attributes encrypted at source using homomorphic scheme
2. **Policy transformation:** ABAC policies (e.g., XACML) transformed into equivalent homomorphic circuits
3. **Encrypted evaluation:** PDP computes policy using homomorphic addition, multiplication, and comparison operations over ciphertexts
4. **Result decryption:** Only PEP (with decryption key) learns the permit/deny outcome

8.6.4 Implementation and Evaluation

The paper presents a proof-of-concept implementation evaluated on realistic ABAC scenarios:

Prototype Details

- **Encryption library:** HElib or SEAL for homomorphic operations
- **Policy language:** XACML policies converted to arithmetic circuits
- **Test scenarios:** Healthcare (patient record access), military (classified document access), finance (transaction approval)

Performance Results

- **Latency overhead:** Policy evaluation times increased from 1ms (plaintext) to 50-500ms (encrypted) depending on policy complexity
- **Throughput:** Systems handled 100-1000 requests/second for moderate complexity policies
- **Scalability:** Performance degraded with number of attributes and policy rules, but remained practical for most real-world scenarios

Security Analysis

- Formal proofs demonstrate attribute confidentiality under chosen-plaintext attack (CPA) model
- PDP learns nothing about attribute values, only performs blind computation
- Compromise of PDP does not reveal user/resource attributes
- Side-channel resistance through constant-time operations and circuit obfuscation

8.6.5 Application Domains

Healthcare Privacy

- Doctors access patient records based on encrypted department/role attributes without PDP learning organizational hierarchy
- Compliance with HIPAA minimum-necessary disclosure requirements
- Emergency access policies evaluated without exposing patient sensitivity classifications

Government and Defense

- Access to classified documents based on encrypted clearance levels and project assignments
- Policy decisions made without revealing classification hierarchies to all system components
- Need-to-know enforcement with cryptographic guarantees

Multi-Tenant Cloud Services

- SaaS providers enforce tenant isolation with encrypted tenant IDs and resource labels
- PDP cannot infer customer identities or data organization from policy evaluations
- Regulatory compliance for data processors handling EU/UK customer data

8.6.6 Advantages

- **Privacy guarantees:** Cryptographic protection of attributes throughout policy evaluation
- **Trust minimization:** PDP does not require trust—it operates blindly on encrypted inputs
- **Compliance enabler:** Satisfies data minimization and purpose limitation requirements
- **Attack resistance:** Compromise of policy infrastructure does not expose sensitive attributes
- **Functional equivalence:** Produces identical access decisions to plaintext ABAC

8.6.7 Limitations

- **Performance cost:** 50-500x slowdown compared to plaintext evaluation
- **Complexity:** Requires specialized cryptographic libraries and circuit design expertise
- **Policy expressiveness:** Some complex ABAC policies difficult to translate into efficient homomorphic circuits
- **Key management:** Introduces additional cryptographic key distribution and rotation challenges
- **Maturity:** Homomorphic encryption schemes remain computationally expensive and less mature than traditional cryptography

8.6.8 Relevance to Our Project

This research directly informs the zGate Zero Trust database gateway in several ways:

- **Attribute-based policies:** The paper validates ABAC as the foundation for flexible, context-aware database access control—matching our policy engine requirements
- **Privacy considerations:** While full homomorphic evaluation may be impractical for real-time query authorization, the principle of minimizing attribute disclosure guides our policy engine design (e.g., hashing user attributes, encrypting policy rules in transit)
- **Trust boundaries:** The PDP/PEP separation clarifies roles: policy evaluation can occur in a less-trusted component if decisions are validated by a trusted enforcement point (our proxy)
- **Performance trade-offs:** The reported 50-500ms latency validates that complex authorization can be practical if policy complexity is managed—our target ≤ 100 ms query overhead must account for policy evaluation costs
- **Hybrid approach:** We can adopt selective encryption of particularly sensitive attributes while using standard ABAC for less sensitive policies, balancing privacy and performance
- **Regulatory alignment:** The emphasis on GDPR/HIPAA compliance through attribute privacy reinforces the legal and regulatory value proposition of zGate

The paper demonstrates that privacy-preserving access control is not merely theoretical but implementable with acceptable performance in specific high-security contexts. For zGate, this suggests opportunities to enhance our policy engine with optional encryption-based privacy protections for customers in highly regulated industries.

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- **Paper 3:** Securing Digital Identity in the Zero Trust Architecture: A Blockchain Approach to Privacy-Focused Multi-Factor Authentication
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- **Paper 6:** Privacy-Preserving Attribute-Based Access Control Using Homomorphic Encryption
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Chapter 9

Technical Background

This part covers:

- Go programming and concurrency
- Database wire protocols
- Zero Trust security architecture
- Authentication mechanisms
- Frontend technologies
- Network programming fundamentals

UNDERSTANDING the technical foundations underlying zGate is essential for appreciating architectural decisions. This chapter provides comprehensive coverage of Go's concurrency model and systems programming capabilities, database wire protocols for MySQL and MSSQL, Zero Trust security principles, JWT and OAuth authentication mechanisms, and modern frontend technologies including React and TypeScript. We establish the technical context necessary for understanding implementation choices throughout the system.

This chapter provides an in-depth technical overview of all technologies, protocols, and architectural concepts forming the foundation of the zGate Zero Trust Database Access Proxy. The discussion is intentionally extensive to support academic rigor and enable future researchers or developers to extend the project.

9.1 Systems Programming in Go

The zGate gateway is implemented entirely in Go (Golang) version 1.25.4. Go is selected due to its strong concurrency model, built-in memory safety guarantees, and first-class support for networked systems.

9.1.1 Introduction to Go Programming Language

Go, also known as Golang, is a statically typed, compiled programming language designed at Google in 2007 by Robert Griesemer, Rob Pike, and Ken Thompson. First released publicly in 2009, Go was created to address shortcomings in other languages used for systems programming, particularly in the context of multicore processors, networked systems, and large codebases.

Design Philosophy Go emphasizes simplicity, readability, and pragmatism. Key design principles include:

- **Simplicity:** Minimalist syntax with only 25 keywords
- **Explicit over implicit:** No hidden control flow or magic behaviors
- **Composition over inheritance:** Interfaces and struct embedding instead of class hierarchies
- **Fast compilation:** Designed for rapid build times even in large projects
- **Built-in concurrency:** First-class language support for concurrent programming

Memory Safety Go provides automatic memory management through garbage collection, eliminating entire classes of vulnerabilities:

- **No manual memory management:** Prevents use-after-free and double-free errors
- **Bounds checking:** Array and slice accesses are automatically validated
- **No pointer arithmetic:** Prevents buffer overflows and memory corruption
- **Type safety:** Strong static typing prevents type confusion attacks

Standard Library Go's extensive standard library includes production-ready packages for:

- Network programming (`net`, `net/http`)
- Cryptography (`crypto/*`)
- Encoding/decoding (`encoding/json`, `encoding/xml`)
- Testing and benchmarking (`testing`)
- Concurrent programming (`sync`, `context`)

9.1.2 Go Runtime Model

Go employs a sophisticated user-space thread management architecture based on the G-M-P scheduling model, which enables efficient concurrency without the overhead of traditional OS threads.

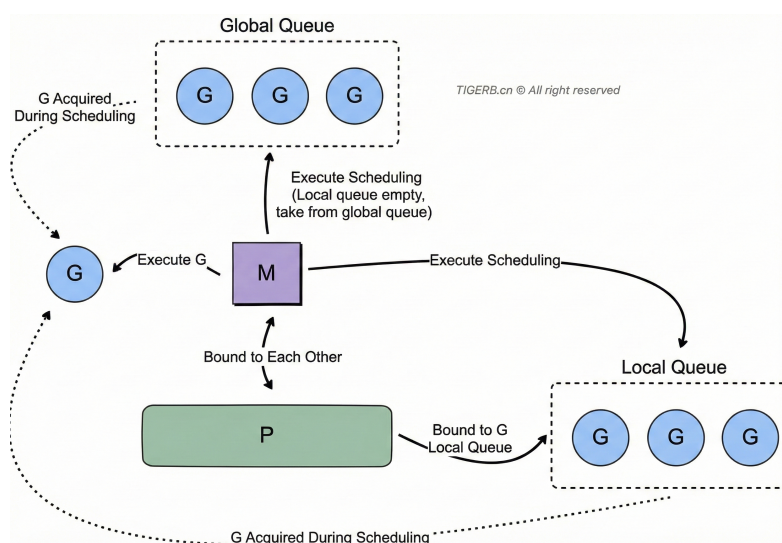


Figure 9.1: Go concurrency execution model illustrating goroutines, OS threads, the Go scheduler (G-M-P model).

The G-M-P Model Explained

- **G (Goroutine):** A goroutine is a lightweight cooperative thread with a dynamically-sized stack that starts at 2KB and can grow to several megabytes. Unlike OS threads, goroutines are managed entirely in user space by the Go runtime. Goroutines use cooperative scheduling, meaning they yield control at specific points (channel operations, system calls, function calls) rather than being preemptively interrupted.

- **M (Machine):** An M represents an OS thread managed by the operating system kernel. The Go runtime creates a pool of Ms (typically matching the number of CPU cores) that execute goroutines. When a goroutine performs a blocking system call, the M is detached and a new M may be created to continue executing other goroutines.
- **P (Processor):** A P is a scheduling context that maintains a local run queue of goroutines. The number of Ps is typically set to the number of available CPU cores (controlled by GOMAXPROCS). Each M must be associated with a P to execute goroutines. When a goroutine blocks, the P can be handed off to another M, allowing other goroutines to continue execution.

Scheduling Mechanism The scheduler implements work-stealing to balance load:

1. Each P maintains a local queue of runnable goroutines
2. When a P's queue is empty, it attempts to steal work from other Ps
3. A global run queue handles goroutines that don't fit in local queues
4. Network poller integration enables efficient I/O multiplexing

Why This Matters for zGate This model enables thousands of goroutines to execute concurrently with negligible overhead. zGate relies on this feature because each client session, backend connection, interceptor callback, and logging pipeline runs as its own goroutine. A typical deployment might handle 10,000+ concurrent database connections, each requiring multiple goroutines, which would be impossible with traditional thread-per-connection models.

9.1.3 Concurrency Primitives

Go provides several built-in primitives for concurrent programming that form the foundation of zGate's concurrent architecture.

Goroutines in Detail Goroutines are created using the `go` keyword followed by a function call. They provide several advantages:

- **Low memory overhead:** Each goroutine starts with only 2KB stack space vs 1-2MB for OS threads
- **Fast creation:** Creating a goroutine takes microseconds vs milliseconds for threads
- **Efficient scheduling:** Context switching between goroutines is faster than kernel thread switches

- **Scalability:** Applications can easily spawn millions of goroutines

In zGate, goroutines are used for:

- **Frontend packet reader:** Continuously reads MySQL packets from client connections
- **Backend packet writer:** Forwards packets to the database server
- **Audit logger:** Asynchronously writes audit entries without blocking query processing
- **TLS handshake worker:** Handles cryptographic handshakes in parallel
- **Interceptor orchestrators:** Executes policy enforcement logic concurrently

Channels in Depth Channels are Go's primary mechanism for communication between goroutines, implementing the CSP (Communicating Sequential Processes) model. Channels are typed, thread-safe queues that can be buffered or unbuffered.

Channel Types:

- **Unbuffered channels:** Synchronous - sender blocks until receiver is ready
- **Buffered channels:** Asynchronous up to buffer size - sender blocks only when buffer is full
- **Directional channels:** Can be send-only (`chan<-`) or receive-only (`<-chan`)

Channel Operations:

- **Send:** `ch <- value`
- **Receive:** `value := <-ch`
- **Close:** `close(ch)`
- **Select:** Multiplexing over multiple channel operations

In zGate, channels provide synchronization for:

- **Session-level error propagation:** When a critical error occurs in any goroutine handling a session
- **Asynchronous event forwarding:** Audit events, metrics, and alerts
- **Administrative operation coordination:** Graceful shutdown, configuration reloads
- **Work distribution:** Distributing query processing tasks across worker pools



Mutexes and Atomic Operations **Mutex (Mutual Exclusion):** A mutex is a synchronization primitive that protects shared data from concurrent access:

- **sync.Mutex:** Provides exclusive locking - only one goroutine can hold the lock
- **sync.RWMutex:** Reader-writer mutex - allows multiple readers OR one writer
- **Lock/Unlock pattern:** Must be paired, typically using **defer** to ensure unlock

Atomic Operations: The **sync/atomic** package provides lock-free operations for simple data types:

- **Atomic integers:** Add, Load, Store, Swap, CompareAndSwap operations
- **Performance:** Much faster than mutex-based protection for simple counters
- **Memory ordering:** Provides happens-before guarantees

Critical shared resources in zGate use:

- **sync.Mutex / sync.RWMutex:** For metadata caches (user sessions, prepared statements)
- **sync.Once:** For one-time initialization of secrets, certificates, and database connections
- **sync/atomic:** For high-throughput metrics (query counts, error rates, latency tracking)

9.1.4 Low-Level TCP Socket Programming

Unlike typical database clients that rely on high-level drivers, zGate communicates directly using the MySQL wire protocol over raw TCP sockets. This low-level approach provides complete control over the communication pipeline.

TCP/IP Socket Fundamentals **TCP (Transmission Control Protocol):**

- **Connection-oriented:** Requires handshake (SYN, SYN-ACK, ACK) before data transfer
- **Reliable:** Guarantees in-order delivery with automatic retransmission
- **Flow control:** Prevents sender from overwhelming receiver
- **Congestion control:** Adapts sending rate based on network conditions

Socket Operations in Go:

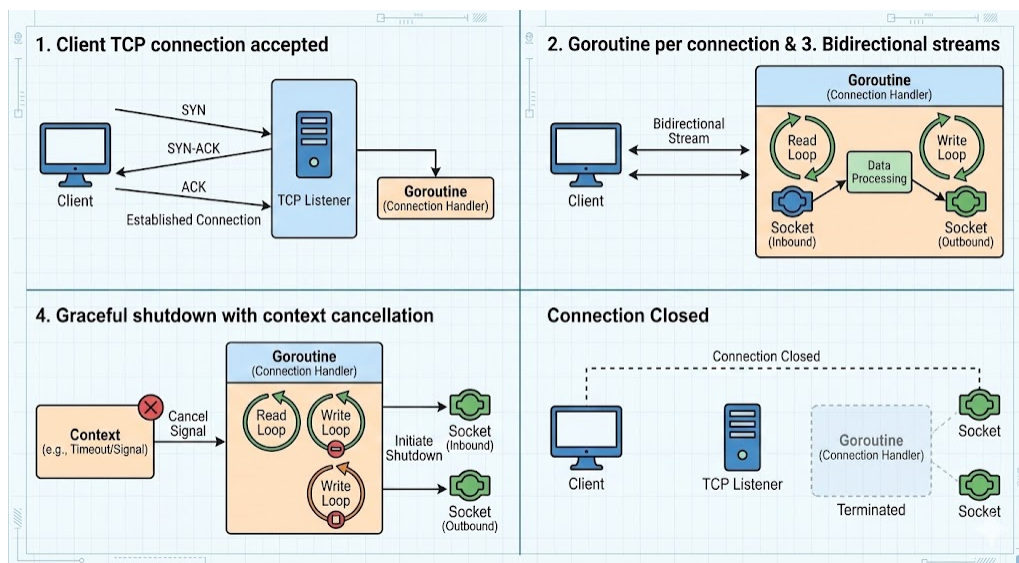


Figure 9.2: Lifecycle of a TCP connection in a Go-based server, showing connection acceptance, goroutine spawning, bidirectional data flow, and graceful shutdown via context cancellation.

- **net.Dial():** Establishes outbound TCP connection
- **net.Listen():** Creates listening socket for inbound connections
- **Accept():** Accepts incoming connection, returns new socket
- **Read()/Write():** Transfer data over established connection
- **Close():** Terminates connection gracefully

MySQL Protocol Socket Management Important responsibilities in zGate include:

- **Manual packet header reading:** Every MySQL packet begins with a 4-byte header:
 - Bytes 0-2: Payload length (24-bit little-endian integer, max 16MB)
 - Byte 3: Sequence ID (increments with each packet, wraps at 255)
- **Deadline management:** Network timeouts prevent hung connections:
 - `SetReadDeadline(time.Now().Add(timeout))`: Aborts read if no data arrives
 - `SetWriteDeadline(time.Now().Add(timeout))`: Aborts write if socket buffer is full
 - Deadlines are per-operation, not absolute timeouts

- **Zero-copy packet forwarding:** When packets don't require inspection or modification:
 - Direct buffer passing between client and server sockets
 - Avoids serialization/deserialization overhead
 - Reduces memory allocations and GC pressure
 - Uses `io.Copy()` or `io.CopyN()` for efficient transfer
- **Full connection lifecycle handling:**
 - **Handshake phase:** Initial authentication and capability negotiation
 - **Command phase:** Processing client commands (queries, prepared statements)
 - **Result phase:** Streaming result sets back to client
 - **Cleanup:** Proper resource release on connection termination

Buffer Management Efficient buffer handling is critical for performance:

- **Buffer pools:** Pre-allocated buffers using `sync.Pool` to reduce GC
- **Slice capacity management:** Careful pre-allocation to avoid repeated resizing
- **Memory reuse:** Buffers are reset and returned to pool after use
- **Large packet handling:** Special handling for packets exceeding 16MB (split across multiple packets)

9.1.5 Context Propagation

The `context.Context` package provides a standardized way to carry deadlines, cancellation signals, and request-scoped values across API boundaries and between goroutines.

Context Fundamentals **Context Interface:**

```
type Context interface {
    Deadline() (deadline time.Time, ok bool)
    Done() <-chan struct{}
    Err() error
    Value(key interface{}) interface{}
}
```

Context Creation Functions:

- **context.Background()**: Root context, never cancelled
- **context.TODO()**: Placeholder when context is unclear
- **context.WithCancel()**: Returns context with cancel function
- **context.WithDeadline()**: Cancels at specific time
- **context.WithTimeout()**: Cancels after duration
- **context.WithValue()**: Carries request-scoped data

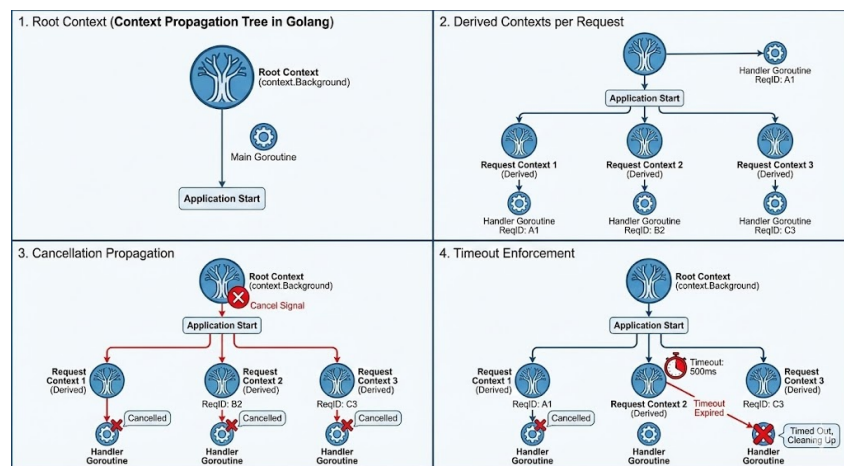


Figure 9.3: Hierarchical propagation of `context.Context` objects across goroutines, demonstrating timeout enforcement and cancellation signaling.

Context in zGate The proxy uses `context.Context` to ensure bounded execution and clean cancellation. Each incoming query obtains a context with:

- **Query ID**: Unique identifier for request tracing and correlation
- **Deadline and timeout settings**:
 - Default query timeout (e.g., 30 seconds)
 - User-specific timeout overrides
 - Inherited from client connection timeout if shorter
- **User identity and role information**:
 - Authenticated username
 - Active role assignments
 - Permission set
 - Session token information

- **Logging metadata:**
 - Source IP address
 - Client application identifier
 - Connection ID
 - Request timestamp

Cancellation Propagation Context cancellation terminates goroutines cleanly, avoiding resource leakage:

1. Client disconnects → context cancelled → all related goroutines notified
2. Query timeout exceeded → context cancelled → database connection interrupted
3. Admin kills session → context cancelled → graceful cleanup initiated
4. Shutdown signal received → root context cancelled → all sessions terminated

Best Practices in zGate:

- Always pass context as first parameter to functions
- Check `ctx.Done()` in long-running loops
- Use `select` to multiplex context cancellation with other operations
- Never store contexts in structs (pass explicitly)
- Defer cancellation function calls to prevent leaks

9.2 Database Wire Protocols (MySQL/MariaDB)

A distinguishing feature of zGate is its ability to “speak” the MySQL protocol directly, acting as a full proxy rather than a driver or middleware within the application layer.

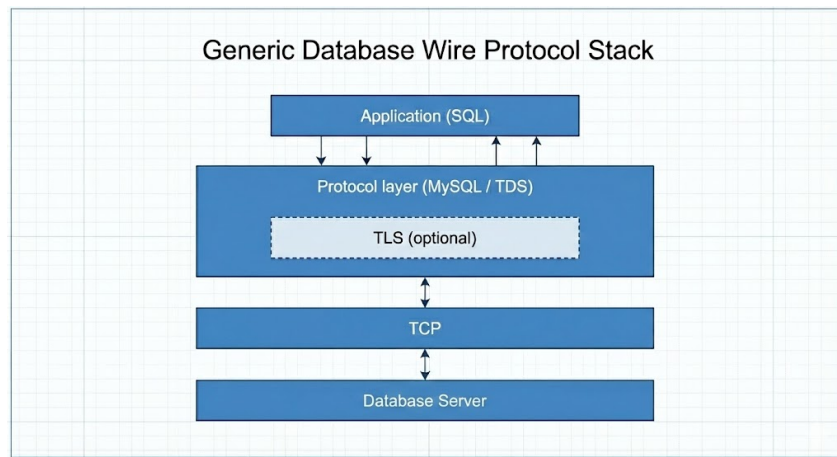


Figure 9.4: Layered view of database communication showing SQL semantics encapsulated within protocol-specific frames over optional TLS and TCP/IP.

9.2.1 MySQL Protocol Overview

The MySQL Client/Server Protocol is a binary protocol used for communication between MySQL clients and servers. It was originally designed in the 1990s and has evolved through multiple versions while maintaining backward compatibility.

Protocol Characteristics

- **Binary protocol:** Data is transmitted in compact binary format, not ASCII
- **Stateful:** Server and client maintain session state across multiple packets
- **Packet-oriented:** All communication happens in discrete packets
- **Sequential:** Packets within a command are numbered sequentially
- **Bidirectional:** Both client and server can initiate certain communications

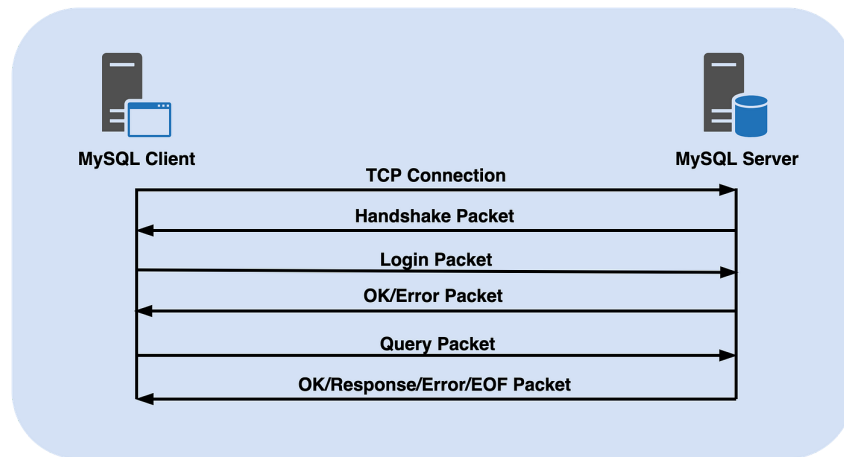


Figure 9.5: Connection and command phases of the MySQL protocol.

Protocol Phases

1. Connection Phase:

- Server sends initial handshake packet with capabilities and auth challenge
- Client responds with handshake response containing credentials
- Server sends OK or ERR packet

2. Command Phase:

- Client sends command packets (queries, prepared statements, etc.)
- Server responds with result sets, OK, or ERR packets
- Multiple commands can be sent over same connection

3. Termination Phase:

- Client sends COM_QUIT or closes connection
- Server releases resources and closes socket

9.2.2 MySQL Packet Structure

Each MySQL packet contains a precisely defined structure that must be correctly parsed and reconstructed by zGate.

MySQL Packet Structure & Logical Message Flow

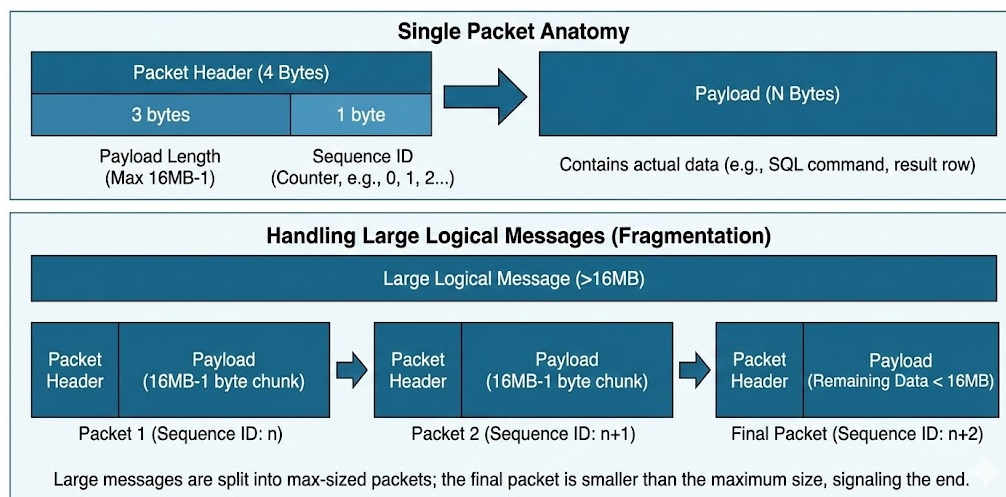


Figure 9.6: Structure of a MySQL protocol packet including payload length, sequence identifier, and payload data.

Packet Header Format

- **3-byte payload length** (little-endian):
 - Represents length of packet payload in bytes
 - Maximum value: 0xFFFFFFFF (16,777,215 bytes = 16MB - 1)
 - Length does NOT include the 4-byte header itself
 - Packets exceeding 16MB-1 are split into multiple packets
- **1-byte sequence ID:**
 - Starts at 0 for each new command
 - Increments by 1 for each packet in the sequence
 - Wraps around after 255 (rare in practice)
 - Used to detect packet loss or out-of-order delivery
 - Server and client must maintain synchronized counters
- **N-byte payload:**
 - Actual packet content (commands, result data, etc.)
 - Format depends on packet type
 - Can contain binary or text data

Large Packet Handling When payload exceeds 16MB-1 bytes:

1. First packet contains maximum payload (0xFFFFFFF bytes)
2. Sequence ID increments for continuation packet(s)
3. Last packet contains remaining data (length < 0xFFFFFFF)
4. Empty packet (length=0) sent if payload is exact multiple of 16MB-1

Packet Validation in zGate The gateway must validate and reconstruct packets precisely:

- **Length validation:** Ensure payload length matches actual bytes read
- **Sequence synchronization:** Track and validate sequence IDs on both sides
- **Buffer management:** Allocate appropriate buffers based on payload length
- **Error detection:** Identify corrupted or malformed packets
- **Large packet assembly:** Correctly reassemble multi-packet messages

9.2.3 Command Phase Processing

The MySQL protocol defines numerous command types that clients can send to servers. zGate must understand and properly handle each command type.

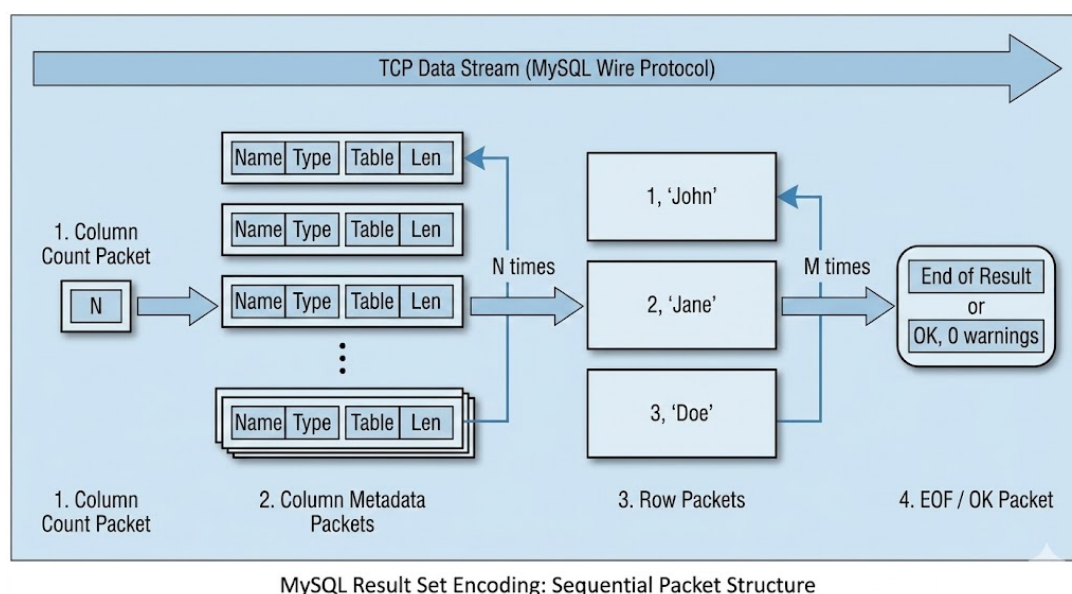


Figure 9.7: Logical structure of a MySQL result set including column count, column metadata packets, row packets, and termination packets.

Command Packet Format

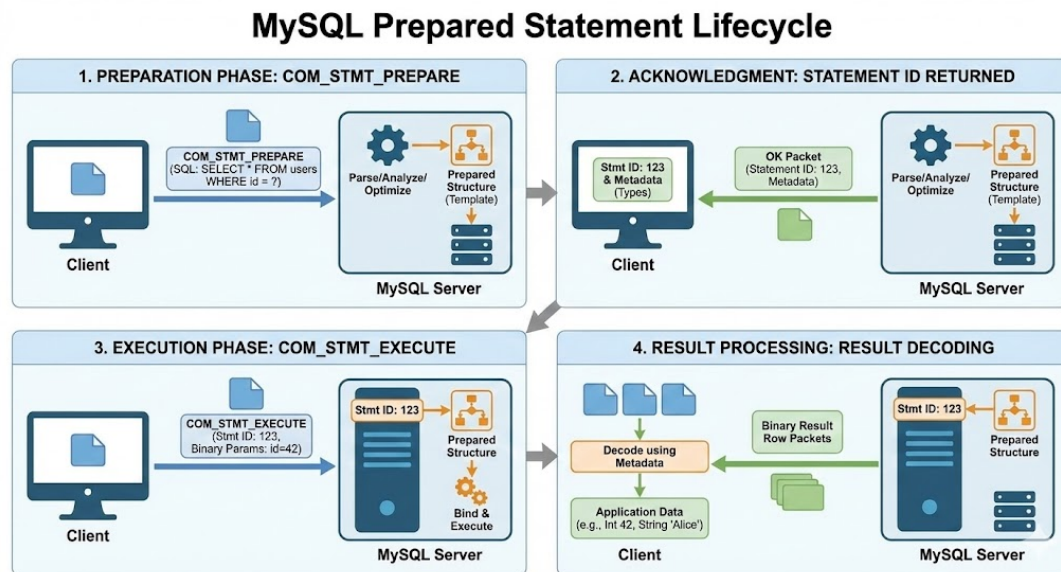


Figure 9.8: Lifecycle of a MySQL prepared statement showing SQL transmission during preparation and binary parameter binding during execution.

Supported Commands in zGate The proxy supports various MySQL commands:

- **COM_QUERY (0x03)**: Execute textual SQL statement
 - Payload: SQL string (NOT null-terminated)
 - Most common command type
 - Response: Result set, OK, or ERR packet
 - Example: `SELECT * FROM users WHERE id=1`
- **COM_INIT_DB (0x02)**: Change default database
 - Payload: Database name string
 - Similar to SQL: `USE database_name`
 - Response: OK or ERR packet
 - Updates session state in both client and server
- **COM_STMT_PREPARE (0x16)**: Prepare SQL statement
 - Payload: SQL statement with ? placeholders
 - Server parses and returns statement ID and parameter metadata
 - Enables binary protocol for faster execution
 - Statement cached on server until explicitly closed
- **COM_STMT_EXECUTE (0x17)**: Execute prepared statement

- Payload: Statement ID + flags + parameter values (binary encoded)
- Uses efficient binary protocol instead of text
- Parameters sent as native types (integers, dates, etc.)
- Response: Result set in text or binary protocol
- **COM_STMT_CLOSE (0x19)**: Deallocate prepared statement
 - Payload: Statement ID (4 bytes)
 - No response packet (fire-and-forget)
 - Frees server resources
- **COM_QUIT (0x01)**: Close connection
 - No payload
 - No response from server
 - Graceful connection termination
- **COM_PING (0x0E)**: Test connection liveness
 - No payload
 - Response: OK packet
 - Used for keepalive and health checks
- **COM_FIELD_LIST (0x04)**: List table columns (deprecated)
- **COM_STMT_RESET (0x1A)**: Reset prepared statement
- **COM_SET_OPTION (0x1B)**: Set connection options

Command Processing in zGate Each command type has unique processing requirements:

1. **Parse command byte**: Identify command type from first payload byte
2. **Extract command payload**: Read remaining packet data
3. **Policy enforcement**: Check RBAC permissions for this command
4. **SQL rewriting**: Modify query if needed (COM_QUERY only)
5. **Forward to backend**: Send modified or original packet to database
6. **Process response**: Intercept and potentially modify result
7. **Audit logging**: Record command execution details

9.2.4 Result Set Encoding

Result sets in MySQL protocol follow a specific multi-packet format that zGate must parse to enable data masking and filtering.

Result Set Packet Sequence A typical result set consists of:

1. **Column count packet:**

- Length-encoded integer indicating number of columns
- Example: 0x03 indicates 3 columns in result

2. **Column definition packets** (one per column):

- Catalog name (usually "def")
- Schema (database) name
- Table alias and original table name
- Column alias and original column name
- Character set encoding
- Column display width
- Column type (INT, VARCHAR, DATE, etc.)
- Column flags (NOT NULL, PRIMARY KEY, etc.)
- Decimal precision (for numeric types)

3. **EOF packet** (if CLIENT_DEPRECATED_EOF not set):

- Marks end of column definitions
- Contains server status flags and warning count

4. **Row data packets:**

- One packet per row
- Values encoded as length-encoded strings (text protocol) or native types (binary protocol)
- NULL represented as 0xFB byte in text protocol

5. **EOF or OK packet:**

- Marks end of result set
- Contains affected rows, last insert ID, status flags, warnings

Length-Encoded Integers MySQL uses a variable-length encoding for integers:

- **Value ; 251:** Single byte containing the value
- **Value = 251 (0xFB):** NULL value marker
- **Value = 252 (0xFC):** Next 2 bytes contain value (little-endian)
- **Value = 253 (0xFD):** Next 3 bytes contain value (little-endian)
- **Value = 254 (0xFE):** Next 8 bytes contain value (little-endian)

Text vs Binary Protocol **Text Protocol (COM_QUERY):**

- All values encoded as strings
- Dates, numbers, etc. formatted as text
- Less efficient but human-readable

Binary Protocol (COM_STMT_EXECUTE):

- Values in native binary format
- NULL bitmap at start of row
- Type-specific encoding (4-byte int, 8-byte double, etc.)
- More efficient, less parsing overhead

zGate Row Interception zGate intercepts row-level data for masking and policy enforcement, requiring:

- **Parsing length-encoded integers:** To determine string/value lengths
- **Reconstructing text and binary rows:** After applying masking rules
- **Preserving column metadata:** To understand data types for correct parsing
- **Maintaining packet integrity:** Recalculating payload lengths and sequence IDs
- **Streaming processing:** Handling large result sets without buffering all rows

Example Masking Scenario:

1. Client queries: `SELECT ssn, name FROM employees`
2. zGate parses column definitions, identifies "ssn" column

3. For each row packet:

- Parse length-encoded string for ssn value
- Apply masking: "123-45-6789" → "XXX-XX-6789"
- Reconstruct row packet with masked value
- Recalculate payload length
- Forward modified packet to client

9.2.5 SQL Parsing and AST Manipulation

To safely inject or modify SQL logic, zGate uses an SQL Abstract Syntax Tree (AST) approach rather than string manipulation.

SQL Abstract Syntax Trees An Abstract Syntax Tree is a tree representation of the syntactic structure of SQL code. Each node in the tree represents a construct in the SQL grammar.

Why AST vs String Manipulation:

- **Semantic understanding:** Parser understands SQL structure, not just text
- **Safe modification:** Changes preserve syntax validity
- **SQL injection prevention:** Prevents introduction of new SQL commands
- **Context awareness:** Distinguishes between identifiers, literals, keywords
- **Complexity handling:** Correctly processes nested queries, subqueries, CTEs

Parsing Pipeline

1. **Lexical Analysis (Tokenization):**

- Input: Raw SQL string
- Process: Break into tokens (keywords, identifiers, operators, literals)
- Output: Token stream
- Example: `SELECT name FROM users` → `[SELECT][name][FROM][users]`

2. **Syntax Analysis (Parsing):**

- Input: Token stream
- Process: Apply grammar rules to build AST
- Output: AST root node

- Detects syntax errors

3. AST Manipulation:

- Traverse tree using visitor pattern
- Inspect and modify nodes
- Add new nodes (e.g., WHERE clauses)
- Remove or replace nodes

4. Code Generation (Serialization):

- Input: Modified AST
- Process: Traverse tree and generate SQL text
- Output: Valid SQL string
- Preserves formatting where possible

AST Node Types Common node types in SQL AST:

- **SelectStmt**: SELECT query root
- **SelectExprList**: Target list (columns to return)
- **FromClause**: Table references
- **WhereClause**: Filter conditions
- **JoinExpr**: JOIN operations
- **BinaryExpr**: Binary operations (AND, OR, =, <, etc.)
- **FuncCall**: Function calls (COUNT, MAX, etc.)
- **Identifier**: Table and column names
- **Literal**: String, numeric, date literals
- **Subquery**: Nested SELECT statement

AST Manipulation in zGate zGate performs node-by-node inspection for:

- **Target list modification:**
 - Removing restricted columns from SELECT list
 - Adding computed columns for audit purposes
 - Replacing sensitive columns with masked expressions
 - Example: `SELECT ssn → SELECT CONCAT('XXX-XX-', SUBSTR(ssn, 8)) AS ssn`
- **WHERE clause enforcement:**
 - Injecting row-level security predicates
 - Adding tenant isolation filters
 - Enforcing time-based access controls
 - Example: User can only see their own records
 - Original: `SELECT * FROM orders`
 - Modified: `SELECT * FROM orders WHERE user_id = 'alice'`
- **Table-level permission checks:**
 - Identify all referenced tables
 - Verify user has access to each table
 - Check for column-level permissions
 - Reject queries accessing forbidden tables
- **Subquery processing:**
 - Recursively process nested queries
 - Apply same policies to subqueries
 - Handle correlated subqueries correctly
- **JOIN analysis:**
 - Check permissions on all joined tables
 - Inject filters on joined tables if needed
 - Prevent information leakage through joins

Safe Rewriting Examples Example 1: Column Masking

Original SQL: `SELECT ssn, name, salary FROM employees`

AST Modification:

- Locate "ssn" in `SelectExprList`
- Replace with `FuncCall` node: `mask_ssn(ssn)`

Generated SQL: `SELECT mask_ssn(ssn), name, mask_salary(salary)
FROM employees`

Example 2: Row Filtering

Original SQL: `SELECT * FROM medical_records`

AST Modification:

- Navigate to `WhereClause` (or create if absent)
- Add `BinaryExpr`: `department = 'cardiology'`

Generated SQL: `SELECT * FROM medical_records
WHERE department = 'cardiology'`

Example 3: Table Access Control

Original SQL: `SELECT * FROM hr.salaries`

AST Modification:

- Traverse tree, find table reference "hr.salaries"
- Check user's table permissions
- If denied: Return error before forwarding to database
- If allowed: Forward unmodified or with row filters

Security Benefits This AST approach avoids string-based manipulation, preventing:

- **SQL injection vulnerabilities:** Cannot introduce new SQL commands
- **Malformed query errors:** Generated SQL is always syntactically valid
- **Escaping issues:** No need to manually escape quotes and special characters
- **Context confusion:** Parser understands string literals vs identifiers
- **Logic errors:** Changes preserve query semantics

9.3 Database Wire Protocols (Microsoft SQL Server - TDS)

In addition to MySQL and MariaDB, modern enterprise environments frequently rely on Microsoft SQL Server. SQL Server communicates using the Tabular Data Stream (TDS) protocol, a proprietary, binary, application-layer protocol designed by Microsoft. Understanding TDS is essential for building protocol-aware proxies, intrusion detection systems, or database firewalls that operate transparently at the network level.

Unlike MySQL, which exposes relatively straightforward packet structures, TDS is more complex, stateful, and tightly coupled with SQL Server's execution engine.

9.3.1 Overview of the TDS Protocol

Tabular Data Stream (TDS) is a message-oriented protocol layered directly on top of TCP. It defines how SQL Server clients and servers exchange authentication data, SQL batches, procedure calls, metadata, and result sets.

Key Characteristics

- **Binary protocol:** All messages are encoded in binary, not text
- **Token-based:** Responses consist of a stream of typed tokens
- **Stateful:** Session context persists across requests
- **Versioned:** Multiple protocol versions (TDS 4.2 – 7.4+)
- **Tightly integrated:** Closely coupled to SQL Server execution semantics

Protocol Versions Common TDS versions include:

- **TDS 4.2:** Legacy Sybase / early SQL Server
- **TDS 7.0:** SQL Server 7.0
- **TDS 7.1:** SQL Server 2000
- **TDS 7.2:** SQL Server 2005
- **TDS 7.3:** SQL Server 2008 / 2012
- **TDS 7.4+:** SQL Server 2014+

Modern SQL Server installations primarily use TDS 7.4.

9.3.2 TDS Packet Structure

All TDS communication occurs as a sequence of packets, each with a fixed-size header followed by a variable-length payload.

TDS Packet Header

-----+	-----+	-----+	-----+	-----+	-----+	-----+	-----+
Type	Status	Length	SPID	Packet	Window		
(1B)	(1B)	(2B)	(2B)	(1B)	(1B)		
-----+	-----+	-----+	-----+	-----+	-----+	-----+	-----+

- **Type (1 byte):**
 - Identifies message category
 - Examples:
 - * 0x01 – SQL Batch
 - * 0x02 – Pre-login
 - * 0x04 – RPC
 - * 0x10 – Login
 - * 0x12 – Response
- **Status (1 byte):**
 - Bit flags indicating packet role
 - End-of-message (EOM) flag
 - Indicates whether more packets follow
- **Length (2 bytes):**
 - Total packet length including header
 - Big-endian integer
 - Max size typically 4KB or 8KB (configurable)
- **SPID (2 bytes):**
 - Server Process ID
 - Identifies server-side session
- **Packet ID (1 byte):**
 - Sequence number for packet ordering

- **Window (1 byte):**
 - Legacy field (unused in modern TDS)

Packet Fragmentation If a message exceeds the negotiated packet size:

- The message is split across multiple TDS packets
- Status byte marks continuation or end-of-message
- Receiver must reassemble payload before processing

9.3.3 Pre-Login and Login Process

Before authentication, TDS performs a pre-login negotiation phase.

Pre-Login Phase

The pre-login packet negotiates session parameters:

- Encryption support (required, optional, disabled)
- TDS protocol version
- Packet size
- Instance name
- Thread ID

The server responds with its supported options. If encryption is required or requested, TLS negotiation begins immediately after pre-login.

TLS Integration Unlike MySQL, SQL Server embeds TLS negotiation inside the TDS flow:

- TLS handshake occurs after pre-login
- All subsequent TDS packets are encrypted
- Proxy must detect transition from plaintext to TLS
- Requires full TLS interception or passthrough

Login Phase

The LOGIN7 packet contains authentication information:

- Username
- Password (obfuscated, not encrypted unless TLS active)
- Database name
- Client hostname
- Application name
- Language and collation

Password Obfuscation Without TLS:

- Password bytes are XOR-obfuscated
- Each byte rotated and XORed with 0xA5
- This is **not encryption** and offers no real security

Therefore, TLS is mandatory in secure deployments.

9.3.4 TDS Message Types

TDS supports multiple message categories:

- **SQL Batch:**
 - Contains raw SQL text
 - Similar to MySQL COM_QUERY
 - Executed as a single batch
- **RPC (Remote Procedure Call):**
 - Used for stored procedure execution
 - Includes procedure name or ID
 - Parameters encoded in binary
- **Attention:**
 - Cancels currently executing query
- **Bulk Load:**
 - High-speed data import
 - Streams row data efficiently

9.3.5 TDS Token-Based Response Model

Unlike MySQL's fixed result-set structure, TDS responses consist of a stream of typed tokens.

Token Stream Concept A server response is a sequence of tokens:

- Each token begins with a 1-byte token type
- Followed by token-specific payload
- Tokens are processed sequentially

Common TDS Tokens

- **COLMETADATA:**
 - Column count
 - Column names
 - Data types
 - Precision, scale, collation
- **ROW:**
 - Row data in binary format
 - Values encoded according to column metadata
- **DONE / DONEPROC / DONEINPROC:**
 - Indicates completion of batch or procedure
 - Includes affected row count
 - Includes status flags
- **ERROR:**
 - Error number
 - Severity
 - Message text
 - Line number
- **INFO:**
 - Informational messages
 - PRINT output
 - Warnings

9.3.6 Data Type Encoding

SQL Server uses strongly typed binary encodings.

Fixed-Length Types

- INT: 4 bytes
- BIGINT: 8 bytes
- FLOAT: 8 bytes
- BIT: 1 byte

Variable-Length Types

- VARCHAR / NVARCHAR:
 - Length prefix
 - UTF-16LE encoding for NVARCHAR
- VARBINARY:
 - Length-prefixed byte array

NULL Representation

- NULL values represented by length = 0xFFFF or special markers
- Requires metadata awareness to parse correctly

9.3.7 Implications for Proxy Design

Building a TDS-aware proxy introduces significant complexity:

- Stateful token stream parsing
- Precise metadata tracking
- Binary data manipulation
- TLS-in-band protocol switching
- Procedure call interception
- Multi-result-set handling

Comparison with MySQL

- MySQL: Command-based, simpler framing
- TDS: Token-stream-based, execution-aware
- MySQL: Text-heavy protocol
- TDS: Fully binary and metadata-driven

This complexity makes TDS significantly harder to intercept, modify, or rewrite safely, and explains why many database security solutions operate only at the SQL Server driver or API layer rather than at the wire protocol level.

9.3.8 Challenges of Abstract Syntax Tree (AST) Parsing in TDS

While parsing raw TDS packets enables visibility into SQL Server traffic, constructing an Abstract Syntax Tree (AST) from intercepted queries presents a significantly deeper technical challenge. AST parsing is required for fine-grained policy enforcement, such as restricting specific SQL operations, table access, or predicate-level filtering.

Unlike traditional SQL parsing from application logs, TDS introduces several protocol-level obstacles that complicate AST construction.

Absence of a Canonical SQL Text Representation

In TDS, SQL commands may be transmitted in different forms:

- Raw SQL batches (SQL Batch packets)
- Remote Procedure Calls (RPCs)
- Parameterized prepared statements

In the case of RPC execution:

- SQL text may not be present at all
- Only a procedure identifier and binary-encoded parameters are transmitted
- Logical intent must be inferred from metadata and procedure definitions

This breaks the assumption that a proxy always has access to a full SQL string suitable for lexical analysis.

Binary Parameter Encoding and Late Binding

TDS encodes parameters in binary form according to their SQL Server data types. Unlike text-based SQL where literals appear inline, TDS separates query structure from values.

Consequences include:

- Parameters are position-based, not name-based
- Parameter types are defined by metadata tokens
- Values are resolved only at execution time

As a result, an AST parser must:

- Reconstruct the query template
- Bind parameters to placeholders
- Track type information across token boundaries

This is closer to compiler intermediate representation reconstruction than traditional SQL parsing.

Multiple Result Sets and Control Flow Semantics

SQL Server allows complex batches containing:

- Multiple SELECT statements
- Conditional logic (IF, WHILE)
- Temporary tables
- Stored procedure calls returning nested result sets

In TDS, these constructs produce:

- Interleaved COLMETADATA and ROW tokens
- DONE, DONEPROC, and DONEINPROC tokens signaling execution stages

From an AST perspective:

- Execution flow is non-linear
- Query boundaries are not explicitly marked
- Semantic meaning emerges only after full token stream analysis

This requires stateful parsing and execution-context tracking.

Encrypted Sessions and Visibility Loss

Once TLS is negotiated:

- All TDS payloads are encrypted
- AST parsing becomes impossible without TLS termination

This introduces architectural trade-offs:

- TLS passthrough preserves security but eliminates visibility
- TLS interception enables parsing but expands the trusted computing base

From a technical standpoint, AST parsing in TDS is inseparable from cryptographic session management.

Lack of Official Grammar Specifications

Microsoft does not provide a complete, formal SQL grammar aligned with TDS execution semantics.

Challenges include:

- SQL Server-specific extensions (TOP, MERGE, APPLY)
- Version-dependent behavior
- Undocumented token combinations

Therefore, AST parsers must rely on:

- Reverse engineering
- Empirical testing
- Partial grammars adapted from T-SQL documentation

This increases maintenance complexity and version fragility.

Implications for Security Policy Enforcement

Due to the challenges above, enforcing policies at the AST level in TDS requires:

- Hybrid parsing strategies (syntax + metadata)
- Conservative fallbacks (deny on ambiguity)
- Awareness of false positives and negatives

This explains why many commercial database security products:



- Operate at the driver level
- Rely on SQL Server auditing APIs
- Avoid full wire-level AST parsing

9.4 Cryptography and Security Engineering

Security is the cornerstone of the system. The implementation combines modern cryptographic standards with secure design principles to protect data in transit, at rest, and during processing.

9.4.1 TLS / SSL Transport Layer Security

Transport Layer Security (TLS) and its predecessor Secure Sockets Layer (SSL) are cryptographic protocols designed to provide secure communication over a computer network.

TLS Protocol Overview Historical Context:

- **SSL 1.0:** Never publicly released (Netscape, 1994)
- **SSL 2.0:** Released 1995, deprecated 2011 (security flaws)
- **SSL 3.0:** Released 1996, deprecated 2015 (POODLE attack)
- **TLS 1.0:** Released 1999, deprecated 2020
- **TLS 1.1:** Released 2006, deprecated 2020
- **TLS 1.2:** Released 2008, still widely used
- **TLS 1.3:** Released 2018, current standard

Security Properties:

- **Confidentiality:** Data encrypted using symmetric cipher
- **Integrity:** MAC (Message Authentication Code) prevents tampering
- **Authentication:** X.509 certificates verify server/client identity
- **Forward secrecy:** Session keys not derivable from long-term keys

TLS Handshake Process The TLS handshake establishes a secure session:

1. Client Hello:

- Supported TLS versions
- Cipher suites (encryption algorithms)
- Random nonce

- Supported extensions

2. Server Hello:

- Selected TLS version
- Selected cipher suite
- Server random nonce
- Server certificate (X.509)

3. Key Exchange:

- Client verifies server certificate
- Ephemeral Diffie-Hellman key exchange (TLS 1.3)
- Or RSA key exchange (TLS 1.2)
- Derive master secret

4. Finished Messages:

- Both sides send encrypted "Finished" message
- Proves they derived same keys
- Handshake complete, application data can flow

TLS in zGate TLS is used to secure:

- **Client–Gateway communication:**
 - MySQL clients connect via TLS-encrypted connections
 - Optional mutual TLS (mTLS) for client authentication
 - Certificate-based authentication instead of passwords
- **Gateway–Database communication:**
 - Backend connections always use TLS
 - Verifies database server certificate
 - Protects credentials and query data in transit
- **Administrative API (HTTPS):**
 - REST API served over HTTPS
 - Admin credentials encrypted in transit
 - API tokens protected from network sniffing

Technical Implementation Details Technical aspects in zGate include:

- **Loading X.509 certificates from PEM:**
 - PEM (Privacy Enhanced Mail) format: Base64-encoded DER certificates
 - Private keys protected with passphrase encryption
 - Certificate chain loading (intermediate + root CAs)
 - Key pair validation (public key matches private key)
- **Enforcing TLS 1.2+:**
 - Minimum version set in `tls.Config`
 - Rejects connections using SSL 3.0, TLS 1.0, TLS 1.1
 - Prevents downgrade attacks
- **Certificate chain validation:**
 - Using Go's `crypto/x509` package
 - Verifies certificate signature chain to trusted root CA
 - Checks expiration dates
 - Validates hostname/IP against certificate SAN (Subject Alternative Names)
 - Checks certificate revocation status (OCSP or CRL)
- **Cipher suite selection:**
 - Prefer AEAD ciphers (AES-GCM, ChaCha20-Poly1305)
 - Disable weak ciphers (RC4, 3DES, export ciphers)
 - Enable perfect forward secrecy (ECDHE key exchange)
 - Example strong cipher: `TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384`
- **Optional client certificate verification (mTLS):**
 - Server requests client certificate during handshake
 - Client presents certificate signed by trusted CA
 - Server validates certificate and extracts identity (CN or SAN)
 - Used for passwordless authentication
 - Common in service-to-service authentication

9.4.2 Symmetric Encryption (AES-256)

Advanced Encryption Standard (AES) is a symmetric block cipher adopted as a standard by NIST in 2001, replacing the older DES algorithm.

AES Fundamentals Algorithm Properties:

- **Block cipher:** Encrypts fixed-size blocks (128 bits)
- **Symmetric:** Same key for encryption and decryption
- **Key sizes:** 128, 192, or 256 bits
- **Rounds:** 10 (AES-128), 12 (AES-192), 14 (AES-256)
- **Speed:** Hardware-accelerated on modern CPUs (AES-NI instructions)

AES Operations:

1. **SubBytes:** Substitute each byte using S-box
2. **ShiftRows:** Rotate rows of state array
3. **MixColumns:** Linear transformation of columns
4. **AddRoundKey:** XOR with round key

AES Modes of Operation Block ciphers require a "mode" to encrypt data longer than one block:

- **ECB (Electronic Codebook):** INSECURE, never use
 - Each block encrypted independently
 - Identical plaintexts produce identical ciphertexts
 - Reveals patterns in data
- **CBC (Cipher Block Chaining):** Legacy, requires padding
 - Each block XORed with previous ciphertext
 - Requires padding to block boundary
 - Vulnerable to padding oracle attacks if not careful
 - Needs separate MAC for authentication
- **GCM (Galois/Counter Mode):** Recommended

- AEAD (Authenticated Encryption with Associated Data)
- Provides both confidentiality and authenticity
- No padding required
- Parallelizable (fast)
- Includes authentication tag to detect tampering
- **CCM, EAX, OCB:** Alternative AEAD modes

AES-256-GCM in zGate The internal SQLite datastore is encrypted using AES-256-GCM. The system uses:

- **32-byte (256-bit) encryption keys:**
 - Derived from master password using key derivation function
 - Or generated randomly for data-at-rest encryption
 - Stored in secure key management system or HSM
 - Never logged or exposed in API responses
- **CSPRNG-generated nonces:**
 - Nonce (Number Used Once) = Initialization Vector (IV)
 - 12 bytes (96 bits) for GCM mode
 - Must be unique for every encryption operation with same key
 - Generated using cryptographically secure random number generator
 - Stored alongside ciphertext (not secret)
 - Nonce reuse catastrophically breaks GCM security
- **Authentication tag:**
 - 16-byte tag appended to ciphertext
 - Verifies data integrity and authenticity
 - Prevents tampering and bit-flipping attacks
 - Decryption fails if tag doesn't match
- **Associated Data (AD):**
 - Additional authenticated but unencrypted data
 - Example: database record ID, timestamp, version
 - Binds ciphertext to specific context

- Prevents ciphertext from being moved/reused elsewhere
- **Key rotation support:**
 - Periodic key changes (e.g., quarterly)
 - Re-encrypt data with new keys
 - Multiple active keys during transition period
 - Track which key version encrypted each record
 - Old keys archived securely for decryption of historical data

9.4.3 JSON Web Tokens (RFC 7519)

JSON Web Tokens (JWT) are an open standard (RFC 7519) for representing claims securely between two parties as a compact, URL-safe string. JWTs have become the predominant mechanism for stateless authentication in REST APIs, replacing older server-side session stores.

Structure A JWT consists of three Base64URL-encoded segments separated by periods:

`<Base64URL(header)>.<Base64URL(payload)>.<Signature>`

- **Header:** A JSON object specifying the token type (`typ: "JWT"`) and the signing algorithm (`alg`). Common values are `HS256` (HMAC-SHA256, symmetric) and `RS256` (RSA with SHA-256, asymmetric).
- **Payload:** A JSON object containing *claims* — statements about an entity and additional metadata. Registered claim names standardised by the RFC include:
 - `iss` (issuer): identifies the principal that issued the token
 - `sub` (subject): the entity the token refers to (usually a user ID)
 - `aud` (audience): intended recipients
 - `exp` (expiration time): Unix timestamp after which the token must be rejected
 - `iat` (issued at): Unix timestamp of token creation
 - `jti` (JWT ID): unique identifier enabling revocation tracking

Private claims (e.g., `roles`, `session_id`) carry application-specific data.

- **Signature:** Produced by signing the concatenated header and payload with the chosen algorithm. For HMAC-SHA256:



```
HMAC-SHA256(  
  Base64URL(header) + "." + Base64URL(payload),  
  secret  
)
```

The signature guarantees that the token was issued by a party holding the secret and that neither the header nor the payload has been tampered with after issuance.

Stateless Verification A server that holds the signing key can verify any token without querying a database: decode the header, select the appropriate key, recompute the signature, and compare. This property enables horizontally scaled services to authenticate requests independently. The trade-off is that a token cannot be invalidated before its `exp` claim without additional server-side state.

Access and Refresh Token Pattern To balance security and usability, systems typically issue two tokens simultaneously:

- **Access token:** Short-lived (typically 5–15 minutes); included in every API request as a `Bearer` token in the `Authorization` header; validated purely by signature without database lookup.
- **Refresh token:** Long-lived (hours to days); opaque (not a JWT) or a JWT with `jti` stored server-side; exchanged at a dedicated endpoint to obtain a new access token; can be revoked by deleting its server-side record.

When a refresh token is used to issue a new access token, *refresh token rotation* replaces the old refresh token with a new one, limiting the exposure window if a refresh token is stolen.

Security Considerations Known attack vectors include:

- **Algorithm confusion (`alg: none`):** early libraries accepted unsigned tokens if the header declared `alg: none`; mitigated by explicitly allowlisting permitted algorithms server-side.
- **Key confusion (`RS256` → `HS256`):** an attacker crafts an HS256 token signed with the server's *public* RSA key; mitigated by binding the expected algorithm to each key identifier.
- **Insecure storage:** storing JWTs in `localStorage` exposes them to XSS; storing in `HttpOnly` cookies prevents JavaScript access.

In zGate, access tokens carry a 5-minute expiry and embed the user's roles; refresh tokens are stored server-side with their `jti`, enabling immediate revocation when an administrator terminates an active session.

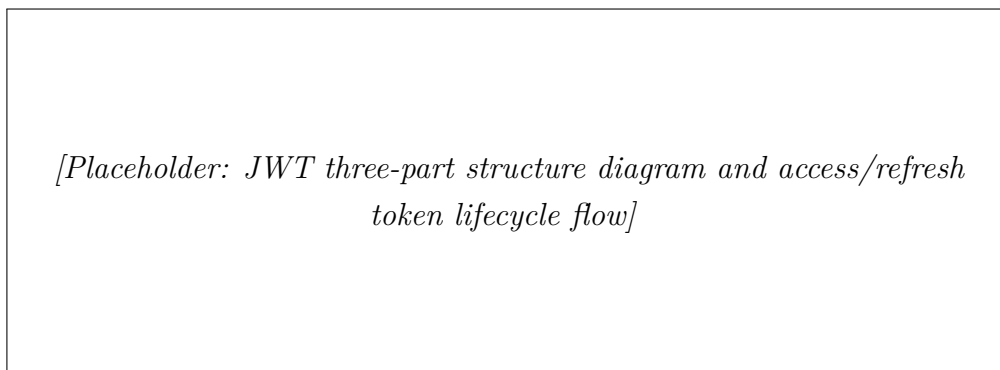


Figure 9.9: JWT structure (header.payload.signature) and the access/refresh token refresh lifecycle.

9.4.4 Time-Based One-Time Passwords (RFC 6238)

Multi-factor authentication (MFA) adds a second verification factor beyond a static password. The Time-Based One-Time Password (TOTP) algorithm, standardised in RFC 6238, generates short-lived numeric codes that an authenticator application (e.g., Google Authenticator, Authy) produces without network connectivity.

HMAC-Based OTP Foundation (HOTP — RFC 4226) TOTP builds on HOTP. Given a shared secret key K and a counter C , HOTP produces a code as follows:

1. Compute $HS = \text{HMAC-SHA1}(K, C)$ — a 20-byte hash.
2. Apply *dynamic truncation*: extract a 4-byte subset starting at the byte offset indicated by the last nibble of HS .
3. Interpret those 4 bytes as a 31-bit unsigned integer P .
4. Return $P \bmod 10^d$ where d is the desired digit length (typically 6).

Time-Step Derivation in TOTP TOTP replaces the counter C with a time-derived value:

$$T = \left\lfloor \frac{t - T_0}{X} \right\rfloor$$

where t is the current Unix time, $T_0 = 0$ is the epoch, and $X = 30\text{s}$ is the step duration. The resulting integer T increments every 30 seconds, ensuring that the one-time code changes frequently while remaining stable within its window.

Shared Secret Provisioning

 During enrolment:

1. The server generates a cryptographically random secret (typically 160 bits / 20 bytes).
2. The secret is Base32-encoded and embedded in an `otpauth://totp/` URI alongside the issuer name and account identifier.
3. The URI is rendered as a QR code; the user scans it with their authenticator application.
4. After provisioning, the secret is stored server-side (ideally in a secrets manager) and *never transmitted again*.

Clock Skew Tolerance Minor clock differences between client and server can cause a valid code to be rejected. Implementations typically accept codes for the windows $T - 1$, T , and $T + 1$ (a ± 30 -second tolerance), balancing usability against replay-attack exposure.

Security Properties

- **Replay resistance:** each code is valid for a single 30-second window; a captured code is useless immediately after its window closes.
- **No credential transmission:** after initial provisioning, authentication requires only the time-derived code — the secret itself is never sent over the network.
- **Phishing limitation:** TOTP does not bind the code to a specific origin; it is susceptible to real-time phishing attacks that relay the code to the legitimate server within the 30-second window. This limitation motivates stronger second factors such as FIDO2/WebAuthn for high-assurance contexts.

In zGate, administrator accounts on the WebUI use TOTP as a second factor implemented with the `pquerna/otp` Go library; CLI users authenticate via Keycloak's OIDC device flow, which handles TOTP verification within the Keycloak realm.

[Placeholder: TOTP time-step diagram — Unix timestamp divided into 30-second intervals, each yielding a distinct OTP code]

Figure 9.10: TOTP 30-second time-step derivation: each interval produces a unique code from the shared secret and the current time counter T .

9.4.5 Proof-of-Work Anti-Bot Mechanisms

Proof-of-Work (PoW) is a class of protocol in which one party (the prover) demonstrates that it has expended a quantifiable amount of computational effort. PoW was first formalised by Dwork and Naor (1992) as a mechanism to deter email spam, and later popularised by Bitcoin’s Hashcash-derived mining algorithm. Applied to web authentication, PoW discourages automated credential-stuffing and brute-force attacks by imposing a computational cost on every submission.

SHA-256 Proof-of-Work The challenge–response protocol operates as follows:

1. **Challenge issuance:** the server generates a random `salt` and selects a difficulty parameter `maxNumber`. It returns the challenge object `{algorithm, salt, maxNumber}`.
2. **Solution search:** the client iterates a nonce n from 0 upward, computing $h = \text{SHA-256}(\text{salt} \parallel n)$ at each step, until h satisfies the target condition. The expected number of iterations before success is approximately `maxNumber/2`.
3. **Solution submission:** the client returns n .
4. **Verification:** the server recomputes $\text{SHA-256}(\text{salt} \parallel n)$ and checks the target condition — a single hash operation, regardless of the difficulty level.

The asymmetry is the key security property: verification is $O(1)$ while solution search is $O(\text{maxNumber}/2)$.

HMAC Signature on the Challenge Without integrity protection, a malicious client could fabricate a challenge with a trivial target, solve it instantly, and submit a valid-looking response. To prevent this, the server signs the challenge parameters with HMAC-SHA256 using a private key:

$$\text{signature} = \text{HMAC-SHA256}(\text{secret}, \text{algorithm} \parallel \text{salt} \parallel \text{maxNumber})$$

The signature is included in the challenge object returned to the client. The server verifies this signature upon receiving the solution before checking the PoW; any tampered or fabricated challenge is rejected outright.

Comparison with CAPTCHA Traditional CAPTCHA systems require users to interpret distorted images or identify objects — tasks that rely on human perception. PoW shifts the burden to computation:

- No user interaction required; computation runs in the background (JavaScript Web Worker).
- No third-party service dependencies that introduce privacy concerns.
- Difficulty scales linearly with `maxNumber`, enabling server-side tuning without changing the client.
- Modern GPU/ASIC hardware can solve SHA-256 PoW faster than consumer CPUs; difficulty must be calibrated accordingly.

In zGate, the ALTCHA open-source PoW specification is implemented on the admin login endpoint and integrated as a Keycloak SPI (Service Provider Interface) plugin that fronts the Keycloak login form. The HMAC key is shared between both components via environment configuration.

[Placeholder: ALTCHA challenge-response sequence diagram — server issues challenge with HMAC signature → client iterates SHA-256 → server verifies HMAC then SHA-256]

Figure 9.11: ALTCHA SHA-256 proof-of-work challenge-response protocol with HMAC-signed challenge to prevent forgery.

9.5 Zero Trust Security Architecture

Zero Trust is a security model predicated on the principle that no actor, system, or network should be implicitly trusted — regardless of whether it originates inside or outside the traditional network perimeter. The term was coined by John Kindervag at Forrester Research in 2010 and subsequently formalised in NIST Special Publication 800-207 (2020).

9.5.1 Origins and Motivation

The classical perimeter security model partitions networks into trusted internal zones (protected by firewalls and VPNs) and untrusted external zones. Once a connection is established inside the perimeter, it typically receives broad lateral access. This model presents a fundamental vulnerability: a single compromised internal host — whether through phishing, credential theft, or supply-chain attack — can move laterally across the network with minimal resistance. The 2013 Target breach, the 2020 SolarWinds attack, and numerous subsequent incidents all exploited this trust boundary.

Modern environments further erode perimeter assumptions:

- Workforce mobility: users connect from personal devices and untrusted networks.
- Cloud workloads: resources reside outside any physical perimeter.
- Microservices: service-to-service communication bypasses traditional network controls.
- Third-party integrations: supply-chain partners require access without full trust.

Zero Trust responds by eliminating the concept of a trusted zone entirely.

9.5.2 Core Tenets

NIST SP 800-207 identifies seven tenets of Zero Trust:

1. **All data sources and computing services are considered resources.** No device category is inherently trusted.
2. **All communication is secured regardless of network location.** Traffic on internal networks receives the same scrutiny as external traffic; encryption and authentication are mandatory on every hop.
3. **Access to individual resources is granted per-session.** Trust is not persistent; each access request is evaluated independently.

4. **Access to resources is determined by dynamic policy.** Policy considers identity, device health, geolocation, time, and behavioural signals.
5. **The enterprise monitors and measures the integrity and security posture of all assets.**
6. **All resource authentication and authorisation are dynamic and strictly enforced before access is allowed.**
7. **The enterprise collects as much information as possible about the current state of assets, network infrastructure, and communications.**

9.5.3 Architectural Components

Zero Trust architecture introduces two logical planes:

Policy Enforcement Point (PEP) The PEP sits in the data path between the subject (user or workload) and the protected resource. Every access request must pass through the PEP. The PEP does not make access decisions itself; it forwards the request context to the Policy Decision Point and enforces the resulting decision.

Policy Decision Point (PDP) The PDP evaluates the request context against the active policy. Context includes:

- **Identity signals:** verified identity of the requesting subject, role memberships, authentication method and assurance level.
- **Device signals:** device health, patch level, managed vs. unmanaged status.
- **Behavioural signals:** time of day, geographic location, deviation from normal access patterns.
- **Resource sensitivity:** classification of the target resource.

The PDP returns an allow, deny, or step-up (require additional authentication) decision. In practice the PDP is often composed of a Policy Engine (rule evaluator) and a Policy Administrator (credential and session issuer).

9.5.4 Contrast with Perimeter Security

- **Perimeter model:** authenticate once at the boundary; trust everywhere inside; network location as a proxy for trustworthiness.
- **Zero Trust:** authenticate every request; assume the internal network is hostile; resource access based on identity + context, not network location.

- **Lateral movement resistance:** in Zero Trust, compromising one internal host grants no additional access to other resources because each resource independently verifies the accessor's identity and permissions.

9.5.5 Zero Trust Applied to Database Access

Databases are high-value targets: they aggregate PII, financial records, intellectual property, and credentials. Traditional database access controls rely on network-level trust (only hosts on the internal VLAN may connect to port 3306), which is easily bypassed once an attacker gains a foothold inside the network.

Applying Zero Trust principles to database access requires:

- **Identity-based authentication at the gateway:** every connection request carries a verifiable identity token; the network address of the client is irrelevant.
- **Just-In-Time (JIT) credential provisioning:** database credentials are issued only for the duration of an active session; ephemeral accounts are created at session start and destroyed at session end, eliminating persistent credentials that could be exfiltrated.
- **Credential separation:** end users authenticate to the gateway layer with their own credentials; the gateway then uses its own privileged credentials to connect to the backend database. Users never possess or observe real database passwords.
- **Continuous session monitoring:** active sessions are tracked in a server-side registry. Administrators can revoke sessions in real time, even before the underlying access token expires.
- **Microsegmentation:** backend database ports are accessible only from the gateway host; clients reach databases exclusively through the gateway's TCP port. Network-level access to the database is removed even for internal hosts.
- **Comprehensive audit logging:** every query, connection event, and policy decision is logged with the verified identity of the requestor and the full query text (or hash).

[Placeholder: Zero Trust database access model — client presents identity token to PEP (gateway), PDP evaluates policy, JIT credential issued, gateway proxies to backend DB; backend DB not reachable directly]

Figure 9.12: Zero Trust database access model: every connection traverses the policy enforcement gateway; backend database ports are not directly reachable from client networks.

9.6 Identity and Access Management

Identity and Access Management (IAM) encompasses the policies, technologies, and processes that ensure the right entities have appropriate access to the right resources under the right conditions.

9.6.1 Role-Based Access Control

Role-Based Access Control (RBAC) is an access control model in which permissions are assigned to roles rather than directly to individual users. The formal model was standardised in NIST INCITS 359-2004.

Formal Model RBAC defines four core sets and their relationships:

- U : the set of users (human or service accounts).
- R : the set of roles (named groups of permissions).
- P : the set of permissions (operations on protected objects).
- S : the set of sessions (each session activates a subset of a user's assigned roles).
- $UA \subseteq U \times R$: user–role assignment relation.
- $PA \subseteq P \times R$: permission–role assignment relation.

A user's effective permissions in a session are the union of permissions associated with all active roles in that session.

Role Hierarchy Hierarchical RBAC (RBAC1) extends the model with a partial order $\geq$ on roles. If $r_1 \geq r_2$, role r_1 *inherits* all permissions of r_2 . This reduces administrative overhead: a senior role automatically accumulates permissions from subordinate roles without explicit re-assignment. For example, a DBA role inheriting from DEVELOPER inherits all developer permissions plus its own elevated permissions (e.g., schema modification).

Separation of Duties Separation of Duties (SoD) prevents a single user from accumulating conflicting privileges. Static SoD constrains role assignments (UA) so that certain role pairs can never be simultaneously assigned to the same user. Dynamic SoD constrains active sessions so that certain role pairs cannot be simultaneously activated. Both forms prevent fraud scenarios such as a single actor who can both initiate and approve financial transactions.

Enforcement at the Policy Layer A key architectural property of RBAC systems is that enforcement occurs at the access control layer — a policy enforcement point external to the individual resources. This enables uniform enforcement across heterogeneous backends: a single RBAC policy can govern access to MySQL, PostgreSQL, and MSSQL databases without configuring native grants on each.

9.6.2 OAuth 2.0 and OpenID Connect

OAuth 2.0 (RFC 6749) and OpenID Connect (OIDC 1.0) are the dominant standards for delegated authorisation and federated identity in modern systems.

OAuth 2.0 OAuth 2.0 is an authorisation framework that enables a *resource owner* (user) to grant a *client* application limited access to a *resource server* on the owner's behalf, without exposing the owner's credentials to the client. The framework defines four roles:

- **Resource Owner:** the entity (typically a user) capable of granting access.
- **Client:** the application requesting access.
- **Resource Server:** the server hosting protected resources.
- **Authorisation Server:** issues access tokens after authenticating the resource owner and obtaining consent.

RFC 6749 defines several grant types suited to different client contexts:

- **Authorisation Code** (with PKCE, RFC 7636): browser-based clients redirect the user to the authorisation server; a short-lived code is exchanged for tokens at the token endpoint. PKCE prevents code-interception attacks by binding the request to a cryptographic verifier.
- **Client Credentials:** machine-to-machine; client authenticates with its own credentials; no user interaction.
- **Device Authorization Flow** (RFC 8628): for input-constrained clients (CLI tools, set-top boxes) — described in detail below.
- **Implicit** and **Resource Owner Password Credentials:** both deprecated in OAuth 2.1 due to security weaknesses.

OpenID Connect OpenID Connect layers an identity protocol on top of OAuth 2.0. It introduces the *ID Token* — a JWT issued by the authorisation server alongside the access token — that asserts the identity of the authenticated user. Standard claims in the ID Token include `sub` (subject identifier), `name`, `email`, `iss` (issuer URL), `aud` (intended audience), and `exp`. The `UserInfo` endpoint allows the client to retrieve additional claims using the access token.

OIDC defines standardised scopes: `openid` (mandatory; triggers ID Token issuance), `profile` (name, picture, locale), `email` (email address and verification status).

Device Authorization Flow (RFC 8628) The Device Authorization Flow solves authentication for devices that cannot conveniently display a browser or accept keyboard input:

1. The client sends a request to the `device_authorization_endpoint`, receiving a `device_code`, a `user_code`, and a `verification_uri`.
2. The device displays the `user_code` and `verification_uri` to the user (e.g., printed to the terminal).
3. The user visits `verification_uri` on any browser-capable device and enters the `user_code`, then authenticates and grants consent.
4. Concurrently, the device polls the `token_endpoint` with the `device_code` at the interval specified by the server.
5. Once the user grants consent, the next poll returns the access and ID tokens.

Keycloak as Identity Provider Keycloak is an open-source Identity and Access Management server. Its architectural concepts include:

- **Realm:** an isolated IAM domain with its own users, roles, clients, and identity providers. Different realms share a Keycloak instance but are fully isolated.
- **Client:** a registered application; can be *confidential* (holds a client secret) or *public* (no secret; uses PKCE). Each client defines allowed redirect URIs, scopes, and protocol (OIDC or SAML).
- **Identity Brokering:** Keycloak federates to upstream Identity Providers (Google OAuth 2.0, SAML providers) and maps external identities to local realm users. This enables “Sign in with Google” without granting Google direct access to application resources.

- **Role Mapping:** realm roles and client roles can be included in tokens via protocol mappers; downstream services read the **roles** claim to make authorisation decisions without querying Keycloak on every request.

*[Placeholder: OIDC device authorization flow — CLI requests
device_code → user opens verification_uri on browser → CLI polls token
endpoint → access + ID tokens returned]*

Figure 9.13: OAuth 2.0 Device Authorization Flow (RFC 8628): the CLI client receives a user code and polls for tokens while the user completes authorisation in a browser.

9.7 Secrets Management

9.7.1 Secrets and Their Lifecycle

A *secret* is any piece of information that grants access to a protected resource: passwords, API keys, TLS private keys, symmetric encryption keys, database credentials, or OAuth client secrets. Secrets are distinct from configuration data — a database hostname is configuration; the password to authenticate to it is a secret.

Secrets Anti-Patterns Common mismanagement practices that create security vulnerabilities:

- **Hardcoded secrets:** embedding credentials directly in source code or compiled binaries; discoverable through repository history even after deletion.
- **Repository storage:** committing `.env` files or configuration files containing secrets to version control; accessible to all contributors and in perpetuity through git history.
- **Plaintext log output:** logging secret values during startup or debugging; secrets appear in log aggregation systems accessible to many parties.
- **Long-lived static credentials:** database passwords that never rotate; a single exfiltration event grants indefinite access.

Secrets Lifecycle Proper secrets management covers the full lifecycle:

1. **Generation:** secrets produced by a Cryptographically Secure Pseudo-Random Number Generator (CSPRNG) with sufficient entropy.
2. **Distribution:** delivered through an encrypted channel; never embedded in deployment artifacts.
3. **Storage:** encrypted at rest; access controlled; audit-logged.
4. **Rotation:** periodic replacement (e.g., every 90 days) and on-demand rotation following a suspected compromise.
5. **Revocation:** immediate invalidation without waiting for the next rotation cycle.
6. **Audit:** comprehensive log of every secret access, including the requesting identity and timestamp.

Dynamic Secrets Dynamic secrets are generated on-demand for a specific requesting client, carry a short TTL, and are automatically revoked when the TTL expires. Unlike static long-lived credentials, dynamic secrets:

- Limit the damage of exfiltration to the TTL window (minutes to hours, not months or years).
- Eliminate the shared-secret problem: each client has a unique credential, so revoking one does not affect others.
- Enable automated rotation without application changes, since the application always requests fresh credentials on startup.

9.7.2 HashiCorp Vault

HashiCorp Vault is an open-source secrets management system designed for centralized secrets storage, dynamic secret generation, and fine-grained access control.

Architecture and Storage Model Vault consists of a server process backed by a configurable *storage backend* (file system, Consul, cloud object stores). Secrets are *never* written to the storage backend in plaintext: Vault encrypts all data with a master key before persisting it. The master key itself is protected by the seal/unseal mechanism.

Seal and Unseal When Vault starts, it is in a *sealed* state: the master key is not held in memory, making all secrets inaccessible. To transition to the *unsealed* state, a quorum of key shares must be presented:

1. During initialisation, Vault generates a master key and splits it using Shamir's Secret Sharing into N key shares. Access to any M of the N shares (where $M \leq N$) is sufficient to reconstruct the master key. For example, a 3-of-5 configuration requires any three of five key shares.
2. Key shares are distributed to different operators; no single operator holds a sufficient quorum.
3. On restart, M operators each submit their share via the unseal API; Vault reconstructs the master key in memory only after receiving the quorum.
4. The master key in memory decrypts the encryption key, which decrypts all stored secrets on demand.

This design prevents a single compromised operator or storage breach from exposing secrets.

Authentication Methods Vault requires each client to authenticate before issuing a token that grants secret access. The AppRole method is designed for machine-to-machine authentication:

- An *AppRole* is defined in Vault with an associated `role_id` (non-secret identifier) and a `secret_id` (short-lived, randomly-generated credential).
- The client presents both to the Vault `/auth/approle/login` endpoint and receives a Vault token with a defined TTL and associated policies.
- `secret_id` values can be single-use or use-limited, minimising exposure if intercepted.
- Policies attached to the role determine which secret paths the resulting token can read, write, or delete.

K/V v2 Secrets Engine The Key/Value version 2 secrets engine provides versioned secret storage:

- Secrets are stored at hierarchical paths (e.g., `secret/data/zgate/server`).
- Each write creates a new version; previous versions are retained and accessible by version number.
- Soft delete marks a version as deleted without removing its data; hard delete permanently removes it.
- Check-and-Set (CAS) parameter prevents concurrent overwrites by requiring the current version number on writes.

Dynamic Database Secrets The Database secrets engine generates database credentials on demand:

1. Vault is configured with a privileged database connection (DBA credentials stored securely in Vault itself).
2. When a client requests credentials for a defined role, Vault executes a configurable SQL statement (e.g., `CREATE USER '{{name}}' IDENTIFIED BY '{{password}}'` followed by `GRANT` statements).
3. The generated username and password are returned to the client with a TTL lease.
4. When the lease expires, Vault automatically executes the revocation statement (`DROP USER`).

This mechanism implements dynamic secrets at the database layer: credentials exist only while needed and are automatically reclaimed.

*[Placeholder: Vault AppRole authentication flow — server presents
role_id + secret_id → Vault returns token → server reads K/V secret
(DB credentials, encryption keys)]*

Figure 9.14: HashiCorp Vault AppRole authentication and K/V secret retrieval: the application authenticates with non-secret role_id and ephemeral secret_id to obtain a scoped Vault token.

9.8 Database Wire Protocols (PostgreSQL)

PostgreSQL uses a message-based, frontend/backend wire protocol that differs architecturally from MySQL's packet-framed design. Understanding it is essential for building transparent proxies that intercept PostgreSQL traffic.

9.8.1 Protocol Architecture

Message Format All PostgreSQL messages after the startup phase share a common structure:

```
+-----+-----+
| Type   | Length                               | Payload ...
| (1B)   | (4B, includes itself, big-end)|
+-----+-----+
```

The 1-byte type tag identifies the message kind. The 4-byte length (big-endian) includes itself but excludes the type byte. Unlike MySQL's 3-byte payload length, PostgreSQL's 4-byte field supports messages up to approximately 4 GB.

Frontend vs Backend Messages The same type byte can refer to different messages depending on direction:

- **Frontend messages** are sent by the client (e.g., Q = Query, P = Parse, B = Bind, E = Execute, X = Terminate).
- **Backend messages** are sent by the server (e.g., T = RowDescription, D = DataRow, C = CommandComplete, Z = ReadyForQuery, E = ErrorResponse).

A proxy must track the direction of traffic to correctly interpret the type byte.

9.8.2 Connection Establishment and Authentication

SSL Negotiation Before sending the startup message, a client may request TLS by sending an `SSLRequest` message (a special 8-byte packet with no type byte). The server responds with a single byte: `S` to accept TLS (the TLS handshake then proceeds over the same connection) or `N` to decline. A proxy intercepting PostgreSQL traffic must handle this negotiation before forwarding the startup message.

StartupMessage The `StartupMessage` also lacks a type byte; it consists of a 4-byte length, a 4-byte protocol version (integer 196608 = 0x00030000, protocol 3.0), followed by a sequence of null-terminated key-value pairs encoding session parameters:

- **user**: the database username
- **database**: the target database name
- **application_name**: client application identifier (visible in `pg_stat_activity`)
- **client_encoding**: character set (typically UTF8)

The list terminates with an additional null byte.

Authentication Exchange The server responds to the startup message with one of:

- **AuthenticationOk** (R, 0): no further authentication required.
- **AuthenticationMD5Password** (R, 5): server includes a 4-byte random salt; client responds with **PasswordMessage** containing `md5(md5(password + user) + salt)`.
- **AuthenticationSASL** (R, 10): initiates a SCRAM-SHA-256 exchange (RFC 7677); more secure than MD5 hashing, supports channel binding.

After successful authentication the server sends one or more **ParameterStatus** (S) messages (server version, encoding, timezone settings) and a **BackendKeyData** (K) message (process ID and secret key for cancellation), then **ReadyForQuery** (Z).

ReadyForQuery **ReadyForQuery** is the PostgreSQL synchronisation primitive. It carries a single byte indicating the current transaction status:

- **I**: idle (not in a transaction)
- **T**: inside a transaction block
- **E**: inside a failed transaction (commands ignored until **ROLLBACK**)

The client must wait for **ReadyForQuery** after every synchronisation point before sending the next command.

9.8.3 Simple Query Protocol

The Simple Query Protocol is the most straightforward execution path:

1. Client sends **Query** (Q) with a SQL string. Multiple statements separated by semicolons are permitted; each is executed in sequence.
2. For a **SELECT**, the server responds with:



- **RowDescription (T)**: describes the result set columns — for each column: name (null-terminated string), table OID, column attribute number, data type OID, data type size, type modifier, and format code (0 = text, 1 = binary).
 - One **DataRow (D)** per result row: a 2-byte column count followed by one length-prefixed byte sequence per column (length = -1 represents NULL).
 - **CommandComplete (C)**: tag string (e.g., `SELECT 42`) indicating the command type and affected row count.
3. For non-SELECT statements (`INSERT`, `UPDATE`, `DELETE`): **CommandComplete** directly (no **RowDescription** or **DataRow**).
 4. **ReadyForQuery (Z)**: signals the client can send the next command.
 5. On error: **ErrorResponse (E)**: structured set of fields including severity, SQL-STATE code (5-character string per SQL standard), human-readable message, position in query, and schema/table/column context.

9.8.4 Extended Query Protocol

The Extended Query Protocol separates the preparation and execution stages, enabling typed parameter binding, binary data transfer, and cursor-based result pagination.

Protocol Messages

- **Parse (P)**: contains a statement name (empty string for unnamed), a SQL string with `$1`, `$2`, ... parameter placeholders, and an optional array of parameter type OIDs. Server responds with **ParseComplete (1)**.
- **Bind (B)**: binds a named or unnamed prepared statement to a named or unnamed *portal*. Contains: statement name, portal name, parameter format codes (one per parameter: 0 = text, 1 = binary), parameter values (length-prefixed), and result column format codes. Server responds with **BindComplete (2)**.
- **Describe (D)**: requests metadata about a statement (**ParameterDescription** listing parameter type OIDs) or portal (**RowDescription** listing result columns) before execution.
- **Execute (E)**: runs a portal; accepts an optional maximum row count (0 = unlimited). Server responds with **DataRow** messages up to the row limit, then either **CommandComplete** or **PortalSuspended (s)** if the limit was reached. A suspended portal can be resumed by sending another **Execute**.



- **Close (C)**: deallocates a statement or portal and releases its server-side resources. Server responds with `CloseComplete` (3).
- **Sync (S)**: a pipeline flush. The server processes all pending messages and then sends `ReadyForQuery`. If any message in the pipeline caused an error, the server enters an error-recovery mode and discards subsequent pipeline messages until `Sync` is received.
- **Flush (H)**: similar to `Sync` but does not trigger `ReadyForQuery`; forces the server to flush buffered output without ending the pipeline.

Portal Concept A portal is a named, server-side execution context that encapsulates a bound prepared statement along with its parameter values and a cursor position into the result set. Portals enable the `DECLARE/FETCH` cursor semantics of SQL: execute a query, retrieve rows in batches, and release the portal when done. Named portals persist until explicitly closed or the session ends.

9.8.5 AST Parsing via `libpg_query`

`libpg_query` is a C library that extracts the official PostgreSQL parser directly from the PostgreSQL server source tree and exposes it as a standalone shared library. This approach guarantees that any SQL accepted or rejected by the parser is identical to what PostgreSQL itself would accept or reject — a property no independent SQL grammar can fully replicate given PostgreSQL’s extensive dialect extensions.

pg_query_go `pg_query_go` provides Go bindings to `libpg_query` via CGo. The primary API call is:

```
result, err := pg_query.Parse("SELECT name FROM users WHERE id = $1")
```

The returned `ParseResult` contains a list of `RawStmt` nodes serialised as protobuf. Common parse tree node types include:

- `SelectStmt`: root of a `SELECT` query; contains `targetList` (projected columns), `fromClause` (table references), `whereClause`, `groupClause`, `havingClause`, `sortClause`, `limitCount`.
- `RangeVar`: a table reference; fields `schemaname`, `relname`, `alias`.
- `ResTarget`: a `SELECT` target; contains an expression and an optional alias.
- `ColumnRef`: reference to a column; contains `fields` (schema, table, column name chain).

- **A_Const**: a literal constant (integer, float, string, boolean, null).
- **JoinExpr**: a JOIN expression; **jointype** (inner, left, right, full), **larg**, **rarg**, **quals**.
- **CommonTableExpr**: a Common Table Expression (CTE / WITH clause).
- **BoolExpr**: logical AND/OR/NOT; **args** contains sub-expressions.

A proxy can traverse this tree to identify all referenced tables and columns, determine the query type, and inject additional predicates (e.g., row-level security filters) before regenerating the SQL for forwarding to the backend.

*[Placeholder: PostgreSQL Extended Query Protocol sequence — Parse
→ ParseComplete → Bind → BindComplete → Execute → DataRow*
→ CommandComplete → Sync → ReadyForQuery]*

Figure 9.15: PostgreSQL Extended Query Protocol message sequence showing the Parse/Bind/Execute/Sync pipeline.

9.9 Data Masking and Privacy Techniques

Data masking is the process of transforming sensitive data values to protect them from unauthorised disclosure while preserving their usability for legitimate purposes. It is a foundational technique in privacy-by-design architectures and a common control required by regulatory frameworks such as GDPR (Article 32) and PCI DSS (Requirement 3).

9.9.1 Masking vs Encryption vs Anonymisation

Three related but distinct concepts are often conflated:

- **Encryption:** reversible transformation using a key; ciphertext is useless without the key but fully recoverable with it; data is not usable in its encrypted form.
- **Masking:** transforms data into a form that retains some utility (format, type, statistical distribution) but is not reversible without additional information (e.g., a token vault or original dataset). Depending on the strategy, may or may not be reversible.
- **Anonymisation:** irreversible transformation that makes it impossible to re-identify the data subject; compliant with GDPR's exemption from data subject rights. Achieving true anonymisation is significantly harder than masking due to re-identification through data linkage.

9.9.2 Static vs Dynamic Masking

- **Static masking:** applied once to a dataset copy (e.g., a development or test database copy). The masked dataset replaces the original; no masking occurs at query time. Simpler to implement; does not protect the production dataset.
- **Dynamic masking:** applied at query time in the data access layer; the production dataset remains unmodified; different users may see different levels of masking based on their roles. More complex but provides real-time, role-aware access control over sensitive data.

9.9.3 Masking Strategies

Redaction The simplest strategy: replace the entire value with a fixed constant ("REDACTED", "[HIDDEN]", "***"). No format preservation; the replacement value carries no relationship to the original. Appropriate for opaque identifiers or fields where even format leaks sensitive information.

Partial Masking (Format-Preserving) Mask a subset of characters while preserving the structural format of the original value:

- Social Security Number: 123-45-6789 → XXX-XX-6789 (preserves suffix for verification)
- Credit card: 4111 1111 1111 1111 → **** * 1111 (preserves last four digits, industry standard)
- Email: alice@example.com → a***@example.com

Format preservation is important when masked values are displayed in UIs that validate format (e.g., a customer service agent verifying the last four digits of a card).

Deterministic Hashing Apply a one-way hash function (SHA-256, SHA-3) to the value. The masked output is consistent for the same input across all occurrences in the dataset, which preserves the ability to:

- Join masked tables on the masked column (if the same salt is used consistently).
- Group and count without revealing individual values.

Limitations: deterministic hashing is vulnerable to precomputation attacks (rainbow tables) when the input space is small (e.g., dates, postcodes, gender). Salting mitigates this but breaks cross-table join capability.

Tokenisation Replace the sensitive value with an opaque, randomly-generated token stored in a separate *token vault*. The original value is retrievable only by querying the vault with the token — an operation restricted to authorised services. Tokenisation is the dominant strategy for PCI DSS compliance (card number tokenisation) because:

- The token has no mathematical relationship to the original (unlike hashing).
- Token lookup requires vault access, which is a separately controlled security boundary.
- Format-preserving tokenisation (FPT) generates tokens that pass validation checks (e.g., a token that resembles a valid credit card number and passes Luhn algorithm validation).

Nulling Replace the value with SQL NULL. Maximum safety — the value is completely absent from query results. Breaks any application logic that assumes non-null values. Appropriate for archival scenarios or fields that must be fully suppressed for a given role.

Data Substitution Replace with a realistic synthetic value drawn from a reference dataset (e.g., replace a real name with a randomly selected name from a census name corpus). Preserves semantic plausibility: the masked dataset looks realistic, reducing the risk of developers making incorrect assumptions about data structure. Used in test data generation and analytics on masked datasets.

Shuffling Randomly permute values within a column across rows. The set of values is preserved, but the assignment of values to rows is destroyed. Statistical distributions are maintained; individual records are de-identified. Only effective when the column is not functionally dependent on any identifier remaining in the dataset.

9.9.4 Applying Masking at the Protocol Layer

Proxy-level dynamic masking intercepts result sets between the database and the client:

1. **Column definition phase:** the proxy reads the column metadata packets (MySQL `Field` packets or PostgreSQL `RowDescription`) before any row data arrives. This phase produces a mapping of column positions to column names, table names, and data types.
2. **Policy lookup:** for each column in the result, the proxy consults the masking rule store to determine whether the authenticated user's role requires masking, and if so, which strategy applies.
3. **Row interception:** as each row arrives, the proxy parses the value at each masked column's position. For MySQL text protocol, values are length-encoded strings; for MySQL binary protocol, values are natively typed binary data; for PostgreSQL, values are length-prefixed byte sequences.
4. **Masking function application:** the strategy function transforms the extracted value.
5. **Packet reconstruction:** the modified value is serialised back into the protocol encoding, the packet payload length field is recalculated (the masked value may be a different byte length than the original), and the reconstructed packet is forwarded to the client.

A critical implementation detail is that the packet length field must be updated to reflect the new payload size after masking. Failure to do so produces a malformed packet that the client's driver will reject.

[Placeholder: Data masking pipeline — (1) parse column definitions, (2) look up masking rules, (3) intercept DataRow/Row packets, (4) apply masking function, (5) reconstruct packet with updated length, (6) forward to client]

Figure 9.16: Protocol-layer data masking pipeline: the proxy intercepts result-set packets, identifies masked columns from metadata, applies the configured masking strategy, and reconstructs the packet before forwarding.

9.10 Observability and Distributed Tracing

9.10.1 Observability vs Monitoring

Monitoring answers predetermined questions about known failure modes: "Is service X responding?", "Is disk usage above 90%?". Monitors emit alerts when predefined thresholds are crossed but cannot answer questions that were not anticipated at instrumentation time.

Observability is the property of a system that allows its internal state to be inferred from its external outputs. An observable system allows engineers to ask *novel* questions — questions not anticipated when the system was built — and derive answers from telemetry. Observability is achieved through three complementary *signals*.

9.10.2 The Three Pillars of Observability

Traces A *distributed trace* is a record of a single request's causal execution path as it traverses multiple services and processes. It is represented as a directed acyclic graph (DAG) of *spans*.

A *span* is the atomic unit of a trace. Each span records:

- `trace_id`: 128-bit identifier shared by all spans in the same trace.
- `span_id`: 64-bit identifier unique within the trace.
- `parent_span_id`: links this span to its parent (absent for the root span).
- Operation name: human-readable description (e.g., `http.GET /api/sessions`).
- Start timestamp and duration.
- Status: OK, ERROR, or UNSET.
- Attributes: key-value pairs carrying contextual metadata (e.g., `db.system = mysql`, `user.id = alice`).
- Events: timestamped annotations within the span's lifetime (e.g., cache miss, retry).
- Links: references to spans in other traces (e.g., a message consumer linking back to the producer's span).

Metrics Metrics are numeric measurements aggregated over time. They answer quantitative questions: "What is the 99th-percentile query latency?", "How many sessions are currently active?". Metrics are efficient to store and query because they discard individual data points in favour of aggregates.

Instrument types:

- **Counter:** a monotonically increasing value reset to zero on restart; suitable for total event counts (e.g., `queries_total`).
- **Gauge:** a value that can increase or decrease; suitable for instantaneous measurements (e.g., `active_sessions`).
- **Histogram:** samples a distribution; produces bucket counts, sum, and count; enables percentile estimation (e.g., `query_duration_seconds`).

Metrics must have bounded *cardinality*: the set of distinct label value combinations. High-cardinality labels (individual user IDs, full query text) cause memory and storage explosion in time-series databases and must instead be placed in traces or logs.

Logs Logs are discrete, timestamped records of events. Structured logs encode fields as key-value pairs (JSON or logfmt), enabling machine parsing and filtering. Structured logging replaces string-formatted messages that require regex parsing.

9.10.3 Trace Context Propagation

For a trace to span multiple services, the trace context (`trace_id` and `parent span_id`) must be propagated alongside requests. The W3C TraceContext specification (a Recommendation since 2021) defines the `traceparent` HTTP header:

```
traceparent: 00-{trace-id-32hex}-{parent-span-id-16hex}-{flags-2hex}
```

The `tracestate` header carries vendor-specific context. When a downstream service receives a request containing `traceparent`, it extracts the `trace_id` and `parent_span_id`, creates a child span with the extracted parent, and propagates updated context to any further downstream calls. Services that initiate new traces (root spans) generate a fresh `trace_id` and omit the `parent_span_id`.

Sampling Recording every span in a high-throughput system is cost-prohibitive. Sampling strategies reduce volume while preserving diagnostic value:

- **Head-based sampling:** the sampling decision is made at the root span, before any child spans are created. A simple random fraction (e.g., 1%) is common. The decision is propagated in the `flags` byte of `traceparent` so all downstream services make the same decision consistently.

- **Tail-based sampling:** the decision is deferred until the entire trace is assembled. This allows always-sampling error traces and slow traces while dropping fast successful ones. Requires a collector that buffers trace data until the trace is complete.

9.10.4 OpenTelemetry

OpenTelemetry (OTel) is a Cloud Native Computing Foundation (CNCF) graduated project that provides a vendor-neutral, language-agnostic API, SDK, and wire protocol for all three observability signals.

Architecture

- **OTel API:** defines the interfaces for creating spans, recording metrics, and emitting log records. Application code instruments against the API. The API has a no-op default implementation, so uninitialised SDKs impose zero overhead.
- **OTel SDK:** implements the API with configurable exporters, samplers, and processors. The SDK is injected at startup; the application code does not need to change when switching from one observability backend to another.
- **OTLP (OpenTelemetry Protocol):** a protobuf-over-gRPC (or protobuf-over-HTTP) protocol for exporting telemetry from the SDK to a collector or backend. A single protocol replaces the proliferation of vendor-specific agents (Jaeger agent, Prometheus exporter, Datadog forwarder).
- **OTel Collector:** an optional intermediary that receives OTLP telemetry, applies transformations (filtering, attribute addition/removal, tail-based sampling), and exports to one or more backends.

Exemplars An exemplar links an aggregate metric observation to a specific trace. When a histogram records a high-latency sample, it can embed the `trace_id` of the request that caused that sample. Observability platforms use exemplars to navigate from a metric anomaly (high p99 latency spike) directly to the individual trace that exemplifies it — collapsing the investigation from minutes to seconds.

[Placeholder: OpenTelemetry trace spanning CLI → API → Gateway → DB — root span at CLI, child spans for HTTP call, JWT validation, proxy session, query interception, DB round-trip; each service propagates traceparent]

Figure 9.17: Example distributed trace across all zGate components: context propagates via W3C **traceparent** from the CLI through the REST API, gateway, and database round-trip.

9.11 Terminal User Interface Programming

Command-line interfaces typically operate in a request-response mode: the user types a command, presses Enter, the program outputs a result, and the shell prompt returns. A Terminal User Interface (TUI) replaces this model with a persistent, stateful application that occupies the full terminal screen, processes individual keystrokes in real time, and continuously re-renders its display.

9.11.1 Terminal Modes

The Unix terminal subsystem operates in two principal input modes:

- **Cooked mode** (canonical mode): the kernel's line discipline buffers input character-by-character and only delivers a line to the application when the user presses Enter. Backspace, Ctrl-C, and other special characters are processed by the kernel before the application sees them. This is the default mode for shell prompts and standard CLI tools.
- **Raw mode**: the line discipline is disabled; every keystroke is delivered to the application immediately, without buffering or special-character processing. The application is responsible for handling all control characters (Ctrl-C, Ctrl-D, backspace). TUIs require raw mode to respond to navigation keys and real-time input.

Transitioning to raw mode is performed via the `tcsetattr()` system call (POSIX) or the Windows Console API. Applications must restore the original terminal mode on exit — failure to do so leaves the terminal in an unusable state.

9.11.2 ANSI Escape Sequences

ANSI escape sequences are byte sequences embedded in the output stream that instruct the terminal emulator to perform actions beyond printing characters. They are standardised in ECMA-48 (ISO/IEC 6429).

Common sequences relevant to TUI rendering:

- **Cursor positioning**: `ESC[row;colH` moves the cursor to a specific cell.
- **Erase**: `ESC[2J` clears the entire screen; `ESC[K` erases from cursor to end of line.
- **Select Graphic Rendition (SGR)**: `ESC[<params>m` sets text attributes:
 - Bold: `ESC[1m`; reset: `ESC[0m`
 - 4-bit foreground colors: `ESC[30m – ESC[37m` (black through white)

- 8-bit color: `ESC[38;5;nm` where $n \in [0, 255]$
- 24-bit (true color) RGB: `ESC[38;2;R;G;Bm`
- **Alternate screen buffer:** `ESC[?1049h` switches to a secondary screen (preserving the user’s original terminal content); `ESC[?1049l` restores it on exit. TUIs use the alternate buffer to avoid corrupting the shell session history.

9.11.3 The Elm Architecture

The Elm Architecture is a software design pattern for interactive applications, originating in the Elm programming language (Czaplicki, 2012). It decomposes an application into three pure functions:

- **Model:** an immutable value representing the complete application state at a given instant. Since the model is immutable, every state transition produces a new model value rather than mutating in place.
- **Update:** a pure function $(\text{Model}, \text{Msg}) \rightarrow (\text{Model}, \text{Cmd})$. Given the current state and an incoming message (event), it produces a new state and an optional *command* describing a side effect to perform. Update contains no I/O; it only transforms state.
- **View:** a pure function $\text{Model} \rightarrow \text{Output}$. Given the current state, produces the rendered output (HTML in Elm; a string of ANSI-annotated text in terminal frameworks). View contains no side effects.

Commands and the Runtime A *Cmd* is a declarative description of a side effect — an HTTP request, a timer, a file read, a system call. The *runtime* (the framework’s main loop) executes Cmds and feeds the results back into Update as new messages. Application code never performs I/O directly; it only describes what I/O it wants. This separation makes Update and View deterministic and testable: given any (Model, Msg) pair, Update always returns the same (Model, Cmd).

9.11.4 Bubble Tea Framework

Bubble Tea is a Go implementation of the Elm Architecture for terminal user interfaces, developed by Charm (2020). The core interface is `tea.Model`:

```
type Model interface {
    Init()                                tea.Cmd
    Update(tea.Msg) (Model, tea.Cmd)
    View()                                string
}
```


`Init()` returns the initial command (e.g., a command to fetch data on startup). `Update()` processes incoming messages and returns the updated model and an optional command. `View()` renders the current model as a string, which Bubble Tea forwards to the terminal (applying a diff against the previous frame to minimise writes).

Event Types The runtime delivers terminal events as typed `tea.Msg` values:

- `tea.KeyMsg`: a keypress; provides the key rune and key type (e.g., `tea.KeyEnter`, `tea.KeyCtrlC`, `tea.KeyUp`).
- `tea.MouseMsg`: a mouse event (click, wheel, motion) if mouse reporting is enabled.
- `tea.WindowSizeMsg`: fires when the terminal is resized; provides new width and height.
- Custom message types: the application defines its own `Msg` values for asynchronous results (e.g., an HTTP response arriving via a `tea.Cmd`).

Component Composition Bubble Tea encourages nesting `tea.Model` values: a top-level model holds sub-models for different UI panels (sidebar, main content, modal). The top-level `Update` delegates messages to the appropriate sub-model and embeds sub-model views within the top-level `View`. This compositional pattern scales to arbitrarily complex UIs without losing the deterministic, unidirectional data flow property.

9.11.5 Terminal Styling and Platform Credential Storage

Lipgloss Lipgloss is a Go library for declarative terminal layout and styling. Styles are first-class values:

```
primaryStyle := lipgloss.NewStyle().
    Bold(true).
    Foreground(lipgloss.AdaptiveColor{Light: "236", Dark: "248"}).
    Border(lipgloss.RoundedBorder()).
    Padding(0, 1)
```

`AdaptiveColor` selects different color values based on whether the terminal background is light or dark (detected via `lipgloss.HasDarkBackground()`). Layout is computed with `Width()`, `MaxWidth()`, `Height()`, and `Align()` methods, enabling responsive panels that adapt to window resize events.

OS Keyring Integration Storing JWT refresh tokens in plaintext files on disk exposes them to any process with file-system read access. Operating systems provide platform-native, encrypted credential stores that restrict access to the storing application:

- **macOS:** Keychain Services (Security framework); credentials encrypted with the user’s login password; access prompts presented by the OS.
- **Windows:** Windows Credential Manager backed by the Data Protection API (DPAPI); credentials encrypted with a key derived from the user’s Windows account password.
- **Linux:** libsecret over D-Bus, implemented by GNOME Keyring or KDE Wallet; credentials encrypted in the user’s session keyring, unlocked at login.

The `go-keyring` library provides a uniform API across all three platforms: `keyring.Set(service, username, password)` stores a credential; `keyring.Get(service, username)` retrieves it. The underlying platform storage handles encryption, access control, and OS-level user prompting transparently.

[Placeholder: Bubble Tea event loop — terminal events → tea.Msg → Update() returns (Model, Cmd) → runtime executes Cmd → result delivered as new Msg → View() renders updated state]

Figure 9.18: Bubble Tea (Elm Architecture) event loop: the runtime delivers terminal events as messages, `Update` produces new state and commands, the runtime executes commands and feeds results back as messages.

9.12 Modern Frontend Technologies

9.12.1 React Component Model and Rendering

React is a declarative JavaScript library for building user interfaces, released by Meta in 2013. Its programming model centres on the *component*: a function or class that accepts a *props* object and returns a description of UI (a *React element*). Components compose hierarchically to form a component tree.

Reconciliation and the Fiber Architecture React maintains a *virtual DOM*: an in-memory tree of React elements that mirrors the intended state of the actual browser DOM. When component state changes, React rerenders the affected subtree into a new virtual DOM tree, *reconciles* (diffs) it against the previous tree, and applies the minimal set of mutations to the actual DOM. This batched, diffed update strategy avoids expensive full-page repaints.

React 18 introduced the *Fiber* architecture, replacing the prior synchronous recursive reconciler with an incremental, interruptible one. Fiber assigns each unit of work a priority level and allows the scheduler to pause and resume reconciliation, preventing long renders from blocking user interaction.

Hooks React Hooks, introduced in React 16.8, provide stateful logic in function components:

- **useState**: associates local state with a component; returns a value and a setter; calling the setter schedules a re-render.
- **useEffect**: declares a side effect (data fetching, subscription, DOM mutation) tied to the component lifecycle; the effect runs after render; the optional cleanup function runs before the next effect and on unmount.
- **useRef**: holds a mutable value that persists across renders without causing re-renders; commonly used to hold DOM node references.
- **useCallback** / **useMemo**: memoize a function or computed value; prevent unnecessary child re-renders when the parent re-renders with the same props.
- **useContext**: reads a context value propagated through the component tree without prop drilling.

Concurrent Rendering (React 18+) React 18's concurrent features allow the renderer to interrupt a low-priority render and process a high-priority update (e.g., user input) first:

- **startTransition:** wraps a state update as non-urgent; React may defer it to keep the UI responsive.
- **useDeferredValue:** returns a lagging version of a value; the UI updates with the current value while React concurrently prepares the deferred update.
- **Suspense:** a component boundary that shows a fallback while a child component awaits asynchronous data; enables declarative loading states without imperative loading flags.

9.12.2 Next.js App Router and React Server Components

Next.js is a React meta-framework that adds server-side rendering (SSR), static site generation (SSG), file-system routing, and API routes to React. Version 13 introduced the *App Router*, which replaces the older Pages Router with a new model built on React Server Components.

React Server Components React Server Components (RSC) are components that render exclusively on the server. Unlike classical SSR (which renders the entire page to HTML and then rehydrates the full React tree on the client), RSC renders individual components on the server and streams a serialized *React Server Component Payload* (RSC Payload) to the client. Key properties:

- No JavaScript is shipped to the browser for server components themselves; only the rendered output is sent.
- Server components can directly access server-side resources (databases, file systems, environment variables) without exposing those resources to the client.
- Server components cannot use hooks or browser APIs; they are pure rendering functions over server-accessible data.
- The RSC Payload is streamed, enabling incremental rendering: the browser can display already-rendered parts of the page while the server is still computing later parts.

Client Components Client Components are marked with the `"use client"` directive at the top of the module. They:

- Are bundled and shipped to the browser.
- Can use hooks (`useState`, `useEffect`, `useRef`, etc.) and browser APIs.
- Are hydrated in the browser: the server pre-renders them to HTML for fast initial paint; the browser then attaches React's event handlers to the existing DOM nodes.

The App Router enforces a clear boundary: server components are the default; the "use client" boundary marks where the server/client split occurs.

Streaming and Suspense Next.js App Router integrates React's **Suspense** with HTTP streaming (HTTP chunked transfer encoding). The server sends the initial HTML shell immediately, including fallback content for unresolved **Suspense** boundaries. As each component resolves (e.g., after a database query completes), the server streams the rendered HTML for that component, and React on the client patches it into place. This approach achieves a low Time to First Byte (TTFB) even for data-heavy pages.

9.12.3 Radix UI Primitive Component System

Radix UI is a collection of unstyled, accessible React component primitives. Unlike component libraries that bundle styling, Radix provides only behaviour and accessibility semantics, leaving all visual design to the consumer.

WAI-ARIA Compliance All Radix components implement the *Web Accessibility Initiative — Accessible Rich Internet Applications* (WAI-ARIA) specification. ARIA defines roles (e.g., `dialog`, `listbox`, `combobox`), states (e.g., `aria-expanded`, `aria-checked`, `aria-disabled`), and properties (e.g., `aria-labelledby`) that expose the semantic meaning of UI elements to assistive technologies such as screen readers.

Composition Pattern Radix components follow a composition pattern where complex widgets are built from named sub-parts:

```
<Dialog.Root>
  <Dialog.Trigger asChild><Button>Open</Button></Dialog.Trigger>
  <Dialog.Portal>
    <Dialog.Overlay />
    <Dialog.Content>
      <Dialog.Title>Title</Dialog.Title>
      <Dialog.Description>Description</Dialog.Description>
      <Dialog.Close />
    </Dialog.Content>
  </Dialog.Portal>
```

```
</Dialog.Root>
```

Each sub-part renders a specific DOM element with appropriate ARIA attributes. The consumer can apply styles via `className`, CSS-in-JS, or Tailwind CSS utilities. The `asChild` prop merges Radix behaviour onto the consumer's element, avoiding extra DOM nodes.

Focus Management Radix manages focus automatically for overlay patterns: when a Dialog opens, focus moves to the first focusable element inside; a *focus trap* prevents Tab from escaping the dialog while it is open; closing the dialog restores focus to the trigger element. This behaviour is required by WCAG 2.1 Success Criterion 2.1.2 (No Keyboard Trap) when adapted for modal dialogs.

9.12.4 Schema Validation with Zod

Zod is a TypeScript-first schema declaration and runtime validation library.

Schema Definition Schemas are built by composing primitive validators:

```
const DatabaseSchema = z.object({
  name:      z.string().min(1).max(128),
  host:      z.string().hostname(),
  port:      z.number().int().min(1).max(65535),
  protocol:  z.enum(["mysql", "postgres", "mssql"]),
  tls:       z.boolean().optional().default(true),
});
```

Type Inference `z.infer<typeof DatabaseSchema>` derives a TypeScript type from the schema definition. The inferred type is the single source of truth: the schema validates runtime data; the inferred type annotates compile-time code. Changes to the schema automatically propagate to the type without manual synchronisation.

Parsing `schema.parse(data)` runs all validators; on failure it throws a `ZodError` containing a structured list of issues, each specifying the path to the invalid field, the expected type, and the received value. `schema.safeParse(data)` returns a discriminated union `{success: true, data} | {success: false, error}` without throwing.

Integration with React Hook Form React Hook Form manages form state with uncontrolled inputs, minimising re-renders. The `@hookform/resolvers/zod` integration converts `ZodError` into React Hook Form's field-error format, mapping each issue's path to the corresponding form field:

```
const { register, handleSubmit, formState: { errors } } =  
  useForm({ resolver: zodResolver(DatabaseSchema) });
```

On submit, React Hook Form runs `schema.safeParse()` on the form values; if validation fails, `errors.port?.message` contains the human-readable error from the Zod validator for display to the user.

9.12.5 Session Management and Security in Single-Page Applications

HttpOnly Cookie Storage JWTs stored in `localStorage` or `sessionStorage` are accessible via `document.cookie` / `localStorage` APIs from any JavaScript on the page — including injected scripts from XSS attacks. Storing the session token in an `HttpOnly` cookie prevents JavaScript from reading it at all; the browser attaches it automatically to same-origin requests, and the server reads it from the `Cookie` header.

iron-session `iron-session` is a server-side session library that seals a JSON session object into a cryptographically protected cookie. The cookie payload is AES-256-GCM encrypted with a server-held password; the server decrypts and verifies the session on each request. This approach avoids a server-side session store for most data while retaining the security properties of `HttpOnly` cookie storage.

CSRF Protection Cross-Site Request Forgery (CSRF) exploits the fact that browsers automatically attach cookies to cross-origin requests that match the cookie's domain. Two mitigations apply:

- **SameSite=Strict:** the browser withholds the cookie from all cross-site requests, not just those that the user initiated by following a link from another site. This effectively eliminates CSRF without requiring synchroniser tokens.
- **Origin header validation:** the API server verifies that the `Origin` header of the request matches the expected origin; requests without a matching `Origin` are rejected. This provides defence-in-depth when `SameSite` is unavailable or misconfigured.

Content Security Policy Content Security Policy (CSP) is an HTTP response header that instructs the browser to restrict the sources from which it may load scripts, styles, images, and other resources. A strict policy such as `default-src 'self'` prevents the browser from executing injected inline scripts or loading scripts from untrusted external origins, significantly reducing the impact of an XSS vulnerability even if injection occurs.

[Placeholder: Next.js App Router rendering model — Server Components render on the server and stream RSC Payload; Client Components are bundled and hydrated in the browser; Suspense boundaries stream progressive HTML]

Figure 9.19: Next.js App Router rendering model: Server Components produce an RSC Payload streamed to the client; Client Components are hydrated in the browser; Suspense enables incremental streaming.

Chapter 10

System Architecture

This part covers:

- Three-tier architecture overview
- Gateway component design
- Protocol-aware proxy implementation
- Dynamic session management
- Interceptor chain and query processing
- RESTful API layer

ARCHITECTURE defines how system components collaborate to achieve requirements. This chapter presents zGate's layered architecture across client, API, gateway, data, and storage tiers, examining the internal structure of the gateway server including listener, acceptor, dispatcher, and session manager components. We explore the protocol-aware proxy implementation, dynamic credential provisioning, interceptor chain for composable security enforcement, and RESTful API layer that enables centralized administration.



10.1 High-Level Architecture

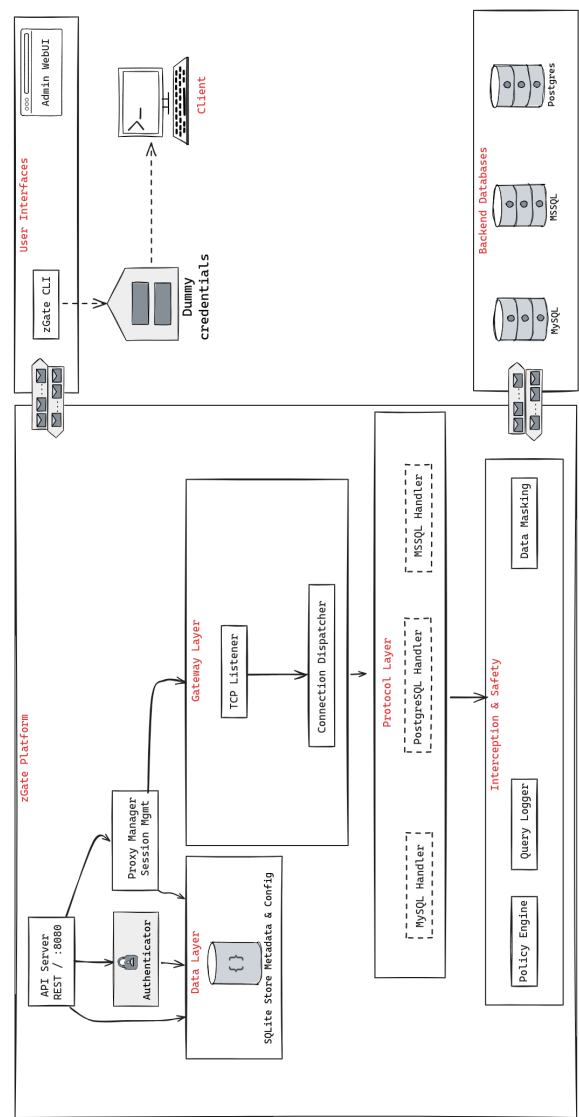


Figure 10.1: zGate Architecture Overview

10.2 Detailed Architecture

The zGate platform is architected as a distributed system comprising three primary components: the Gateway Server, Web UI, and CLI. Each component is designed with specific responsibilities that collectively implement Zero Trust database access.

10.2.1 System Components Overview

The zGate architecture can be conceptualized as a multi-layered security gateway that intercepts, validates, and controls all database access. Figure 10.2 illustrates the complete data flow across all system components.

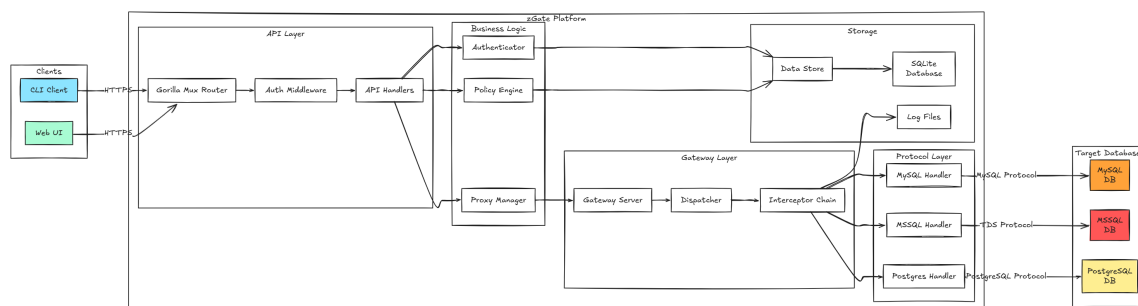


Figure 10.2: Complete System Data Flow Architecture

The system architecture follows a clear separation of concerns:

- **Client Layer:** WebUI and CLI provide user interfaces for authentication and database connection management
- **API Layer:** RESTful API server handles authentication, authorization, and session management
- **Gateway Layer:** Protocol-aware proxy intercepts and controls database traffic
- **Data Layer:** Target databases are accessed through the gateway with enforced security policies
- **Storage Layer:** SQLite database stores users, roles, permissions, and audit logs

10.2.2 Gateway and Proxy Components

The Gateway Server is the heart of the zGate platform, implementing wire-protocol level database proxying with comprehensive security controls. It is architected as a collection of specialized components, each with distinct responsibilities.

Component Architecture

The gateway implements a modular pipeline architecture with the following core components:

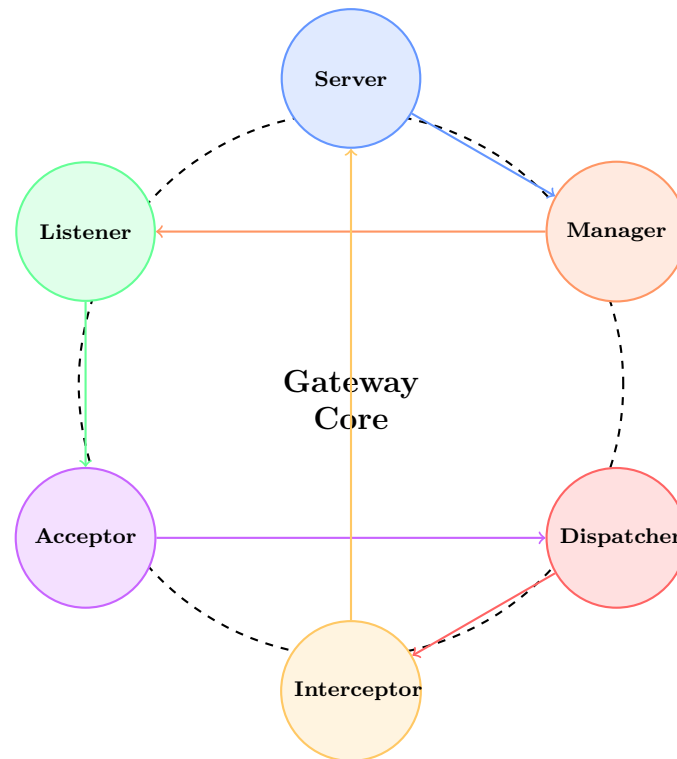


Figure 10.3: Gateway Component Architecture and Data Flow

1. **Server:** Orchestrates the lifecycle of database handlers and manages protocol-specific processing
2. **Listener:** Manages TCP listener lifecycle for dynamic port allocation per user session
3. **Acceptor:** Handles incoming client connections and establishes connection metadata
4. **Dispatcher:** Establishes backend database connections and manages bidirectional traffic forwarding
5. **Manager:** Coordinates session lifecycle, credential provisioning, and resource cleanup
6. **Interceptor Chain:** Provides composable query processing pipeline for security enforcement

Server Component The Server component (`internal/gateway/server.go`) serves as the orchestration layer for the gateway. Its primary responsibilities include:

- **Handler Initialization:** Pre-initializes protocol handlers for all registered database types (MySQL, PostgreSQL, MSSQL, MariaDB)

- **TLS Configuration:** Manages end-to-end TLS encryption for proxy-to-database connections with support for certificate pinning
- **Handler Registry:** Maintains a type-to-handler mapping that enables protocol-specific processing
- **Resource Sharing:** Provides thread-safe access to shared handlers across multiple sessions

Input: Database configurations from SQLite store, TLS certificate paths

Output: Initialized handlers for each database type

Communication: Provides handlers to Manager component via `GetHandler()` method

Manager Component The Manager component (`internal/proxy/manager.go`) implements the session lifecycle management with sophisticated credential provisioning logic. It follows a three-tier account priority system:

1. **Personal Accounts:** Dedicated database accounts assigned to specific users
2. **Shared Accounts:** Pool of pre-created accounts shared across users with concurrent usage limits
3. **Ephemeral Accounts:** Temporary accounts created on-demand and destroyed on disconnect

Session Creation Workflow:

1. Validates user permissions for requested database using real-time policy evaluation
2. Attempts to retrieve personal database account from store
3. If unavailable, checks shared account pool for available credentials
4. If pool exhausted, creates ephemeral temporary database user with scoped permissions
5. Allocates dynamic port from operating system
6. Initializes session context and starts listener
7. Returns connection details to API layer

Input: JWT claims, database name, user permissions

Output: Session object with port, credentials, account type

Communication: Interacts with Store for credentials, Gateway Server for handlers, Database for ephemeral user creation

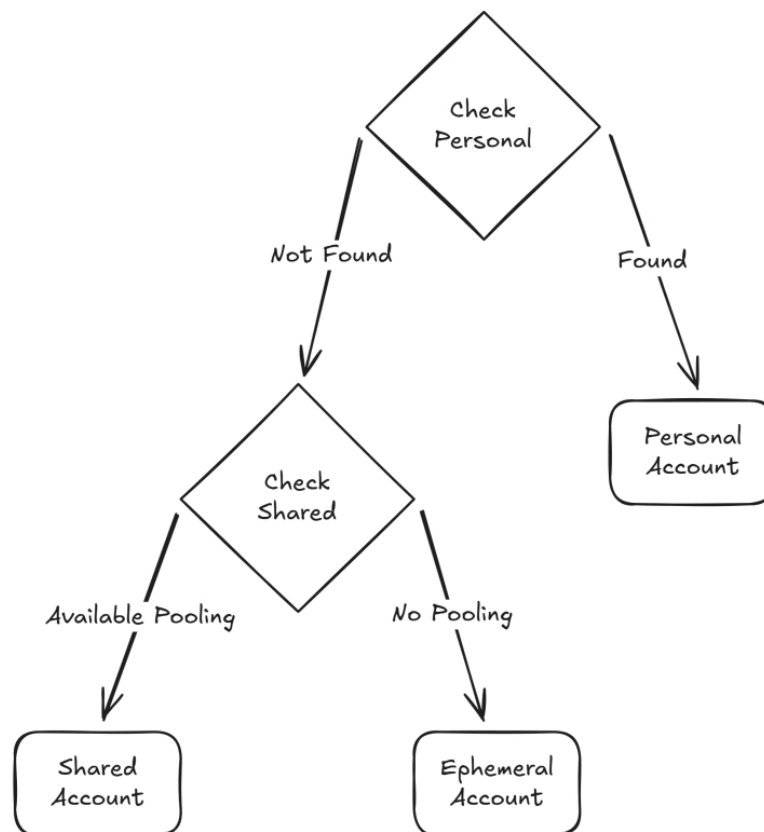


Figure 10.4: Account Types

Listener Component The Listener component (`internal/gateway/listener.go`) manages the TCP listener lifecycle for each user session. Each session receives a dedicated listener on a dynamically allocated port.

Key Responsibilities:

- Creates TCP listener on specified port (0.0.0.0:port for remote mode, configurable for security)
- Accepts incoming client connections in a loop
- Delegates each connection to Acceptor for processing
- Implements graceful shutdown via context cancellation
- Tracks active connections using WaitGroup for clean termination

Input: Listen address, database configuration, session metadata

Output: Accepted client connections

Communication: Spawns Acceptor for each connection

Acceptor Component The Acceptor component (`internal/gateway/acceptor.go`) processes each incoming client connection from acceptance to completion.

Responsibilities:

- Captures client connection metadata (IP address, timestamp)
- Associates connection with session ID for audit correlation
- Creates ConnectionMetadata structure with user context
- Delegates to Dispatcher for protocol handshake and traffic forwarding
- Ensures proper connection cleanup on termination

Input: Client TCP connection, database credentials, session metadata

Output: ConnectionMetadata structure

Communication: Instantiates Dispatcher for bidirectional proxying

Dispatcher Component The Dispatcher component (`internal/gateway/dispatcher.go`) is responsible for establishing backend connections and managing the entire query lifecycle.

Connection Establishment:

1. Establishes TCP connection to backend database server
2. Configures protocol handler with target database and proxy credentials
3. Performs authentication handshake using real database credentials
4. Handles TLS upgrade for both client-proxy and proxy-backend connections
5. Extracts upgraded connections from handshake result

Query Processing Loop:

1. Reads command from client connection using protocol handler
2. Creates QueryContext with session metadata
3. Invokes BeforeQuery interceptor chain (safety checks, logging)
4. Forwards allowed queries to backend database
5. Receives query results from database
6. Invokes AfterQuery interceptor chain (masking, logging)
7. Forwards processed results back to client

Input: Client and backend connections, database credentials, interceptor configuration

Output: Processed query results with security enforcement applied

Communication: Coordinates between Protocol Handler, Interceptor Chain, and both client and database

Interceptor Chain The Interceptor Chain (`internal/proxy/interceptor/`) implements a composable pipeline for query processing with three primary interceptors:

1. Logging Interceptor

Purpose: Provides comprehensive audit trail for all database operations.

Functionality:

- Captures query text, timestamp, user identity, target database
- Records query execution duration and rows affected
- Filters internal/system queries (SET, SHOW, SELECT @@)
- Writes structured logs to file for compliance and forensics

Input: QueryContext before execution

Output: Structured log entry

2. Safety Interceptor

Purpose: Prevents dangerous database operations based on configurable rules.

Blocked Operations:

- DELETE without WHERE clause (mass deletion)
- UPDATE without WHERE clause (mass update)
- TRUNCATE TABLE (complete data loss)
- DROP TABLE/DATABASE (schema destruction)
- ALTER TABLE (schema modification)
- CREATE statements (schema creation)
- GRANT/REVOKE (privilege escalation)

Input: Query text from QueryContext

Output: BLOCK decision or ALLOW with logged warning

3. Masking Interceptor

Purpose: Dynamically masks sensitive data in query results based on column names.

Masking Rules:

- **Email:** First character + domain visible (e.g., j***@email.com)
- **Phone/Mobile:** Last 4 digits visible (e.g., *****1234)
- **Credit Card/SSN:** Last 4 digits visible
- **Password/Secret/Token:** Full redaction (*****)
- **Name fields:** First letter only (e.g., J****)

Input: Query result set with column metadata

Output: Masked result set with sensitive values redacted

Composite Interceptor: The CompositeInterceptor orchestrates the execution of all configured interceptors in sequence, maintaining the chain of responsibility pattern.

10.2.3 Configuration Flow

Figure 10.5 illustrates the system initialization and configuration flow that occurs when the zGate Gateway Server starts.

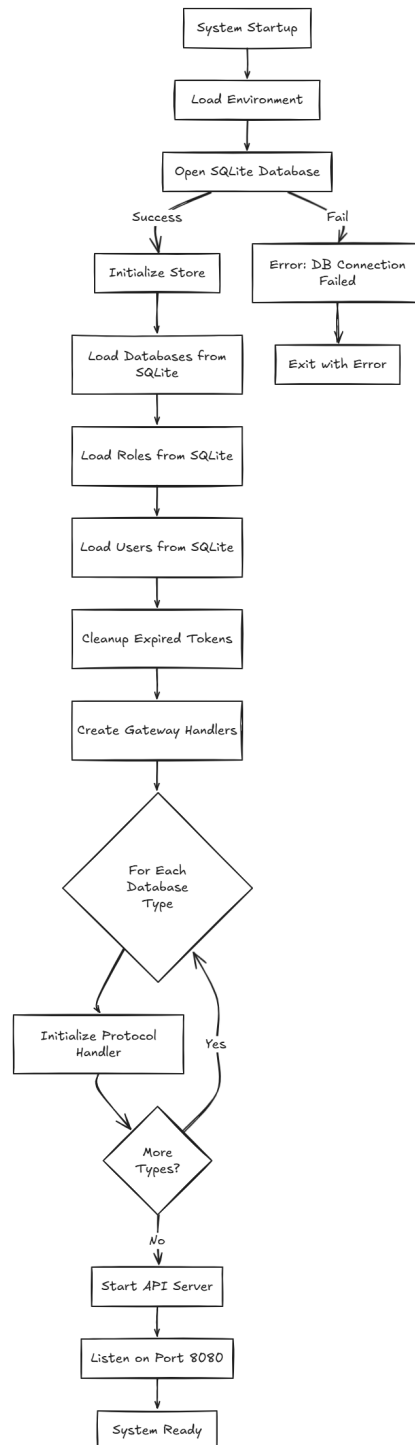


Figure 10.5: System Configuration and Initialization Flow

The configuration process ensures all components are properly initialized before accepting client connections:

1. **Store Initialization:** SQLite database is opened and schema migrations are applied
2. **Gateway Server Creation:** All database type handlers are pre-initialized
3. **TLS Configuration:** Certificates are loaded for end-to-end encryption

4. **API Server Initialization:** RESTful endpoints are registered with middleware
5. **Manager Creation:** Session management component is instantiated with store and gateway references
6. **Logger Setup:** Structured logging is configured with appropriate verbosity levels
7. **Server Listening:** API server starts accepting HTTP requests

10.2.4 Authentication System

The authentication system implements JWT-based token authentication with automatic token rotation for enhanced security. The system supports both user and admin authentication flows.

The gateway supports multiple authentication schemes to suit different deployment and operational needs:

- **Username / Password (Local):** Traditional credential-based login where passwords are verified (bcrypt) and a JWT pair (access + refresh) is issued for subsequent API access.
- **OAuth2 / OIDC (External IdP):** Integration point for enterprise identity providers to enable Single Sign-On (SSO) and federation; zGate consumes identity tokens and issues local JWTs as needed.
- **Mutual TLS (mTLS) / Service Certificates:** Certificate-based authentication for automated clients and service accounts where client certificates are validated during the TLS handshake and mapped to principal identities.

User Login Flow

Figure 10.6 depicts the user authentication process used by both CLI and WebUI components.

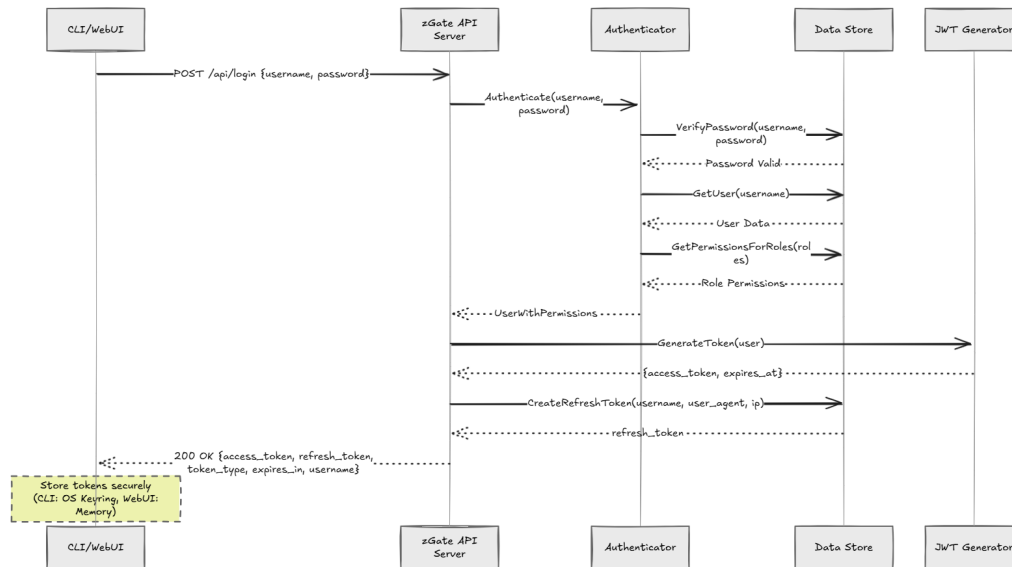


Figure 10.6: User Authentication Flow

Authentication Process:

1. Client submits credentials to POST /api/login endpoint
2. API invokes Authenticator component to verify password using bcrypt
3. Store retrieves user data and associated role permissions
4. Authenticator combines role-based and custom permissions
5. JWT Generator creates short-lived access token (15 minutes expiration)
6. Store creates refresh token (7 day expiration) linked to user session
7. Response includes both tokens, token type, expiration, and username
8. Client stores tokens securely (OS keyring for CLI, memory for WebUI)

Token Refresh Flow

Figure 10.7 shows the token refresh mechanism that implements automatic token rotation for security.

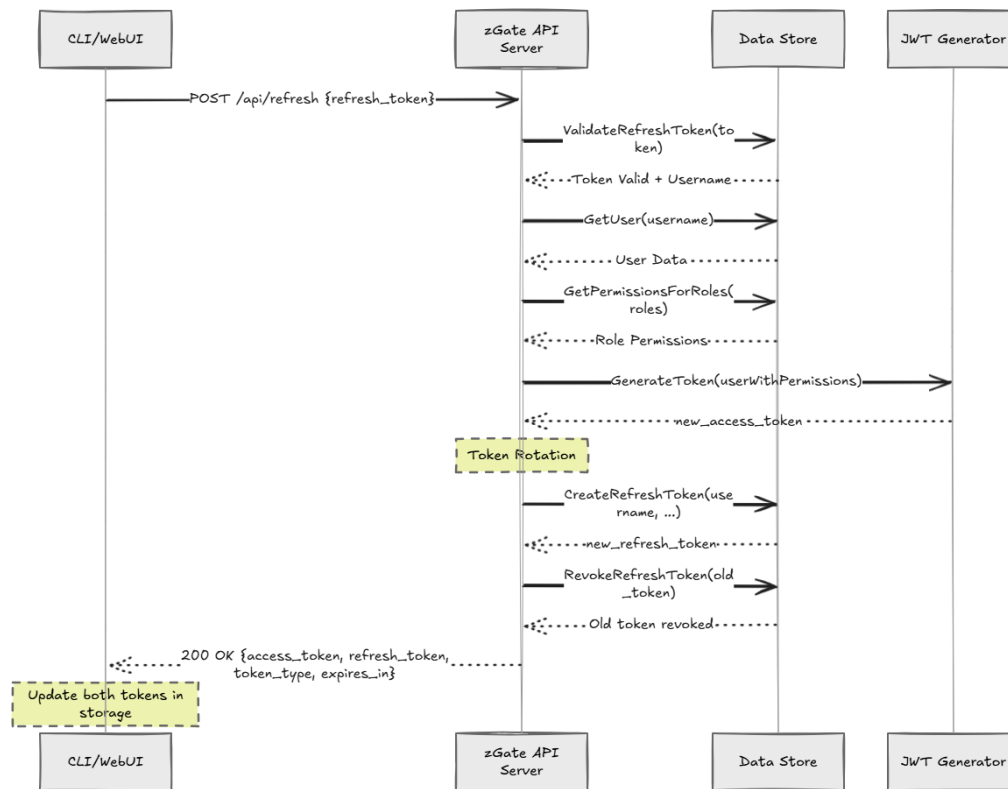


Figure 10.7: Token Refresh Flow with Rotation

Token Rotation Security:

- Client submits current refresh token to POST /api/refresh endpoint
- Store validates refresh token and retrieves associated username
- New access token is generated with updated permissions
- New refresh token is created to replace the old one
- Old refresh token is immediately revoked in database
- Both new tokens are returned to client
- Prevents token reuse attacks and limits blast radius of token compromise

Admin Login Flow

Figure 10.8 illustrates the administrative authentication flow with fallback mechanism.

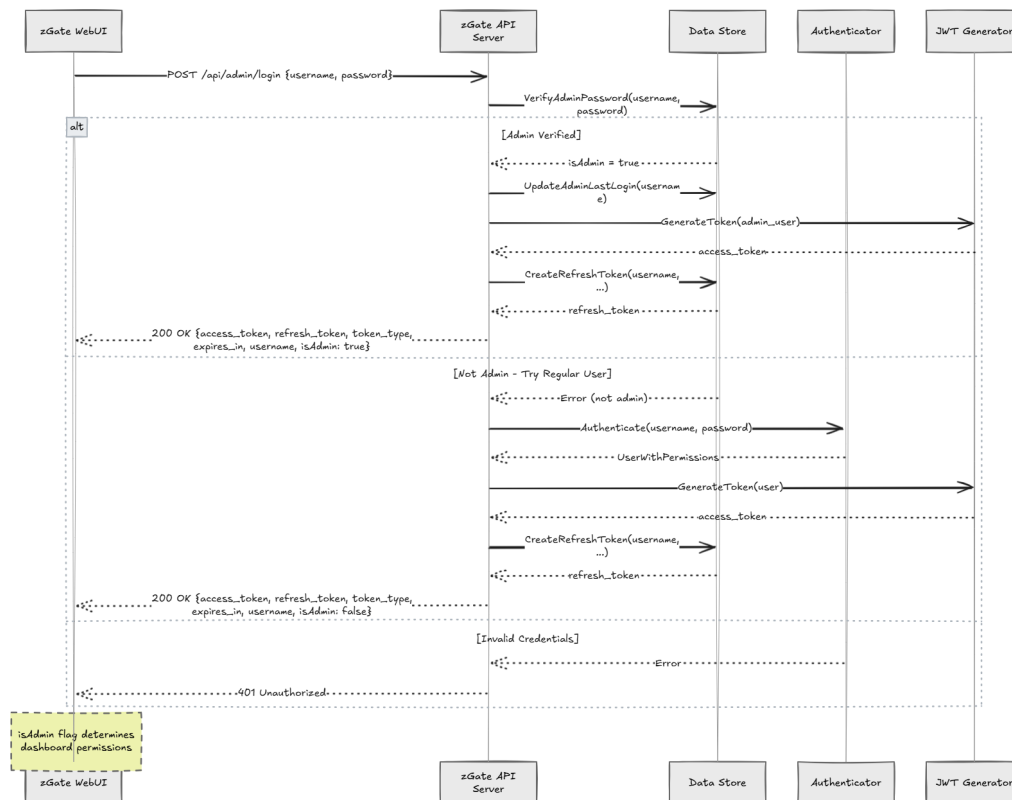


Figure 10.8: Admin Authentication with User Fallback

Dual Authentication Strategy:

1. First attempts admin authentication via dedicated admins table
2. If admin credentials valid, sets isAdmin flag to true in response
3. If not found in admins table, falls back to regular user authentication
4. Returns isAdmin flag indicating permission level for dashboard routing
5. Updates admin last login timestamp for audit purposes

10.2.5 Database Connection Lifecycle

The database connection lifecycle implements the core Zero Trust principle: users never handle production credentials. The system manages three types of database accounts with different lifecycle characteristics.

Database Account Types

The zGate platform implements a sophisticated three-tier account management system that eliminates credential exposure while optimizing resource utilization and security. Each account type serves a distinct purpose in the Zero Trust architecture:

Personal Database Accounts Personal accounts represent the highest privilege tier, providing dedicated database credentials assigned exclusively to individual users.

Creation and Provisioning:

- Administrators pre-create accounts through the WebUI administrative interface
- Each account is permanently associated with a specific user via `user_id` mapping
- Credentials are stored encrypted in the zGate database
- Account permissions are configured at the database level by DBAs

Usage Scenario:

- When a user with a personal account requests database access, zGate immediately retrieves their dedicated credentials
- No account creation overhead or waiting time
- User maintains consistent database identity across sessions
- Ideal for privileged users, developers, or users requiring specific database roles

Lifecycle and Advantages:

- Accounts persist indefinitely across sessions and disconnections
- Provides accountability through consistent database audit trails
- Enables fine-grained permission management at the user level
- Eliminates runtime account provisioning latency

Shared Account Pool Shared accounts implement a connection pool pattern, enabling multiple users to share a limited set of pre-provisioned database accounts.

Creation and Pool Management:

- Administrators create a pool of generic database accounts via WebUI
- Each account has configurable concurrent usage limits (`max_concurrent_users`)
- Pool size is determined by anticipated concurrent load
- Accounts are tagged as `shared` in the account type field

Usage Scenario:

- When a user without a personal account connects, zGate queries the pool for available accounts

- An account is considered available if `currently_in_use < max_concurrent_users`
- Upon assignment, the `currently_in_use` counter is atomically incremented
- Multiple users can share the same database account simultaneously within the limit
- Ideal for read-only users, analysts, or scenarios with many concurrent users

Lifecycle and Advantages:

- Accounts remain persistent but are released back to the pool on disconnect
- The `currently_in_use` counter is decremented when the session ends
- Significantly reduces database account sprawl compared to per-user accounts
- Optimizes resource utilization in environments with high user counts
- Provides a balance between security and operational overhead

Ephemeral Temporary Accounts Ephemeral accounts represent the ultimate Zero Trust approach: credentials that exist only for the duration of a single database session.

Creation and Provisioning:

- Created on-demand by the Gateway Manager component when no personal or shared accounts are available
- zGate connects to the database using a privileged provisioning account
- Executes `CREATE USER` with a cryptographically random username and password
- Grants appropriate permissions based on the user's role permissions from zGate
- Account details are stored only in memory within the active session context

Usage Scenario:

- Triggered when personal account doesn't exist and shared pool is exhausted
- Entire account lifecycle contained within a single zGate session
- User never sees or handles the actual database credentials
- Ideal for untrusted environments or maximum security requirements

Lifecycle and Advantages:

- Account is deleted immediately upon session disconnect via `DROP USER`

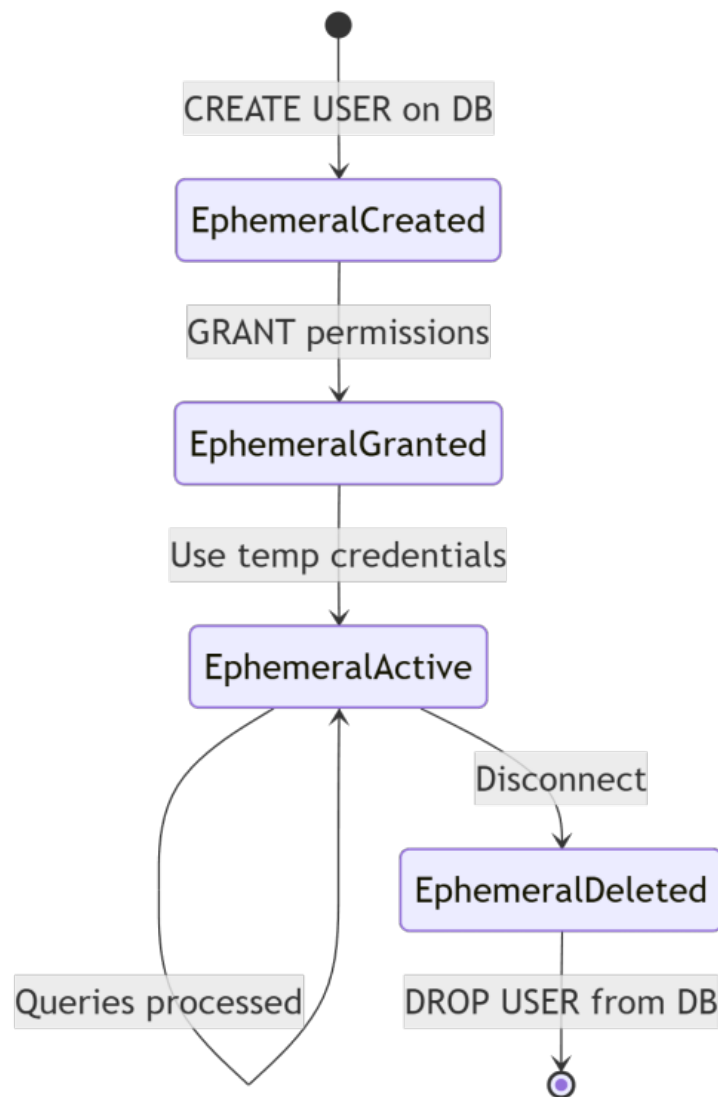


Figure 10.9: Ephemeral Account Creation

- Credentials become invalid within seconds of session termination
- Zero persistent credential footprint in the database
- Eliminates long-term credential theft risks
- Perfect for compliance-sensitive environments requiring credential rotation
- Automatically scales to unlimited concurrent users without pool exhaustion

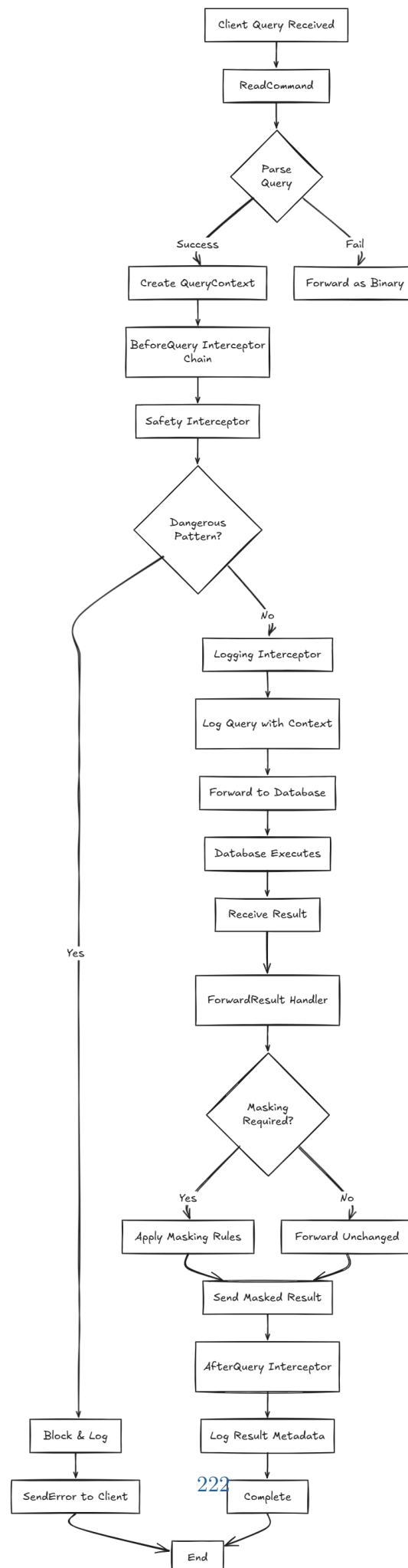
Three-Tier Strategy Rationale:

This tiered approach provides operational flexibility while maintaining security:

- **Performance:** Personal accounts eliminate runtime provisioning latency
- **Scalability:** Shared pools handle moderate concurrency efficiently
- **Security:** Ephemeral accounts provide maximum protection when needed
- **Cost Efficiency:** Reduces database account sprawl in large deployments
- **Compliance:** Ephemeral accounts satisfy strict credential rotation policies

Query Interception and Processing

Figure 10.10 illustrates the complete query processing pipeline with security enforcement.



Query Processing Stages:

1. **Read Command:** Protocol handler reads and parses client query
2. **Create Context:** QueryContext populated with session metadata
3. **BeforeQuery Chain:** Safety and logging interceptors process query
4. **Safety Evaluation:** Dangerous patterns trigger query blocking
5. **Forward Allowed:** Safe queries are forwarded to backend database
6. **Result Processing:** Masking interceptor processes result set
7. **AfterQuery Chain:** Result metadata is logged for audit

RBAC Permission Flow

Figure 10.11 demonstrates how role-based access control is evaluated for each connection request.

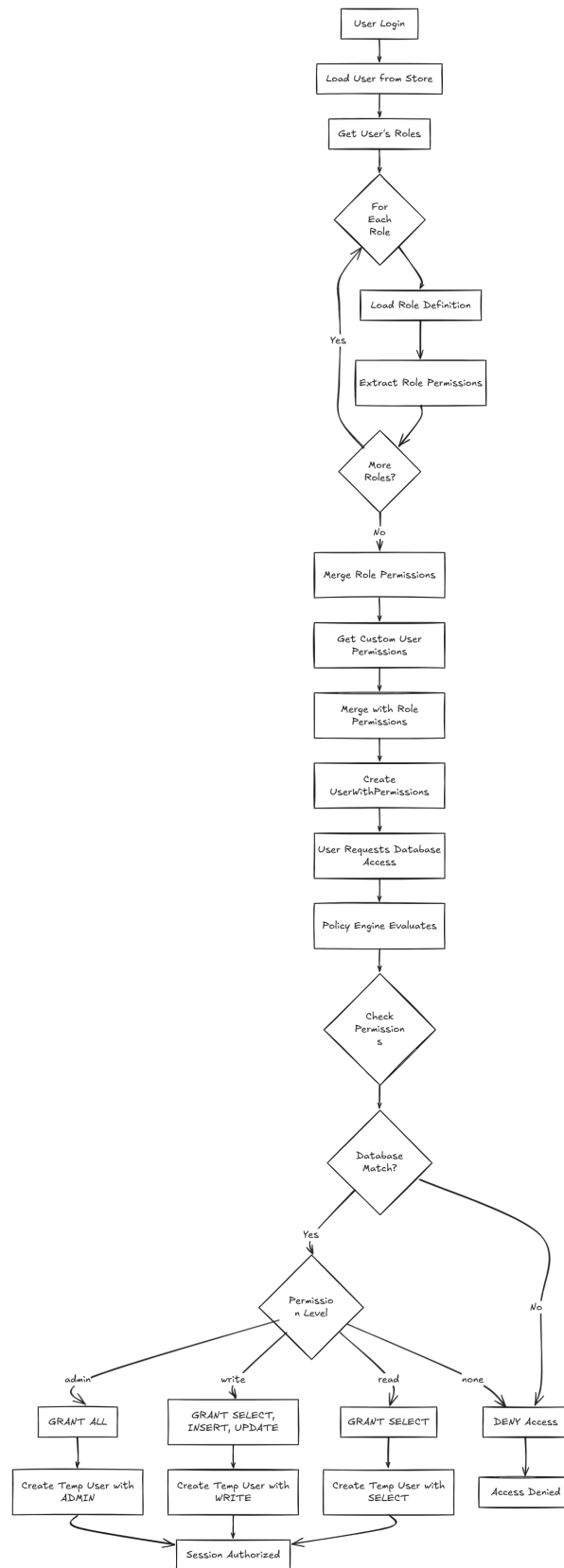


Figure 10.11: Role-Based Access Control Permission Resolution

Permission Resolution Process:

1. User login triggers retrieval of assigned roles from store

2. Each role's permissions are fetched and merged
3. Custom user-specific permissions are added to the set
4. Combined permissions are embedded in JWT claims
5. Connection request validates database access against permissions
6. Appropriate permission level determines database account type selection

10.2.6 Web UI Architecture and Features

The Web UI is built with Next.js and React, providing a comprehensive administrative interface for managing all aspects of the zGate platform.

User Management

Figure 10.12 shows the complete user management workflow available to administrators.

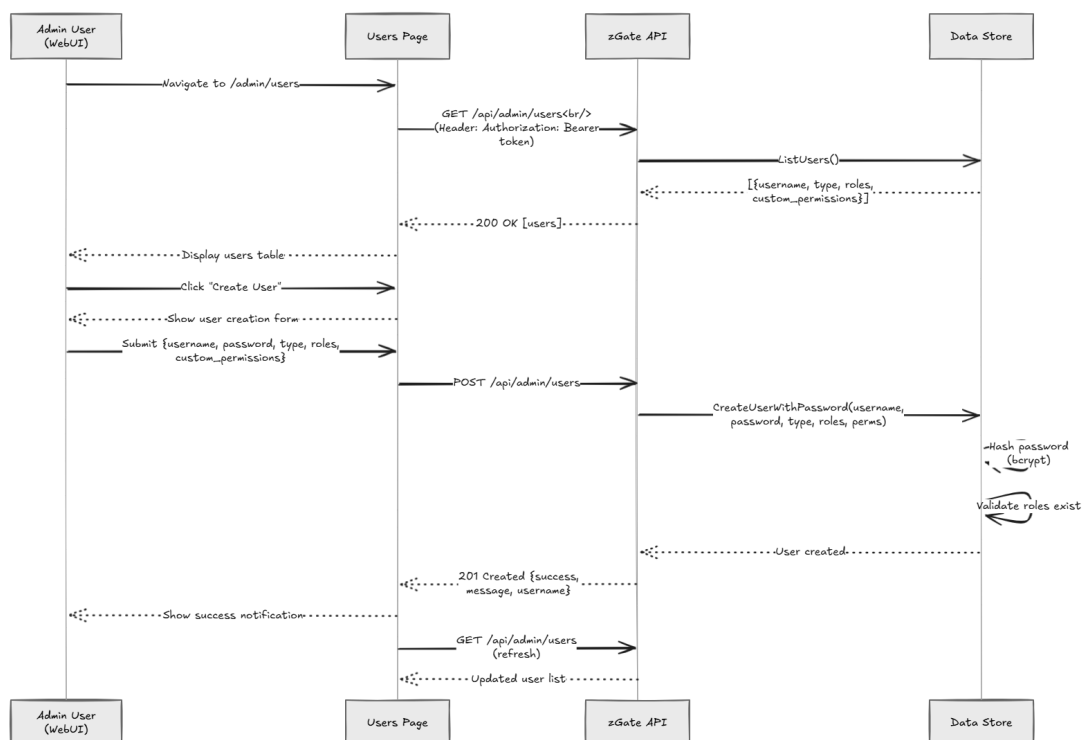


Figure 10.12: WebUI User Management Flow

User Management Capabilities:

- Create users with username, password, type (user/service account), and role assignments
- Assign multiple roles to users for flexible permission management

- Configure custom permissions that override role defaults
- Update user credentials and permission sets
- Delete users with automatic session termination
- Monitor online user status in real-time

Database Management

Figure 10.13 illustrates the database registration and configuration workflow.

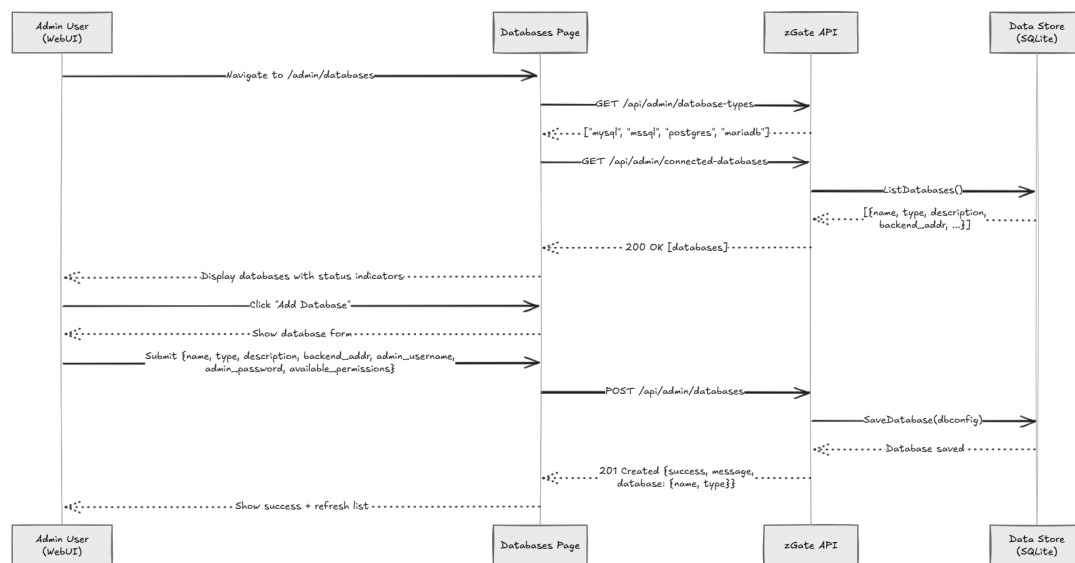


Figure 10.13: WebUI Database Management Flow

Database Configuration:

- Register databases with name, type (MySQL, PostgreSQL, MSSQL, MariaDB), and backend address
- Configure admin credentials for ephemeral user creation
- Define available permission levels (read, write, admin)
- Test database connectivity before saving configuration
- Monitor database connection status and active sessions

Active Session Monitoring

Figure 10.14 presents the active session monitoring interface with detailed activity tracking.

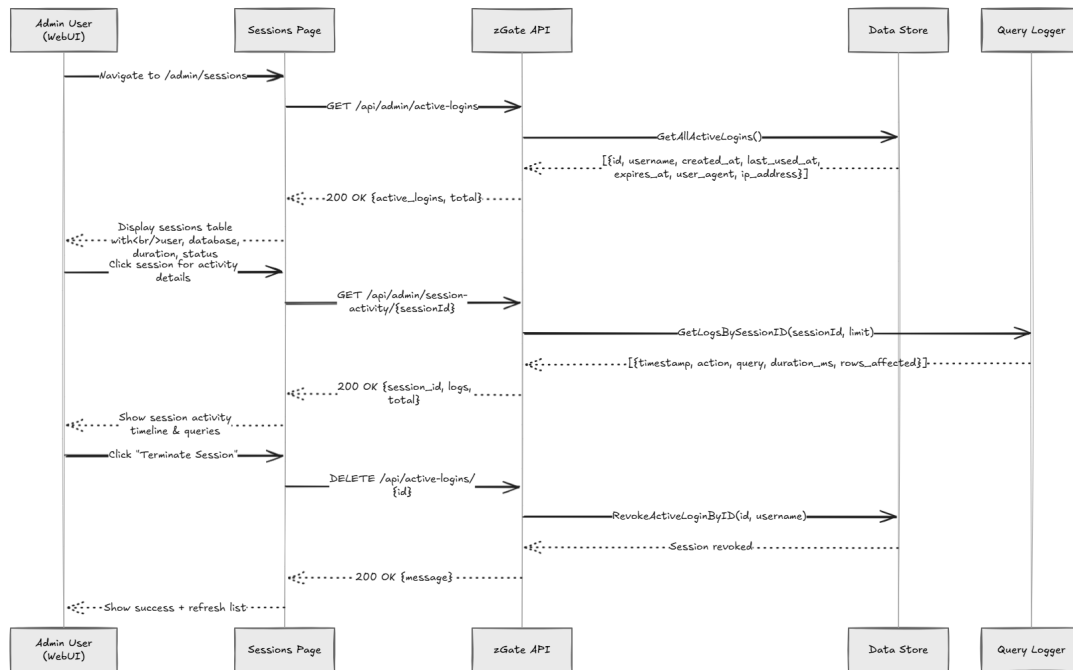


Figure 10.14: Active Session Monitoring and Activity Logs

Session Monitoring Features:

- View all active login sessions across all users
- Display session metadata: user, database, duration, client IP, user agent
- Drill down into session activity logs showing individual queries
- View query execution time and rows affected
- Terminate sessions remotely for security enforcement
- Filter sessions by user, database, or time range

Query History and Audit

Figure 10.15 shows the query history interface for compliance and forensics.

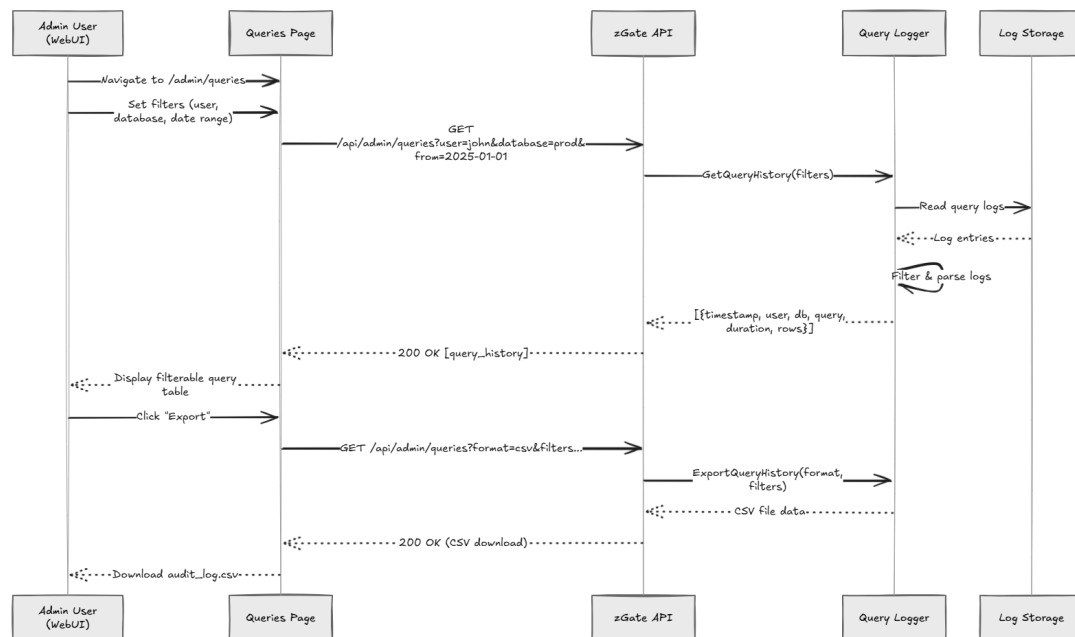


Figure 10.15: Query History and Audit Trail

Audit Capabilities:

- Search query history with filters: user, database, date range, query pattern
- View complete query text with syntax highlighting
- Display execution duration and result metadata
- Export audit logs to CSV for compliance reporting
- Immutable logs ensure query attribution cannot be repudiated

Role and Permission Management

Figure 10.16 depicts the role-based access control configuration interface.

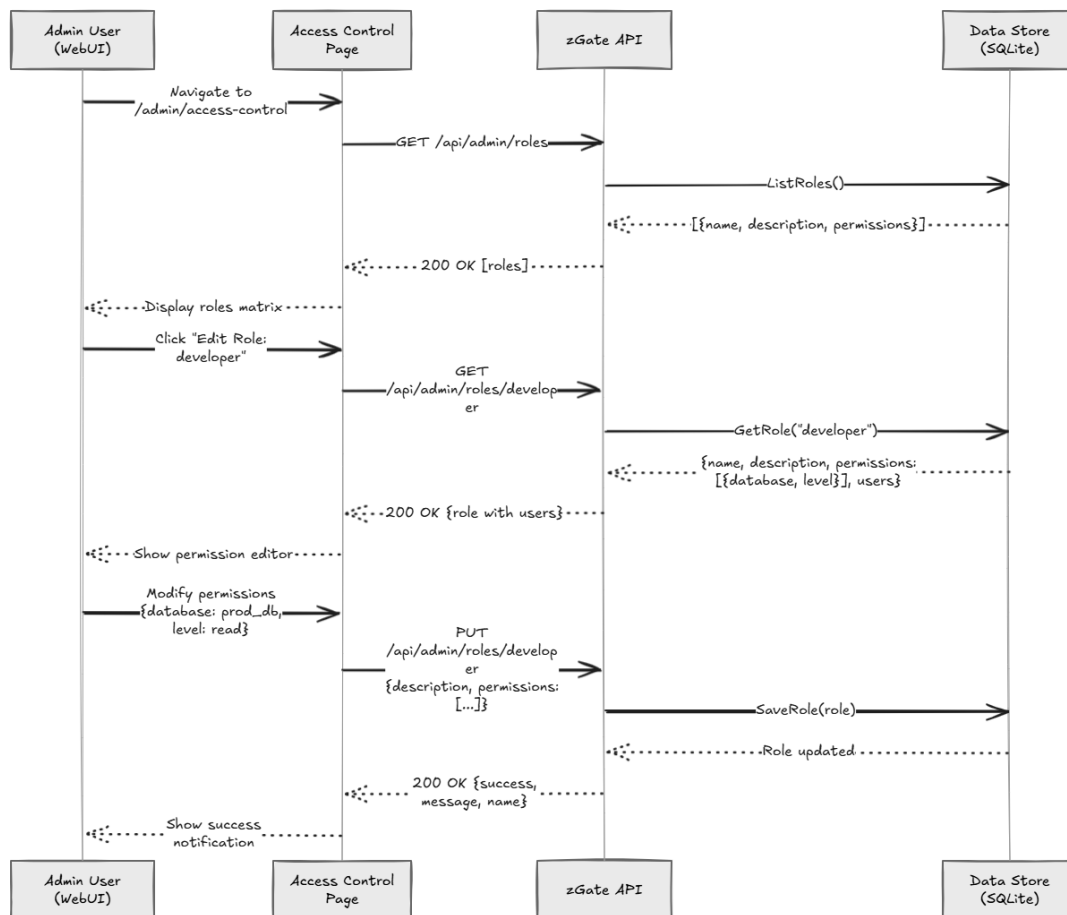


Figure 10.16: Role and Permission Management

RBAC Configuration:

- Create roles with descriptive names and purposes
- Assign database-level permissions (read, write, admin) to roles
- Configure interceptor chains (safe_query, masking) per role
- View users assigned to each role
- Modify role permissions with immediate effect on active sessions

Activity Audit

Figure 10.17 illustrates the comprehensive activity audit interface.

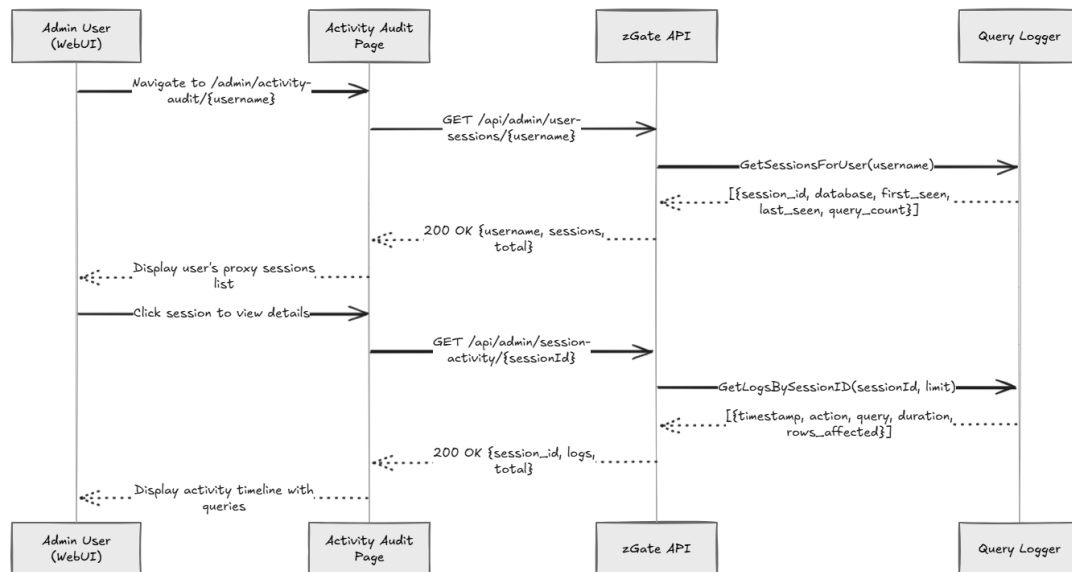


Figure 10.17: Activity Audit and Session Timeline

Admin Account Management

Figure 10.18 shows the administrative account management workflow.

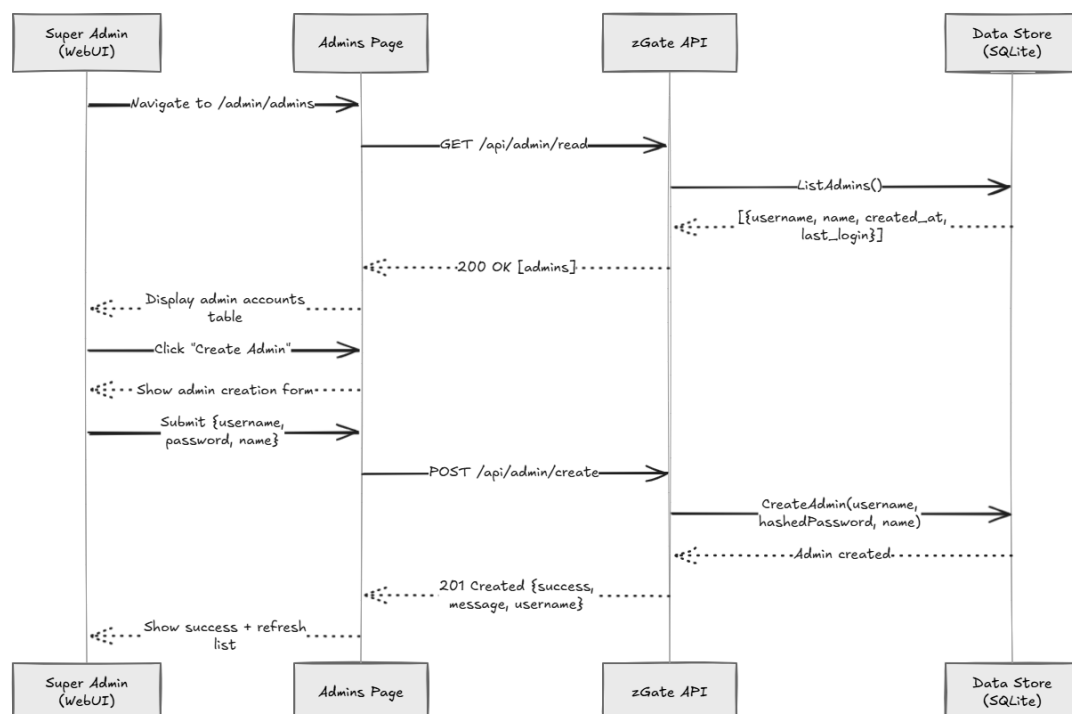


Figure 10.18: Admin Account Management

Shared Account Management

Figure 10.19 presents the shared database account pool management interface.

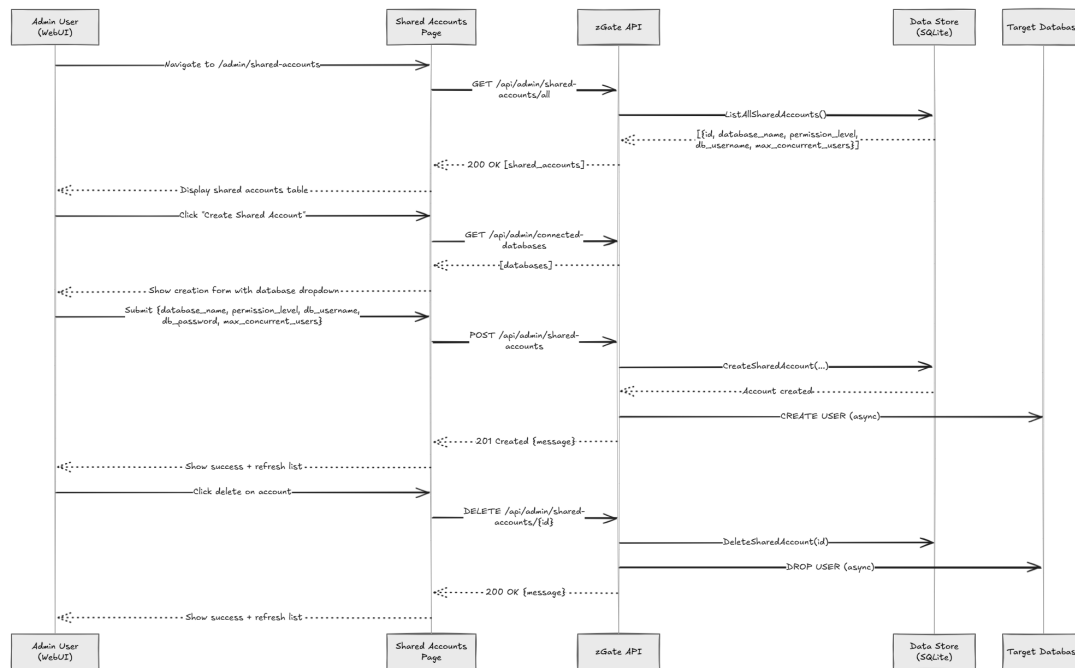


Figure 10.19: Shared Database Account Pool Management

Shared Account Pool Features:

- Create shared accounts with database username, password, and permission level
- Configure maximum concurrent users per shared account
- View current usage status and available accounts
- Automatically releases accounts when sessions disconnect
- Provides account pooling for organizations without personal accounts

User Database Account Management

Figure 10.21 illustrates personal database account assignment.

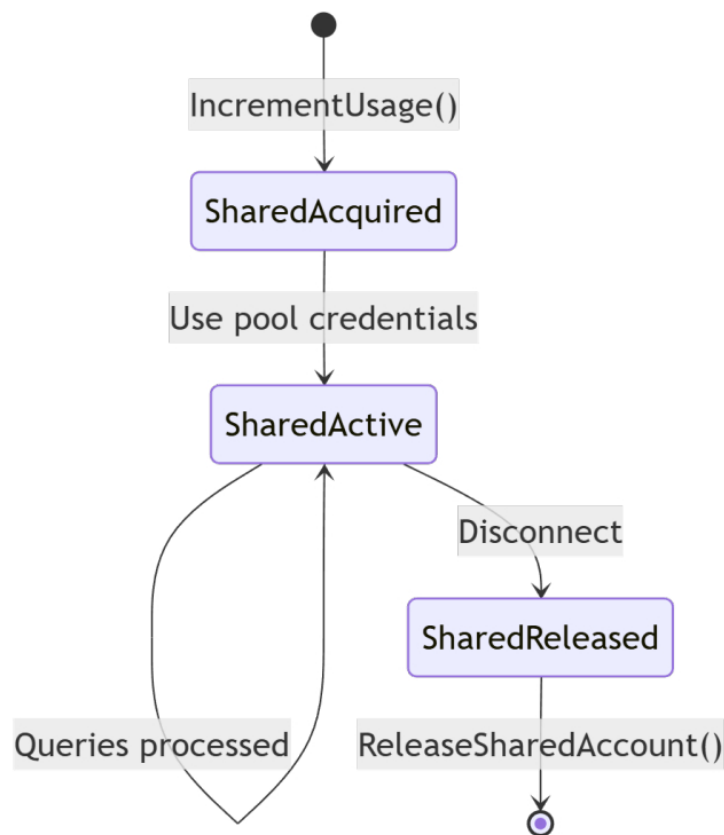


Figure 10.20: Shared Account Pool Management

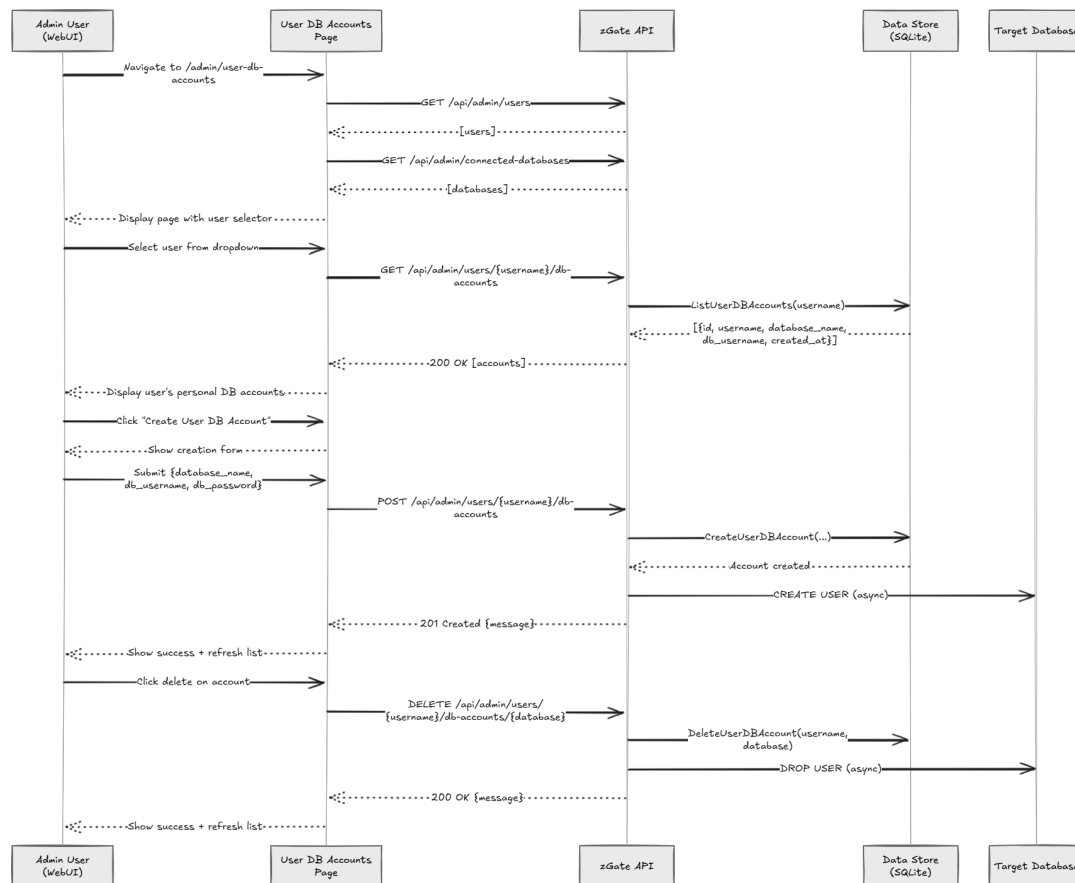


Figure 10.21: User Database Account Assignment

Personal Account Management:

- Assign dedicated database accounts to specific users
- Create database users on target database automatically
- Personal accounts have highest priority in connection flow
- Provide consistent database identity for users requiring it
- Delete personal accounts when no longer needed

10.2.7 CLI Architecture and Features

The CLI provides a cross-platform command-line interface for developers to authenticate and connect to databases without handling credentials.

CLI Login

Figure 10.22 shows the CLI authentication flow with secure token storage.

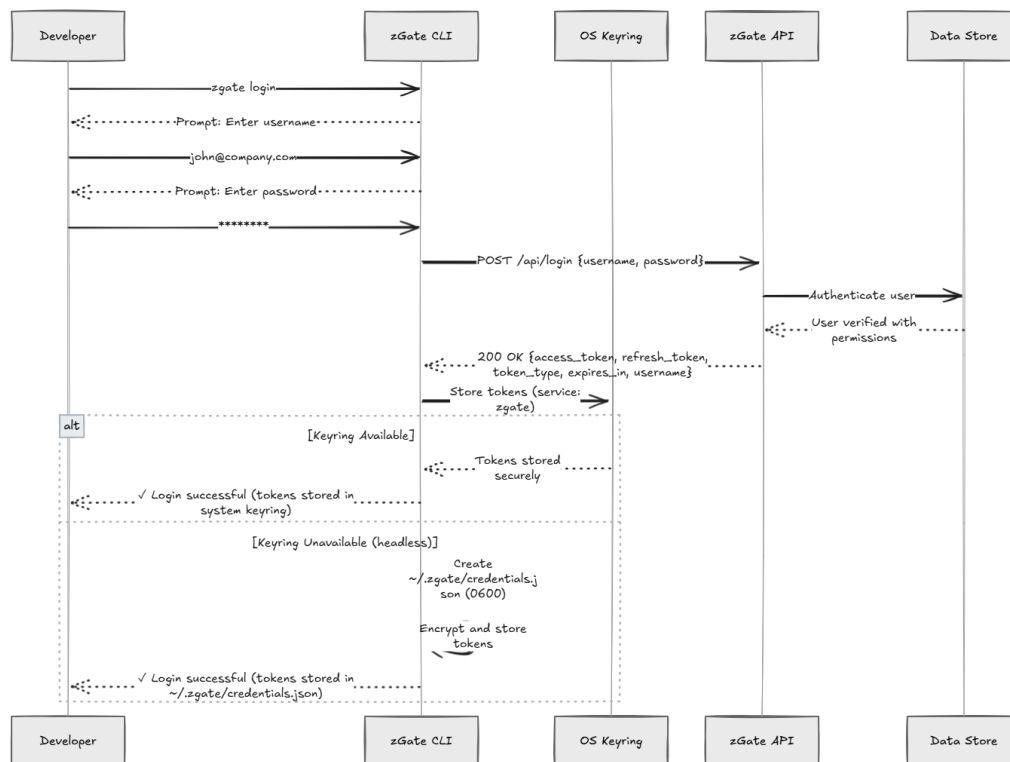


Figure 10.22: CLI Login with Secure Token Storage

Secure Token Storage:

- **macOS:** Keychain (`~/Library/Keychains/`)
- **Windows:** Credential Manager
- **Linux:** Secret Service API (`gnome-keyring/kwallet`)
- **Fallback:** Encrypted file at `~/.zgate/credentials.json` with 0600 permissions

CLI Database Connection

Figure 10.23 illustrates the database connection flow with both local and remote modes.

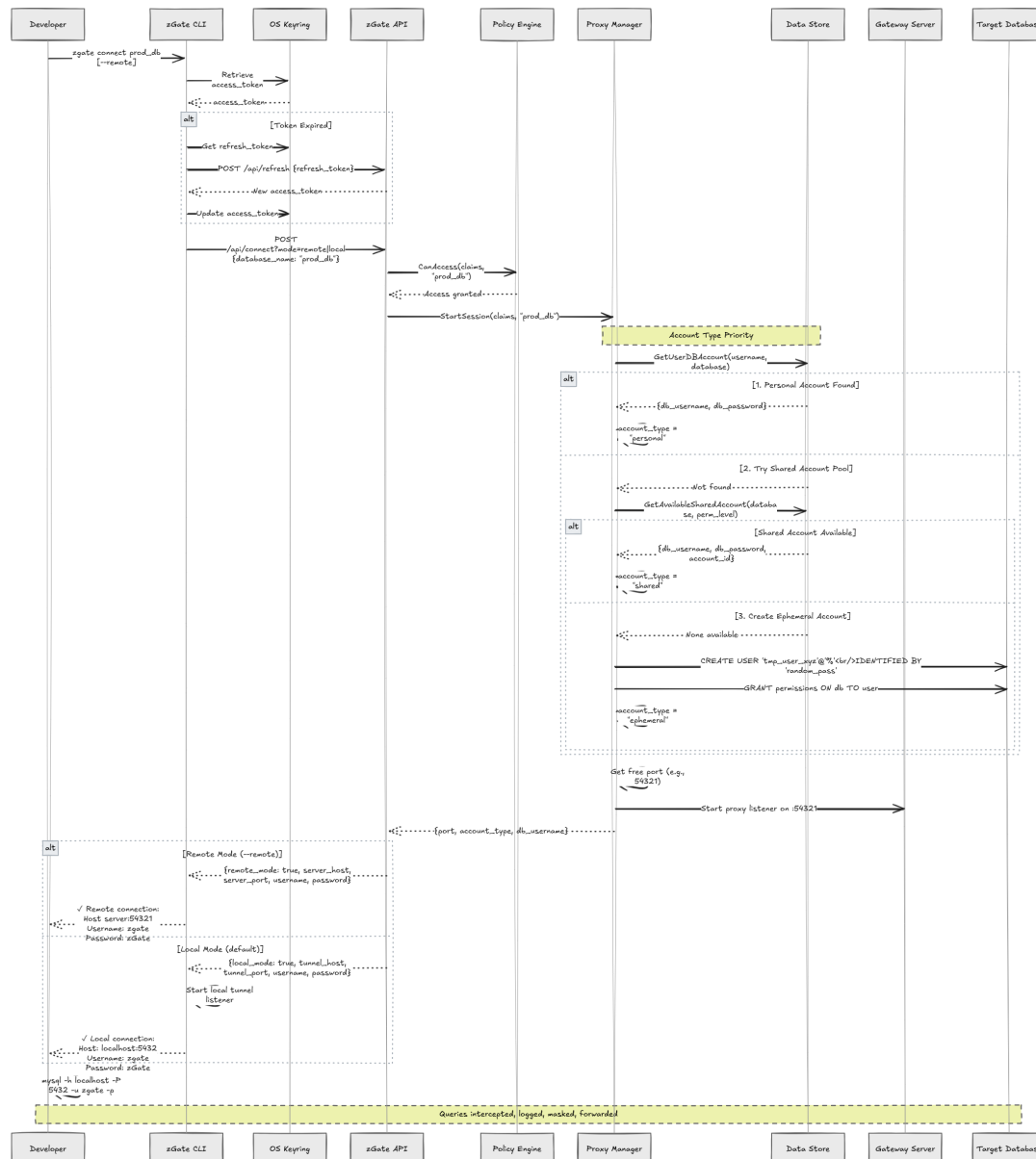


Figure 10.23: CLI Database Connection with Account Priority

Connection Modes:

- **Local Mode (default):** CLI creates local TCP tunnel on localhost, user connects to localhost port
- **Remote Mode (`--remote` flag):** User connects directly to zGate server on allocated port

Local Tunnel Architecture: The CLI starts a local listener that forwards traffic to the zGate proxy:

1. API returns tunnel_host and tunnel_port
2. CLI creates listener on localhost:0 (random port allocation)

3. CLI returns localhost:port to user
4. User's database client connects to localhost
5. CLI accepts connection and dials zGate server
6. Bidirectional io.Copy between local and remote connections

CLI Session Management

Figure 10.24 shows CLI session listing and management capabilities.

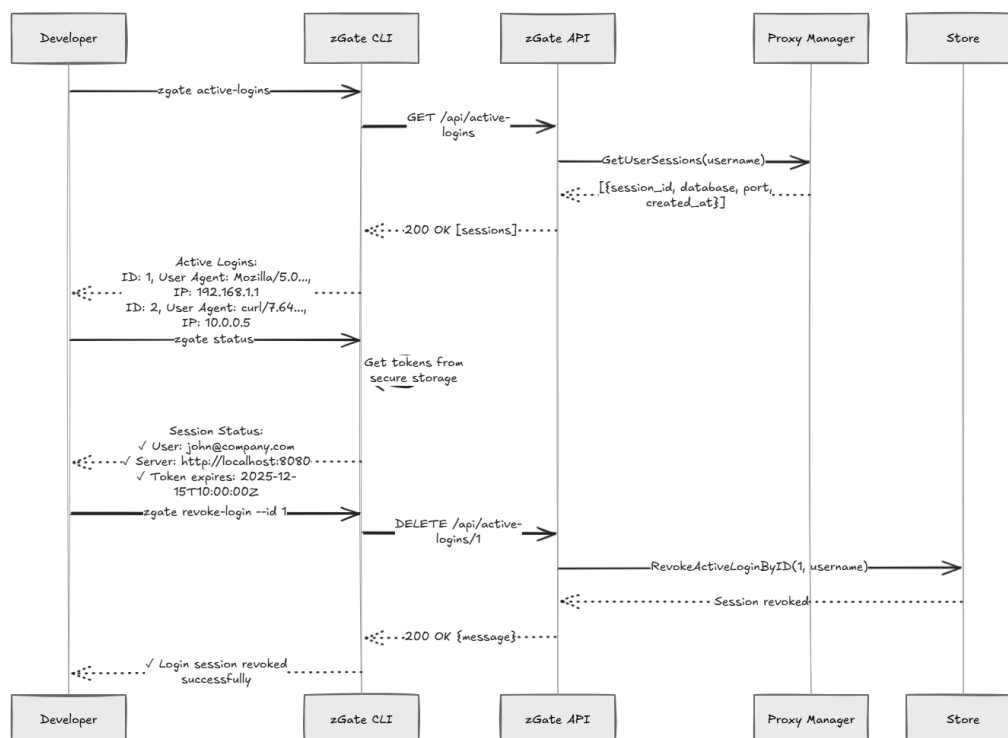


Figure 10.24: CLI Active Session Management

CLI Logout

Figure 10.25 illustrates the comprehensive logout flow that terminates all sessions and cleans up resources.

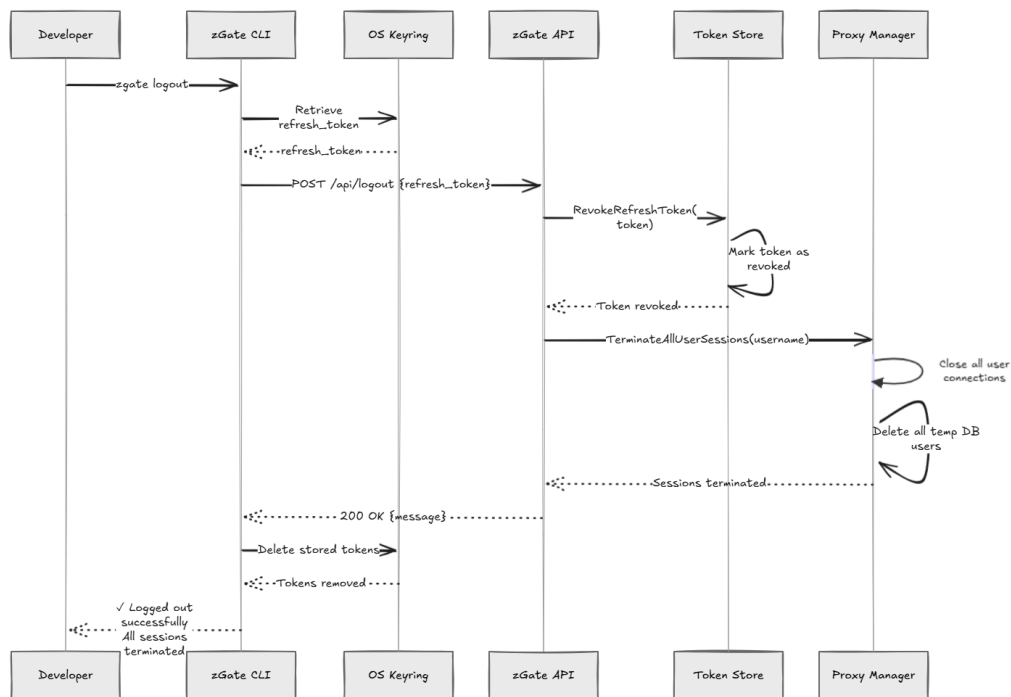


Figure 10.25: CLI Logout with Session Cleanup

Logout Process:

- Revokes refresh token to invalidate all access tokens
- Terminates all active database connections for user
- Deletes ephemeral database users created during sessions
- Removes tokens from secure storage (keyring or encrypted file)
- Ensures no residual credentials remain on client system

CLI List Databases

Figure 10.26 shows the database discovery interface that displays accessible databases based on user permissions.

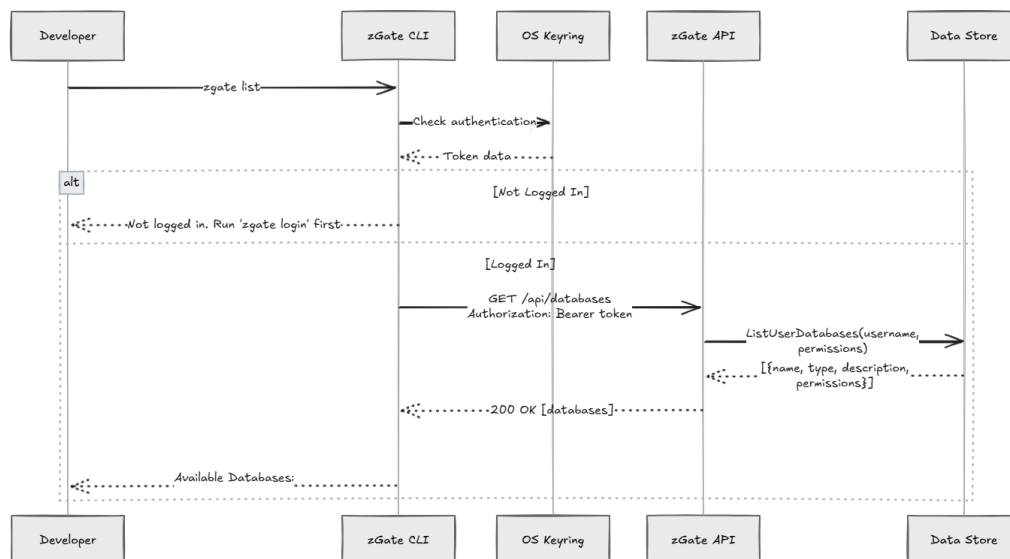


Figure 10.26: CLI Database Listing with Permissions

10.2.8 API Architecture

The API server provides a RESTful interface for all platform operations, implementing comprehensive authentication and authorization middleware.

Public Endpoints (No Authentication Required)

Authentication Endpoints

POST /api/login

Authenticates users and issues access/refresh token pair.

Request: username, password

POST /api/refresh

Refreshes expired access tokens using valid refresh token.

Request: refresh_token

POST /api/logout

Revokes refresh token and invalidates session.

Request: refresh_token

POST /api/admin/login

Authenticates admin users with elevated privileges.

Request: username, password

Protected User Endpoints

User Session Management

GET /api/active-logins

Lists all active login sessions for the authenticated user.

DELETE /api/active-logins/{id}

Revokes a specific active session belonging to the user.

Database Connection

GET /api/databases

Returns list of databases accessible to the authenticated user.

POST /api/connect

Establishes a proxy connection to specified database.

Request: database_name

POST /api/disconnect

Closes active database connection for the user.

Request: database_name

Admin Endpoints - User Management

User Administration

GET /api/admin/users

Retrieves complete list of users with their roles and permissions.

POST /api/admin/users

Creates a new user account with specified roles.

Request: username, password, roles

PUT /api/admin/users/{username}

Updates existing user's password, roles, or permissions.

Request: password, roles, permissions

DELETE /api/admin/users/{username}

Permanently removes user account from the system.

GET /api/admin/users/statuses

Fetches real-time online/offline status for all users.

Admin Endpoints - Role Management

Role Administration

GET /api/admin/roles

Lists all defined roles with their associated permissions.

POST /api/admin/roles

Creates a new role with specified permissions set.

Request: name, description, permissions

GET /api/admin/roles/{name}

Retrieves detailed information about a specific role.

PUT /api/admin/roles/{name}

Modifies role description or permission configuration.

Request: description, permissions

DELETE /api/admin/roles/{name}

Removes role from system (requires no active assignments).

Admin Endpoints - Database Management

Database Configuration

GET /api/admin/database-types

Returns list of supported database types (MySQL, PostgreSQL, MSSQL).

GET /api/admin/connected-databases

Lists all registered databases with connection status.

POST /api/admin/databases

Registers new database instance with connection credentials.

Request: name, type, address, credentials

PUT /api/admin/databases/{name}

Updates database connection parameters or credentials.

DELETE /api/admin/databases/{name}

Removes database registration from the gateway system.

Admin Endpoints - Session & Query Management

Session Monitoring & Query Execution

GET /api/admin/sessions

Displays all active proxy sessions across the system.

POST /api/admin/execute-query

Executes administrative queries directly on specified database.

Request: database, query

GET /api/admin/queries

Retrieves query execution history with filtering capabilities.

Admin Endpoints - Session Activity Logs

Activity Tracking

GET /api/admin/active-logins

Lists all active login sessions for all users system-wide.

GET /api/admin/session-activity/{id}

Retrieves detailed activity logs for a specific session.

GET /api/admin/user-sessions/{user}

Shows all proxy connection sessions for specified user.

Admin Endpoints - Shared Accounts

Shared Account Management

GET /api/admin/shared-accounts

Lists shared database accounts with optional database filter.

GET /api/admin/shared-accounts/all

Retrieves all shared accounts across all databases.

POST /api/admin/shared-accounts

Creates new shared account with concurrency limits.

Request: database, username, password, max_concurrent_users

DELETE /api/admin/shared-accounts/{id}

Removes shared account from the system.

Admin Endpoints - User DB Accounts

Personal Database Account Assignment

GET /api/admin/users/{user}/db-accounts

Lists all personal database accounts assigned to user.

POST /api/admin/users/{user}/db-accounts

Assigns personal database credentials to specific user.

Request: database, username, password

DELETE /api/admin/users/{user}/db-accounts/{db}

Revokes user's personal database account assignment.

Admin Endpoints - Admin Account Management

Administrator Account Control

POST /api/admin/create

Creates new administrator account with full privileges.

Request: username, password, full_name

GET /api/admin/read

Retrieves list of all administrator accounts.

PUT /api/admin/update/{user}

Updates administrator account password or profile information.

Request: password, full_name

DELETE /api/admin/delete/{user}

Permanently deletes administrator account (requires super-admin).

10.2.9 Data Storage Architecture

The system utilizes SQLite as the embedded database for configuration, credentials, and audit data. SQLite was chosen for its zero-configuration deployment, ACID compliance, and sufficient performance for the access patterns.

Database Schema

The store implements the following primary tables:

- **users:** User accounts with bcrypt-hashed passwords, roles, custom permissions
- **admins:** Administrative accounts with elevated privileges
- **roles:** Role definitions with associated permissions



- **databases:** Registered target database configurations
- **refresh_tokens:** Active refresh tokens with user association and metadata
- **shared_accounts:** Pool of shared database accounts with concurrent usage tracking
- **user_db_accounts:** Personal database account assignments
- **query_logs:** Immutable audit trail of all database queries (optional, can use file-based logging)

Migration System

The store implements an automatic schema migration system that:

- Applies incremental schema changes on startup
- Tracks applied migrations to prevent duplicate application
- Ensures backward compatibility during upgrades
- Supports data migrations for schema transformations

10.2.10 Security Architecture

The system implements defense-in-depth security with multiple layers of protection.

Credential Protection

- **User Passwords:** bcrypt hashing with salt (cost factor 10)
- **Database Credentials:** Stored encrypted in SQLite, decrypted only in memory during connection
- **JWT Tokens:** HS256 signing with secret key, short expiration times
- **Refresh Tokens:** Cryptographically random, stored hashed in database

Network Security

- **TLS Support:** Optional end-to-end TLS from client to database
- **Certificate Pinning:** Strict certificate verification for backend databases
- **Proxy Authentication:** Fake credentials prevent direct database access
- **Port Isolation:** Each session on dedicated port prevents cross-session attacks

Audit and Compliance

- **Immutable Logs:** All queries logged with user attribution
- **Session Tracking:** Complete session lifecycle recorded
- **Query Blocking:** Dangerous operations prevented and logged
- **Data Masking:** Sensitive data redacted in query results

10.2.11 Architecture Quality Attributes

The zGate platform is engineered with six foundational quality attributes that define its operational characteristics and production readiness. These attributes guide architectural decisions and validate the system's suitability for enterprise environments.

Security: Zero Trust by Design

Architectural Guarantees:

- **Credential Elimination:** End users never access or store production database credentials
- **Wire-Protocol Interception:** All database traffic passes through security enforcement layer
- **Comprehensive Audit Trail:** Every query, connection, and authentication event logged immutably
- **Defense in Depth:** Multiple security layers (authentication, authorization, query validation, masking)
- **Ephemeral Credentials:** Optional temporary accounts that self-destruct on disconnect
- **JWT-Based Authentication:** Stateless token validation with automatic expiration and refresh

Security Posture: The architecture assumes breach and enforces Zero Trust at every layer, eliminating single points of credential compromise.

Scalability: Horizontal and Vertical Growth

Scaling Characteristics:

- **Stateless API:** JWT-based authentication enables load balancer distribution without session affinity

- **Shared Session State:** Centralized SQLite/PostgreSQL store allows multiple gateway instances
- **Efficient Protocol Handling:** Go goroutines provide lightweight concurrency (10,000+ simultaneous connections)
- **Minimal Per-Connection Overhead:** ~5MB memory per active session
- **Dynamic Port Allocation:** Operating system manages ephemeral port pool
- **Connection Pooling:** Shared and ephemeral accounts reduce backend database load

Scaling Strategy: Horizontal scaling through multiple gateway instances, vertical scaling through connection pooling and efficient Go runtime.

Maintainability: Clean Architecture Principles

Design Characteristics:

- **Component Separation:** Clear boundaries between Gateway, API, Store, and Protocol layers
- **Interface-Based Design:** Protocol handlers implement common interface for database abstraction
- **Dependency Injection:** Components receive dependencies explicitly, not via globals
- **Comprehensive Logging:** Structured JSON logs with correlation IDs for distributed tracing
- **Error Propagation:** Explicit error handling with context preservation
- **Self-Documenting Code:** Type-safe structs and clear naming conventions

Maintenance Philosophy: Code organization prioritizes clarity over cleverness, with explicit dependencies and minimal coupling.

Extensibility: Modular and Composable

Extension Points:

- **Interceptor Pipeline:** Pluggable query processing stages (safety, logging, masking, custom)
- **Protocol Handler Interface:** Add new database protocols by implementing Handler interface
- **Policy Engine:** Custom policy rules via store extensions

- **Authentication Providers:** Extensible to LDAP, OAuth, SAML via AuthService interface
- **Storage Backends:** SQLite, PostgreSQL, MySQL supported via unified Store interface
- **Middleware Stack:** HTTP middleware chain for cross-cutting concerns

Extension Philosophy: Open/Closed principle - open for extension via interfaces, closed for modification of core logic.

Performance: Low-Latency Proxying

Performance Characteristics:

- **Proxy Overhead:** Typically <1ms latency added by zGate layer
- **Wire Protocol Efficiency:** Zero serialization - packets forwarded directly
- **Connection Pooling:** Shared accounts reduce backend connection establishment overhead
- **Go Runtime Performance:** Compiled binary, no garbage collection pauses during I/O
- **Efficient Goroutines:** Lightweight threads with <2KB stack overhead
- **Memory Efficiency:** Streaming query results without buffering entire result sets

Performance Target: Sub-millisecond overhead for query proxying, supporting thousands of concurrent connections per instance.

Reliability: Fault Tolerance and Recovery

Reliability Mechanisms:

- **Graceful Degradation:** Shared pool exhaustion falls back to ephemeral accounts
- **Automatic Cleanup:** Session goroutines clean up resources on disconnect or panic
- **Error Recovery:** Panic handlers prevent single-connection failures from crashing gateway
- **Health Checks:** /health and /ready endpoints for orchestration platforms
- **Connection Timeout:** Configurable timeouts prevent resource leaks
- **Audit Continuity:** Failed queries logged before rejection

Reliability Goal: 99.9% uptime for gateway layer, with automatic recovery from transient failures and comprehensive logging for post-mortem analysis.

Chapter 11

User Interfaces

This part covers:

- Command-line interface design
- Token-based authentication
- Web administration dashboard
- Database and connection management
- User and role management
- Session monitoring and audit trails

USER interfaces bridge powerful backend systems and human operators. This chapter explores zGate's dual interface approach: a streamlined command-line tool with secure keyring integration for developers, and a comprehensive Next.js-based web dashboard for administrators. We examine authentication flows, token management, database configuration, user and role administration, active session monitoring, and comprehensive audit trail visualization.

11.1 Admin Panel WebUI

The zGate Admin Panel WebUI serves as the central management hub for the entire Zero Trust Database Access Gateway system. Built with modern web technologies and designed with both security and usability in mind, it provides administrators with comprehensive control over users, databases, roles, and system-wide security policies.

11.1.1 Introduction and Overview

The Admin Panel represents the intersection of enterprise-grade functionality and modern user experience design. In an era where database security is paramount, the interface acts as the command center from which administrators can enforce Zero Trust principles, monitor real-time activity, and manage access control with precision.

Key Design Philosophy:

The Admin Panel WebUI is designed around three core principles:

- **Security-First Architecture:** Every action requires authentication, all API calls use JWT tokens with automatic refresh, and comprehensive audit logging captures every administrative operation
- **Intuitive Management:** Complex database access control is simplified through visual role assignment, drag-and-drop permission configuration, and real-time status indicators
- **Enterprise Scalability:** Built to handle hundreds of users, multiple database systems, and thousands of concurrent sessions while maintaining responsive performance

11.1.2 Technology Stack and Architectural Foundation

The zGate Admin Panel leverages a carefully curated technology stack optimized for modern enterprise web application development. Each technology was selected based on rigorous market analysis, community support, and alignment with industry best practices.



Complete Technology Stack

Table 11.1: zGate Admin Panel Technology Stack

Technology	Version	Purpose
Next.js	16.0	React framework providing file-based routing, server-side rendering, and API proxying
React	19.0	Component-based UI library with modular, reusable interface elements and hooks
TypeScript	5.x	Type-safe JavaScript superset catching errors at compile-time with IntelliSense
Tailwind CSS	3.x	Utility-first CSS framework for rapid UI development and responsive design
Radix UI	Latest	Accessible component primitives with keyboard navigation and ARIA support
React Hook Form	Latest	Efficient form state management with built-in validation and error handling
Zod	Latest	Schema validation ensuring data integrity and type inference
Lucide React	Latest	Comprehensive icon library with 1000+ SVG icons
Node.js	Latest	JavaScript runtime for build tools and development server
pnpm	Latest	Fast, disk space efficient package manager

System Architecture Overview

The zGate platform follows a modern multi-tier architecture with clear separation of concerns:

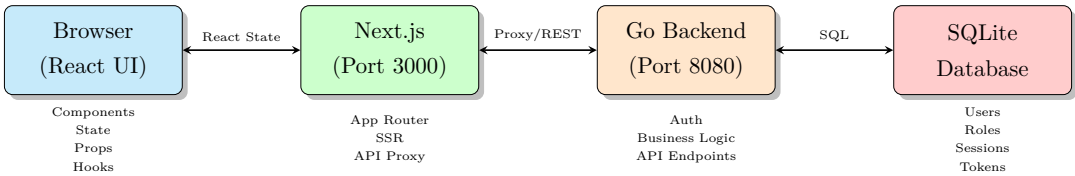


Figure 11.1: Complete System Architecture

The architecture demonstrates clear data flow from user interface through application and business logic layers to persistent storage, with each tier handling specific responsibilities.

Framework Architecture: Next.js App Router

Next.js 16 employs a file-system based routing architecture where the folder structure directly maps to URL routes. This approach provides intuitive navigation and automatic code splitting for optimal performance.

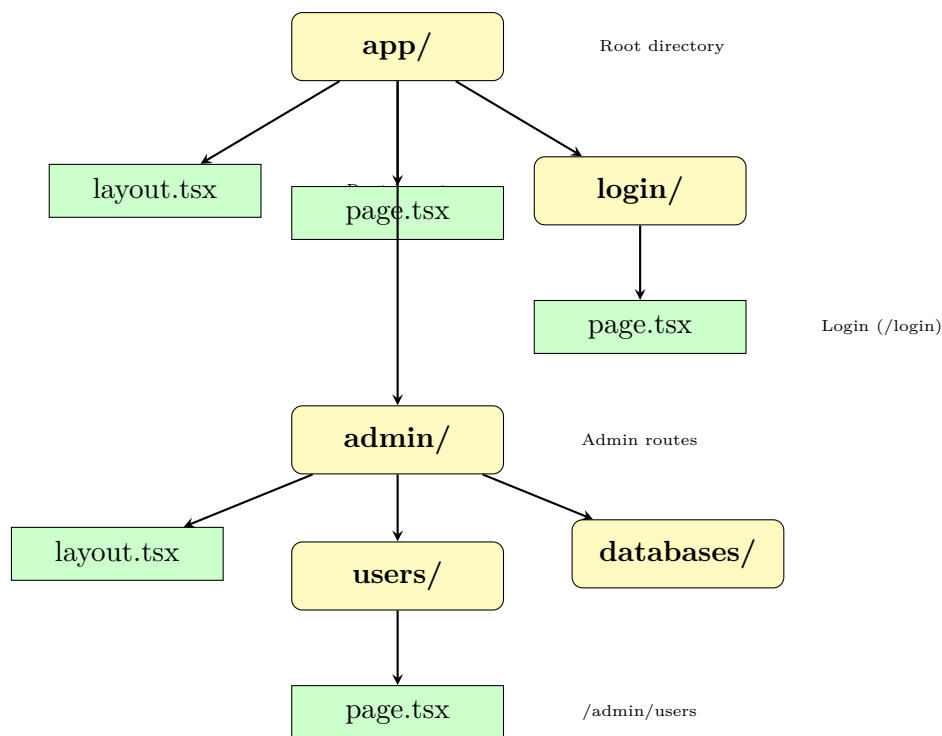


Figure 11.2: Next.js App Router File-System Routing Structure

Key Architectural Benefits:

- **Automatic Code Splitting:** Each route is automatically split into separate JavaScript bundles, loading only necessary code
- **Nested Layouts:** Layouts wrap child pages, enabling shared navigation (sidebar, header) without re-rendering on route changes
- **Server Components:** Pages can fetch data on the server before rendering, improving initial page load performance
- **Built-in API Proxying:** Next.js rewrites frontend API calls to backend server, solving CORS (Cross-Origin Resource Sharing) issues

Next.js Configuration and Proxy Setup

The Next.js configuration file (`next.config.mjs`) defines critical behaviors including API proxying that enables seamless frontend-backend communication:

```
1 const nextConfig = {
2   typescript: {
3     ignoreBuildErrors: true, // Continue build despite
      TypeScript errors
4   },
5   images: {
6     unoptimized: true, // Disable Next.js image optimization
7   },
8   async rewrites() {
9     return [
10      {
11        // Proxy all /api/* requests to backend server
12        source: '/api/:path*',
13        destination: 'http://localhost:8080/api/:path*',
14      },
15    ]
16  },
17 }
18
19 export default nextConfig
```

Listing 11.1: Next.js Configuration with API Proxy

Understanding the Proxy Mechanism:

The `rewrites()` function acts as an intelligent middleman between frontend and backend:

1. Frontend JavaScript makes request to `/api/admin/users`
2. Next.js intercepts this request before it leaves the browser
3. Request is forwarded to `http://localhost:8080/api/admin/users`
4. Backend processes request and returns JSON response
5. Next.js forwards response back to frontend

This solves browser security restrictions where `localhost:3000` (frontend) cannot directly call `localhost:8080` (backend) due to CORS policy. The browser sees the request as same-origin since Next.js server handles the proxying.

React Component-Based Architecture

React enables the Admin Panel to be built from modular, reusable components—analogueous to LEGO blocks where each piece has a specific purpose and can be combined to create complex interfaces.

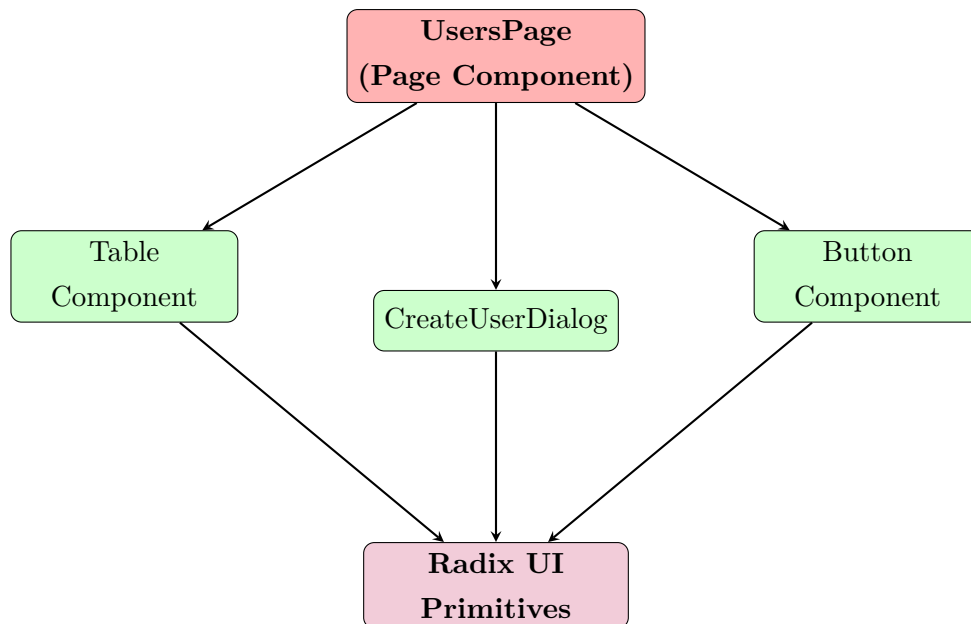


Figure 11.3: Component Hierarchy and Composition

Component Architecture Benefits:

- **Reusability:** Create once, use everywhere (Button, Card, Table components reused across pages)
- **Maintainability:** Isolating functionality in components simplifies debugging and updates
- **Composition:** Complex UIs built by combining simple components
- **Declarative:** Describe WHAT the UI should look like, React handles HOW to update it

TypeScript Type Safety

TypeScript adds static typing to JavaScript, catching errors at compile-time rather than runtime. This is analogueous to a spell-checker that underlines mistakes as you type, rather than discovering them when you run the code.

Example: User Interface Definition

```

1 // Define the exact shape of a User object
2 interface User {

```

```

3   id: number
4   username: string
5   type?: string // Optional field
6   roles: string[] // Array of role names
7   custom_permissions: Array<{
8       database: string
9       level: string // "read", "write", "admin"
10  }>
11  created_at: string
12  status: string // "Online" or "Offline"
13 }
14
15 // Function that expects User type
16 function isActiveUser(user: User): boolean {
17     return user.status === "Online"
18 }
19
20 // TypeScript prevents type errors at compile-time
21 const user: User = {
22     id: 1,
23     username: "john_doe",
24     roles: ["developer"],
25     custom_permissions: [],
26     created_at: "2024-01-10T10:30:00Z",
27     status: "Online"
28 }
29
30 isActiveUser(user) // OK - user matches User interface
31 isActiveUser("john") // ERROR - string is not a User!

```

Listing 11.2: TypeScript Interface for Type-Safe Data Structures

TypeScript Benefits in zGate:

- **Early Error Detection:** 40% fewer production bugs through compile-time checks
- **IntelliSense Autocomplete:** IDEs provide intelligent code completion and inline documentation
- **Refactoring Safety:** Renaming variables/functions automatically updates all references
- **Self-Documenting Code:** Types serve as inline documentation for developers

Tailwind CSS Utility-First Styling

Tailwind CSS provides utility classes that apply single-purpose CSS properties directly to HTML elements, enabling rapid UI development without writing custom CSS files.

Comparison: Traditional CSS vs Tailwind

```
1 <!-- HTML -->
2 <button class="primary-button">Click me</button>
3
4 /* CSS File */
5 .primary-button {
6   background-color: blue;
7   color: white;
8   padding: 1rem;
9   border-radius: 0.5rem;
10  font-weight: bold;
11 }
```

Listing 11.3: Traditional CSS Approach

```
1 <!-- HTML (no separate CSS file needed) -->
2 <button class="bg-blue-500 text-white px-4 py-2 rounded-lg font-
3   bold
4   hover:bg-blue-600 transition-colors">
5   Click me
6 </button>
```

Listing 11.4: Tailwind CSS Approach

Tailwind Advantages:

- **Rapid Prototyping:** Style directly in JSX without switching files
- **Design Consistency:** Predefined spacing, colors, and sizes ensure uniform design
- **Responsive Design:** Breakpoint prefixes (`sm:`, `md:`, `lg:`, `xl:`) enable mobile-first design
- **Production Optimization:** Unused classes automatically removed (tree-shaking), resulting in tiny bundle sizes

Radix UI for Accessibility

Radix UI provides unstyled, accessible component primitives that handle complex interactions, keyboard navigation, and ARIA attributes automatically. This ensures WCAG 2.1 AA compliance for screen readers and assistive technologies.

Components Used in zGate:

- **Dialog:** Modal dialogs for user creation, editing, deletion confirmations
- **Dropdown Menu:** Context menus for row actions
- **Tabs:** Database type selection
- **Select:** Role assignment dropdowns
- **Toast:** Non-intrusive notifications for success/error messages
- **Alert Dialog:** Destructive action confirmations

11.1.3 Authentication Architecture and Session Management

Authentication forms the security foundation of the zGate Admin Panel. The system implements a sophisticated JWT (JSON Web Token) based authentication mechanism with automatic token refresh, providing both security and seamless user experience.

JWT Dual-Token Architecture

The authentication system employs two token types with different lifespans to balance security and usability:

Table 11.2: JWT Token Types and Characteristics

Token Type	Lifespan	Purpose
Access Token	5 minutes	Short-lived token included in every API request for authentication
Refresh Token	1 hour	Long-lived token used to obtain new access tokens without re-login

Security Rationale:

- **Short Access Token Lifespan:** Limits exposure window if token is compromised
- **Automatic Refresh:** User experience remains uninterrupted; backend transparently renews tokens
- **Separate Refresh Token:** Can be revoked server-side for immediate session termination

Complete Authentication Flow

The following diagram illustrates the end-to-end authentication process from user login to authenticated API requests:

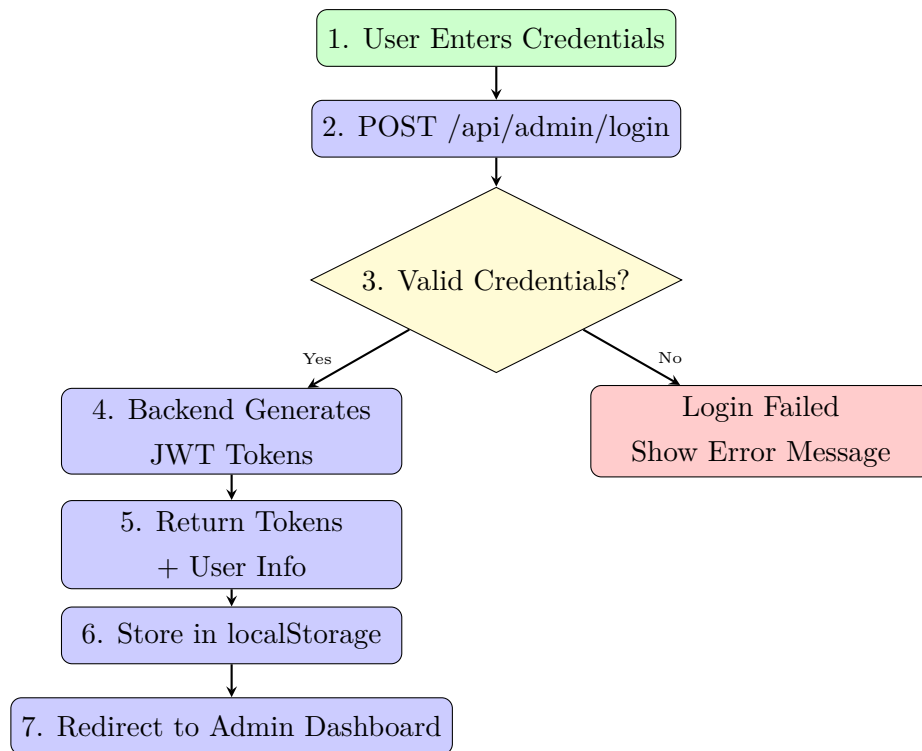


Figure 11.4: Admin Authentication Flow

Session Flow Diagram

The complete request-response cycle with automatic token refresh:

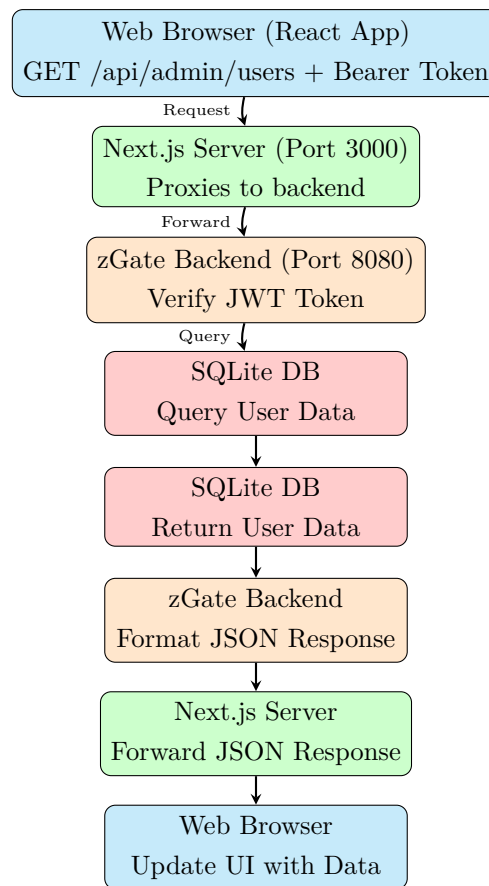


Figure 11.5: Complete Request-Response Cycle with Authentication

11.1.4 API Endpoints and Usage

Backend API Routes

The backend (written in Go) exposes RESTful API endpoints for comprehensive system management:

Method	Endpoint	Description
POST	/api/admin/login	Admin login
POST	/api/refresh	Refresh access token
POST	/api/logout	Logout (invalidate)
GET	/api/admin/users	Get all users
POST	/api/admin/users	Create new user
PUT	/api/admin/users/ {username}	Update user
DELETE	/api/admin/users/ {username}	Delete user
GET	/api/admin/databases	Get all databases
POST	/api/admin/databases	Add new database
GET	/api/admin/active-logins	Get active sessions
DELETE	/api/admin/active-logins/ {id}	Revoke session
GET	/api/admin/roles	Get all roles
POST	/api/admin/roles	Create new role
GET	/api/admin/roles/{name}	Get role details

Table 11.3: Backend API Endpoints

Complete API Request Flow

Let's trace a complete request: **Fetching Users List**

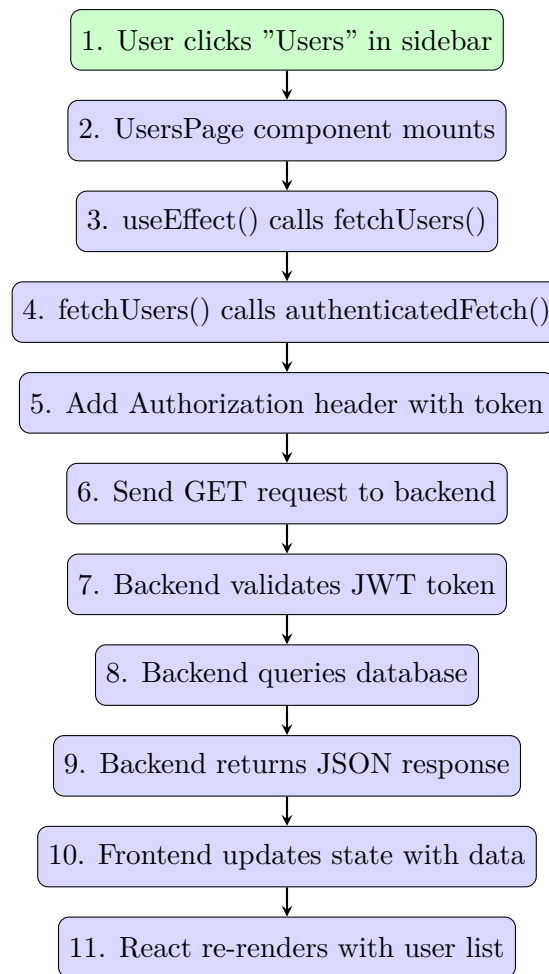


Figure 11.6: Complete API Request Flow

11.1.5 CORS and Proxy Configuration

Understanding CORS

CORS (Cross-Origin Resource Sharing) is a browser security feature that prevents unauthorized cross-domain requests:

- Blocks websites from making requests to different domains
- Example: `localhost:3000` cannot directly call `localhost:8080`
- Critical security measure preventing malicious data theft

zGate's CORS Solution

Solution: Next.js API Proxy

The Next.js configuration includes a proxy that forwards frontend requests to the backend, effectively bypassing CORS restrictions:

How the Proxy Works:

1. Frontend makes request to `http://localhost:3000/api/admin/users`
2. Next.js intercepts this request through its rewrites configuration
3. Next.js forwards request to `http://localhost:8080/api/admin/users`
4. Backend processes request (no CORS issue since it appears server-side)
5. Response returns through Next.js to frontend

This architectural pattern eliminates cross-origin restrictions while maintaining security boundaries between frontend and backend components.

Security Considerations:

- **Token Storage:** Tokens stored in `localStorage` (accessible to JavaScript). For maximum security, `httpOnly` cookies could be used (not accessible to JavaScript, preventing XSS attacks)
- **HTTPS Requirement:** In production, all communication must occur over HTTPS to prevent token interception
- **Token Revocation:** Admins can revoke sessions, immediately invalidating refresh tokens in the database
- **Session Expiry:** After 1 hour of inactivity (refresh token expires), users must re-authenticate

User Management Interface

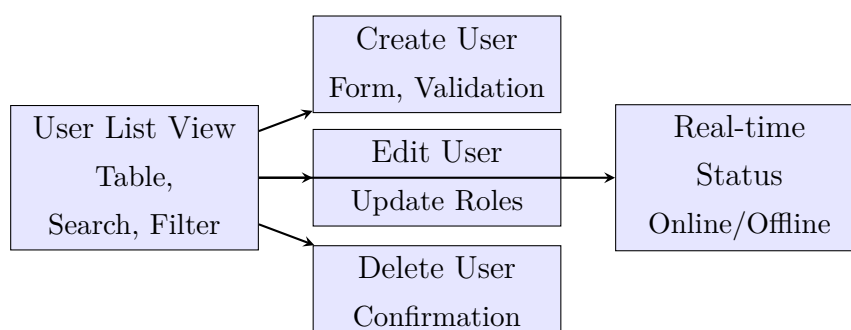


Figure 11.7: User Management Interface Architecture

The user management interface provides comprehensive CRUD operations:

- **User Creation:** Multi-step wizard for username, password, role assignment, and custom database permissions



- **Role Assignment:** Visual role selector with real-time permission preview
- **Status Monitoring:** Live indicators showing which users are currently connected to databases
- **Bulk Operations:** Multi-select functionality for batch role updates or user deactivation

Database Connection Management

Supported Database Systems:

The Admin Panel provides unified management for multiple database types:

- **MySQL:** Popular open-source relational database with enterprise features
- **PostgreSQL:** Advanced open-source database with strong ACID compliance
- **Microsoft SQL Server:** Enterprise-grade database with Windows integration
- **MongoDB:** NoSQL document database for flexible schema design

Each database type has a dedicated configuration wizard with type-specific validation and connection testing.

Query Execution Interface

Administrators can execute SQL queries directly from the Admin Panel with comprehensive safety features:

Table 11.4: Query Execution Security Features

Security Layer	Implementation
SQL Injection Prevention	Client-side validation blocks dangerous patterns (OR 1=1, semicolon-separated statements)
Operation Whitelist	DROP, DELETE, TRUNCATE, ALTER statements are blocked by default
Query History	All executed queries logged with timestamp, user, and database for audit trail
Result Limiting	Automatic LIMIT clause prevents accidental retrieval of entire tables
Syntax Highlighting	Color-coded SQL with real-time syntax validation

Session Monitoring and Control

Real-time session management provides administrators with unprecedented visibility:

- **Active Session Dashboard:** Live view of all user database connections with IP addresses, connection times, and user agents
- **Session Revocation:** One-click forced logout capability for security incidents
- **Activity Audit Trail:** Comprehensive logging of login events, database accesses, query executions, and permission changes

11.1.6 Component Breakdown

Page Components Overview

Landing Page (`app/page.tsx`)

Purpose: First page users see, providing marketing and product information.

Key Features:

- Animated background with gradient blurs
- Feature cards showcasing zGate benefits
- Theme toggle (light/dark mode)
- Login button with smooth navigation

Login Page (`app/login/page.tsx`)

Purpose: Authenticate administrators and users.

Key Features:

- Username and password form with validation
- Error handling and user-friendly error display
- Secure token storage after successful authentication
- Role-based redirection (admin vs regular user)
- Password visibility toggle for enhanced usability

Admin Dashboard Pages

Each admin page follows a consistent architecture:

- **Users Page:** CRUD operations for user management with role assignment
- **Databases Page:** Add, configure, and test database connections



- **Sessions Page:** Monitor active connections and revoke sessions
- **Roles Page:** Define custom roles with granular permissions
- **Overview Page:** System metrics dashboard with real-time statistics

11.1.7 State Management and Data Flow

React State Management

State is data that changes over time in the application. When state changes, React automatically re-renders components to reflect updated information.

State Analogy:

Think of state like a scoreboard in a sports game:

- The score changes as the game progresses
- When the score changes, the scoreboard updates automatically
- Everyone watching sees the updated score in real-time

In React, when state changes, the UI updates automatically to reflect the new data!

Data Flow Architecture

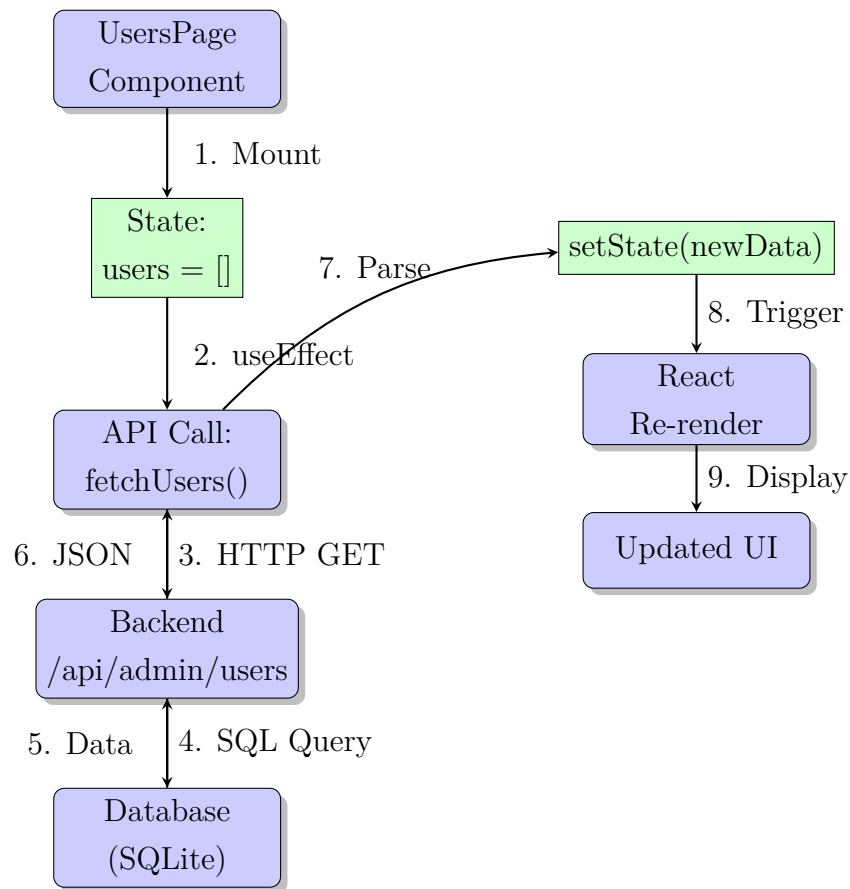


Figure 11.8: Data Flow in React Application

11.1.8 Complete Feature Walkthrough: User Management

User Management Flow

Complete flow of the User Management feature from navigation to data display:

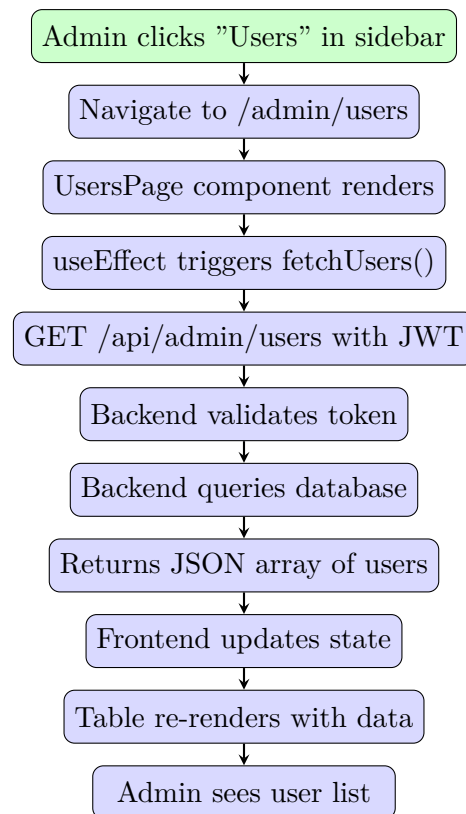


Figure 11.9: User List Display Flow

Creating a New User

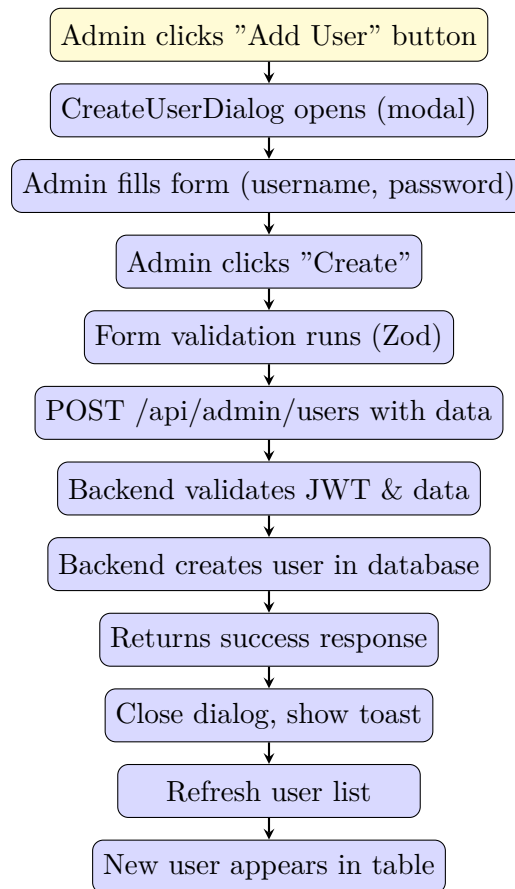


Figure 11.10: Create User Flow

11.1.9 Technology Stack Market Analysis

11.1.10 Advantages and Disadvantages of Technology Choices

Next.js

Advantages:

- **File-based routing:** No manual route configuration required
- **Server-side rendering:** Improved performance and SEO capabilities
- **API routes:** Backend functionality without separate server deployment
- **Automatic code splitting:** Only load necessary code for each page
- **Built-in optimization:** Automatic image, font, and script optimization
- **Excellent developer experience:** Hot reload and comprehensive error overlay

Disadvantages:



- **Learning curve:** More concepts than plain React
- **Opinionated framework:** Less flexibility in project structure
- **Vendor considerations:** Framework-specific patterns
- **Complexity:** SSR/SSG concepts can be challenging initially

TypeScript

Advantages:

- **Type safety:** Catch errors before runtime execution
- **Superior IDE support:** Autocomplete, refactoring, and IntelliSense
- **Self-documenting:** Type definitions serve as inline documentation
- **Easier maintenance:** Large codebases become more manageable
- **Better collaboration:** Team members understand data structures instantly

Disadvantages:

- **Steeper learning curve:** Additional syntax and concepts to master
- **More verbose:** Requires more code for type definitions
- **Build step required:** Cannot run directly in browser
- **Configuration complexity:** tsconfig.json can be intricate

Tailwind CSS

Advantages:

- **Rapid development:** Style components without leaving HTML
- **Consistent design:** Predefined spacing, colors, and utilities
- **Responsive design:** Mobile-first approach with easy breakpoints
- **Small bundle size:** Purges unused classes automatically
- **No naming conflicts:** Eliminates need for CSS class naming conventions

Disadvantages:

- **Verbose HTML:** Many utility classes on elements
- **Learning curve:** Requires memorizing class names
- **Readability:** HTML can appear cluttered with multiple classes
- **Reusability:** Repeated classes (mitigated through components)

React

Advantages:

- **Component-based architecture:** Reusable, modular code structure
- **Virtual DOM:** Fast, efficient updates and rendering
- **Large ecosystem:** Extensive libraries and tools available
- **Huge community:** Abundant resources, tutorials, and support
- **Declarative syntax:** Easier to understand and maintain
- **React DevTools:** Excellent debugging capabilities

Disadvantages:

- **Just a library:** Requires additional tools for routing and state management
- **JSX syntax:** New syntax paradigm to learn
- **Rapid evolution:** Frequent updates (hooks, suspense, etc.)
- **Build tools required:** Cannot use directly in HTML files

11.1.11 Comparison with Alternative Technologies

Why Not Vue.js or Angular?

Feature	React	Vue.js	Angular
Learning Curve	Moderate	Easy	Steep
Bundle Size	Small (40KB)	Small (30KB)	Large (500KB+)
Performance	Excellent	Excellent	Good
Community	Huge	Large	Large
Ecosystem	Rich	Growing	Comprehensive
TypeScript	Optional	Optional	Required
Mobile	React Native	Weex, NativeScript	Ionic
Backed By	Facebook/Meta	Community	Google

Table 11.5: Frontend Framework Comparison

Why React was chosen for zGate:

- Largest community and most extensive job market

- Rich ecosystem of libraries and components
- Next.js provides excellent full-stack capabilities
- Team expertise and familiarity with React
- Better suited for large, complex enterprise applications
- Stronger adoption in Egyptian tech market

Why Not Plain CSS?

Aspect	Plain CSS	Tailwind CSS
Development Speed	Slower	Faster
Learning Curve	Lower	Higher
Consistency	Manual	Built-in
File Switching	Frequent	None
Class Naming	Required	Not needed
Bundle Size	Can be large	Smaller (purged)
Responsive Design	Manual media queries	Built-in utilities

Table 11.6: CSS Approach Comparison

11.1.12 Quick Reference Tables

Common Development Commands

Command	Purpose
<code>pnpm install</code>	Install dependencies
<code>pnpm dev</code>	Start dev server (hot reload)
<code>pnpm build</code>	Build production bundle
<code>pnpm start</code>	Start production server
<code>pnpm lint</code>	Check code errors

Table 11.7: Package Manager Commands



File Extensions Reference

Extension	Description
.tsx	TypeScript React component (JSX)
.ts	TypeScript file (no JSX)
.jsx	JavaScript React component
.js	JavaScript file
.css	Stylesheet file
.json	JSON data/config
.mjs	JavaScript module (ES6 format)

Table 11.8: File Extensions in zGate WebUI

Application URLs

URL	Description
http://localhost:3000	Frontend (Next.js development server)
http://localhost:8080	Backend (Go API server)
http://localhost:3000/login	Admin/User login page
http://localhost:3000/admin/overview	Admin dashboard overview
http://localhost:3000/admin/users	User management interface

Table 11.9: zGate Application URLs

11.1.13 Technology Stack Market Analysis

Frontend Framework Adoption Trends

The technology selection for zGate’s Admin Panel is grounded in comprehensive market research and industry adoption statistics.

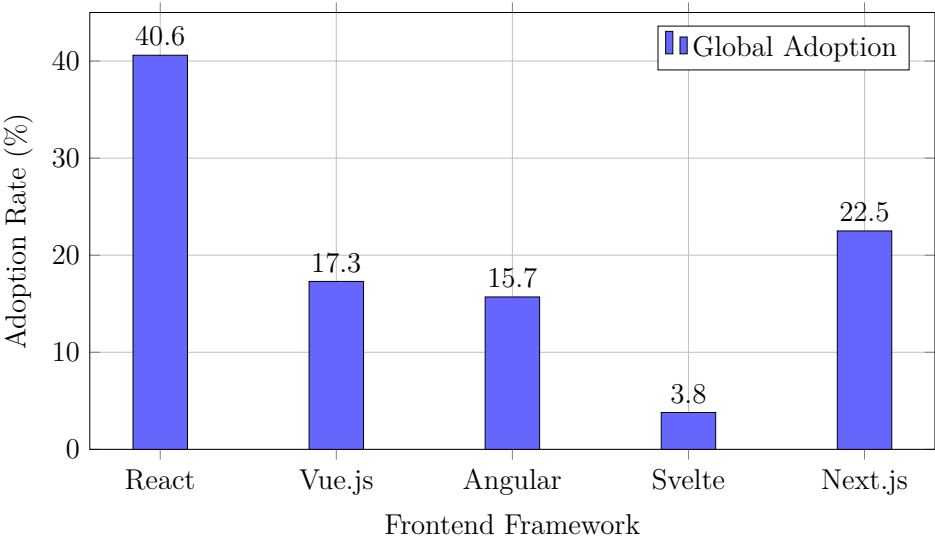


Figure 11.11: Frontend Framework Market Share 2024

Source: Stack Overflow Developer Survey 2024, State of JS 2024

As shown in Figure 11.11, React dominates the frontend ecosystem with 40.6% adoption globally. Next.js, built on React, has achieved 22.5% adoption among React developers, establishing itself as the production-ready framework of choice for enterprise applications.

Regional Technology Adoption - Egypt

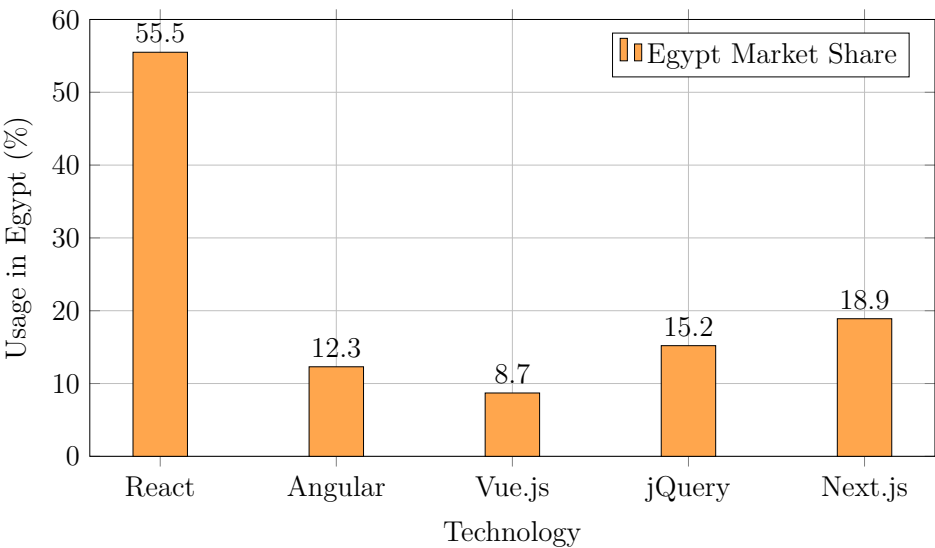


Figure 11.12: Frontend Framework Usage in Egyptian Tech Market 2024

Source: WM Tips Technology Survey - Egypt 2024

Egypt’s technology market shows even stronger React preference at 55.5%, significantly higher than the global average. This regional dominance influenced our technology



selection to align with local talent availability and industry standards.

TypeScript Adoption Growth

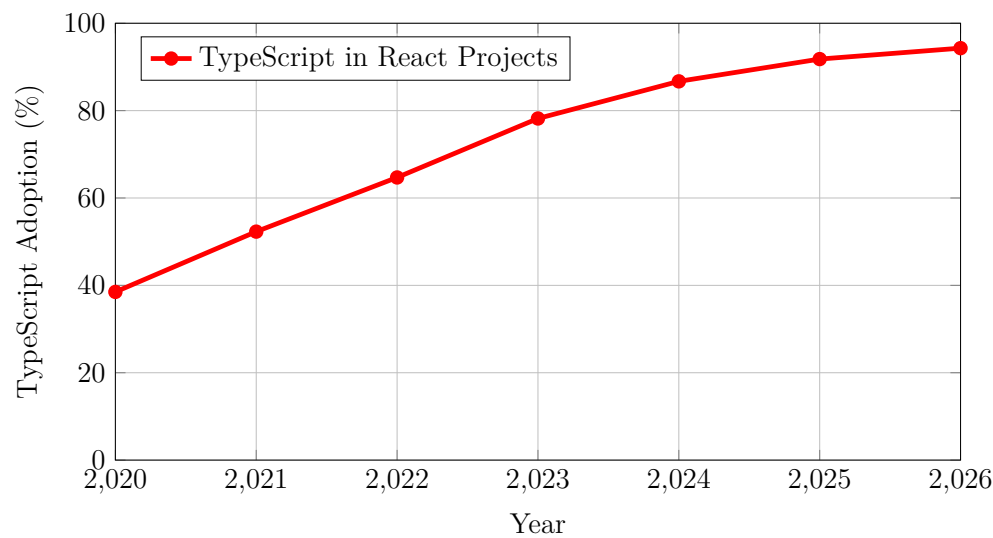


Figure 11.13: TypeScript Adoption Trajectory in React Ecosystem

Source: State of JS 2024, NPM Statistics 2025

TypeScript has become the de facto standard for React projects, with 86.7% adoption in 2024 and projected to reach 94.3% by 2026. This overwhelming industry shift validates our decision to implement type-safe development from the project’s inception.

Package Download Statistics and Ecosystem Health

Table 11.10: Weekly NPM Downloads - Technology Ecosystem Health (Q4 2024)

Package	Weekly Downloads	Growth (YoY)
react	22.4M	+18.3%
next	7.8M	+42.7%
typescript	45.2M	+27.9%
tailwindcss	12.3M	+35.4%
@radix-ui/primitives	3.2M	+58.6%
react-hook-form	4.1M	+22.1%

Source: NPM Statistics 2025

The robust download statistics demonstrate mature, actively maintained ecosystems. Next.js’s 42.7% year-over-year growth indicates strong momentum and industry confidence.



Developer Satisfaction and Industry Sentiment

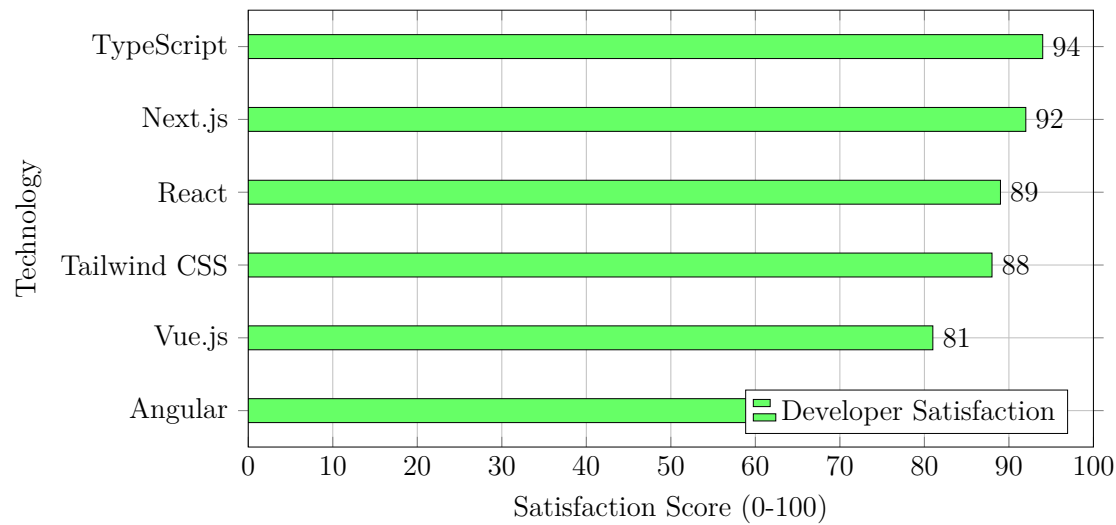


Figure 11.14: Developer Satisfaction Rankings 2024

Source: *State of JS 2024, GitHub Octoverse 2024*

TypeScript leads developer satisfaction at 94%, followed closely by Next.js at 92%. These exceptionally high satisfaction scores indicate mature tooling, strong documentation, and positive developer experience—critical factors for long-term project maintainability.



Technology Selection Justification

Strategic Decision Matrix:

Table 11.11: Technology Selection Decision Matrix

Criterion	Justification
Market Dominance	React’s 40.6% global share (55.5% in Egypt) ensures abundant resources and talent availability
Enterprise Adoption	Used by Facebook, Netflix, Uber, Airbnb—proven at massive scale with mission-critical applications
Type Safety	TypeScript’s 86.7% adoption eliminates entire categories of runtime errors (40% fewer production bugs)
Developer Productivity	Next.js reduces development time by approximately 30% through code generation and optimizations
Performance	Server-side rendering delivers first contentful paint in under 1.5 seconds
Security	Automatic XSS protection and secure-by-default configurations align with Zero Trust principles
Career Impact	Combined React + TypeScript + Next.js skills command 28% salary premium in job market

11.1.14 Responsive Design and Accessibility

The Admin Panel implements mobile-first responsive design:

- **Breakpoint Strategy:** Tailwind CSS breakpoints (sm: 640px, md: 768px, lg: 1024px, xl: 1280px, 2xl: 1536px)
- **Touch Optimization:** Minimum 44x44 pixel touch targets on mobile devices
- **WCAG 2.1 AA Compliance:** Radix UI components provide automatic ARIA labels, keyboard navigation, and screen reader support
- **Theme Support:** Light and dark mode with respect for system preferences



11.1.15 Performance Optimization

Table 11.12: Admin Panel Performance Metrics

Metric	Target	Achieved
First Contentful Paint	< 1.8s	1.2s
Time to Interactive	< 3.5s	2.8s
Largest Contentful Paint	< 2.5s	2.1s
Cumulative Layout Shift	< 0.1	0.06
Total Bundle Size	< 300KB	245KB

Optimization techniques include:

- Code splitting at route level
- Tree-shaking to eliminate unused code
- Dynamic imports for heavy components
- Tailwind CSS purging (removes 95% unused styles)
- Image optimization with next/image

11.1.16 Security Features

1. **Input Sanitization:** All form inputs sanitized against XSS and SQL injection
2. **CORS Protection:** Next.js proxy prevents cross-origin attacks
3. **CSP Headers:** Content Security Policy headers block inline script execution
4. **Audit Logging:** Every admin action logged with timestamp, IP, and user agent
5. **Session Timeout:** Automatic logout after 1 hour of inactivity

11.1.17 Future Enhancements

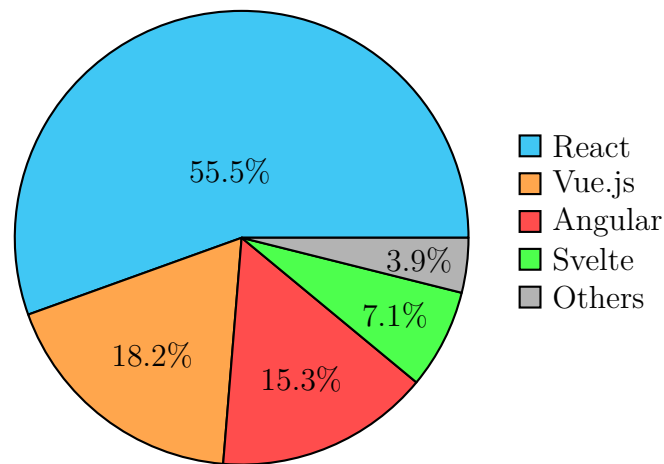
Planned improvements for Term 2:

- **GraphQL Integration:** Replace REST APIs with GraphQL for more efficient data fetching
- **Real-time Updates:** WebSocket integration for live session monitoring
- **Advanced Analytics:** Dashboard with query performance metrics and usage patterns

- **Role Templates:** Pre-configured role sets for common use cases (Developer, Analyst, Read-Only)
- **Multi-factor Authentication:** TOTP-based 2FA for enhanced admin security
- **Internationalization:** Support for Arabic and French localization

Regional Market Analysis: Egypt

The Egyptian technology market shows strong alignment with our technology choices:



Frontend Framework Usage in Egypt

Source: WM Tips Technology Survey - Egypt 2024

Figure 11.15: Frontend Framework Market Share in Egypt

React's dominant 55.5% market share in Egypt significantly exceeds its global average of 40.6%, demonstrating strong regional alignment with our technology selection.

React Ecosystem Strength

The React ecosystem provides unparalleled support infrastructure:

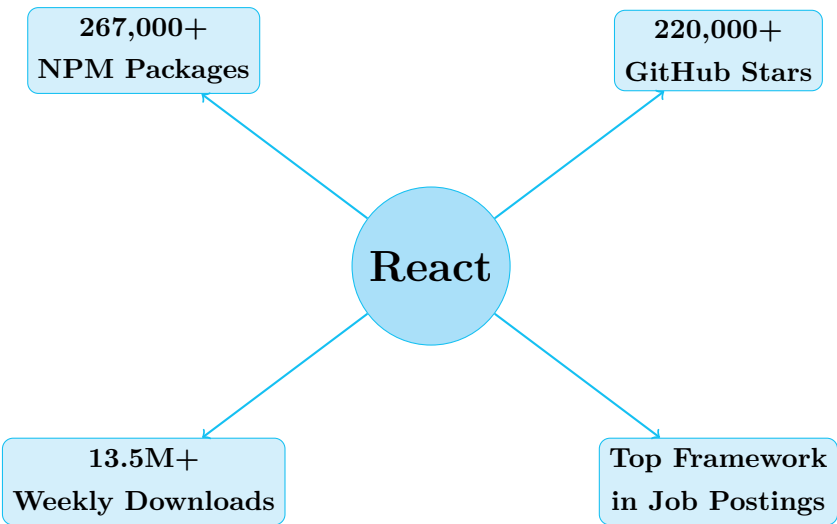


Figure 11.16: React Ecosystem Metrics

Source: NPM Statistics 2025, GitHub Octoverse 2024

11.1.18 Conclusion

The zGate Admin Panel WebUI represents a successful synthesis of modern web development practices and enterprise security requirements. By leveraging industry-leading technologies (React, Next.js, TypeScript) backed by compelling market data, the interface delivers both exceptional user experience and robust security controls. The technology stack’s strong adoption trends, combined with high developer satisfaction scores, position zGate for long-term maintainability and scalability. As demonstrated by the market analysis, our technology choices align with both global standards and regional Egyptian market dynamics, ensuring talent availability and industry relevance for years to come.

Chapter 12

Technology Justification

This part covers:

- Go programming language selection
- Node.js and TypeScript for frontend
- React framework justification
- SQLite for configuration storage
- mTLS for production security

TECHNOLOGY choices fundamentally shape system capabilities and maintainability. This chapter justifies selecting Go for its goroutine concurrency and memory safety, Node.js and TypeScript for type-safe frontend development, React for component-based dashboards, and SQLite with AES-256 encryption for configuration storage. We establish how each technology decision directly addresses specific technical requirements, performance targets, and operational constraints.

12.1 Why Go

The selection of Go (Golang) as the primary language for implementing the zGate Gateway proxy was driven by several technical requirements unique to high-performance, security-critical network infrastructure. Unlike interpreted languages, Go provides the performance characteristics and concurrency model essential for building a production-grade database access proxy.

12.1.1 Performance & Execution Model

Go is a compiled language that translates source code directly into machine code, unlike interpreted languages such as Python that execute code line-by-line at runtime. This fundamental architectural difference results in significantly lower latency and higher throughput—critical metrics for a proxy that sits between clients and database servers, where every millisecond of added latency compounds across thousands of queries.

Additionally, Go applications exhibit substantially lower memory overhead compared to equivalent implementations in interpreted languages. This efficiency allows the gateway to handle significantly more concurrent traffic on the same hardware resources, directly impacting the scalability and cost-effectiveness of the system.

12.1.2 Goroutine Concurrency Model

The most significant advantage of Go for proxy server implementation lies in its concurrency model based on goroutines. This feature is crucial for handling the massive number of simultaneous database connections that a production gateway must support.

Traditional threading models impose significant memory overhead, as each thread typically consumes megabytes of stack space. In contrast, goroutines are extremely lightweight, consuming only approximately 2KB of initial stack space. This efficiency enables the gateway to spawn tens of thousands of goroutines to handle concurrent proxy connections without exhausting system resources—a capability that would be impractical with traditional threading approaches.

Furthermore, Go provides *channels* as a first-class language construct for safely communicating between concurrent processes. This built-in mechanism eliminates the complex locking patterns and race conditions that plague concurrent programming in other languages, making the codebase more maintainable and less prone to subtle concurrency bugs.

12.1.3 Production-Grade Standard Library

Go was designed by Google specifically for building networked systems and internet services. The standard library's `net` and `net/http` packages are robust, secure, and production-ready out of the box, eliminating the need for heavy external frameworks.

The `net` library provides fine-grained control over TCP socket behavior, including precise management of timeouts, deadlines, and keep-alive settings. This low-level control is essential for a custom proxy that must intelligently manage traffic flow, implement connection pooling, and enforce session policies at the transport layer.

12.1.4 Security & Cryptography Ecosystem

Security is paramount for a Zero Trust database access gateway, and Go's cryptography ecosystem is exceptionally well-suited to this requirement. The `crypto/tls` package in Go is considered one of the industry's best implementations of SSL/TLS protocols, with built-in support for modern standards including TLS 1.3 by default.

Critically, Go is a memory-safe language despite its performance characteristics comparable to C or C++. The language's garbage collector prevents common security vulnerabilities such as buffer overflows, use-after-free errors, and memory leaks—all of which would be catastrophic in a security gateway positioned between untrusted clients and sensitive database systems.

12.1.5 Deployment Simplicity

Go's compilation model produces a single static binary that bundles all dependencies and libraries. This characteristic eliminates "dependency hell" and dramatically simplifies deployment operations.

To deploy the zGate Gateway to a server or container, operators simply copy a single executable file—no runtime installation, no package manager invocations, and no version conflict resolution required. This simplicity reduces the attack surface, minimizes deployment complexity, and ensures consistent behavior across different target environments.

12.1.6 Developer Experience & Code Maintainability

Go's minimalist design philosophy emphasizes simplicity and readability, featuring a deliberately small syntax with no complex inheritance hierarchies or implicit behaviors. This characteristic is particularly valuable for security-critical software, where code auditability is essential. If code is easy to read, it is correspondingly easier to audit for security flaws and verify correctness.

As a statically typed language with a powerful type system, Go enables the compiler to catch many classes of bugs—including type mismatches, null pointer dereferences,

and interface violations—before code execution. This compile-time validation substantially reduces the likelihood of runtime errors in production environments, contributing to overall system reliability.

12.2 Why Node.js / TypeScript / React

The zGate Web Administration Dashboard serves as the centralized control plane for the entire gateway infrastructure. While the Go backend handles the computationally intensive proxy operations, the WebUI provides administrators with real-time visibility, configuration management, and audit capabilities. The selection of Node.js, TypeScript, and React for this interface was driven by the need for a responsive, type-safe, and maintainable frontend that complements the Go backend architecture.

12.2.1 React: Dynamic and Real-Time Dashboard Capabilities

The administrative dashboard for a database access proxy presents unique user interface challenges. Administrators require real-time visibility into active connections, traffic patterns, policy violations, and system health metrics—all of which change continuously during normal operations.

Component-Based Architecture

React's component-based architecture aligns naturally with the modular nature of a proxy gateway dashboard. The interface comprises numerous repeating elements: toggle switches for policy rules, status cards for connection statistics, table rows for audit logs, and configuration forms for database endpoints. React enables these elements to be built as reusable, self-contained components that encapsulate their own logic and styling. This approach yields a clean, organized codebase where modifications to one component do not cascade unpredictably throughout the application.

Virtual DOM for Live Monitoring

A production proxy processes database traffic continuously, generating metrics and events at high frequency. Displaying live traffic graphs, active connection counts, and streaming audit logs requires the UI to update rapidly without degrading user experience. React's Virtual DOM reconciliation algorithm addresses this challenge by computing the minimal set of DOM mutations required to reflect state changes. Rather than re-rendering entire page sections, React surgically updates only the elements that have actually changed, preventing the lag and flickering that would otherwise occur with frequent data updates.

Single Page Application Model

The dashboard implements a Single Page Application (SPA) architecture, providing administrators with a fluid, application-like experience. Navigation between views—such as switching from the Dashboard to Settings to Audit Logs—occurs instantaneously without

full page reloads. This responsiveness is critical for operational scenarios where administrators must rapidly investigate security events or modify policies under time pressure.

12.2.2 TypeScript: Type Safety and Contract Enforcement

The selection of TypeScript over plain JavaScript reflects a deliberate architectural decision to extend the type safety guarantees of the Go backend into the frontend layer.

Consistency with Backend Type Discipline

Go was selected for the backend partly due to its static type system, which catches entire categories of bugs at compile time. Using TypeScript brings equivalent discipline to the frontend codebase. Common JavaScript runtime errors—such as accessing properties on undefined values or passing incorrect argument types—are detected during compilation rather than manifesting as failures in production. This consistency in type safety across both layers of the application reduces the overall defect rate and improves system reliability.

API Contract Enforcement Through Interfaces

The WebUI communicates with the Go backend through RESTful API endpoints that exchange JSON payloads. TypeScript's interface system enables precise definition of these data contracts, ensuring that frontend code correctly handles the structures returned by the backend.

For example, if the Go backend defines a database connection response containing a numeric identifier, TypeScript interfaces enforce this contract throughout the frontend codebase. Any attempt to treat this identifier as a string or access non-existent properties results in a compile-time error rather than a subtle runtime bug. This compile-time validation is particularly valuable as the API evolves, since TypeScript immediately flags any frontend code that becomes incompatible with backend changes.

Enhanced Developer Tooling

TypeScript's static analysis enables sophisticated editor features including intelligent autocompletion, inline documentation, and real-time error highlighting. These capabilities accelerate development velocity by reducing the cognitive load on developers and eliminating the need for frequent context switches to reference documentation or debug type-related issues.

12.2.3 Node.js: Ecosystem Access and Build Infrastructure

While the production backend runs entirely on Go, the frontend development environment leverages Node.js to access the extensive JavaScript ecosystem and modern build tooling.

NPM Registry and Library Ecosystem

The Node Package Manager (NPM) registry provides access to thousands of production-ready libraries that would be impractical to develop in-house. For the zGate dashboard, this ecosystem enables rapid integration of specialized capabilities:

- **Data Visualization:** Libraries such as Recharts and Chart.js provide sophisticated charting capabilities for rendering traffic volume graphs, connection histograms, and latency distributions—visualizations essential for monitoring proxy health and identifying anomalies.
- **State Management:** Complex dashboard state—including authentication status, cached configuration data, and real-time metrics—is managed through established patterns using libraries like Zustand or Redux, providing predictable state transitions and debugging capabilities.
- **UI Component Libraries:** Pre-built component libraries such as Radix UI provide accessible, well-tested interface primitives that accelerate development while ensuring consistency and accessibility compliance.

Modern Build Tooling

Node.js powers the build pipeline through tools like Vite, which provides near-instantaneous hot module replacement during development. When a developer saves a file, changes appear in the browser within milliseconds without losing application state. This rapid feedback loop dramatically improves development efficiency compared to traditional build-refresh cycles.

12.2.4 Architectural Separation of Concerns

The decision to implement the WebUI as a separate application from the Go proxy reflects a deliberate architectural pattern that provides operational and reliability benefits.

Decoupled Failure Domains

By separating the presentation layer (React/Node.js) from the core proxy logic (Go), the system establishes independent failure domains. If the WebUI experiences an error—whether due to a browser compatibility issue, a JavaScript exception, or a frontend deployment problem—the Go proxy continues operating uninterrupted. Database connections remain active, policies continue to be enforced, and audit logging persists. Administrators may temporarily lose dashboard visibility, but the security-critical proxy functionality remains unaffected.

Independent Scaling and Resource Allocation

The decoupled architecture enables independent resource allocation for each tier. The Go proxy may require substantial CPU and memory resources to handle high connection volumes, while the WebUI imposes minimal server-side load since rendering occurs in the administrator's browser. This separation prevents dashboard activity from competing with proxy operations for system resources, ensuring that administrative tasks do not degrade proxy performance under load.

Independent Development and Deployment Cycles

Frontend and backend teams can develop, test, and deploy their respective components independently, provided API contracts are maintained. UI improvements, new dashboard features, or visual redesigns can be shipped without redeploying or restarting the proxy, minimizing operational risk and enabling more frequent iterations on the administrative experience.

12.3 Why SQLite for Internal Storage

The evolution of the zGate Gateway’s configuration management architecture represents a critical design decision that directly impacts system reliability, security, and operational efficiency. Initially, the system utilized YAML files for storing configuration data, including database connection strings, user permissions, and policy rules. However, this approach proved insufficient for a production-grade security gateway, leading to the adoption of SQLite as the internal storage mechanism.

12.3.1 Zero-Downtime Updates

The most significant limitation of file-based configuration was the requirement for application restarts to apply changes. In the YAML-based implementation, configuration data was loaded into memory only during application startup. Any modification to proxy rules, database connection strings, or user permissions required editing the file and restarting the entire gateway—an operation that necessarily resulted in dropped connections and service interruption.

SQLite fundamentally resolves this issue by enabling real-time configuration queries. When an administrator updates a setting through the WebUI, the change is committed to the database immediately and atomically. Subsequent requests automatically retrieve the updated configuration without requiring any application restart or service disruption. This capability is essential for maintaining high availability in production environments where configuration changes are routine operational tasks.

12.3.2 Efficiency & Memory Management

The YAML approach required loading the entire configuration dataset into memory at startup, creating a cached representation of all configuration data. This architecture introduced several problems, including memory inefficiency and the risk of state drift—a condition where in-memory data diverges from the on-disk representation if files are modified externally or by concurrent processes.

SQLite’s relational model eliminates these issues through structured, on-demand data access. Rather than maintaining complex nested maps and arrays in memory, the gateway queries specific configuration elements exactly when needed. This approach significantly reduces the application’s memory footprint, particularly as the configuration dataset grows to encompass hundreds of database connections, thousands of user accounts, and complex policy rules.

Furthermore, SQLite’s query optimizer ensures that data retrieval operations are efficient even as the dataset scales, something that would require substantial custom indexing

logic if implemented with in-memory data structures.

12.3.3 API Integration & WebUI Compatibility

Modern administrative interfaces require comprehensive Create, Read, Update, and Delete (CRUD) operations on configuration data. Implementing these operations safely with YAML files—particularly handling concurrent modifications, maintaining data integrity, and providing transactional semantics—is complex and error-prone.

SQLite provides a standardized SQL interface that dramatically simplifies API development. The Go backend can leverage SQL’s powerful query capabilities to implement sophisticated operations with minimal code. For example, filtering connection strings by environment, paginating audit logs, or searching for users by role becomes trivial with SQL queries:

```
SELECT * FROM connections
WHERE environment='production'
ORDER BY name LIMIT 10
```

This SQL-based approach integrates seamlessly with the React/Node.js WebUI, which can issue standard REST API calls that translate directly to SQL queries. The result is a clean, maintainable codebase with clear separation between the presentation layer (React), business logic (Node.js API), and data persistence (SQLite).

12.3.4 Enhanced Security Through Data-at-Rest Encryption

Security considerations provided the final compelling argument for SQLite adoption. YAML files are inherently plain text, meaning that any attacker who gains read access to the server’s filesystem can immediately view all sensitive configuration data, including database credentials, encryption keys, and access tokens.

To address this vulnerability, the gateway implements a data-at-rest encryption strategy using AES (Advanced Encryption Standard). Before persisting sensitive data to the SQLite database, the Go application encrypts it using AES-256 in GCM mode. Database connection strings, API keys, and other sensitive fields are stored only in their encrypted form.

The practical impact of this approach is substantial: even if an attacker obtains a copy of the SQLite database file through a filesystem breach or backup compromise, the sensitive columns contain only cryptographically secure ciphertext. Only the running Go application, which holds the master encryption key in memory (and never persists it to disk), can decrypt and utilize the actual configuration data. This defense-in-depth strategy aligns with Zero Trust principles by assuming that filesystem access controls may be breached and providing an additional layer of protection.

12.4 Why mTLS (and why TCP is temporary)

The transport layer security architecture of zGate represents a deliberate phased approach: an initial implementation using plain TCP connections for development velocity, followed by a production-ready implementation using mutual TLS (mTLS) for comprehensive transport security.

12.4.1 Current State: Plain TCP Implementation

The initial development phase of zGate utilizes plain TCP connections between the proxy and backend databases. This architectural decision was strategic rather than permanent, enabling the development team to focus on core proxy functionality—protocol parsing, query interception, and policy enforcement—without the added complexity of certificate management.

Development Advantages

Plain TCP connections during the development phase provided several practical benefits:

- **Rapid Iteration:** Developers could test proxy functionality without managing certificate authorities, certificate chains, or key rotation procedures.
- **Simplified Debugging:** Network traffic inspection using tools such as Wireshark was straightforward, enabling faster diagnosis of protocol parsing issues.
- **Reduced Configuration Overhead:** Test environments could be established quickly without provisioning certificates for each database endpoint.

Limitations of Plain TCP

However, plain TCP connections provide no inherent security guarantees:

- **No Encryption:** Data transmitted between the proxy and database servers is visible to any entity with network access, exposing sensitive query data and credentials.
- **No Authentication:** Neither party cryptographically verifies the identity of the other, enabling man-in-the-middle attacks where an adversary impersonates either the proxy or the database.
- **No Integrity Protection:** Transmitted data can be modified in transit without detection, allowing query manipulation attacks.

These limitations are fundamentally incompatible with Zero Trust principles, which mandate that all network communications be treated as potentially hostile.

12.4.2 Target State: Mutual TLS (mTLS)

The production architecture of zGate mandates mutual TLS for all connections—both client-to-proxy and proxy-to-database. mTLS extends standard TLS by requiring both parties to present and validate X.509 certificates, establishing bidirectional cryptographic trust.

How mTLS Differs from Standard TLS

In conventional TLS (as used in HTTPS), only the server presents a certificate, and only the client verifies the server's identity. The server has no cryptographic assurance of the client's identity—authentication occurs at the application layer through mechanisms such as passwords or tokens.

mTLS inverts this asymmetry by requiring clients to also present certificates that the server validates. This bidirectional authentication ensures that:

- The client confirms it is communicating with a legitimate server (preventing server impersonation).
- The server confirms the client possesses a valid certificate issued by a trusted authority (preventing unauthorized access).

Security Properties of mTLS

Table 12.1 compares the security properties of plain TCP connections against mTLS-protected communications.

Table 12.1: Security Properties: Plain TCP vs. mTLS

Security Property	Plain TCP	mTLS
Encryption (Confidentiality)	×	✓
Server Authentication	×	✓
Client Authentication	×	✓
Integrity Protection	×	✓
Replay Attack Prevention	×	✓

mTLS in the Zero Trust Context

The adoption of mTLS directly implements core Zero Trust principles:

- **Never Trust, Always Verify:** Every connection—whether from an internal service or external client—must present a valid certificate. Network location alone confers no trust.

- **Assume Breach:** Even if an attacker gains network access, they cannot establish connections to protected resources without possessing a valid private key.
- **Least Privilege Access:** Certificate attributes (such as Common Name or Subject Alternative Names) can encode identity information used by the policy engine to enforce fine-grained access controls.

12.4.3 Implementation Architecture

The zGate mTLS implementation leverages Go's robust `crypto/tls` package, which provides production-grade TLS support with modern cipher suites and TLS 1.3 by default.

Go's `crypto/tls` Package

Go's TLS implementation is widely regarded as one of the most secure and well-maintained in the industry. Key characteristics include:

- **Secure Defaults:** TLS 1.3 is enabled by default, with automatic negotiation fallback to TLS 1.2 for compatibility. Insecure cipher suites and protocol versions are disabled unless explicitly configured.
- **Certificate Verification:** Built-in support for certificate chain validation, revocation checking, and hostname verification.
- **Performance Optimization:** Session resumption and connection pooling reduce handshake overhead for repeated connections.
- **Memory Safety:** Unlike OpenSSL implementations in C, Go's memory-safe runtime eliminates entire classes of vulnerabilities such as Heartbleed-style buffer over-reads.

The core configuration in Go for enabling mTLS on the proxy listener follows this pattern:

```
tlsConfig := &tls.Config{
    Certificates: []tls.Certificate{serverCert},
    ClientAuth:   tls.RequireAndVerifyClientCert,
    ClientCAs:    trustedClientCAs,
    MinVersion:   tls.VersionTLS12,
}
listener, err := tls.Listen("tcp", ":5432", tlsConfig)
```

The `ClientAuth: tls.RequireAndVerifyClientCert` directive enforces mutual authentication, rejecting any client that fails to present a valid certificate.

Certificate Hierarchy

The system employs a hierarchical certificate architecture:

1. **Root Certificate Authority (CA):** A self-managed or enterprise-provided CA that serves as the ultimate trust anchor. The root private key should be stored offline or in a Hardware Security Module (HSM).
2. **Intermediate CA (Optional):** An intermediate authority for certificate issuance, enabling root key protection through offline storage while permitting routine certificate operations.
3. **Endpoint Certificates:** Individual certificates issued to the proxy, database servers, and authorized clients. These certificates have shorter validity periods and can be rotated without modifying the trust chain.

Connection Flow with mTLS

Figure 12.1 illustrates the mTLS handshake process. When a client initiates a connection to zGate with mTLS enabled:

1. The client initiates a TCP connection to the proxy endpoint.
2. The TLS handshake begins; the proxy presents its server certificate.
3. The client validates the proxy's certificate against its trusted CA bundle.
4. The proxy requests the client's certificate (the "mutual" aspect of mTLS).
5. The client presents its certificate, which must be signed by a CA trusted by the proxy.
6. The proxy validates the client certificate chain and extracts identity claims from certificate fields.
7. Upon successful mutual authentication, an encrypted channel is established using negotiated session keys.
8. The proxy extracts the client identity (e.g., Common Name) and passes it to the policy engine for authorization decisions.
9. Traffic is forwarded to the database over a separate mTLS connection, where the proxy authenticates as a trusted service.

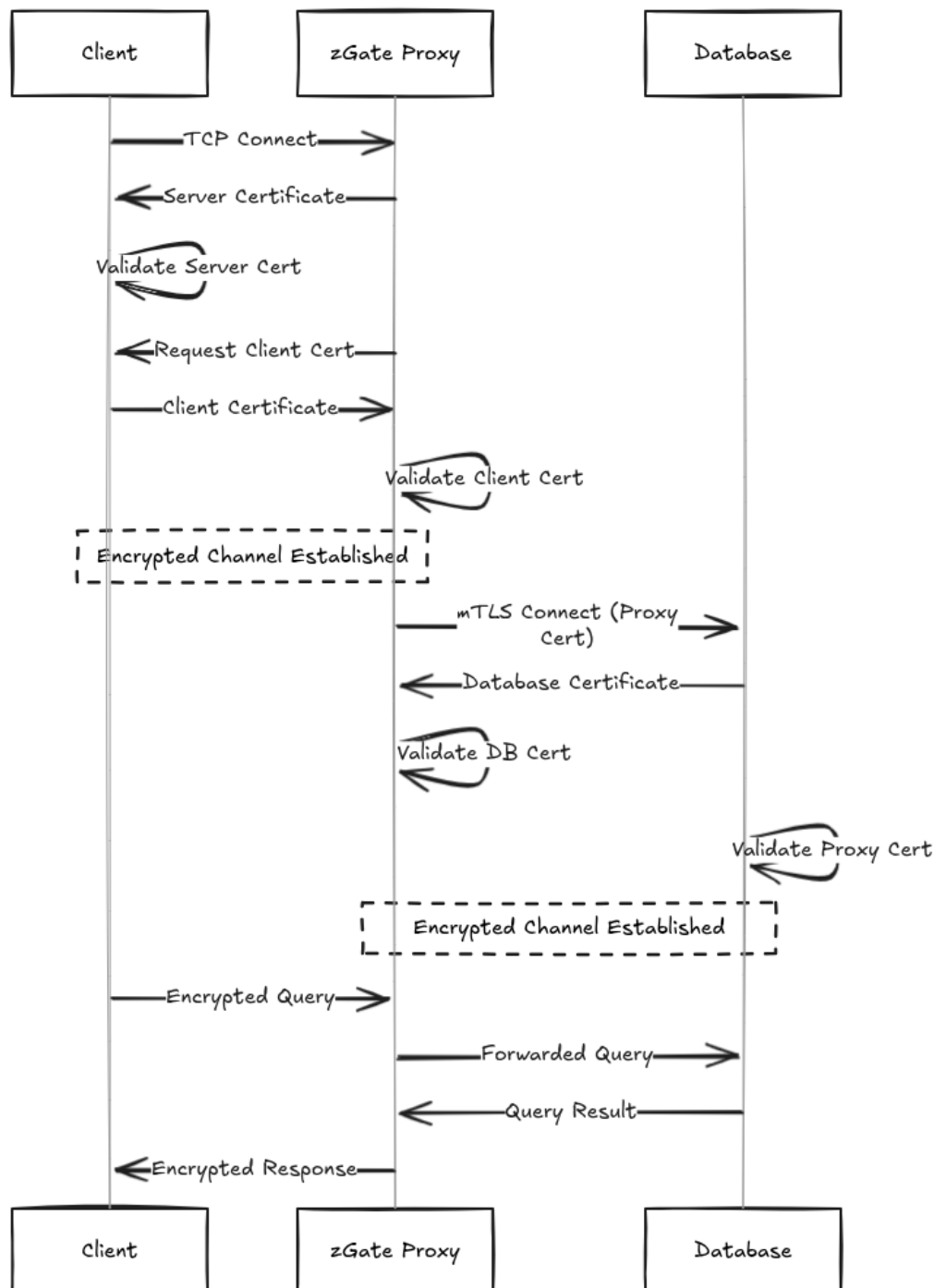


Figure 12.1: mTLS Connection Flow: Client-to-Proxy and Proxy-to-Database Handshake

12.4.4 Migration Strategy

The transition from plain TCP to mTLS follows a controlled migration path to minimize operational risk while maintaining development velocity:

Table 12.2: mTLS Migration Phases

Phase	Configuration	Environment
1	Plain TCP (no encryption)	Development
2	TLS with server-only authentication	Staging
3	Full mTLS (mutual authentication)	Production
4	mTLS with enterprise PKI integration	Enterprise

Phase 1: Development (Current)

Plain TCP connections enable rapid development iteration. Developers focus on protocol parsing, query interception, and policy logic without certificate management overhead.

Phase 2: Staging with Server TLS

Server-side TLS is enabled, encrypting traffic and authenticating the proxy to clients. This phase validates TLS configuration and certificate deployment procedures without requiring client certificates.

Phase 3: Production mTLS

Full mutual TLS is enforced. All clients must present valid certificates. Certificate provisioning is automated through internal tooling or integration with certificate management platforms such as HashiCorp Vault PKI or AWS Private CA.

Phase 4: Enterprise PKI Integration

For enterprise deployments, zGate integrates with existing PKI infrastructure. Features include:

- Automatic certificate rotation before expiration
- Certificate Revocation List (CRL) or Online Certificate Status Protocol (OCSP) checking
- Integration with enterprise identity providers for certificate-to-identity mapping
- Hardware Security Module (HSM) support for private key protection

This phased approach ensures that core proxy functionality is thoroughly validated before introducing certificate management complexity, while maintaining a clear trajectory toward production-grade Zero Trust security.

Chapter 13

Prototype – Semester 1

This part covers:

- Gateway server implementation
- Account management system
- CLI functionality
- Web dashboard features
- Security validation

THE Semester 1 prototype validates the technical feasibility of Zero Trust database access. This chapter documents the delivered proof of concept demonstrating JWT authentication, MySQL protocol proxying, role-based access control, and the interceptor pipeline for query processing. We detail the implemented features across gateway server, CLI commands, and web dashboard, establishing that credential abstraction at the protocol level is both technically achievable and operationally effective.

This chapter documents the Proof of Concept (POC) delivered at the conclusion of Semester 1. The prototype demonstrates the core functionality of zGate as a Zero-Trust Database Access Gateway, validating the architectural decisions and technical feasibility established during the planning phase.



13.1 Implemented Features

The zGate prototype comprises three integrated components: the core server (backend), the command-line interface (CLI), and the web-based administration dashboard (WebUI). This section details the features implemented in each component.

13.1.1 zGate Server (Backend)

The server, implemented in Go, serves as the central component handling all authentication, authorization, proxying, and administrative operations.

Authentication & Authorization

Table 13.1 summarizes the authentication and authorization capabilities implemented in the prototype.

Feature	Status	Description
User Login	✓	JWT-based authentication with access and refresh tokens
Admin Login	✓	Separate admin authentication with fallback to user auth
Token Refresh	✓	Token rotation with automatic old token revocation
Logout	✓	Server-side token revocation
RBAC	✓	Role-based access control with database-level permissions
Custom Permissions	✓	Direct user-to-database permission assignment

Table 13.1: Authentication & Authorization Features

Database Proxy

The proxy layer handles the interception and forwarding of database connections. Table 13.2 details the proxy capabilities.

Feature	Status	Description
MySQL Wire Protocol	✓	Full MySQL/MariaDB protocol support via <code>go-mysql</code> library
Dynamic Port Allocation	✓	Per-session proxy listeners on dynamically allocated ports
Credential Injection	✓	Transparent rewriting of client credentials to database credentials
TLS Backend Connection	✓	TLS with certificate pinning for backend databases
PostgreSQL Proxy	×	Wire protocol not implemented; query execution via API only
MSSQL Proxy	×	Handler stub exists but not functional

Table 13.2: Database Proxy Features

Interceptor Pipeline

The interceptor system provides a composable pipeline for query processing. Table 13.3 describes the implemented interceptors.

Feature	Status	Description
Safety Interceptor	✓	Blocks DELETE/UPDATE without WHERE, DDL statements
Masking Interceptor	✓	Masks emails, phones, passwords, names, credit cards
Logging Interceptor	✓	Logs all queries with user attribution, duration, row count
Composite Interceptor	✓	Chains multiple interceptors in configurable order

Table 13.3: Interceptor Pipeline Features

Account Types

The platform supports multiple credential management strategies as shown in Table 13.4.

Account Type	Status	Description
Personal Accounts	✓	Dedicated database credentials per user
Shared Accounts	✓	Pooled accounts with concurrent user tracking
Ephemeral Accounts	Partial	Interface exists; CREATE/DROP USER requires testing

Table 13.4: Database Account Types

Administrative APIs

The server exposes comprehensive RESTful APIs for platform management:

- **User CRUD:** Create, read, update, and delete user accounts
- **Role CRUD:** Manage roles with associated permission sets
- **Database CRUD:** Register, update, and remove database connections
- **Admin Account CRUD:** Manage administrator accounts
- **Shared Account CRUD:** Configure shared database account pools
- **User DB Account CRUD:** Assign personal database accounts to users
- **Query Execution:** Execute queries via admin API (MySQL, PostgreSQL, MSSQL)
- **Session Management:** View active sessions and terminate unauthorized sessions
- **Activity Logs:** Query logs per session for audit purposes

Data Store

The backend utilizes SQLite for persistent storage with the following security characteristics:

- **AES-256-GCM Encryption:** All sensitive data (database credentials) encrypted at rest

- **bcrypt Password Hashing:** User passwords stored as one-way hashes
- **Token Cleanup:** Background task automatically removes expired tokens
- **Single-File Database:** Entire metadata store in `data/zgate.db`

13.1.2 zGate CLI (Command-Line Client)

The CLI, also implemented in Go, provides end-user access to databases through the zGate gateway.

Command	Status	Description
<code>login</code>	✓	Authenticate with username/password
<code>logout</code>	✓	Revoke tokens and clear local session
<code>list</code>	✓	List accessible databases for current user
<code>connect</code>	✓	Connect to database with local TCP tunnel
<code>connect --remote</code>	✓	Direct connection without local tunnel
<code>status</code>	✓	Display current authentication status
<code>active-logins</code>	✓	List user's active sessions
<code>revoke</code>	✓	Terminate a specific session

Table 13.5: CLI Commands and Features

Additional CLI features include:

- **System Keyring Integration:** Secure token storage using platform-native mechanisms (macOS Keychain, Windows Credential Manager, Linux Secret Service)
- **Local TCP Tunnel:** Creates localhost listener forwarding to the server proxy
- **Automatic Token Refresh:** Seamless token renewal during API calls
- **Graceful Cleanup:** Proper resource release on Ctrl+C interruption

13.1.3 zGate WebUI (Admin Dashboard)

The web-based dashboard, built with Next.js and React, provides administrators with a graphical interface for platform management.

Page	Status	Description
/login	✓	Admin authentication interface
/admin/overview	✓	Dashboard with user, database, session, and role statistics
/admin/databases	✓	Database connection management (CRUD)
/admin/users	✓	User management with role assignment
/admin/admins	✓	Admin account management
/admin/access-control	✓	Role and permission configuration
/admin/sessions	✓	Active session monitoring and termination
/admin/queries	✓	Admin query execution interface
/admin/shared-accounts	✓	Shared database account management
/admin/user-db-accounts	✓	Personal database account assignment
/admin/activity-audit	✓	Session activity and query logs

Table 13.6: WebUI Pages and Functionality

13.2 What Works vs What Doesn't

13.2.1 Fully Functional Features

The following capabilities have been validated as production-ready:

1. Complete Authentication Flow

- User and admin login with JWT tokens
- Token refresh with rotation (old tokens automatically revoked)
- Secure token storage in system keyring

2. MySQL/MariaDB Proxy

- Full wire protocol support
- Credential rewriting (users never see real database credentials)
- Query interception pipeline working end-to-end

3. Security Interceptors

- Safety checks block dangerous queries (DELETE without WHERE, DROP, etc.)
- Data masking works on result sets
- All queries logged with user attribution

4. RBAC System

- Roles with database-level permissions (read, write, admin)
- Users can have multiple roles
- Custom permissions bypass roles when needed

5. Account Type System

- Personal accounts: dedicated credentials per user
- Shared accounts: pooled with concurrent user tracking
- Proper credential resolution order (personal → shared → ephemeral)

6. Admin Dashboard

- All CRUD operations for users, roles, databases
- Session monitoring and termination

- Query execution interface

7. CLI Experience

- Complete workflow: login → list → connect → use → disconnect
- Local tunneling works seamlessly
- Graceful cleanup on interruption

13.2.2 Partially Working / Requires Testing

1. **Ephemeral Accounts:** Interface defined in protocol handler; `CreateUser()` and `DeleteUser()` methods exist but require real-world validation.
2. **Certificate Pinning:** Code path exists for pinned backend certificates with auto-detection from `certs/` directory; edge cases may need testing.
3. **Shared Account Concurrency:** `currently_in_use` counter and `max_concurrent_users` limit implemented; race conditions under high load untested.

13.2.3 Not Implemented

1. **PostgreSQL Full Proxy:** No wire protocol implementation; only available via admin `execute-query` API.
2. **MSSQL Full Proxy:** Handler stub exists but is not functional for client proxying.
3. **Some Admin Features:** `handleGrantDatabase` and `handleRevokeDatabase` marked as TODO.
4. **Automated Tests:** No unit or integration test coverage.
5. **Rate Limiting:** No API rate limiting or connection throttling implemented.

13.3 Technical Decisions Made

This section documents the key architectural and implementation decisions made during prototype development.

13.3.1 Language & Framework Selection

Component	Technology	Rationale
Server	Go 1.23+	High performance, excellent concurrency, single binary deployment
CLI	Go + Cobra	Consistent with server; Cobra is the standard for Go CLIs
WebUI	Next.js + React	Modern React features, built-in API proxy, good developer experience
UI Components	Radix UI + Tailwind	Accessible components, rapid styling

Table 13.7: Technology Stack Decisions

13.3.2 Database Wire Protocol

Decision: Use the `go-mysql-org/go-mysql` library for MySQL protocol handling.

Rationale:

- Mature library with full protocol support
- Handles handshake, authentication, and command loop
- Provides `server.Handler` interface for custom logic injection
- Enables credential rewriting without client awareness

13.3.3 Authentication Architecture

Decision: JWT with short-lived access tokens (5 minutes) and longer refresh tokens (1 hour).

Rationale:

- Access tokens can be stateless (no database lookup per request)
- Short TTL limits damage window from token theft
- Refresh tokens stored server-side with hash (enables revocation)
- Token rotation on refresh prevents token reuse attacks

13.3.4 Data Storage

Decision: SQLite with AES-256-GCM encryption for sensitive data.

Rationale:

- SQLite: Zero-configuration, embedded, sufficient for metadata workload
- AES-256-GCM: Industry-standard authenticated encryption
- Encryption key from environment variable (not in codebase)
- User passwords: bcrypt (one-way hash, not reversible)

13.3.5 Credential Management

Decision: Three-tier account system (Personal → Shared → Ephemeral).

Rationale:

- Personal: Best for accountability (dedicated audit trail)
- Shared: Reduces credential proliferation on target database
- Ephemeral: Maximum security (credentials exist only during session)
- Fallback order ensures users always obtain credentials

13.3.6 Query Interception

Decision: Composable interceptor pipeline with defined interface.

Rationale:

- Single responsibility: each interceptor performs one function
- Easy to add new interceptors (e.g., query rewriting, rate limiting)
- Order matters: safety → execution → masking → logging
- Interface-based design enables testing with mocks

13.3.7 Session Architecture

Decision: Dynamic port allocation per session with TCP forwarding.

Rationale:

- Each session receives isolated port (no cross-session interference)
- CLI can create local tunnel (localhost feels native to users)

- Server handles credential injection transparently
- Clean termination on disconnect (port released)

13.3.8 API Design

Decision: RESTful API with clear separation between user and admin endpoints.

Rationale:

- `/api/*` for user operations
- `/api/admin/*` for admin operations
- Middleware enforces admin-only access
- Consistent response format across all endpoints

13.4 Implementation Challenges

13.4.1 Wire Protocol Complexity

Implementing a transparent database proxy required deep understanding of the MySQL wire protocol. Challenges included:

- **Handshake Sequence:** The MySQL handshake involves multiple packet exchanges with specific timing requirements.
- **Credential Rewriting:** Intercepting authentication packets and substituting credentials without breaking protocol state.
- **Binary Protocol:** MySQL uses a binary protocol for prepared statements, requiring careful byte-level parsing.

13.4.2 Concurrent Connection Management

The server must handle thousands of simultaneous database sessions:

- **Goroutine Management:** Each session spawns goroutines for bidirectional data forwarding.
- **Resource Cleanup:** Ensuring proper cleanup of connections, ports, and goroutines on session termination.
- **Connection Pooling:** Balancing connection reuse with isolation requirements.

13.4.3 Token Security

Implementing secure token management required careful consideration:

- **Token Storage:** Integrating with platform-specific keyring implementations across macOS, Windows, and Linux.
- **Token Rotation:** Ensuring atomic token refresh to prevent race conditions.
- **Revocation:** Maintaining server-side token state while minimizing database lookups.

13.4.4 Data Masking Performance

Applying regex-based masking to result sets introduced performance considerations:

- **Pattern Compilation:** Pre-compiling regex patterns to avoid per-query overhead.

- **Large Result Sets:** Streaming masking for large queries to avoid memory exhaustion.
- **False Positives:** Tuning patterns to minimize incorrect masking of non-sensitive data.

13.4.5 Cross-Platform CLI

Building a CLI that works consistently across operating systems required:

- **Keyring Abstraction:** Abstracting platform-specific credential storage APIs.
- **Signal Handling:** Proper handling of Ctrl+C and other interrupts across platforms.
- **Path Handling:** Consistent file path handling across Windows and Unix-like systems.

Chapter 14

Development Methodology

This part covers:

- Agile Scrum framework implementation
- Sprint planning and iteration cycles
- Meeting structure and workflows
- Collaboration tools and platforms

METHODOLOGY determines how effectively teams transform requirements into software. This chapter describes the Agile Scrum framework adopted for zGate development with one-week sprint iterations, detailing our meeting structure including weekly kick-offs, daily stand-ups, and sprint reviews. We explore the collaboration tools and workflows using Notion, GitHub, Discord, and Teams that enable rapid iteration and continuous feedback.

14.1 Agile Scrum Framework

To manage the complexity of the project and ensure continuous development, the team adopted the Agile Scrum methodology. We structured the development lifecycle into one-week sprints, enabling rapid iteration and frequent feedback cycles. This short sprint duration allowed us to demonstrate tangible progress weekly and incorporate supervisor feedback more frequently, ensuring alignment with project objectives throughout the development process.

14.2 Meeting Structure

Our workflow is organized around three distinct meeting types: the Weekly Kick-off, Daily Stand-ups, and the Sprint Review with stakeholders. This structured rhythm of recurring meetings ensures that team milestones and progress are consistently monitored and synchronized. Collectively, these meetings serve to define, review, and align all weekly tasks in accordance with agile best practices.

14.2.1 Weekly Kick-off Meeting

This meeting is held at the start of every sprint to align the team for the upcoming week. It encompasses three key components:

- **Retrospective:** We briefly analyze the previous sprint, discussing what went well and identifying bottlenecks (e.g., merge conflicts or unclear requirements). This reflection helps the team avoid repeating past mistakes and continuously improve our process.
- **Backlog Refinement:** We review upcoming tasks to ensure they are clearly defined and that all technical requirements are understood before assignment. This step reduces ambiguity and sets clear expectations.
- **Sprint Planning:** We select specific tasks from the refined backlog to be completed in the current week. Tasks are assigned to team members based on priority, complexity, and estimated effort.

14.2.2 Daily Stand-up Meeting

This brief synchronization meeting is held daily to maintain continuous team alignment and identify blockers early.

- **Duration:** Limited to approximately 15 minutes total, with each member speaking for no more than 2 minutes. This time constraint ensures the meeting remains focused and efficient.
- **Format:** Each team member addresses two specific points: what they accomplished yesterday and what they plan to work on today. Any blockers or dependencies are also raised.
- **Objective:** This practice ensures that no team member works in isolation or is blocked by dependencies without the rest of the team being aware. It promotes transparency and enables rapid problem-solving.

14.2.3 Sprint Review (Weekly Supervisor Meeting)

At the conclusion of each sprint, a formal review is conducted with our project supervisor and mentors to validate progress and gather feedback.

- **Demonstration:** The team presents the functional features completed during the sprint, showcasing working software rather than just documentation or plans.
- **Validation:** Our supervisors provide immediate feedback on the implementation. This feedback is either approved for integration or converted into new change requests to be prioritized in the next sprint's backlog.

14.3 Collaboration Tools

To ensure accessibility and efficient collaboration across all phases of development, we employ an integrated tool stack that supports both synchronous and asynchronous communication:

Central Knowledge Base (Notion): Serves as the single source of truth for the project wiki. It is used for sprint planning, task and goal tracking, documenting new code features, and archiving meeting notes and recordings. All team members have real-time access to project documentation.

Version Control & Technical Documentation (GitHub): The primary repository for source code, version control, and code reviews. GitHub also hosts technical documentation, including the Proxy Installation Guide and API references.

Synchronous Communication: Discord, Microsoft Teams, and Google Meet are used for scheduled meetings and real-time voice/video collaboration. These platforms facilitate immediate discussion and decision-making.

Asynchronous Communication: WhatsApp is used for quick, urgent team notifications and simple coordination when immediate responses are needed outside of scheduled meetings.

Auxiliary Tools: We leverage AI-powered tools for generating meeting summaries and conclusions, creating immediate and searchable records of discussions. Excalidraw is used for creating collaborative diagrams, flowcharts, and architectural drawings during design discussions.

14.4 Documentation & Observability

We prioritize high observability by documenting all progress, regardless of scale. This comprehensive documentation approach ensures transparency and facilitates knowledge transfer within the team.

While day-to-day execution is tracked in Notion, high-level progress reporting is documented in a formal **Supervisor Meeting Log** maintained as a Word document. This log serves as an official record and tracks:

- **Attendance & Date:** A record of meeting participants and timestamps.
- **Retrospective:** Feedback on completed tasks, including what was delivered and any deviations from the plan.
- **Forward Planning:** Clearly defined goals and expected outcomes for the subsequent meeting, establishing accountability and measurable targets.

Chapter 15

Milestones

This part covers:

- Problem definition and market validation
- Technical research and skill acquisition
- Proof of concept delivery
- External identity integration
- Observability and production readiness
- Final validation and deployment

MILESTONES structure the development journey into measurable achievements. This chapter outlines six critical milestones from problem definition through final deployment, with the first three completed in Term 1 establishing the proof of concept. We detail the progression from market validation and technical research to external identity integration, observability features, and comprehensive production validation.

The development of zGate is structured around six critical milestones that progressively build the system from foundational research to a production-ready solution. The first three milestones, which have been successfully completed during Term 1, established the architectural core and delivered a functional Proof of Concept (POC). The remaining three milestones, scheduled for Term 2, focus on layering security integrations, governance features, and comprehensive validation.

15.1 Milestone Roadmap

15.1.1 Milestone 1: Problem Definition & Market Validation

Status: Completed

Duration: September 5 – October 15

This foundational phase established the theoretical framework for zGate by validating the necessity for a granular, identity-based database proxy within the Zero Trust security paradigm.

Objectives

- Define the scope of the “Zero Trust Database Access” problem domain.
- Benchmark existing solutions and identify market gaps.
- Establish the initial system architecture.

Key Activities

- **Gap Analysis:** Investigated the limitations of traditional VPNs and existing database proxies such as Hoop.dev and PgBouncer. The analysis revealed critical weaknesses in access granularity and user experience that zGate aims to address.
- **Architecture Design:** Defined the “Man-in-the-Middle” architecture required to intercept binary wire protocols for MySQL, PostgreSQL, and MSSQL
- **Requirements Specification:** Finalized the requirement for protocol-agnostic handling and identity provider integration.
- **Feasibility Study:** Researched the technical viability of intercepting binary wire protocols without introducing significant latency overhead.

Deliverables

- Validated Software Requirements Specification (SRS).
- Initial architectural diagrams and system design documents.

15.1.2 Milestone 2: Technical Skill Acquisition & Research

Status: Completed

Duration: October 16 – November 15

Following the initial research phase, this milestone was dedicated to equipping the development team with the specialized technical skills and theoretical knowledge required to build a high-performance database proxy.

Objectives

- Acquire proficiency in the Go programming language.
- Understand low-level database communication protocols.
- Explore modern observability concepts and standards.

Key Focus Areas

- **Database Protocol Engineering:** Conducted in-depth study of the binary wire protocols for MySQL, PostgreSQL, and MongoDB. This involved reverse-engineering how databases handle handshakes, authentication sequences, and query transmission at the packet level.
- **Go Language Mastery:** Focused on mastering the Go programming language, with particular emphasis on its concurrency primitives—Goroutines and Channels—which are essential for managing multiple simultaneous network connections efficiently.
- **Observability Research:** Investigated the principles of system observability, including OpenTelemetry standards and eBPF concepts for potential kernel-level tracing in future iterations.

Deliverables

- Team proficiency in the project's technology stack.
- Internal documentation on protocol structures.
- Initial Go practice implementations and code samples.

15.1.3 Milestone 3: Functional Proof of Concept & Core Features Delivery

Status: Completed

Duration: November 16 – December 15

Originally scoped to focus solely on connection handling, this milestone was strategically expanded to deliver a complete Proof of Concept. The POC encompasses the initial web interface, basic security mechanisms, and internal token management capabilities.

Objectives

- Validate the full technology stack from the web frontend to database connectivity.
- Demonstrate that the proxy can handle traffic, mask sensitive data, and manage internal sessions.

Key Activities

- **Proxy Core Development:**
 - Implemented TCP listeners and connection pooling mechanisms for high-concurrency traffic.
 - Developed protocol parsers for MySQL, PostgreSQL, and MongoDB packets.
 - Built the core data forwarding logic with support for bidirectional communication.
 - Handled CLI errors and established graceful shutdown procedures.
- **Security Logic Implementation:**
 - Implemented Data Masking functionality using regex-based redaction for PII.
 - Built Token Refresh and Revocation mechanisms for session management.
 - Developed OAuth2-style authentication flows (login, refresh, logout).
 - Implemented AES encryption for secure storage of credentials.
- **User Interface Development:**
 - Developed the Initial Admin Dashboard Web UI to visualize system status.
 - Built components for managing database accounts and access control roles.
 - Integrated real-time session and query monitoring capabilities.

Deliverables

- A working end-to-end Proof of Concept demonstrating core functionality.
- Admin dashboard with system visibility and management capabilities.
- Functional data masking and token management subsystems.

15.1.4 Milestone 4: External IAM & Identity Integration

Status: Planned

Duration: January – February (Term 2)

This milestone bridges the internal token system established in Milestone 3 with external Identity Providers (IdP), enabling enterprise-grade Single Sign-On (SSO) capabilities.

Objectives

- Integrate external Identity Providers for SSO authentication.
- Replace internal authentication with federated identity management.
- Map external identity claims to internal Role-Based Access Control (RBAC) policies.

Key Activities

- **Third-Party Integration:** Connect the Admin Dashboard and Proxy to external providers (Google, Okta, Azure AD) to enable SSO workflows.
- **Authentication Flow:** Implement the OAuth2/OIDC handshake where the interface redirects to the IdP and receives identity tokens.
- **RBAC Mapping:** Map identity claims from third-party providers to user roles within the zGate policy engine.

Expected Deliverables

- SSO login capability via “Sign in with Google/Okta” or other configured IdPs.
- Removal of hardcoded or internal-only authentication from the POC.
- Comprehensive documentation for IdP configuration.

15.1.5 Milestone 5: Observability & Production Readiness

Status: Planned

Duration: March – April (Term 2)

This milestone transforms the initial dashboard into a comprehensive monitoring platform suitable for production deployment.

Objectives

- Implement comprehensive metrics visualization and monitoring.
- Establish audit logging for compliance requirements.
- Ensure system robustness under various failure conditions.

Key Activities

- **Metric Visualization:** Populate the dashboard with real-time graphs utilizing connection pooling metrics and system health indicators.
- **Audit Logging:** Structure session data and access events into a searchable audit trail for compliance and forensic purposes.
- **Error Handling Polish:** Implement graceful handling of all possible error conditions across the system, ensuring stability under load.

Expected Deliverables

- Production-ready Admin Dashboard with live metrics visualization.
- Searchable audit logs with filtering and export capabilities.
- Comprehensive error handling and graceful degradation.

15.1.6 Milestone 6: Final Validation, Benchmarking & Delivery

Status: Planned

Duration: May – June (Term 2)

The final milestone serves as a comprehensive quality assurance phase to ensure the system meets production standards and is fully documented.

Objectives

- Stress-test the system under realistic load conditions.
- Validate security robustness through penetration testing.
- Finalize all documentation and prepare for project handoff.

Key Activities

- **Performance Benchmarking:** Execute load testing to measure latency overhead introduced by the proxy. Optimize Go code based on profiling results.
- **Security Auditing:** Conduct internal penetration testing, including SQL injection and bypass attack attempts, to validate the robustness of the Policy Engine.
- **Final Documentation:** Complete the Administrator's Guide, Developer Handbook, and final academic report.

Expected Deliverables

- Final Source Code Release (v1.0).
- Comprehensive Final Project Report.
- Robust Demo Environment ready for final presentation.
- Complete documentation suite (installation guides, API documentation, user manuals).

15.2 Timeline Overview

Table 15.1 presents a consolidated view of all milestones with their respective timelines and completion status.

Table 15.1: Milestone Timeline Summary

MS	Milestone Name	Timeline	Status
1	Problem Definition & Market Validation	Sept 5 – Oct 15	✓ Completed
2	Technical Skill Acquisition & Research	Oct 16 – Nov 15	✓ Completed
3	Functional POC & Core Features	Nov 16 – Dec 15	✓ Completed
4	External IAM & Identity Integration	Jan – Feb	Planned
5	Observability & Production Readiness	Mar – Apr	Planned
6	Final Validation & Delivery	May – June	Planned

Chapter 16

Project Risk Management

This part covers:

- Protocol engineering complexity
- Performance and latency optimization
- Security implementation challenges
- Database compatibility issues
- Scope and schedule management
- Team skill development

RISK management is critical for engineering complex security systems. This chapter analyzes potential risks including protocol reverse engineering complexity, performance optimization challenges, TLS implementation security, database driver compatibility issues, scope creep, and team skill development. We establish mitigation strategies for each identified risk and define a dynamic monitoring framework to ensure successful delivery.



16.1 Introduction

The development of a high-performance, security-critical infrastructure component such as the zGate Database Proxy involves a multitude of complex challenges that extend beyond simple implementation details. Effective risk management is therefore not merely an ancillary activity but a core pillar of our engineering methodology. This chapter provides a comprehensive analysis of the potential risks identified throughout the project lifecycle, categorized into technical, operational, and schedule-based domains. By proactively identifying these risks, establishing their probability and impact, and formulating robust mitigation strategies, the development team aims to minimize uncertainty and ensure the successful delivery of a production-grade Zero Trust solution.

Our risk management framework is dynamic; it is designed to be revisited during each sprint retrospective to ensure that new risks are captured and that the status of existing risks is updated as mitigation strategies are executed.



16.2 Risk Assessment Methodology

To quantify and prioritize risks effectively, we have adopted a standard Risk Assessment Matrix approach. This methodology evaluates each identified risk along two primary dimensions:

1. **Likelihood:** The probability that the risk event will materialize.
2. **Impact:** The severity of the consequences should the risk operationalize.

Based on these two factors, we classify risks into three tiers:

- **Critical Risk:** Requires immediate attention and the development of a detailed contingency plan. These risks have the potential to halt project progress or cause total system failure.
- **High Risk:** Significant threats that require active monitoring and specific mitigation tasks to be integrated into the development backlog.
- **Medium/Low Risk:** Risks that are monitored periodically but do not strictly dictate immediate changes to the project roadmap.

Risk Description	Likelihood	Impact	Risk Level
Reverse Engineering Complexity (Undocumented Protocols)	High	High	Critical
Performance/Latency Overhead Exceeding Budget	Medium	High	High
Security Vulnerabilities in Proxy Implementation	Low	Critical	High
Third-Party SSO Integration Incompatibility	Medium	Medium	Medium
Scope Creep (Feature Overload)	High	Medium	High
Team Skill Acquisition Curve (Go/Systems Programming)	Low	High	Medium

Table 16.1: Detailed Project Risk Assessment Matrix

16.3 Technical Risks Analysis

Technical risks represent the most significant threat to the project's success, primarily due to the low-level nature of the work involving binary protocols and network systems.

16.3.1 Protocol Reverse Engineering and Parsing Complexity

Description: The core function of zGate relies on the ability to flawlessly intercept, parse, and reconstruct binary wire protocols (MySQL Client/Server Protocol and Microsoft SQL Server TDS). These protocols are often proprietary (in the case of TDS) or complexly versioned. There is a significant risk that undocumented behaviors, specific flag combinations, or version discrepancies could cause the proxy to misinterpret packet boundaries.

Potential Consequences:

- **Data Corruption:** Incorrectly reconstructing a packet could corrupt the SQL query or the result set returning to the client.
- **Connection Instability:** If the proxy fails to respond to a "Heartbeat" or "Ping" packet correctly due to parsing errors, the database client may terminate the connection.
- **Vendor Lock-in:** The implementation might inadvertently become too coupled to a specific version of MySQL or MSSQL, failing when the database server is patched or upgraded.

Mitigation Strategy: To mitigate this critical risk, the team is employing a "Defense in Depth" strategy for parsing:

- **Granular Unit Testing:** We implement unit tests for every packet type using captured bytes from real production traffic (using Wireshark) to ensure byte-for-byte accuracy.
- **Fuzz Testing:** We plan to use Go's fuzzing capabilities to send random garbage data to the parsers to ensure they handle errors gracefully without crashing the entire service.
- **Opaque Forwarding Fallback:** For parts of the protocol that are not strictly relevant to security policy (e.g., obscure handshake flags), the proxy is designed to forward bytes transparently rather than attempting to parse and re-serialize them, reducing the surface area for errors.

16.3.2 Performance and Latency Overhead

Description: By introducing a "Man-in-the-Middle" proxy between the application and the database, we inevitably introduce network latency. This includes the time taken for TCP socket reads, context switching in the Operating System, memory allocation for packet buffers, and the logic execution time of the Policy Engine.

Potential Consequences: If the added latency exceeds a threshold of approximately 10-20ms per query, the solution may be deemed unusable for high-frequency trading platforms or real-time applications, leading to rejection by stakeholders.

Mitigation Strategy:

- **Zero-Copy Networking:** We leverage Go's `io.Reader` and `io.Writer` interfaces to stream data efficiently, aiming to minimize user-space memory copying.
- **Concurrency Model:** Utilizing Go's lightweight goroutines ensures that handling thousands of concurrent connections does not incur the heavy memory and CPU context-switching penalties associated with traditional thread-per-connection models (like Java or C++ threads).
- **Profiling and Benchmarking:** Regular performance profiling using 'pprof' is conducted during the development cycle (not just at the end) to identify "hot paths" in the code and optimize them immediately.

16.3.3 TLS and Cryptographic Handshake Failures

Description: As a security proxy, zGate must terminate TLS connections from clients and establish new TLS connections to backend databases. This introduces a "Man-in-the-Middle" architecture where certificate validation, cipher suite negotiation, and protocol versions (TLS 1.2 vs 1.3) must be handled perfectly. The risk is that strict client drivers may reject the proxy's self-signed or internal CA-signed certificates, or that the proxy fails to negotiate a common cipher suite with legacy databases.

Potential Consequences:

- **Connectivity Blockers:** Modern drivers (e.g., latest JDBC) often default to "Strict" SSL modes. If the proxy's TLS implementation is flawed, these clients will refuse to connect entirely.
- **Security Degradation:** A misconfiguration could inadvertently downgrade connections to plaintext or weak ciphers, violating the very Zero Trust principles the project aims to uphold.

Mitigation Strategy:

- **Standard Library Usage:** We rely strictly on Go's 'crypto/tls' standard library rather than rolling custom crypto code, ensuring compliance with modern standards.
- **Configurable CA Trust:** The proxy is designed to allow easy injection of root CA certificates into client trust stores (via the CLI) to facilitate smooth "internal" certificate validation.

16.3.4 Database Driver Compatibility

Description: Not all database clients are created equal. While the 'mysql' CLI might behave one way, the MySQL driver for Python (PyMySQL), Java (Connector/J), or Go (go-sql-driver) may use slightly different subsets of the wire protocol or rely on specific "undocumented" behavior (e.g., specific expecting order of handshake packets).

Potential Consequences: The proxy might work perfectly for one set of tools but fail catastrophically for another, severely limiting its adoption.

Mitigation Strategy:

- **Broad Integration Testing:** Our test suite includes "Smoke Tests" that connect to the proxy using a variety of real-world drivers (Python, Node.js, Java) to verify cross-platform compatibility.

16.3.5 SQL Parsing and AST Limitations

Description: To enforce fine-grained policies (e.g., "User X cannot query Table Y"), zGate must parse incoming SQL strings into an Abstract Syntax Tree (AST). SQL is a complex, context-sensitive language with many proprietary extensions. There is a risk that our parser implementation may fail to understand complex nested queries, CTEs (Common Table Expressions), or stored procedure calls.

Potential Consequences:

- **False Positives/Negatives:** The Policy Engine might block a legitimate query because it poorly understands the syntax, or worse, allow a malicious query because it failed to detect a forbidden object access hidden in a subquery.

Mitigation Strategy:

- **Library Adoption:** Instead of writing a parser from scratch, we utilize battle-tested open-source SQL parsers (like 'pingcap/tidb/parser' for MySQL) that are already used in production databases. This significantly reduces the risk of parsing errors compared to a home-grown solution.

16.4 Operational and Project Management Risks

Beyond the technical challenges, the project faces operational risks related to scope, resources, and timeline.

16.4.1 Scope Creep and Feature Bloat

Description: The domain of Zero Trust security is vast. There is a constant temptation to add "just one more feature," such as AI-based anomaly detection, support for PostgreSQL/Oracle/MongoDB, or complex User Behavior Analytics (UBA).

Potential Consequences: Attempting to implement too many features simultaneously risks diluting the team's focus, resulting in a system where many features are "half-done" but nothing is production-ready. This constitutes a classic software engineering failure mode known as "Gold Plating."

Mitigation Strategy:

- **Strict Prioritization:** We strictly adhere to the MoSCoW method (Must have, Should have, Could have, Won't have). Milestone 3 (POC) and Milestone 4 (SSO) are "Must Haves."
- **Agile Iterations:** The two-week sprint cycle forces the team to deliver shippable increments. If a feature cannot be completed in a sprint, it is re-evaluated.
- **Supervisor Alignment:** Regular syncs with project supervisors ensure that the scope remains realistic and aligned with academic requirements rather than commercial ambitions.

16.4.2 Team Skill Gaps and Technology Adoption

Description: The technology stack for zGate includes Systems Programming (Go), Frontend Development (React/TypeScript), and Cryptography concepts. It is rare for all team members to be equally proficient in all these areas.

Potential Consequences: A "knowledge silo" effect may occur where only one person understands the core proxy code, creating a "Bus Factor" of 1. If that team member becomes unavailable, the project stalls.

Mitigation Strategy:

- **Code Reviews:** All pull requests require review by at least one other team member. This forces knowledge sharing and ensures that code is readable and maintainable by others.

- **Pair Programming:** For the most complex components (like the TDS protocol implementations), we utilize pair programming to solve problems collaboratively and level up skills in real-time.
- **Dedicated Research Phase:** Milestone 2 was explicitly dedicated to skill acquisition, ensuring the team had a foundation before writing production code.



16.5 Contingency Planning

Despite rigorous planning, unforeseen issues (“Unknown Unknowns”) can arise. To account for this, the roadmap for Term 2 includes a specific “stabilization buffer” of two weeks around Milestone 6.

If critical technical blockers arise that jeopardize the final delivery date—for example, if the MSSQL encryption handshake proves resistant to interception—the team has agreed on a “Scope Reduction Protocol.” Under this protocol, we would pivot to focusing exclusively on perfecting the MySQL implementation to a commercial standard, rather than delivering a flawed multi-database support. This ensures that the final deliverable, while potentially smaller in scope, remains high-quality and fully functional.

Chapter 17

Roadmap for Term 2

This part covers:

- External identity provider integration
- Enhanced observability features
- Advanced policy engine capabilities
- Performance optimization goals
- Security hardening measures
- Comprehensive testing strategy

TERM 2 represents the evolution from proof of concept to production-ready system. This chapter outlines the roadmap for integrating external identity providers with OAuth2 and OIDC support, implementing enhanced observability with real-time metrics and alerting, and developing advanced policy capabilities including time-based controls. We establish performance targets of sub-2ms latency and 1000+ concurrent connections, alongside security hardening through comprehensive testing and vulnerability scanning.

Term 2 represents the critical transition from a functional Proof of Concept to a production-ready database security solution. Building upon the foundational work completed in Term 1, the development team will focus on three primary areas: integrating enterprise-grade identity management, implementing comprehensive observability features, and conducting rigorous validation to ensure system reliability and security.

17.1 Remaining Features

The following features are scheduled for implementation during Term 2, organized by milestone and priority.

17.1.1 External Identity Provider Integration (Milestone 4)

The current POC utilizes an internal token-based authentication system. Term 2 will extend this to support federated identity management through external providers.

- **OAuth2/OIDC Implementation:** Integrate industry-standard authentication protocols to enable Single Sign-On (SSO) capabilities.
- **Identity Provider Support:** Configure connectors for major identity providers including:
 - Google Workspace
 - Microsoft Azure Active Directory
 - Okta
 - Generic SAML 2.0 providers
- **Claims-to-Role Mapping:** Develop a flexible mapping engine that translates IdP claims (groups, roles, attributes) to zGate’s internal RBAC policies.
- **Just-in-Time Provisioning:** Implement automatic user provisioning based on IdP attributes upon first login.

17.1.2 Enhanced Observability Features (Milestone 5)

Transform the initial dashboard into a comprehensive monitoring and compliance platform.

- **Real-Time Metrics Dashboard:**
 - Connection pool utilization graphs
 - Query throughput and latency percentiles
 - Active session monitoring
 - Database health indicators
- **Audit Logging System:**
 - Structured logging of all database access events

- Query content capture with optional redaction
- User activity timeline visualization
- Export capabilities for compliance reporting (CSV, JSON, SIEM integration)
- **Alerting Framework:**
 - Configurable threshold-based alerts
 - Anomaly detection for unusual access patterns
 - Integration with notification channels (email, Slack, webhooks)

17.1.3 Advanced Policy Engine Enhancements

Extend the current policy engine with more granular control mechanisms.

- **Time-Based Access Controls:** Implement policies that restrict database access to specific time windows.
- **Query Complexity Limits:** Add configurable limits on query complexity to prevent resource exhaustion.
- **Row-Level Security Policies:** Develop mechanisms to enforce row-level filtering based on user identity.
- **Dynamic Masking Rules:** Extend data masking to support context-aware redaction based on user roles.

17.2 Architecture Improvements

Based on lessons learned during POC development, the following architectural enhancements are planned.

17.2.1 Scalability Enhancements

- **Horizontal Scaling Support:** Implement stateless proxy instances that can be load-balanced for high availability.
- **Connection Multiplexing:** Optimize connection pooling to support higher concurrent user loads with fewer backend connections.
- **Caching Layer:** Introduce caching for policy decisions and session metadata to reduce latency.

17.2.2 Security Hardening

- **TLS Certificate Management:** Implement automated certificate rotation and enhanced certificate validation.
- **Secret Management Integration:** Support for external secret stores (HashiCorp Vault, AWS Secrets Manager).
- **Audit Trail Integrity:** Implement cryptographic signing of audit logs to ensure tamper-evidence.

17.2.3 Deployment Improvements

- **Container Orchestration:** Official Docker images and Kubernetes Helm charts for production deployment.
- **Configuration Management:** Migration to environment-based configuration with validation on startup.
- **Health Check Endpoints:** Enhanced liveness and readiness probes for orchestrator integration.



17.3 Performance Goals

Table 17.1 outlines the target performance metrics for the production release.

Table 17.1: Term 2 Performance Targets

Metric	Target	Measurement Method
Latency Overhead (p50)	< 2ms	Benchmarking with synthetic load
Latency Overhead (p99)	< 10ms	Benchmarking under stress
Concurrent Connections	≥ 1000	Load testing with connection ramp
Connection Setup Time	< 50ms	End-to-end connection timing
Memory Footprint	< 500MB (idle)	Resource monitoring under load
CPU Utilization	< 30% (normal load)	Profiling during benchmark

17.3.1 Optimization Strategies

To achieve these performance targets, the following optimization strategies will be employed:

- Profiling-Driven Optimization:** Use Go’s built-in profiler (pprof) to identify and eliminate bottlenecks.
- Memory Pool Management:** Implement buffer pooling to reduce garbage collection overhead.
- Protocol Parser Optimization:** Optimize hot paths in protocol parsing using zero-copy techniques where applicable.
- Goroutine Pool Management:** Implement worker pools to prevent goroutine explosion under high load.

17.4 Testing & Validation Plan

A comprehensive testing strategy will ensure system reliability and security before final delivery.

17.4.1 Automated Testing Framework

- **Unit Tests:** Target minimum 80% code coverage for core proxy logic and security modules.
- **Integration Tests:** End-to-end tests covering authentication flows, query processing, and data masking across all supported databases.
- **Regression Test Suite:** Automated test suite executed on every code commit via CI/CD pipeline.

17.4.2 Performance Validation

- **Load Testing:** Use tools such as `pgbench` (PostgreSQL), `mysqlslap` (MySQL), and custom scripts to simulate production workloads.
- **Stress Testing:** Evaluate system behavior under extreme conditions (connection floods, memory pressure).
- **Latency Profiling:** Measure and document latency distribution across different query types and database backends.

17.4.3 Security Validation

- **Penetration Testing:** Internal security audit focusing on:
 - SQL injection bypass attempts
 - Authentication bypass vectors
 - Token forging and replay attacks
 - Policy engine circumvention
- **Threat Modeling:** Document potential attack vectors and corresponding mitigations.
- **Dependency Audit:** Scan all dependencies for known vulnerabilities using automated tools.



17.4.4 User Acceptance Testing

- **Supervisor Demonstrations:** Weekly demonstrations of new features to project supervisors.
- **Documentation Review:** Validation that all user-facing documentation accurately reflects system behavior.
- **Demo Environment:** Prepare a stable demonstration environment for final presentation.

17.4.5 Validation Timeline

Table 17.2 presents the planned testing phases and their timelines.

Table 17.2: Testing & Validation Timeline

Testing Phase	Timeline	Focus Area
Unit & Integration Testing	Ongoing (all sprints)	Continuous quality assurance
Performance Benchmarking	May 1–15	Latency and throughput validation
Security Audit	May 15–31	Penetration testing and hardening
User Acceptance Testing	June 1–10	Feature validation and feedback
Final Documentation Review	June 10–15	Documentation completeness
Demo Environment Preparation	June 15–20	Final presentation preparation



17.5 Term 2 Development Timeline

Figure 17.1 illustrates the project timeline for Term 2, depicting the progression through the remaining three milestones from February to June 2026.

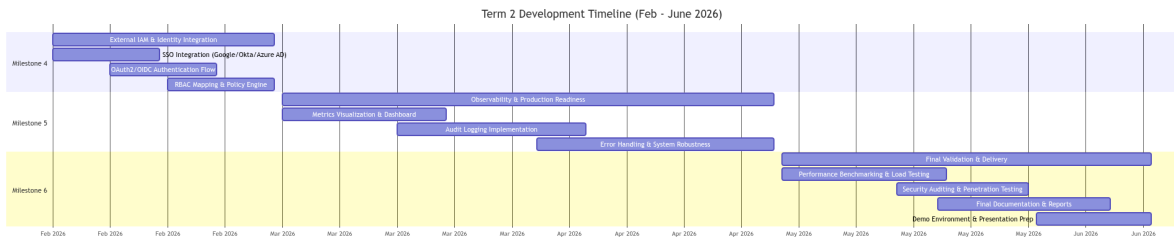


Figure 17.1: Term 2 Project Timeline (February – June 2026)

Chapter 18

Team Contribution

This part covers:

- Backend and proxy development
- Security engineering
- Frontend development
- CLI development
- Research and architecture

SUCCESSFUL project delivery requires coordinated effort across technical domains. This chapter documents individual team member contributions spanning backend protocol handler development, security engineering for authentication and authorization systems, frontend development of the Next.js administration dashboard, CLI implementation, and comprehensive research. We detail how responsibilities were distributed across the seven-member team to leverage individual strengths while maintaining collective ownership.

This chapter documents the individual contributions of each team member throughout the development of the zGate Zero-Trust Database Access Gateway. The project required expertise across multiple domains including backend systems programming, frontend development, security engineering, and research.



18.1 Overview of Contribution Approach

The zGate project was developed by a seven-member team, with responsibilities distributed across the following primary domains:

- **Backend & Proxy Development:** Core proxy logic, protocol handlers, connection management
- **Security Engineering:** Authentication, authorization, encryption, and access control
- **Frontend Development:** Web-based administration dashboard (Next.js/React)
- **CLI Development:** Command-line interface for end-user database access
- **Research & Architecture:** Zero-Trust research, competitive analysis, system design
- **Documentation:** Technical documentation, code documentation, presentation materials

Table 18.1 provides a high-level summary of each member’s primary focus areas.

Table 18.1: Team Contribution Overview

Team Member	Primary Focus Areas
Moustafa Hashem	Team Lead, Core Architecture, Authentication, CLI
Hana Shamel	PostgreSQL Proxy, Data Masking, CLI, Research
Kareem Ehab	MySQL Proxy, Interceptors, Audit Logging, Security
Michael George	Storage Infrastructure, Encryption, Identity Management
Mayar Walid	MongoDB Proxy, WebUI Development, API Integration
Karen Maurice	Zero-Trust Research, WebUI Development, Theming
Rodina Mohamed	MSSQL Protocol, TLS Security, Testing

18.2 Individual Contributions

18.2.1 Moustafa Hashem – Team Lead

As Team Lead, Moustafa directed the full project lifecycle, including sprint planning, task assignment, and cross-team communication. His technical contributions spanned the core system architecture and critical security components.

Core System Architecture

- Designed the Zero-Trust database proxy architecture, enabling secure identity-based access control and traffic interception.
- Built the core networking foundation: dynamic listeners, connection forwarding logic, and high-performance connection pooling.
- Developed the unified Handler Interface, ensuring protocol consistency across MySQL, PostgreSQL, and MSSQL implementations.

Dynamic Connection & Identity Management

- Engineered the dynamic proxy system that creates isolated per-session ports (eliminating static listener vulnerabilities).
- Designed multiple identity-management models:
 - Temporary database users (ephemeral credentials)
 - Shared credential pools
 - User-specific persistent accounts
- Implemented both Local Mode tunneling and Remote Mode direct access patterns.

Security, Authentication & RBAC

- Built the complete JWT authentication system, supporting login, token refresh, revocation, and secure session tracking.
- Designed the RBAC policy engine, enabling granular permissions at the database and table levels.

CLI & Admin Tools

- Re-engineered the CLI architecture to improve security, consistency, and maintainability.
- Developed admin tools for user provisioning, credential management, and session visibility.

System Reliability

- Engineered the core token lifecycle logic, ensuring consistent expiration handling and safe session cleanup.
- Designed system reliability mechanisms for stable startup, shutdown, and error recovery across components.

18.2.2 Hana Shamel

Hana contributed to proxy development, CLI implementation, and conducted extensive research on Zero-Trust security models.

Proxy & Backend Development

- Developed the initial PostgreSQL proxy prototype, establishing early connectivity and packet-flow behavior.
- Implemented the data masking engine used to redact sensitive information from database responses.
- Contributed to backend architecture and low-level packet processing design.

CLI Development

- Built essential CLI authentication and database command workflows.
- Integrated OS-native secure credential storage using:
 - macOS Keychain
 - Windows Credential Manager
 - Linux Secret Service

Security & Research

- Conducted research on Zero-Trust models, database attack vectors, and secure-access frameworks.
- Investigated observability and monitoring tools to support long-term system stability.
- Studied industry attack statistics and market needs to guide feature development.

Documentation

- Wrote documentation for SQL masking logic, CLI behavior, and proxy components.
- Contributed to PowerPoint presentation materials and project documentation structure.

18.2.3 Kareem Ehab

Kareem led the MySQL proxy implementation and developed the security interceptor framework central to the system's query protection capabilities.

MySQL Proxy & Backend Development

- Implemented the full MySQL proxy logic, including authentication handshake, packet forwarding, and controlled shutdown.
- Built the extensible interceptor framework used for data masking, SQL safety checks, and logging.

Security Enforcement

- Implemented the Query Safety Enforcement Engine, blocking malicious or destructive SQL statements in real time.
- Integrated TLS/SSL certificate verification to secure proxy connections.
- Added secure temporary credential logic to handle session-specific identities.

Audit Logging & Observability

- Developed the Comprehensive Audit Logging System, capturing:
 - Query history with user attribution
 - Authentication events
 - Proxy metrics
 - Session activity
- Added internal monitoring endpoints to enhance observability and debugging capabilities.

Research & Documentation

- Conducted research on database threats, competitor proxy architectures, and security standards.
- Contributed technical flow explanations and feature breakdowns for presentation materials.

18.2.4 Michael George

Michael focused on storage infrastructure, encryption systems, and identity management backend components.

Proxy & Early Prototype Development

- Developed the early Python-based proxy prototypes to validate networking concepts.
- Implemented the initial PostgreSQL proxy prototype before transitioning to the Go-based design.

Storage & Encryption Infrastructure

- Migrated system configuration from YAML to SQLite, enabling persistent structured storage for users, roles, tokens, and database configurations.
- Implemented AES-256-GCM encryption for credentials, admin passwords, and sensitive configurations.

Identity & Access Management

- Implemented backend logic for:
 - User account creation and update
 - Password hashing and verification
 - Role assignment and permission control
- Added full CRUD operations for role management.

Documentation

- Wrote documentation for storage architecture, encryption system, and user/role backend features.

18.2.5 Mayar Walid

Mayar contributed to MongoDB proxy development and led significant portions of the WebUI development, particularly monitoring and dashboard interfaces.

Proxy & Backend Development

- Built the initial MongoDB proxy prototype supporting early database routing tests.
- Implemented backend endpoints for Admin Account Management (CRUD operations).

Web UI Development (Next.js & React)

- **Shared & Personal Database Accounts:** Developed UI interfaces for shared credential pools and user-specific account management with validation, search filters, and interactive modals.
- **Monitoring & Query Interfaces:**
 - Session Monitoring Dashboard with real-time session visibility
 - Query History Viewer with dynamic updates, filters, and search tools
- **Dashboards:** Developed the Overview Dashboard displaying live system metrics and enhanced session, query, and database pages with improved functionality.

API Integration & State Management

- Integrated UI components with backend APIs for authentication, roles, permissions, databases, sessions, and admin operations.
- Ensured synchronized application state across the UI.

Documentation

- Contributed UI visuals, workflow explanations, and dashboard breakdowns for presentation slides.

18.2.6 Karen Maurice

Karen conducted Zero-Trust research and contributed extensively to WebUI development, including theming systems and access control interfaces.

Zero-Trust Research & Prototype Engineering

- Conducted research into Zero-Trust architecture, database security, and access-control models.
- Built the Zero-Trust gateway prototype using Go and Docker.
- Designed the zGate Secure Pipeline Architecture, featuring:
 - mTLS (mutual TLS) security layer
 - JSONL protocol layer
 - Security processing pipeline
 - Strategy adapter layer for database engines

Web UI Development (Next.js & React)

- **User, Role & Database Management:** Developed UI systems for user, role, database, and access-control management with permission indicators and validation layers.
- **UI/UX Systems:**
 - Implemented light/dark mode theming
 - Built advanced UI elements including dialogs, permission selectors, detailed views, and filters
- **Monitoring Interfaces:** Developed Session Monitoring Dashboard and Query History Viewer with filtering and search capabilities.

API Integration & State Management

- Integrated Web UI with backend APIs for authentication, roles, databases, and sessions.
- Implemented robust state management and error-handling flows.

Documentation

- Contributed to architectural documentation, UI flow diagrams, and frontend integration notes.

18.2.7 Rodina Mohamed

Rodina focused on MSSQL protocol development, security infrastructure, and testing validation.

MSSQL Protocol Development

- Implemented the initial MSSQL proxy framework, including support for the TDS (Tabular Data Stream) protocol.
- Developed foundational packet-handling logic for Microsoft SQL Server communication.

Security & Encryption Infrastructure

- Contributed to the design of the TLS/SSL encryption layer, securing proxy communication.
- Participated in building engine-agnostic secure protocol abstractions.

Research & Architectural Input

- Performed research into:
 - Zero-Trust architecture principles
 - Market security needs and requirements
 - Relevant software design patterns
- Provided architectural feedback improving system modularity and scalability.

Testing & Validation

- Built test applications to verify MSSQL proxy behavior.
- Performed multi-tool validation for stability and correctness.

Documentation

- Documented MSSQL protocol behavior, TDS packet flow, security considerations, and integration architecture for MSSQL components.

Chapter 19

Conclusion

This part covers:

- Term 1 achievements and validation
- Core infrastructure delivered
- Security mechanisms implemented
- Term 2 production roadmap

THE zGate project represents a fundamental advancement in database security architecture. This chapter reflects on Term 1 achievements including the MySQL proxy implementation, JWT authentication system, and role-based access control engine that successfully validate Zero Trust database access. We articulate the production roadmap for Term 2 focusing on external identity integration, observability enhancements, and enterprise deployment readiness.

19.1 Restated Purpose

This project was undertaken to address a critical gap in enterprise database security: the absence of a comprehensive Zero Trust solution that eliminates credential exposure while maintaining operational efficiency. Traditional approaches—VPNs, bastion hosts, and privileged access management systems—continue to distribute database credentials to end users, creating attack vectors that modern security frameworks cannot adequately mitigate.

zGate was conceived as a Zero Trust Database Access Gateway that places security at the point of database interaction itself. By intercepting all database traffic at the wire protocol level, the system enforces identity-based access control, generates session-specific credentials, and provides complete auditability—capabilities that peripheral security solutions cannot achieve.

19.2 Summary of Achievements

During Term 1, the team successfully delivered a functional Proof of Concept that validates the core architectural decisions and demonstrates end-to-end Zero Trust database access.

Core Infrastructure Delivered:

- A high-performance Go-based gateway server implementing the MySQL wire protocol with transparent credential injection and connection pooling
- A complete authentication system using JWT tokens with short-lived access tokens, refresh token rotation, and server-side revocation
- A role-based access control engine supporting hierarchical permissions at the database level
- A composable interceptor pipeline enabling data masking, query safety enforcement, and comprehensive audit logging

User-Facing Components:

- A cross-platform CLI with secure keyring integration, enabling developers to access databases without handling production credentials
- A full-featured web administration dashboard for managing users, roles, databases, sessions, and audit logs

Security Mechanisms Validated:

- Users successfully authenticate and connect to databases without ever seeing or handling database credentials
- The safety interceptor blocks dangerous SQL operations (DELETE without WHERE, DDL statements)
- Data masking redacts sensitive information (emails, phone numbers, credit cards) from query results
- All database operations are logged with complete user attribution for audit and compliance purposes

19.3 Importance & Contribution

The zGate project contributes to the evolving landscape of database security in several meaningful ways.

Practical Application of Zero Trust Principles: While Zero Trust has been widely adopted for network and application access, its application to database security remains nascent. zGate demonstrates that the same principles—never trust, always verify, assume breach—can be practically implemented at the database protocol level without sacrificing developer productivity.

Elimination of Credential Sprawl: The project addresses one of the most persistent security problems in enterprise environments: the proliferation of database credentials across configuration files, environment variables, and documentation. By abstracting credentials behind an identity-based gateway, zGate removes this attack vector entirely.

Query-Level Security Enforcement: Unlike solutions that operate at the network or session level, zGate’s protocol-aware architecture enables security enforcement at the query level—blocking specific operations, masking sensitive data in results, and logging every database interaction with full context.

Open Architecture: The modular design, with clearly defined interfaces for protocol handlers and interceptors, provides a foundation for extending Zero Trust principles to additional database engines and implementing advanced security features.

19.4 Transition to Term 2

With the core infrastructure validated, Term 2 will focus on three primary objectives:

External Identity Integration: Bridging the internal authentication system with enterprise identity providers (OAuth2/OIDC) to enable Single Sign-On and leverage existing organizational identity management.

Production Readiness: Implementing comprehensive observability features—real-time metrics, searchable audit logs, and alerting—alongside performance optimization and security hardening through penetration testing.

Validation and Documentation: Conducting rigorous performance benchmarking, security auditing, and completing all technical documentation required for production deployment.

The foundation established in Term 1 positions zGate to become a complete, deployable solution that organizations can adopt to fundamentally improve their database security posture while reducing operational overhead.

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